Who Runs Short in a Drought? Water Rights and Canal Delivery in Arizona*

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Abstract

In Arizona, water rights and irrigation district rules determine how much water each farm may use, while the water reaches farms through canals shared with other farms. Along a canal, water reaches upstream farms first, so most of what they use is no longer available to farms farther down. I use recent droughts to ask whether farms with the same rights share water shortages equally. I rebuild Arizona's canal network from public maps, link 54,946 farm fields to it, and track their crops each year from 2016 to 2023. In drought years, fields with more farmed land upstream show larger declines in a satellite yield proxy and crop moisture. Farmed land far downstream, which cannot take water first, shows no such effect. Outside drought, the difference is near zero. The gap is clearest in junior-rights districts, which lose surface water first. In these districts, the gap is smaller where fields have more access to groundwater. Within these districts, the gap in water use is larger in months when crops need more water. Under an assumed conversion from the satellite proxy, the gap is equivalent to about 1.4–2.7 percent of crop revenue, or 14–28 percent of net returns in representative farm budgets. An earlier, drier drought, when major deliveries largely held, shows no such gap. In 2022, farms in the Central Arizona Project's Agricultural Settlement Pool lost about two-thirds of their project water, even after replacement water. Fields still farming used less water in the late-season months, and cotton fields lost greenness as bolls formed, when more land upstream than downstream stayed in crop. In 2023, when the pool received no water, fields near the end of the canal with little access to groundwater were more likely to stop farming. Farms with the same rights do not share the costs of delivery cuts equally.

1 Introduction

Laws and rules often decide how much of a shared resource each user may take. In many cases, the resource also reaches those users one after another. When the resource is plentiful, the order

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in which users are served makes little difference. When it is scarce, the order can matter. If each user's use could be measured and controlled exactly, the order would not matter even then, because the rules alone would decide who gets what. In practice, use is rarely measured that precisely, so the users the resource reaches first could affect how much is left for everyone after them.

Irrigation water is a clear example of this problem. Water rights and the rules of irrigation districts, the local organizations that run the canals, determine how much water each farm is entitled to. But when there is not enough water for everyone, the amount a farm actually receives can be different from what it is entitled to. Water reaches farms through branching canals shared with other farms, and along a canal, delivery is sequential: water passes the upstream farms before it reaches the farms below. Such a system is hard to monitor and to operate exactly, so farms upstream can use more water than the rules intend. A district may also serve the upstream farms first when water is short. Any extra water they use is water the farms downstream do not receive. In ordinary years these shortfalls are small, and reservoirs, wells, and adjustments to delivery timing cover them. When a drought lasts several years, these buffers run low. Every field in crop upstream then draws on the same shrinking flow, and a farm's crop can start to depend on how much of the land upstream of it is being farmed, even though the farm's own land and rights have not changed.[1]

Suppose ten farms on one canal are each entitled to a tenth of its water, and a long drought cuts the canal's flow by 30 percent. On paper, each farm loses 30 percent. In practice, the water reaches the upstream farms first. If each of them consumes even a little more than its reduced share, that water is no longer in the canal for the farms below. These small amounts add up along the canal, so the farms at the end lose much more than 30 percent. If some of the upstream farms leave their fields fallow that year, more water stays in the canal for the farms at the end. Even where water rights are well developed, the order in which water reaches farms, and not only their rights, can decide who faces a shortfall.

In the United States, this problem has received little attention. The only direct test, in small ditches in Colorado, found no difference between farms at the head and at the end of a canal (Cody 2018a). This paper asks: when farms hold formal water rights, does a long drought make a farm's crop more dependent on how much of the land upstream of it is being farmed? Using data from Arizona, I find that it does. Farms with the same rights therefore do not share a shortage equally. In drought years, farms with more farmed land upstream of them have weaker crops. The effect follows the direction of water flow along the canals, not simple distance on the map. Its timing follows water deliveries, not the weather.

Arizona is a good place to study this question, for two reasons. First, its water rules are well developed. Water rights and contracts determine which users lose water first when the Colorado River runs short, and irrigation districts deliver the water to farms through their own canals. Second, Arizona has faced a long and severe drought. The years 2000–2021 were the driest 22-year period in southwestern North America since at least 800 CE (Williams, Cook, and Smerdon 2022). In 2022 the shortage reached Arizona's farms directly. That year brought the first shortage ever declared on the lower Colorado River, known as Tier 1 (Reclamation 2021). The Central Arizona

[1] Appendix A25 documents delivery constraints outside the Central Arizona Project: historical lower-end losses and contemporary gate losses in the Gila Valley, capacity pressure in senior Yuma districts, and downstream storage operations in Roosevelt Irrigation District. Tables A28–A30 give source citations, dates, sample coverage, and interpretive limits.

Project (CAP) sets aside a low-priority block of its water, the Agricultural Settlement Pool, for central Arizona's irrigation districts. In 2022 these districts lost about two-thirds of their CAP water, even after replacement water (ADWR and CAP 2021). In 2023 the pool received no water at all (CAP 2023). These cuts were made at the source and followed the rules, so the users with the weakest claims lost water first.

The simplest approach would compare farms at the top of a canal with farms at its end. But such a comparison can be misleading. Farms at the end often get less water every year, and over time they adjust to it. They may grow different crops or rely more on wells, and the weakest fields may have stopped farming long ago. And in years with enough water, the farms at the end may not look worse at all. So instead of comparing different farms, I follow the same fields over time. I ask whether a field's crop does worse in drought years when more of the land upstream of it is farmed that year.

Measuring the farmed land upstream of a field requires two things: a map of the whole canal network and a yearly record of which fields are in crop. Distance alone is not enough. A main canal splits into smaller canals, and those split again, so the water reaching a field has passed only the fields on the canals that feed its own branch. Two fields at the same distance from the top of the canal can therefore have very different amounts of land above them. How much of that land is in crop also changes from year to year, while the distance stays the same.

I build both for Arizona. Using public maps of canals and land elevation, I reconstruct the state's network of canals that deliver surface water, including the direction in which water flows through it. I then link 54,946 farm fields (called plots in the rest of the paper) to this network. For each field, the network shows which fields take water before it (upstream) and which take water after it (downstream). Yearly satellite crop maps from the U.S. Department of Agriculture show which of these fields are in crop and which are fallow in each year from 2016 to 2023. Together, they show how much land upstream of each field is farmed in each year. To measure how well crops do, I also use satellite data. The main outcome is the Green Chlorophyll Vegetation Index (GCVI), which measures how green a crop is and is used here as a proxy for yield. Crop moisture and evapotranspiration (ET), which measures the water a crop uses, are secondary outcomes. Fields that grow no crop in a year are left out of these crop measures for that year. I also study whether fields are left fallow.

The regressions compare each field with itself across years, and with fields growing the same crop in the same year. A field's water rights do not change from year to year, so this comparison holds its rights fixed. About 72 percent of fields do not match any surface-water right in the state register, and most of them likely have no right of their own. Inside irrigation districts, farms get water under the district's rules, so I also study these districts separately. I count a year as a drought year for a field when the area around the fields upstream of it has been unusually dry over the previous two years, based on a standard drought index that compares rainfall with how much water the air could evaporate (SPEI-24). The regressions also control for local drought and for the share of nearby fields left fallow.

The main concern is that farmed land upstream may simply reflect local conditions. Nearby fields share the same weather and other local conditions, so they tend to be farmed or left fallow together. The same conditions also affect a field's own crop. I address this in two ways. First, I leave out nearby upstream fields and keep only distant ones that are still connected to the field through the canals. Distant fields share fewer local conditions with the field, but they still take water before it. Second, I also measure the farmed land downstream of the field and include it in

the same regression. Downstream fields sit on the same canals and share the same weather and water supply, but they cannot take water before it reaches the field. If the effect works through water delivery, drought should make upstream land matter more, but not downstream land. If it works through something the whole canal shares, both should change together.

I report six main findings. First, in drought years, fields with more farmed land upstream have weaker crops. With both GCVI and the number of farmed fields upstream measured in logs, the drought-year change in this relationship is −0.015 when all upstream fields are used and −0.035 when only fields more than 10 km away in a straight line are used. In other words, in drought years, a field with 10 percent more farmed fields upstream has a GCVI about 0.15 to 0.35 percent lower, compared with the same difference in other years. The estimate is negative at every distance cutoff I try and largest at the widest one. If shared local conditions drove the result, leaving out nearby fields should weaken it. Instead, it becomes larger. A similar estimate appears within the largest canal systems alone. Crop moisture and ET show the same pattern, but GCVI is the most consistent across specifications (Section 7.1).

Second, for GCVI, distant farmed land upstream of a field matters in drought years, but distant farmed land downstream does not. When both enter the same regression, the upstream estimate beyond 10 km stays negative, while the downstream estimate is close to zero. The difference between them is −0.032.[2] Farmed land just downstream of a field also has a negative estimate, but it fades as nearby fields are left out, while the distant upstream estimate remains. This pattern is hard to explain with a shock that hits both sides of a field alike, such as local weather, because such a shock would affect upstream and downstream land in the same way (Section 7.1).

Third, the effect is clearest in the ten junior-rights districts, whose surface water is cut first in a shortage. Here I measure farming upstream as the share of the farmland upstream of a field that is in crop that year. Inside these districts it has two parts. The first part is long-standing. Some fields have always had more farmed land upstream of them, because of the canal branch they sit on and how much land above them can be farmed, not because of their distance from the top of the canal. In drought years these fields do worse: a field whose share of farmed land upstream in 2016–2019 was one standard deviation higher has GCVI 0.034 log points lower. The second part changes from year to year. In a drought year, when more land than usual is farmed upstream of a field, its crop is weaker: GCVI is 0.014 log points lower for every ten percentage points of extra upstream farmed land. Outside drought, these yearly changes make no difference to the crop. This yearly part stays negative when I also control for each field's recorded water rights and wells. Section 7.2 shows that both parts are robust.

These gaps are economically large. In 2021–22, the crop difference between a field with a lot of farmed land upstream (the 75th percentile) and one with little (the 25th percentile) is worth about 1.4–2.7 percent of crop revenue, under an assumed link between GCVI and yield. That is about $21–42 per acre of cotton and $30–59 per acre of alfalfa, the two largest crops (Appendix A13). This gap comes on top of the drought's effect on all fields, which the estimates hold fixed. It is also large compared with farm profits. In representative University of Arizona budgets, returns over total costs are about a tenth of gross income, so a gap of this size equals roughly 14–28 percent of

[2]The far-cutoff upstream-minus-downstream interaction has $p = 0.026$ under two-way clustering by field and crop-year, $p = 0.075$ under network-component clustering, and $p = 0.106$ under canal-block clustering. The contrast concerns changes in slopes between drought and other years; total drought-year slopes and inference for the contrast are in Appendix A20 (Table A24).

a field's net return.

Fourth, the timing follows water deliveries, not the weather. Compared with 2016, the yearly estimates of this relationship are close to zero in 2017–2019, negative in 2020–2022, and close to zero again in 2023. The negative estimates begin in 2020, before the drought index itself signals a drought. I also extend the data back to 2008 to look at an earlier drought. That drought was more severe by the same drought index, yet it shows no similar negative estimate. In those years, major water deliveries largely held: CAP deliveries to the Agricultural Settlement Pool stayed close to schedule over 2008–2014, and the available Salt River Project schedules show no cut. Because the earlier years use an older set of field boundaries, this is a historical comparison rather than an experiment (Section 7.2.3). Timing within the year points the same way. In drought years, upstream water use measured early in the year, in January–April, goes with lower water use on the field later, in July–October, although this estimate is less precise.

Fifth, in the junior-rights districts, fields with better access to groundwater are hurt less by farmed land upstream. I measure this access by the number of wells drilled before 2016 on the groundwater right that covers a field. Few new large wells were drilled after that, so this number shows the access each field had before the shortage. In drought years, for a right with one standard deviation more wells, the GCVI loss is smaller by 0.022 log points for every ten percentage points of extra upstream farmed land. Most of this comparison comes from the three districts in Pinal County, where groundwater rights can be matched to fields.[3]

Sixth, when CAP water was cut, farms left much more land fallow, and the fields still farmed showed the same pattern as in drought years. Four of the junior districts receive CAP water. Between 2019 and 2022, the share of their fields left fallow rose 13 to 19 percentage points more than in other irrigation districts. In 2022, CAP water was cut sharply, but the canals still carried some water. That year, fields still farmed used less water when more land than usual was farmed upstream of them, net of the same change downstream of them. The loss came in the late-season months, when deliveries were lowest and crops needed the most water, and it was larger for fields with crops at peak demand. Cotton fields lost greenness in the weeks when bolls form. In 2023 the pool received no water at all. Across the whole state, the yearly estimate for 2023 is close to zero. Inside the CAP districts the 2023 evidence is mixed. Fields far down the canals still show the penalty, but across all fields it is close to zero. Two of the four districts still moved pumped groundwater through their canals that year, so their canals were not empty.[4]

The fallowing data also show which fields stopped farming. In the three CAP districts in Pinal County, CAP water for farms ran out in 2023. That year, fields near the end of the canal were

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[3]Under Arizona's 1980 Groundwater Management Act, farms in this area may irrigate with groundwater only on land that already holds a groundwater right. In the Pinal Active Management Area, the state well registry lists 1,260 large wells installed before 2016 and one installed in 2016–2018. The district wild-bootstrap p-value is 0.10.

[4]The fallowing range comes from two ways of assigning fields to districts. Differences between the districts before the cut, and the choice of comparison districts, make these changes descriptive. The water table in districts that lost CAP water also fell, by about 8 feet more than elsewhere by 2023 ($p = 0.008$). The estimates for fields still farmed come from the crop water demand design of Section 7.3.2. That design measures the crop water demand of the land linked to a field through its district's canals, and it leaves out land within a distance cut set from district size. In 2022, compared with 2017–2019, ten points more demand upstream lowered the ET of a field at peak crop demand by 2.7 percent more than the same demand downstream (95 percent interval −3.9 to −1.5). For cotton, the same contrast lowered ET in July–September by 2.2 percent (interval −4.0 to −0.4) and greenness over the boll window by 3.7 percent (interval −8.1 to +0.8); over 2020–2022 the greenness contrast is 4.3 percent (interval −7.9 to −0.6) (Appendix Table A36). The cotton intervals use the t distribution with six or seven degrees of freedom, one less than the number of crop-year clusters. Measured with the logged count of farmed fields upstream, the 2022 gradient is not reliably steeper (Appendix A28).

more likely to stop farming, but only where groundwater was hard to get: the field's groundwater right had few wells and the water table was deep. There, a field one standard deviation farther down the canal was 6.2 percentage points more likely to be left fallow. In 2020 and 2021, before CAP water ran out, position along the canal made no such difference. A field's own well did not explain who stopped farming (Section 7.3).[5]

These findings have a simple interpretation. When there is enough water, any extra water used upstream is small, and reservoirs and wells cover it. When water is scarce, the same extra use leaves less water for the fields below, and their crops suffer. Water rights and contracts determine how much water each district receives at the source. But inside the district, how much land is farmed upstream along the canal still affects which fields face the shortage. The findings do not show who causes this. A district may deliver water in order down the canal, so fields at the end are cut first. Or farms upstream may use more water than the rules intend. Both would produce the same pattern, and either way the order in which water reaches farms affects who faces the shortage. The estimates also have limits. I observe crops from satellites, not the water delivered to each field. The estimates show how the link between upstream farming and a field's crop changes in drought. They do not show how much one more farmed acre upstream would hurt a field below.

The paper contributes to the literature on common-pool resources with sequential access, in which irrigation is the main example (Ostrom 1990; Ostrom and Gardner 1993). Theory and field studies show that farms at the head of a canal have an advantage over farms at the end, and that enforcement and local institutions affect how large this advantage is (Ray and Williams 2002; Akram and Mendelsohn 2017; Jacoby and Mansuri 2020; Jacoby, Mansuri, and Fatima 2021; Das and Dutta 2026). Most of this evidence comes from developing countries, where water rules are often weakly enforced. In the western United States, Cody (2018a) finds no clear difference in crop growth between the head and the end of 60 small ditches in Colorado's San Luis Valley. These ditches do not branch, and the farms on them take turns using the water. It has therefore been unclear whether the head-end advantage matters where water rights are well developed. I find that it does when a drought lasts several years. To my knowledge, this is the first study in the United States to test for this advantage field by field across the canal networks of a whole state.

The paper also contributes a way to measure who is upstream and who is downstream of each farm field. Studies of irrigation rarely observe which fields take water before which others. They usually compare farms near the head of a canal with farms near its end, which misses how canals branch. I build the measure from public data available across the United States: maps of canals, land elevation, and USDA field boundaries. The method finds the direction of flow in each canal from the land elevation and links each field to the canal that most likely serves it. It then gives, for every field, the set of fields upstream and downstream of it. The flow directions it finds agree with the flow records of the national canal map for 95.5 percent of canal segments. Because the data are public, the method can be applied to other irrigated regions. Combined with yearly crop maps, it shows how much land upstream of each field is farmed in each year.

Finally, the paper adds to recent work on how water is allocated among farms. Rafey (2023) measures the value of reallocating water through markets in Australia's Murray–Darling Basin, and Rafey (2026) measures water misallocation in California. Harris (2024) studies how Arizona

[5]The sample is fields farmed in every earlier year. Harquahala is left out because it falls under different groundwater rules. The 2023 estimate has a permutation p-value of 0.004. One standard deviation more wells on the groundwater right lowers fallowing in 2022 or 2023 by 5.7 percentage points ($p = 0.019$).

farmers changed their crops and planted acreage when CAP deliveries were cut. These studies look at how water is divided among rights holders, or at how farms respond to a cut at the source. This paper looks inside a district's canals. It asks which fields face a shortage once the water has reached the district.

The problem studied here is not limited to water. Wherever rules decide each user's share of a resource that reaches its users one after another, reading the rules is not enough to know who faces a shortfall. Rules that seem to work while the resource is plentiful may stop deciding the outcome once it becomes scarce, so they need to be judged by how they work in a shortage. Natural gas pipelines and electricity grids that cut supply during shortages are possible examples. The method used here could be applied to them as well, because it maps which users are connected to each location and tracks how their use changes from year to year.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents a simple model of how water is shared along a canal. Section 4 describes the data, Section 5 explains how I build the canal network, and Section 6 sets out the empirical design. Section 7 reports the results, and Section 8 concludes.

2 Literature Review

2.1 Sequential Delivery and Long-Run Spatial Equilibrium

Because canal water diverted upstream is unavailable downstream, and monitoring extraction along kilometers of open canal is costly and incomplete, canal systems combine rivalry with costly exclusion, the defining features of a common-pool resource. The literature anchored by Ostrom (1990) studies how communities govern such resources. Ostrom and Gardner (1993) and Ray and Williams (2002) make sequential head-end/tail-end asymmetry, unequal access for users near the source and users at the end of the canal, central to irrigation governance. Their core insight is directly relevant here: allocation rules operate within a physical system in which the order of access can shape realized incidence, that is, who actually absorbs a shortfall. The distinction is the one between legal and economic property rights (Barzel 1989): an entitlement is what the rule assigns, realized access is what enforcement delivers, and in a sequentially delivered resource the gap between them has a spatial order. In western water that order is layered on prior appropriation, whose emergence and enforcement Leonard and Libecap (2019) and Libecap (2011) study; this paper asks what the priority layer leaves ungoverned below the headgate.

Field evidence establishes both the spatial gradient and the importance of implementation. Wade (1988) shows that tail-end scarcity induces stronger collective organization in South India. Manero et al. (2019) finds that position along secondary canals predicts yields and income in Tanzania. Akram and Mendelsohn (2017) pairs GPS position with farm-level flow measurements in Pakistan and finds that head farms receive more water than tail farms. Jacoby and Mansuri (2020) provides the most direct institutional comparison, contrasting the proportional allocation on paper with measured water availability across hundreds of channels in Pakistan's Indus Basin. Political influence tilts both water and land values toward head-end users, demonstrating that formal formulas and realized delivery can diverge inside an administered system.

Institutions do not move the gradient in a uniform direction. Decentralization increased upstream water appropriation and shifted wealth toward head-end users in Pakistan (Jacoby, Mansuri, and Fatima 2021). Longer exposure to locally elected canal bodies reduced distance-based mis-

allocation and raised revenue for distant farmers in Odisha (Das and Dutta 2026). Water-user associations disproportionately raised productivity at tail-end farms in Pakistani Punjab (Mekonnen, Channa, and Ringler 2015). These studies establish that monitoring, political power, and governance can weaken or deepen head-to-tail differences along a canal; the district characteristics examined in Section 7.2.1 are observational counterparts rather than identified institutional treatments.

For this paper, the key implication is that canal position is itself an equilibrium object: an outcome of long-run choices rather than an assigned treatment. Most irrigation studies compare users at different fixed locations. Land development, infrastructure, rights, groundwater investment, crops, and exit can all adjust around a persistent hierarchy. A static crop-outcome gradient consequently combines physical delivery order with sorting and long-run adaptation. Direct flow measurements such as those in Akram and Mendelsohn (2017) and Jacoby and Mansuri (2020) observe the allocation margin more closely than satellite outcomes do. They do not turn changes in cropped acreage over a branching upstream set into a within-user exposure. The design here holds position fixed and uses within-plot variation in upstream cropping instead.

2.2 Western Water Allocation and Delivery Under Scarcity

Burness and Quirk (1979) formalize the efficiency properties of prior appropriation (the Western rule that serves older, senior water rights before newer, junior rights during shortage) under stochastic flows, clearly defined priorities, and costless enforcement. Rafey (2023) shows that market reallocation on river networks can generate substantial gains: water trading in Australia's Murray–Darling Basin increased output by 4–6%, with the largest gains during drought. This literature concerns who holds and reallocates claims to water. A complementary empirical literature shows how shortage risk and adaptation vary with those claims: drought reduces agricultural yields (Kuwayama et al. 2019), groundwater buffers the damage (Mieno et al. 2024), and junior rights holders absorb more drought risk through fallowing and crop adjustment (Li, Xu, and Zhu 2019; Cobourn et al. 2022).

Two papers by Cody come closest in the Western United States, at two scales. Cody (2018b) compares ditches in Colorado's snowmelt-fed San Luis Valley: seniority and catchment size, not a ditch's upstream rank on the stream, predict diverted volume, months of diversion, and irrigated share.[6] Cody (2018a) moves inside 60 of these systems and relates field-level irrigation and peak greenness to distance from the diversion in four annual cross-sections.[7] Rotational delivery, with or without flexible allocation, leaves no significant head-to-tail difference in crop growth. Both studies stop at small, unbranched ditches in which a field's position is a fixed distance to a single headgate; neither observes a branching network, an upstream set, or within-plot variation over time. And in Cody's extreme-drought year position stops mattering and seniority bites harder, the opposite of the sustained-drought pattern examined here.

This paper changes the comparison rather than reopening that result. Cody asks which system-level rules make outcomes at fixed positions more equal; I hold a plot and its mapped network fixed and ask whether its association with annual upstream cropping shifts in sustained drought.

[6] The sample is 1984–2015 averages for 402 ditches.

[7] The sample is 6,711 fields in four annual cross-sections spanning the 2011–2014 drought; the shortage-sharing and rotational-delivery rules come from a 2013 survey of system managers.

Arizona-specific evidence further separates source-level allocation from last-mile delivery to individual fields. Bautista et al. (2000) documents operational service inside one district, studying arranged service with volumetric pricing and turnout meters, meters at the gates through which water leaves the canal for a field, in the Maricopa-Stanfield Irrigation and Drainage District. Farmer interviews associate long travel times along laterals, the branch canals that carry water from a main canal toward fields, and a common service window with delivery-rate and timing problems reported more often downstream, although 85% of respondents describe service as equitable. Harris (2024) identifies average adjustment to a source-level policy shock, studying Central Arizona Project policies that reduced Colorado River deliveries during 2015–2021: farmers reduce cropped acreage, change crop mix, and respond in ways consistent with substituting groundwater for surface water.

Recent network-based water studies ask related questions. Rafey (2026) combines field microdata, historical rights, remotely sensed evapotranspiration, and California's hydrological network to value rights and evaluate reallocation. Garcia and Belmar (2025) combines administrative, hydrological, geographic, and satellite data to study whether irrigator organizations limit upstream extraction capacity in Chilean water markets, finding downstream gains. This paper asks a different question of similar data: how upstream cropping relates to the same plot's outcome across drought and other years.

2.3 Dynamic Exposure in Directed Networks

Section 5 reconstructs a directed, plot-linked representation from which ancestor and descendant sets follow. It is not an observed hydraulic network: mapped canals stand in for operating ones, flow direction is inferred, and most plot-to-plot links join spatial neighbors, oriented by canal structure. Its value is the plot-specific measure of upstream cropped acreage it permits. In the language of network spillovers, the regressor is an exposure mapping over a directed graph (Aronow and Samii 2017): each plot's exposure is a function of other plots' cropping, weighted by canal ancestry. Direction is what the standard peer-effects setting lacks (Manski 1993): it supplies a reverse-direction comparison, the descendant set, that an undirected network cannot.

Direction-based empirical designs are established outside managed irrigation. Garg et al. (2018) map upstream and downstream river relationships and use upstream sanitation exposure to study downstream health. Keiser and Shapiro (2019) attach monitoring sites to national hydrography and compare outcomes along water flow. These studies demonstrate how a directed graph can distinguish exposures that travel with a network from ordinary spatial association. Satellite measurement of canal performance is established as well: Peña-Arancibia and Ahmad (2021) studies head-to-tail equity and groundwater buffering across 46 Indus canal commands, and BenYishay et al. (2025) measures greenness around rehabilitated Afghan canals. Remote sensing supplies the outcomes here.

The empirical challenge is that canal-connected farms are also geographic neighbors. Weather, pests, soil, and local economic shocks can affect both upstream cropped acreage and the focal plot's outcome. Murgai et al. (2002) shows that even informal water trade is spatially localized, illustrating how proximity can generate both an economic relationship and its confounders. The design in Section 6 separates canal connection from straight-line distance by excluding nearby ancestors and by entering a parallel descendant measure; these steps narrow specific confounding channels without making upstream cropped acreage exogenous.

3 A Model of Positional Rationing

This section develops a simple model of how physical position, water scarcity, and institutional design interact to disadvantage downstream users. Arizona agriculture operates within a layered formal allocation regime: state-law rights and priority, federal and Central Arizona Project contracts, district entitlements, grandfathered groundwater rights, and local delivery rules govern different nodes between source and field.[8] The model takes one node of that structure (a canal serving users in sequence) and abstracts from the rest; it assumes no farmer holds an observed annual volumetric entitlement, and the empirical work never observes one. Some districts deliver on rotational schedules and apply pro-rata curtailment protocols, equal proportional cuts, during shortage; Appendix S6 reports how many have adopted these arrangements.

The model is deliberately minimal. It formalizes a single accounting identity: the water physically available to a plot depends on diversions upstream of it, never downstream. The water a downstream plot loses because users above it divert more than their entitlements is the *wedge*. The identity signs five patterns: a head-to-tail level gradient, drought amplification of the wedge, a directional upstream-only structure, canal-size scaling, and, through comparative statics on its parameters, institutional weakening of the wedge. Table 1 collects them.[9] The identity does the work. The behavioral story attached to it below, undetected over-diversion, is one illustrative polar mechanism. Rotation whose turns shrink or end early when the canal runs short, pro-rata rationing with undetected over-diversion on top of it, branch scheduling, conveyance loss, and directional district management generate the same residual-supply arithmetic and the same reduced-form signs. A shortage shared exactly equally generates no wedge. The analysis observes none of the leak, the deliveries, the shortfall, or the entitlement at any plot. Appendix A1 works the pro-rata case through explicitly.

3.1 Position and Priority

Consider a sub-canal serving two farmers, one senior and one junior in rights. When the junior sits upstream, the downstream senior can invoke a "call on the river," so legal priority restrains upstream extraction.[10] Reverse the positions and the downstream junior has neither physical control over the flow nor legal recourse. If the upstream senior extracts their full allocation, or slightly more without detection, the junior receives only the residual. That undetected excess is the *leak*, a behavioral margin distinct from the physical seepage losses treated below. Junior rights and downstream position coinciding is the configuration in which the wedge bites hardest. What priority reorders is the entitlement queue, not the physics: the leak burdens whoever sits downstream in either configuration, because no legal instrument recovers water already diverted.

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[8] Prior appropriation ranks surface-water rights by historical seniority ("first in time, first in right") and sets the curtailment order among them during shortage (Arizona Department of Water Resources and Central Arizona Project 2023). It governs one layer. District distribution, canal capacity, and turnout scheduling are distinct operational objects, and roughly 72% of plots here match no recorded surface-water right (Section 4).

[9] The head-to-tail mechanism is well documented in the irrigation literature reviewed in Section 2 (Ostrom and Gardner 1993; Ray and Williams 2002; Akram and Mendelsohn 2017; Jacoby and Mansuri 2020). "Positional rationing" is a descriptive label for that mechanism; the paper's contribution is the plot-year upstream cropped-acreage measure and the directional design built on it.

[10] A "call on the river" is rooted in the prior appropriation doctrine: senior rights holders can demand that upstream junior appropriators curtail diversions until the senior allocation is met (The Nature Conservancy Arizona and Friends of the Verde River 2020; Feldman 2004).

When water is abundant, the gap between what would reach a farmer if no one over-diverted and that farmer's own entitlement (the *cushion*) absorbs upstream leaks invisibly; when supply tightens, the accumulated leak lands on whoever sits last in line. Where water is allotted per acre and delivered by order, the seasonal gradient can vanish and the gap moves into the months a canal is at capacity (Appendix A2). The next subsection makes this accounting explicit for a canal serving N users.

3.2 Residual Supply and the Positional Wedge

Consider a single irrigation season t and a directed canal serving users $i = 1, \ldots, N$ ordered from head to tail. System supply S_t is exogenous, and drought corresponds to a low realization of S_t. Each user holds a formal entitlement w_i^* and is either senior or junior in rights, a distinction that matters for the priority-queue comparative static of Section 3.3. Each upstream diversion carries a non-negative leak $\varepsilon_{jt} \geq 0$: *undetected over-diversion*, extraction above entitlement that monitoring cannot detect, whether because meters have finite precision or because turnouts are unmetered. Monitoring intensity μ governs its scale: the leak is decreasing in μ, with $\varepsilon_{jt} \to 0$ as $\mu \to \infty$.[11] The empirical counterpart of μ varies widely across Arizona's districts, from metered Bureau of Reclamation turnouts to smaller systems that allocate per acre with no volumetric measurement at all (Appendix A1).[12] Applied water is not fully consumptive: a fraction $\rho \in [0, 1]$ returns to the hydrologic system, and $\lambda_{ij} \in [0, 1]$ is the share of user j's return flow recoverable at downstream user i.

The residual supply physically available at user i is

$$A_{it} \;=\; S_t \;-\; \sum_{j<i} \big(w_j^* + \varepsilon_{jt}\big) \;+\; \rho \sum_{j<i} \lambda_{ij} \big(w_j^* + \varepsilon_{jt}\big), \tag{1}$$

and delivery is $w_{it} = \big[\min\{w_i^*, A_{it}\}\big]_+$. Let A_{it}^0 denote residual supply under perfect monitoring ($\varepsilon_{jt} = 0$ for all j). The gap between the two is the *positional wedge*,

$$\Delta_{it} \;=\; A_{it}^0 - A_{it} \;=\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big)\, \varepsilon_{jt}, \tag{2}$$

the accumulated upstream leak net of return-flow recovery. The wedge harms user i only once it exhausts the cushion $\ell_i(S_t) = A_{it}^0 - w_i^*$: the realized shortfall is $q_{it} = \big[\Delta_{it} - \ell_i(S_t)\big]_+$.[13] Crop outcomes follow $Y_{it} = Y_i^0 - \beta\, q_{it}/A_{it}^0 + u_{it}$ with $\beta > 0$, where Y_{it} stands in for the satellite indices (NDMI, GCVI, ET), Y_i^0 is plot i's baseline, and u_{it} is mean-zero. The shortfall is scaled by the water that would have been available rather than by the entitlement, since the same acre-foot shortfall costs more when little water is on the canal. Normalizing by w_i^* would bound the ratio.

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[11] The leak is modeled as an exogenous shock rather than a strategic choice; its one-sidedness ($\varepsilon_{jt} \geq 0$) already encodes that users exploit, rather than fall short of, the detection margin. A fully strategic treatment of over-diversion would require equilibrium monitoring machinery without changing the sign predictions used here. Appendix A1 records this and the other parameterization choices.

[12] Delivery losses from evaporation and seepage are not modeled separately; they would enter equation (1) as a deterministic depletion term that compounds along the canal, reinforcing the head-to-tail gradient without changing any sign prediction.

[13] Where the cushion is negative, q_{it} pools the leak-driven wedge with the shortfall entitlement claims alone would produce; both cumulate over upstream users $j < i$, and the empirical measures do not separate them (Appendix A1).

It would also remove the second amplification channel noted below.

Four sign predictions follow. *First*, a head-to-tail **level gradient**: the wedge in equation (2) cumulates over upstream users $j < i$, so expected shortfall $E[q_{it}]$ rises toward the tail, under maintained assumptions on return-flow recovery and the entitlement profile (Appendix A1). *Second*, **drought amplification**: a low draw of S_t lowers residual supply A^0_{it} and with it the cushion $\ell_i(S_t)$. The *same* wedge then binds more often and weighs more heavily against the water that remains. Holding participation and supplemental water fixed, the reduced-form association between upstream cropped acreage and outcomes is more negative in drought years than outside them. *Third*, **directional asymmetry**: the wedge sums only over upstream users $j < i$, so diversions by downstream users cannot enter plot i's residual supply through the physical delivery channel. This identity is what the downstream comparison of Section 6.3 tests. It bounds the physical delivery channel only; factors that propagate in the same direction as water, such as branch releases or head-to-tail scheduling, remain possible. *Fourth*, **canal-size scaling**: $E[\Delta_{it}]$ grows with the number of upstream extractors, so the wedge should be larger in bigger canal systems.[14]

The timing of shortfalls matters for which institutions bite. When shortfalls are anticipated before delivery begins, prior appropriation operates as designed (junior rights holders are curtailed first, regardless of position), and districts apply the rotational schedules and pro-rata reductions described above. When shortfalls are realized *during* delivery, upstream users can maintain planned extraction, and the tail absorbs the residual shock: a delivery scheduled for ten hours that ends after nine leaves downstream users whatever remains of the final hour. Equation (1) formalizes that residual-claiming regime; Appendix A1 shows the wedge and its predictions survive under faithful pro-rata rationing too, making residual claiming a transparent polar case rather than a necessary assumption. The distinction also seeds the enforcement asymmetry behind the next subsection's seniority prediction. Appendix A2 extends the identity to the months of a season. Central Arizona and Maricopa–Stanfield ration water by a seasonal allotment per acre and charge each farm's deliveries against it (Section 7.3.12). Under such allotments the fixed head-to-tail gradient disappears and the wedge works within the season, so the first prediction (the level gradient) is not what the CAP districts show.

3.3 Institutions in the Model

In this stylized model, two extreme mechanisms would remove the wedge. One is frictionless trade, in which downstream farmers buy allocations from upstream users who value the water less, the standard Coasian solution. The other is perfect detection of diversions beyond entitlement. Both are benchmarks that Arizona meets only partly.

Markets remain thin: volumes are low, cross-district transfers occur at the district rather than individual level, and conservation-based trading often compensates below opportunity cost (Harris 2024). Rights historically operated under "use it or lose it" provisions, creating an incentive to extract the full entitlement even at low marginal value (Brocious 2021); the 2021 reform relaxing this covers only the sample's final three years. And enforcement is limited by the same measurement margins that generate the leak. Institutions therefore enter the model through its ingredients (the leak ε, the cushion ℓ_i, or the crop-relevant cost of the shortfall q_{it}):

[14] The model signs the *level* of the wedge in the number of upstream extractors. The empirical canal-size test (Appendix Table A8) is on the drought *interaction*, which requires an auxiliary assumption about the cushion in larger systems; Appendix A1 makes it explicit.

- **Groundwater access** caps the crop cost of a surface-water shortfall. A farmer with a well replaces undelivered surface water by pumping at marginal cost c_g, bounding the shadow cost of q_{it}. A slower recharge channel, upstream return flows raising the recovery share λ_{ij}, works in the same direction with multi-year lags. Both push the crop-outcome wedge toward zero where groundwater is a feasible substitute. Section 7.3.5 finds the 2022 penalty in upstream land with no district pump between it and the field, and the split by water source does not order with source. Fields that share a district pump compete for its water instead.

- **Volumetric pricing** forces measurement. Charging per acre-foot requires metering deliveries, which raises monitoring intensity μ and shrinks the leak, and it puts a marginal cost on every unit diverted, counteracting "use it or lose it" incentives. Pricing that also reduces runoff and recharge lowers the recovery term $\rho\lambda_{ij}$, so the net effect on the wedge is ambiguous in theory, and district records are too sparse to estimate it here.

- **Water rights seniority** operates through the priority queue. Where it is enforced, senior entitlements are filled before juniors receive residual delivery, so senior shortfalls are pushed toward zero ($q_{it}^{\text{sen}} \approx 0$). Juniors compete over a thinner residual pool; the queue reorders who absorbs the leak rather than removing it. Two assumptions qualify this: enforcement is asymmetric, so realized mid-season shocks can reach seniors before curtailment operates; and seniors must not sit systematically downstream of juniors. Appendix A1 formalizes the queue and records the Arizona enforcement mechanics, including that the binding queue runs at the river level, with districts as its users.

- **Irrigation district membership** bundles coordinated scheduling, maintenance, and administrative oversight, and could work through any of the three parameters above: tightening μ, smoothing the cushion ℓ_i, or capping the crop cost of a shortfall. The model leaves these unseparated, so district membership enters empirically rather than as a derived comparative static.

Together these comparative statics predict the steepest interaction in surface-connected, junior-rights districts, the configuration in which priority supplies less protection and pumping need not fully cap the wedge.

3.4 Predictions and Scope

Table 1 collects the model's sign predictions, the corresponding estimates, and where each is tested. Section 7.3 tests the same accounting more directly inside the four CAP districts, whose canal water is recorded by month. Seniority and district policies are recorded for districts, not plots. The model's predictions about priority within a canal, junior–senior equality, and pricing therefore cannot be tested here; Section 7.2.1 reports a district-level profile instead.

The model has four limits. It signs directions rather than magnitudes. It takes fallowing as given, so the fallow-contagion controls (Section 6.1) and the magnitudes (Appendix S13) lie outside it, and its regressor marks potential demand rather than observed diversion. It has no spatial-correlation structure for the leak, which Section 6 handles empirically. And its canal is a line. The water graph branches: plots on sibling laterals that share a trunk compete for trunk water without being canal ancestors, so the upstream count omits some competition, whose sign I leave open.

Table 1: Model Predictions and What the Data Show

Prediction	**Predicted sign**	**Supported?**	**Result**	**Where tested**
Motivating fact the model rationalizes				
Head-to-tail level gradient	Lower downstream	Descriptive	GCVI −0.032***, NDMI −0.007**; ET null; none in the CAP districts (App. A2). Not identified: position is time-invariant	Cross-section: Sec. 5.7, Table 3
Predictions taken to the panel				
Drought amplification	More negative in drought	✓	GCVI interaction −0.015** over the full path, −0.035*** beyond 10 km	Panel: Sec. 7.1, Table A22
Directional amplification	Upstream interaction more negative than downstream	✓	Upstream minus downstream interaction −0.032 beyond 10 km (GCVI; 95% CI excludes zero under two-way clustering, Fig. 5)	Directional: Sec. 7.1 (Table A23); App. Table A6
Institutional scope	Weakening possible; pricing ambiguous	Descriptive	District drought reveals both a network-level baseline and year-to-year departures in upstream cropped acreage inside junior-rights districts; senior-inside estimates are imprecise; the split by water source does not order with source	Sec. 7.2, Tables 5–A11
Canal-size pattern (support-limited)	Steeper in larger systems	partial	GCVI −0.037*** beyond 10 km in systems with >300 plots; small and medium bins hold 4,168 of 75,826 far-cutoff rows, no cross-bin test	App. Table A8

Notes: Signs are stated for crop outcomes. For the cross-sectional row, negative means worse NDMI, GCVI, or ET at more downstream positions. For the panel rows, negative means the association between upstream cropped acreage and the outcome is more negative in drought years than outside them, and positive means the interaction weakens; these are interaction coefficients, not drought-year slopes. Results report the headline estimate from the table named in the final column; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. ✓ marks a prediction supported on the paper's preferred specification; "partial" marks support that holds only in part of the parameter space or under some weightings. The drought-amplification prediction is tested in the panel; the cross-sectional Position × Drought interaction is under-powered by design (Section 5.7). Appendix A1 reports the derivations, auxiliary assumptions, the model objects behind each prediction, and the full model-to-empirics map.

4 Data

4.1 Plot Panel and Geographic Setting

This study assembles a plot-level panel from nine data blocks. The overview below lists each block, its main source, and its role in the design; the rest of this section takes them in turn, and

Appendix B documents each source, variable, and construction step in full.

Data Inputs and Role in the Empirical Design

Data block	Main source	Role in the paper
Plot panel	USDA Crop Sequence Boundaries	Defines 54,946 plots from 2016–2023.
Crop outcomes	Sentinel-2 and OpenET	Measures NDMI, GCVI, and ET.
Crop/fallow status	USDA Cropland Data Layer	Identifies active/fallow plot-years and generates the upstream cropped-acreage measure.
Drought	SPEIbase	Measures SPEI-24 and SPEI-3 drought exposure.
Canal infrastructure	NHDPlus and DEM	Links plots to canals and constructs water-graph position.
Water access	ADWR rights, groundwater rights, well registry, USGS withdrawals	Measures priority, groundwater access, wells, and water-source dependence.
Institutions	Irrigation-district records	Measures pricing, seniority, shortage protocols, rotation schedules, and district governance.
Soils	SSURGO and Sentinel-2 bare-soil indices	Controls for plot quality (soil salinity and clay content).
Geography	OpenStreetMap; protected-area, tribal, state-trust, and federal-land layers; community-water-system	Controls for market access and land status.

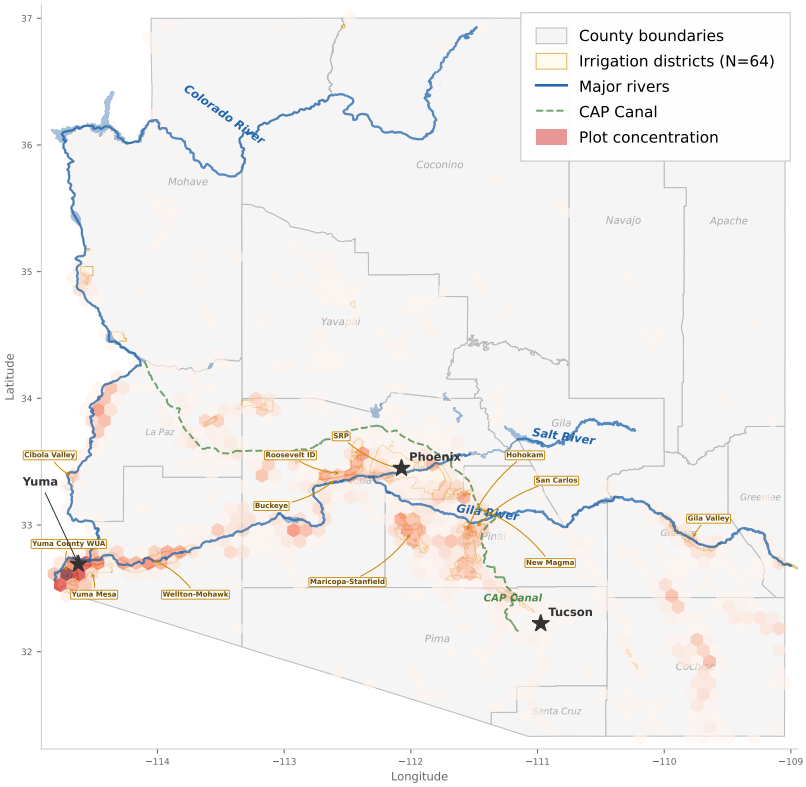
Note: A few further inputs feed controls within these blocks. District-institution variables draw on ADWR records and Central Arizona Project allocation documents alongside district records. Census and watershed (HUC12) boundaries enter only as base layers for mapping and spatial joins. Appendix B gives the complete source list and construction steps.

The unit of observation is an agricultural plot in a given year. Plot geometries come from the USDA Crop Sequence Boundaries, which define stable field boundaries from the satellite record; the same 54,946 Arizona plots are observed in every year of 2016–2023, a panel balanced by construction with 439,568 plot-years ($54{,}946 \times 8$). Row-specific counts differ across variables because satellite outcomes, drought measures, and network-position variables are observed on different supports.

Annual crop-type classifications from the USDA Cropland Data Layer record each plot's crop and whether it is cropped or fallow in a given year; approximately 75% of observed plot-years are actively cropped and 25% are fallow. This classification does double duty. It drives the crop-by-year fixed effects, and it defines the active/fallow measure whose year-to-year changes among upstream plots generate the time-varying upstream cropped acreage at the center of the panel design.

Canal geometries come from the USGS National Hydrography Dataset (NHDPlus). Filtering out ephemeral streams leaves 147,892 canal, ditch, and aqueduct segments out of 547,842 raw Arizona flowlines. A digital elevation model (DEM) supplies the elevations that orient flow along them. Arizona's canal network saw no major construction or realignment during the sample period, so the mapped structure and plot–canal associations are time-invariant, a fixed backbone along which the upstream cropped-acreage measure moves.

Figure 1 shows the setting. Irrigated agriculture concentrates along three surface-water corridors: the Colorado River and its diversion canals in the western border counties (Yuma, La Paz), the Salt–Gila system feeding Phoenix (Maricopa, Pinal), and the upper Gila Valley (Graham). The 540 km Central Arizona Project aqueduct connects the Colorado to central Arizona. Plot density tracks the canal network closely: surface-water infrastructure, not rainfall, determines



54,946 agricultural plots (2016–2023); 24,923 (45%) inside an irrigation district, 30,023 (55%) outside

Figure 1: Study area: Arizona agricultural plots and canal infrastructure. Plot locations (dots) cluster along the Colorado River corridor (west), the Salt–Gila system (central), and the upper Gila Valley (east). Major irrigation districts and the Central Arizona Project aqueduct are shown.

where agriculture occurs here.

Section 5 builds the two variables used throughout from this mapped network: each plot's fixed position along its canal, and the count of its upstream plots in crop in a given year beyond d kilometers, which moves within a plot as upstream plots enter and leave fallow (Section 5.5).

4.2 Outcomes and Drought

I measure crop outcomes with three satellite indices, all transformed by $\ln(1+x)$ before analysis. The retained extraction code defines vegetation outcomes as May–September pixelwise maxima followed by plot means, and ET as the plot mean of May–October monthly sums. *NDMI* measures vegetation water content and is the water-stress proxy; *GCVI* is sensitive to chlorophyll and is the primary outcome, used as a yield proxy; *ET*, from the OpenET ensemble, measures crop water consumption. Using three traces the signal from water stress through chlorophyll to water consumption and guards against any single index reflecting a sensor artifact; satellite vegetation indices are validated yield proxies against ground-truth harvest data (Lobell et al. 2020). Unless explicitly labeled raw, NDMI, GCVI, and ET below refer to these transformed outcomes; their coefficients are not changes in raw index units or percentage changes in harvested yield. Appendix S4 gives the index formulas, spatial and temporal aggregation, and seasonal windows.

To separate irrigation-infrastructure effects from meteorological variability, I use the Standard-

ized Precipitation Evapotranspiration Index (SPEI), which accounts for both precipitation and evaporative demand, a first-order constraint in arid agriculture. The primary drought measure is the 24-month index (SPEI-24), which proxies reservoir storage and multi-year surface-water reliability (the slow-moving scarcity that governs canal deliveries) rather than short-run rainfall. The 3-month index (SPEI-3) marks acute meteorological dry spells and enters only as a robustness contrast.

The headline estimates of Table A22 use a plot-specific *upstream-path* indicator, Drought_{it}^{up}, which equals one when the acreage-weighted mean SPEI-24 across the actively cropped plots on plot i's upstream path falls below -0.6, an operational dryness cutoff. Because that active set changes as upstream plots enter or leave fallow, the indicator changes with which upstream plots are in crop each year; the cross-section of Section 5.7 uses the same indicator. Three variants serve other tables. The priority profile in Table A11 averages SPEI-24 over every eligible plot in the full upstream set, in crop or fallow. The decomposition and timing tables (Table 5 and Appendix Table A10) use a district-level indicator: the acreage-weighted SPEI-24 mean over the fixed set of all sample plots inside each district boundary, at the same -0.6 threshold. It is common within a district-year and independent of which upstream plots are in crop; its continuous version is reversed and standardized so that larger values mean greater scarcity. The outcome-year fallowing diagnostic in Appendix A22 uses the fixed-upstream-set indicator. None of these constructions is a delivery or flow measure, and the plot-centroid indicators in Table 2 are descriptive. Appendix S5 gives each construction in full and the case for SPEI-24 as the primary timescale.

Because the index cumulates over 24 months, Drought_{it}^{up} marks a *sustained* water-balance deficit rather than a single dry season. Farm-level scarcity runs through institutions. A multi-year deficit draws reservoirs down until the allocation rules tied to reservoir levels bind on administered systems, and it thins the flow left in unadministered gravity canals. A short dry spell is less likely to do either. SPEI proxies those supply conditions, not the allocation itself, and as an atmospheric index it also embeds evaporative demand, which affects crop outcomes directly. The indicator marks years in which surface-water constraints bind, not a measured allocation cut; Appendix Table A34 tests the measured margin directly.

4.3 Water Access, Rights, and Institutions

The heterogeneity analysis uses district classifications verified against district records: water source, priority, and mapped boundaries are coded from each district's own documentation rather than from proxies. The district set is fixed at the 30 districts accounting for 95.6% of service-supported member plots. Physical supply is classified from district evidence as groundwater only, surface water only, mixed, or unresolved; a broader-system definition resolves six cases in which a district belongs to a larger encompassing system as mixed, without imputing a groundwater share. The resulting comparison contains 3 groundwater-only, 5 surface-water-only, and 22 mixed systems.

Groundwater fallback access is measured by grandfathered groundwater rights and the count of registered wells serving each plot, an alternative supply during surface-water shortage. The scope-condition checks additionally use irrigation-well applications and reported installation dates from the ADWR Wells55 registry, as recorded through 2014, and direct depth-to-water readings from the ADWR Groundwater Site Inventory. The well records measure predetermined infrastructure; the depth-to-water readings measure subsequent aquifer conditions rather than pumping volume.

Priority is classified within each district's own allocation system rather than forced onto a

common 1–7 ladder: 6 districts are senior, 10 junior, 5 mixed, 6 unknown, and 3 not applicable.[15] Pricing and governance records distinguish mechanisms documented at some point in 2016–2023 from those documented for the whole period; unknown, conflicted, and period-ineligible entries remain missing. District boundaries define the inside/outside geography used below; an outside plot carries a nearest-system label for spatial comparison without being coded as a member or customer.

Surface-water rights come from the Arizona Department of Water Resources; roughly 72% of plots match no recorded right and are assigned the lowest possible seniority. The seniority control is therefore a coarse proxy rather than a precise legal ranking. State-law priority, district entitlement, CAP contract position, and grandfathered groundwater rights are distinct legal objects, and none is a measured annual delivery entitlement for a given field; they enter as controls and in observational comparisons. Appendix S6 details the rights-matching procedure, the groundwater and well registries, and the district-level construction of each institutional variable; Appendix S7 documents the market-access controls.

4.4 Summary Statistics

Table 2 reports summary statistics separately for active and fallow plots. It is organized around the variables central to the design: the three crop outcomes, network position, upstream cropped acreage beyond the exclusion radius with its downstream mirror, the drought measures, the two fallow types, and the physical and water-access covariates. Position is normalized within each weakly connected component (WCC), the largest set of plots joined to one another by canal paths, effectively one irrigation sub-network. A crop-sequence classifier separates two kinds of fallow. Stress fallow, outside any rotation, reflects water constraints or economic stress and is the kind that threatens identification. Rotation fallow is planned as part of the crop cycle (Section 6.1; classifier detail in Appendix S8).

Two patterns anchor the table. Active plots outnumber fallow by roughly three to one, and fallow plot-years show markedly lower NDMI, GCVI, and ET; among them stress and rotation types are of comparable prevalence and together cover just over half. Elevation difference to intake and distance to canal are kept in the main table as physical inputs that let readers sanity-check the plot–canal linkage and the gravity-fed-delivery assumption. Full statistics are in Appendix S9.

[15] Within a district, surface priority is usually held by the district rather than by individual farms. Among the 30 audited districts, those in which all member farms' surface deliveries fall under one district-held priority contain 77.8% of audited plots (63.9% verified from Reclamation, CAP, district, and decree records, and 14.0% inferred). Farm-level priority differences, at least on paper, are documented in nine districts with 15.8% of plots: the Salt River Project (Kent Decree land classes), San Carlos and five Upper Gila canal companies (Globe Equity decree schedules), Mohave Valley (individual present perfected rights), and Buckeye (the Benson–Allison decree). The remaining 6.4% are in three groundwater-only systems and one single-farm entity. Where a rate schedule was located, members are billed under one district schedule, except in the Salt River Project, Mohave Valley, and Unit B, which use differentiated charge classes; many retail schedules were not located.

Table 2: Summary Statistics: Active vs. Fallow Plots

		Active Plots			Fallow Plots		
	TV	Mean	SD	N	Mean	SD	N
Crop Outcomes							
NDMI (max)	✓	0.253	0.156	327,999	0.022	0.138	107,154
GCVI (max)	✓	1.78	0.629	327,999	0.930	0.493	107,154
ET Ensemble (log)	✓	6.29	0.552	327,999	5.26	0.809	107,154
Network Position & Upstream Competition							
Network Position (0–1)		0.499	0.268	331,327	0.503	0.286	108,241
log(1 + upstream active >1km)	✓	3.49	1.65	272,464	2.79	1.77	85,112
log(1 + downstream active >1km)	✓	3.43	1.72	262,102	2.63	1.86	82,786
Canal Network Size (plots)		4330.8	4691.8	331,327	2692.8	3903.9	108,241
Drought							
Plot drought$_{24M}$ (binary)	✓	0.160	0.366	328,297	0.174	0.379	107,351
Upstream drought$_{24M}$ (binary)	✓	0.150	0.357	331,327	—	—	—
SPEI 24-Month	✓	-0.108	0.512	328,297	-0.092	0.601	107,351
SPEI 3-Month	✓	-0.303	0.545	328,297	-0.355	0.514	107,351
Fallow							
Stress fallow (non-rotation)	✓	0.000	0.000	331,327	0.279	0.448	108,185
Rotation fallow	✓	0.000	0.000	331,327	0.260	0.439	108,185
Plot & Water Access							
Plot Size (acres)		21.39	36.12	331,327	14.58	40.89	108,241
Well Count		1.48	4.51	331,327	1.59	4.84	108,241
Distance to Canal (m)		727.1	970.2	331,327	1021.3	1355.0	108,241
Elevation Diff. to Intake (m)		-0.103	4.84	331,327	0.463	7.16	108,241
Pumping (dummy)		0.418	0.493	331,327	0.503	0.500	108,241
GW Share (HUC12)		0.436	0.310	323,967	0.521	0.302	101,465

Notes: TV = time-varying; a check mark indicates variation at the plot-year level. Panel years: 2016–2023 (Section 4.1); all plots have canal network positions. Row-specific counts vary by variable support. Active ≈ 75%, Fallow ≈ 25%. NDMI, GCVI, and ET all enter as $\ln(1+x)$, so the reported standard deviations use the units of the estimated outcomes. Network position normalized 0–1 within each weakly connected component (WCC). The tabulated upstream drought$_{24M}$ row is the baseline active-set measure and is therefore shown only for the active estimation sample; its fallow cells are omitted (—). Table A11 uses the separately constructed fixed-upstream-set drought indicator; Table 5 and Appendix Table A10 use the district-level drought indicator. This is the compressed main-text panel; full summary statistics, including soil, slope, upstream fallow shares, and other secondary controls, appear in Appendix S9 (Table S4).

5 Measuring Position in the Water Network: The Water-Graph Approach

5.1 Overview

The water-graph is a reconstructed directed representation of the canal delivery network. Nodes are agricultural plots, and an edge from plot A to plot B records that water passes A's canal intake before reaching B in the mapped infrastructure. It is assembled from mapped canals, elevation, inferred flow directions, and algorithmic plot-to-plot links rather than observed from operations. The algorithm takes three public datasets as inputs (NHDPlus canal geometries, USDA Crop Sequence Boundaries, and digital elevation models) and produces a normalized position measure $\in [0, 1]$ for each plot, where 0 denotes the canal headgate and 1 the tail end.

It serves two purposes. It fixes each plot's static position, which the descriptive pattern of Section 5.7 uses. And it fixes the *set* of plots upstream of each plot, so the number of those in crop in a year becomes a time-varying measure defined on a structure that stays fixed (Section 5.5).

The construction proceeds through four stages: (i) building a directed canal graph from NHDPlus flowlines, (ii) linking each plot to its nearest canal segment, (iii) assembling a directed plot graph from canal-structure and spatial-neighbor edges, and (iv) computing positions via least-cost signed-hop traversal from canal-graph roots. The final graph covers 54,946 plots linked to 147,881 canal graph nodes connected by 201,079 directed edges. They fall into 3,362 WCCs, ranging from 2 to over 12,000 plots, and 1,162 canal blocks, each a connected canal network or, for the 6 largest networks, a compact cluster of plots within one (Appendix S1). Position follows mapped canal structure rather than Euclidean distance wherever a canal path exists, and spatial-neighbor links order only pairs that no mapped path connects. Two plots 100 m apart may therefore have very different positions if they sit on different canal branches.

5.2 Canal Graph Construction

Figure 2 visualizes the four construction stages for one WCC. I extract canal, ditch, and aqueduct segments from the NHDPlus flowline layer, excluding ephemeral streams, which retains 147,892 of the 547,842 raw Arizona flowlines. For each segment I infer the direction of flow from the slope of elevation along it, measured on high-resolution DEM rasters, and correct these inferences wherever the NHDPlus Value Added Attributes (VAA) flow table records a connection between segments. Segments whose ends meet within a short tolerance are then joined into a directed canal graph, G_{canal}. Appendix S1 gives the regression rule, the endpoint tolerance, and the merging of parallel segments and splitting at junctions.

5.3 Plot–Canal Linking

I match each of the 54,946 agricultural plots to its nearest canal segment, penalizing assignments that would require pumping uphill: each meter of lift from canal to plot counts as 30 meters of added distance. Where the canal sits at or above the plot, the match is by distance alone, and no maximum plot–canal distance is imposed. In practice 43% of plots sit above their nearest canal, and for large lifts the penalty is enough to redirect the match to a nearby canal at a more compatible elevation. The match identifies the segment and the plot's *intake point*, the location along the canal geometry closest to the plot centroid; plots sharing a segment are ordered by projecting their intake points along the flow direction. Appendix S1 gives the cost function and the size of the penalty in practice.

5.4 Plot Graph and Position Computation

I construct the directed plot graph G_{plot} from two types of edges. *Canal-structure edges* connect plots whose canal segments are linked in G_{canal}: if a directed path exists from $canal_A$ to $canal_B$, all plots on A are upstream of all plots on B. For plots on the same canal, their ordering follows the intake-point projection along flow direction. *Spatial-neighbor edges* connect plots identified as geographic neighbors via raster flood-fill at approximately 11 m resolution, providing local connectivity where canal paths are indirect.

Position is computed within each WCC. From a root plot on the component's highest-elevation source canal, I count signed hops along the cheapest path to every other plot, where a step forward along the flow adds one and a step backward subtracts one. The cost hierarchy is the substantive part: canal-structure edges are cheap and spatial-neighbor edges two orders of magnitude dearer,

so ordering follows the canal wherever a canal path exists, and a spatial link orders a pair only when no canal path does. Hop counts are then scaled to $[0, 1]$ within the WCC, with 0 at the head and 1 at the tail. Appendix S1 gives the root-plot rule, the edge prices, the shortest-path algorithm, and the normalization. Figure 2 illustrates the four construction stages: canal network with flow directions, intake-point linking, directed plot graph, and the resulting plot positions.

5.5 From Graph to Regression: Two Exposure Measures

The graph is a single static structure. The empirical strategy needs two measures built on it, serving different identification purposes.

The *time-invariant* measure, network position, is each plot's least-cost signed-hop position normalized to $[0, 1]$ within its WCC: *where* a plot sits on the canal. Being fixed, its main effect is absorbed by plot fixed effects, so the level gradient can only be estimated in the cross-section, comparing plots at different positions within the same canal-year cell. Relative to the model of Section 3 it is a within-WCC proxy for eligible upstream acreage: hop depth orders plots by ancestor count. But a unit of normalized position corresponds to different upstream acreage in WCCs of different sizes, so the pooled cross-sectional coefficient averages heterogeneous per-unit gradients. Identifying comparisons are within WCC, where the ordering is preserved.

The *time-varying* measure, $U_{it}^{>d} = \ln(1+N_{it,>d}^{\text{active, upstream}})$, counts active upstream plots beyond Euclidean distance d in year t. It moves as plots enter or exit fallow, creating within-plot variation that *survives* plot fixed effects: how crowded the canal is above a plot, not where it sits. This is the main panel regressor.

The distinction matters for interpretation. A negative coefficient on position says plots further from the headgate have worse outcomes. A negative coefficient on $U_{it}^{>d}$ says that more mapped ancestors in crop are conditionally associated with worse downstream outcomes, a different regressor on variation the cross-section does not use. Throughout the panel I call the second upstream cropped acreage beyond the exclusion radius rather than position, and reserve "gradient" for the cross-sectional position coefficient.

5.6 Validation

I validate the graph along two dimensions: the flow directions that orient it, and the provenance of the edges that order plots within it. Inferred flow directions agree with the NHDPlus VAA flow table in 95.5% of the 105,744 retained segments where both are defined, before any reconciliation against that table. Agreement rises to 98.6% among the 88,653 segments with reliable elevation fits, and a manual check of 100 randomly drawn canals shows 100% agreement with start-point coordinates. Appendix A35 reports full coverage, agreement, and fallback statistics.

Of the graph's 132,971 edges, 30% are read directly off the canal map, linking plots that a single canal orders consecutively or that sit either side of a mapped junction. The other 70% link spatially adjacent plots, and the canal network supplies their direction too, by reachability between the two plots' canals or, where both draw from one canal, by intake position along it. Spatial adjacency determines which pairs are considered; canal structure determines which is upstream. Edge counts therefore do not translate into influence over position, because spatial links are priced two orders of magnitude above canal links in the traversal. Removing them tests the claim. Recomputing ancestor sets on the canal-structure subgraph alone, 39,764 of 132,971 edges, leaves the GCVI interaction more negative than in the main specification: -0.023^{***} at $d > 1\,\text{km}$ and -0.025^{**} at

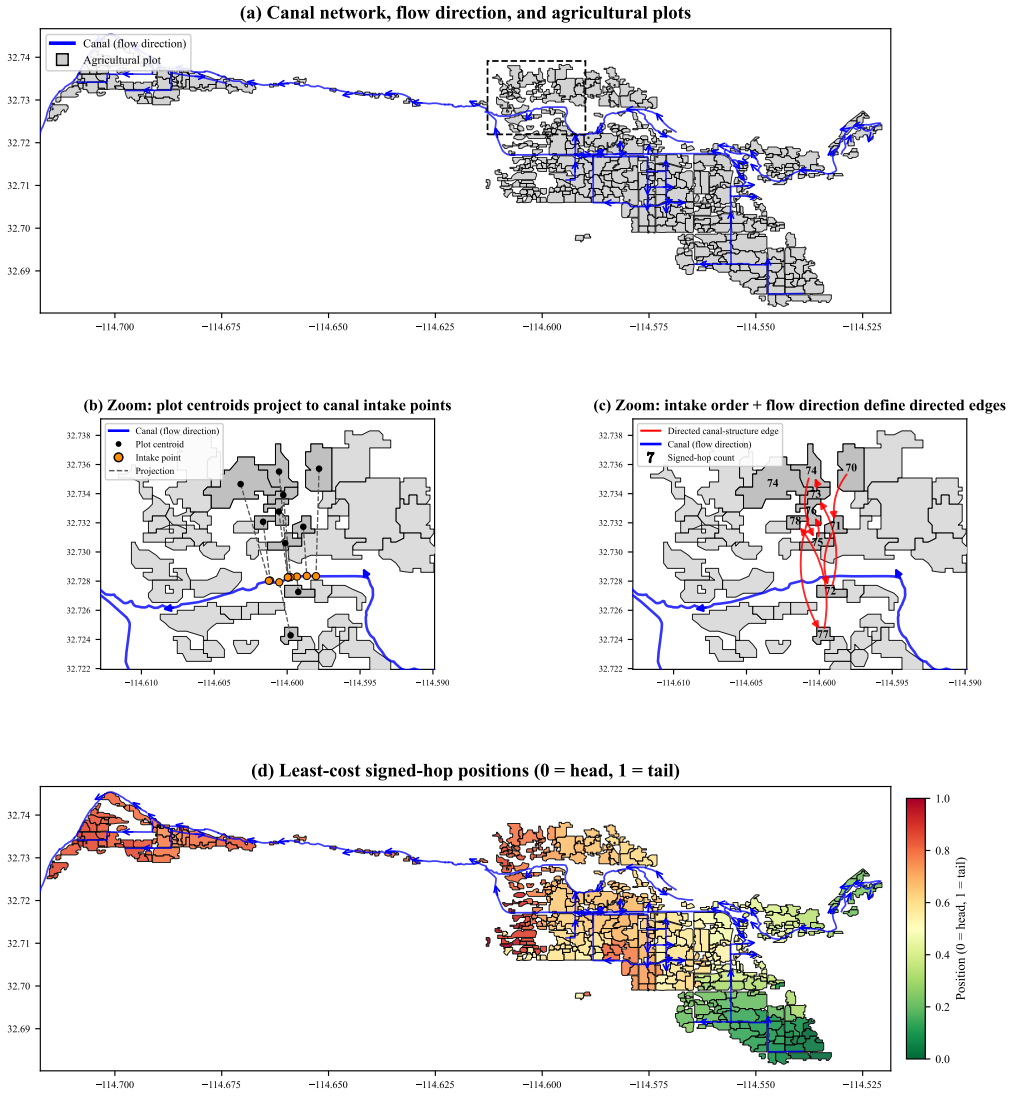


Figure 2: Water-graph construction for an example WCC in the Yuma area (493 plots). (a) Identify NHDPlus canal segments serving the component's plots (blue; arrows show inferred flow direction) and agricultural plots (gray); the dashed box marks the zoom window of panels (b)–(c). (b) Link plots to canals: each centroid (black dot) projects onto its nearest canal, defining an intake point (orange). (c) Order intake points along flow: adjacent intakes generate the directed canal-structure edges of the plot graph (red arrows); numbers are signed-hop counts. (d) Compute position: least-cost signed-hop traversal on the directed plot graph gives normalized WCC position, with Head/0 at the green end and Tail/1 at the red end.

$d > 5\,\text{km}$ (Appendix Table S5). The comparison is not like-for-like on sample, because restricting to canal-structure ancestors drops plots with none; the table note reports the sample reduction and the correlation with the measure used in the main results.

What the reconstruction makes possible is this paper's empirical object: a plot-year measure of upstream cropped acreage, the set of a plot's mapped canal ancestors and how many of them are in crop in a given year, computed reproducibly across an entire state. System-level and single-channel proxies cannot deliver it. The graph is mapped infrastructure, and four limitations follow. First, it represents mapped infrastructure rather than delivery: seasonal closures, gate operations, branch releases, and delivery orders are unobserved. Second, accuracy depends on NHDPlus completeness and DEM precision, which vary regionally. Third, gravity-fed delivery is assumed, so control gates, check structures, and complex pumping are unrepresented. Fourth, 466 plots in grid-loop structures receive valid positions and are flagged as lying on a cycle; excluding them leaves the Table A22 estimates essentially unchanged (Appendix A35).

5.7 A Descriptive Head-to-Tail Pattern

The graph makes one descriptive pattern visible before any modelling, and it motivates the design rather than testing it. Panel A of Figure 3 shows it raw: tail-quartile plots fallow more often than head-quartile plots in every year, the gap widening once the Tier 1 shortage binds.

A conditional version sharpens it. I regress crop outcomes on network position among plots in comparable canal systems and years:

$$Y_{irt} = a_1 \cdot \text{Position}_{ir} + a_2 \cdot \left(\text{Position}_{ir} \times \text{Drought}_{it}^{up}\right) + a_3 \cdot \text{Drought}_{it}^{up} + X_{it}'\pi + \gamma_{\text{crop},t} + \delta_{m,c,t} + e_{irt} \quad (3)$$

where i indexes plots, r indexes canal networks, and t indexes years. Y_{irt} is a satellite-derived crop outcome (NDMI, GCVI, or ET). $\text{Position}_{ir} \in [0, 1]$ is the time-invariant normalized network position (0 = headgate, 1 = tail). Drought_{it}^{up} is the upstream drought indicator constructed in Section 4.2; it varies across plots within the same canal network and year because upstream sets differ by plot, and a_3 is its main effect. X_{it} includes 42 plot-level controls, several time-varying: plot size, slope, well count, the SPEI indices, soil characteristics, groundwater access, water-rights seniority, market access, network structure, protected-area distances, and canal-type interactions.[16] The network-structure controls include head and tail dummies, which absorb endpoint plots so that the position coefficient reflects interior-of-canal variation. $\gamma_{\text{crop},t}$ denotes crop × year fixed effects. $\delta_{m,c,t}$ denotes irrigation district (m) × WCC (c) × year fixed effects, which absorb all time-varying institutional heterogeneity at the district-canal level, including district-specific drought responses.[17] Standard errors are clustered two-way by crop-year and plot.

Table 3 reports the estimates. Position level effects are negative for NDMI (-0.007^{**}) and GCVI (-0.032^{***}), with ET smaller and insignificant (-0.003). Moving from headgate to tail is therefore associated with lower crop water content and lower GCVI within the same district, canal network, crop type, and year. The full head-to-tail GCVI reduction of 0.032 is 5.1% of the active-plot standard deviation, and a one-standard-deviation increase in position 1.4%.

[16] The exact 42-variable list is given by the cross-sectional estimation routine in the replication package.

[17] These fixed effects are the full interaction of irrigation district, weakly connected component, and year.

The district × WCC × year fixed effects absorb the variation that would identify a_2, so this specification cannot test drought amplification. The table reports the interaction for completeness; it is small and insignificant for all three outcomes. The panel of Section 6 estimates a different regressor on within-plot variation this specification lacks.

Position is time-invariant, so the cross-section cannot separate canal position from unobserved characteristics correlated with it. And the gradient depends on the specification. Across the six variants in Appendix Table A20, Panel A, the GCVI level gradient is significant in the baseline and when re-clustered by plot. It is significant in none of the other four columns; the ET level is null in all six. The ordering is descriptive and the conditional version fragile.

The rest of the paper leaves this pattern as motivation and turns to a within-plot test estimated on variation the cross-section does not use.

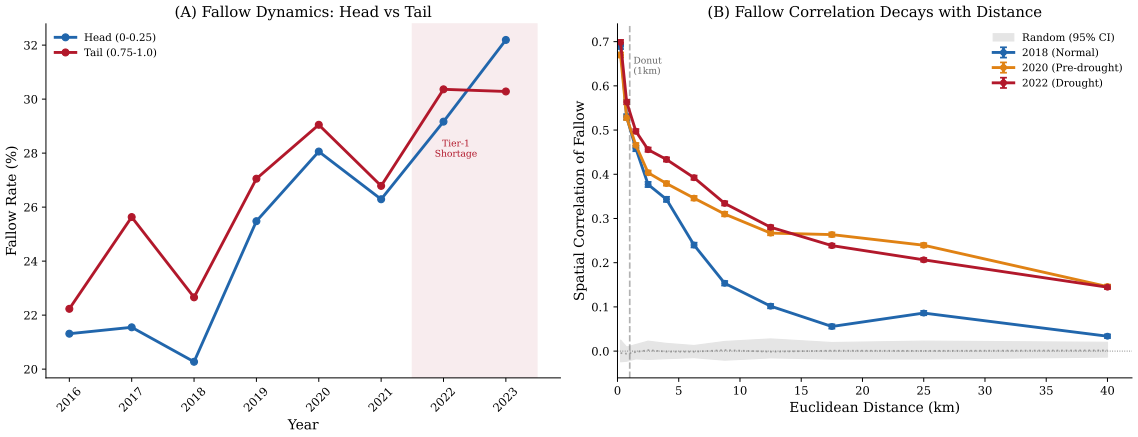


Figure 3: Fallow dynamics and spatial correlation. (A) Descriptive head-to-tail pattern: mean fallow rate for the bottom and top quartiles of normalized network position (0–0.25 and 0.75–1.0), 2016–2023, with the shaded band marking the Tier 1 shortage years. Position is time-invariant, so this comparison is between plots and is not identified. (B) Spatial autocorrelation of fallow status vs. distance: correlation persists far above random (gray band) and extends beyond 30 km during drought years. The 1 km marker is one point in the exclusion-cutoff grid used below, not the only exclusion radius.

6 Identification Strategy: Distant Upstream Cropping

The design uses a difference in geography: local shocks spread across nearby fields, and canal-linked exposure follows the directed graph. Fallowing is spatially clustered, so a nearby upstream plot going fallow may reduce measured upstream demand and also reflect the same local shock that affects the downstream plot. I therefore control for nearby fallowing in geographic space and use cropped acreage among distant upstream canal neighbors as the focal variation.

The interaction is not identified from year-to-year changes alone. Plot fixed effects absorb fixed exposure levels. A persistent exposure multiplied by a drought indicator that changes over time remains in the interaction, so the baseline coefficient can combine annual changes in upstream cropped acreage with drought revealing persistently different network environments. This is an observational network design: the decomposition in Section 7.2, the spatial exclusions, and the descendant comparison narrow and diagnose classes of confounding, and none assigns upstream cropped acreage.

Table 3: A Descriptive Head-to-Tail Pattern: Conditional Cross-Sectional Gradient

	NDMI (1)	GCVI (2)	ET (3)
Network position	−0.007**	−0.032***	−0.003
	(0.003)	(0.011)	(0.012)
× Drought_{24M}	−0.007	−0.025	−0.029
	(0.005)	(0.020)	(0.019)
Drought_{24M}	0.005*	0.014	0.023*
	(0.003)	(0.013)	(0.012)
FE: Crop × Year + Irr. Dist. × WCC × Year	✓	✓	✓
Observations	326,732	326,732	326,732
R^2	0.448	0.488	0.510

Notes: Descriptive, not identified: position is time-invariant, so these comparisons are between plots, not within them. Active plots only. Network position is signed-hop position normalized to $[0, 1]$ within each WCC, with 0 at the headgate and 1 at the tail. $\text{Drought}_{24M} = 1$ if upstream SPEI-24 < -0.6. The Position × Drought row is a within-system diagnostic: the district×WCC×year fixed effects absorb nearly all within-cell drought variation, so this specification is not a test of drought amplification. The level gradient is specification-dependent; see Appendix Table A20, Panel A. FE: Crop×Year + Irrigation District×WCC×Year. Full plot-level controls. Two-way clustered SE (Crop×Year + Plot). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

6.1 The Endogeneity Problem: Neighbors Fallow Together

Moving to a panel specification with plot fixed effects requires a time-varying measure of upstream pressure. The natural candidate is the log count of active upstream plots, which changes year to year as plots enter or exit fallow. But this creates an endogeneity problem. When a nearby upstream plot goes fallow, two things happen at once. (1) The downstream plot's measured upstream competition decreases. (2) The same local shock that caused the neighbor to fallow (drought, pest pressure, market conditions) likely affects the downstream plot as well. The observed correlation between upstream cropped acreage and downstream outcomes confounds directional water competition with shared local shocks.

The concern is concrete. Panel B of Figure 3 shows fallow status spatially correlated far above random chance, persisting beyond 30 km in drought years, which is why the head-to-tail gap of Section 5.7 cannot be read as a water-competition effect on its own. Fallowing reflects shared local conditions, so nearby fallow changes respond to the same local conditions as the downstream plot.

Distance alone does not capture that clustering. Among plots with both neighbor types inside 1 km, a neighbor sharing the same canal-block label predicts own fallowing at 0.679^{***} (SE 0.012), compared with 0.131^{***} (SE 0.006) for an equally distant neighbor without that label. The comparison holds the distance band, year and plot fixed effects, and weather controls constant (Appendix Table A3). The contrast is a same-block local association: it can reflect shared operations, programs, classification, or other block-level conditions and does not establish a verified hydraulic link. Fallowing is nonetheless symmetric in direction: upstream and downstream fallow shares predict own fallow status with comparable magnitude (Appendix Table A2). So the design treats near fallowing as a local-shock control and reserves the directional upstream margin, beyond the exclusion radius, as the regressor of interest.

Appendix Table A4 and Supplement Table S1 make the distance, direction, and inference comparisons explicit inside the mapped boundaries of junior-rights districts. They split the activation gap, adjusted for eligible upstream acreage, into four parts: the share of nearby and of distant plots in crop, each averaged over the upstream and downstream directions, and the upstream-minus-

downstream difference at each distance. They then shift the four parts backward and forward one year in separate regressions on common support. The same-year association is concentrated in nearby cropped acreage averaged over both directions; the distant parts are indistinguishable from zero, the future terms are generally imprecise under baseline, district, and WCC inference, and the one-year-lag nearby estimates are weaker and clustering-sensitive. This diagnostic motivates the 5 km exclusion as a way to weaken the symmetric tendency of nearby fields to move together; it does not identify a causal water-flow or planting-response mechanism.

Fallow correlation persists past 30 km, beyond the widest exclusion radius the design uses (10 km). Identification rests on residual rather than raw correlation: crop × year fixed effects absorb statewide drought-year shocks and each plot's own SPEI controls absorb local meteorological conditions, so what must decay at long range is the residual outcome-relevant correlation. A shared canal- or district-level allocation shock that reduces both upstream cropped acreage and the focal outcome would bias a negative interaction *toward zero*, an argument that covers that class of shocks only. The descendant comparison of Section 6.3 is the direct test for whatever symmetric correlation survives, and the estimates plateau at 15 and 20 km (Appendix A10), consistent with contamination fading with distance, though short of establishing that all confounding is gone.

Not all fallow is equally problematic. A classifier trained on 8-year crop-sequence patterns splits it into *non-rotation fallow*, which reflects water constraints or economic stress, and *rotation fallow*, which is planned; the middle band is omitted from both classifier-based shares (Appendix S8). In the baseline Table A22 specification, these two spatial controls separate the channels. The later timing/incidence specifications use the broader CDL fallow share described below.

6.2 Distant Upstream Cropped Acreage and the Exclusion Radius

The baseline panel design uses distant upstream cropped acreage as a proxy for network load or potential diversion pressure, not as observed water throughput. For plot i in year t, let $\mathcal{A}_i$ denote the set of plots upstream of i in the directed canal graph. For a geographic cutoff d, define

$$U_{it}^{>d} = \ln\left(1 + \sum_{j \in \mathcal{A}_i} \mathbf{1}\{Active_{jt} = 1\}\mathbf{1}\{Dist(i,j) > d\}\right), \tag{4}$$

where $Dist(i,j)$ is Euclidean centroid distance. The variable $U_{it}^{>d}$ differs from static network position: it is a time-varying count of cropped upstream plots located more than d kilometers from plot i, and it changes from year to year as upstream plots enter or exit fallow; Appendix A15 defines the acreage-weighted versions. Each cutoff redefines the eligible ancestor set, its total eligible acreage, and the supported plot population; each column therefore uses a different set of ancestor plots, so the columns are not steps along a single distance gradient.

Nearby upstream cropped acreage is excluded because those plots' active/fallow decisions are most likely to reflect the same local shocks affecting plot i; I control for those shocks with fallow shares within radius d (Figure 4).

The specification is:

$$\begin{aligned} Y_{it} = {} & b_1 U_{it}^{>d} + b_2\big(U_{it}^{>d} \times \text{Drought}_{it}^{up}\big) + b_3 \text{Drought}_{it}^{up} \\ & + \theta_{nr} \cdot \text{Fallow}_{it}^{nr,\leq d} + \theta_{rot} \cdot \text{Fallow}_{it}^{rot,\leq d} + X_{it}'\pi + \alpha_i + \gamma_{\text{crop},t} + e_{it} \end{aligned} \tag{5}$$

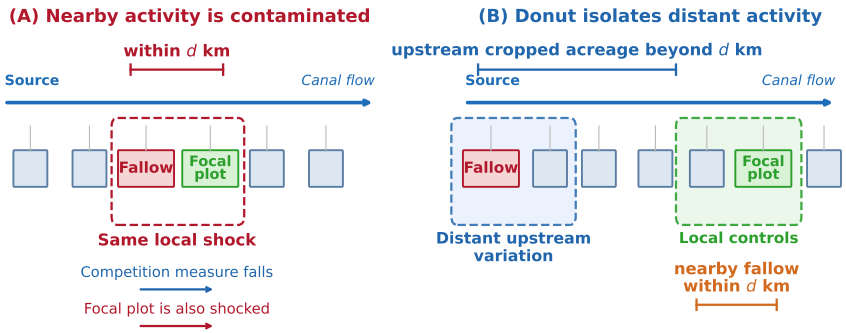


Figure 4: Distant upstream cropped acreage and local-shock controls. The design excludes upstream cropped acreage within cutoff d from the treatment, retains upstream cropped acreage beyond d, and controls for nearby fallowing in geographic space.

Here $\text{Fallow}_{it}^{nr,\leq d}$ and $\text{Fallow}_{it}^{rot,\leq d}$ are the two classifier-based shares used by the Table A22 specification. They are computed in geographic space, regardless of canal direction, because the shocks they absorb (weather, pests, salinity, and market conditions) are spatially correlated rather than carried by the water. By contrast, $U_{it}^{>d}$ is computed only over plots upstream of plot i, because the water-competition channel runs from upstream to downstream through the canal network. The later priority-and-geography crop models of Section 7.2.1 use one cutoff-matched broad CDL fallow share; their own-fallowing models omit a spatial fallow control because fallowing is the dependent variable. Drought_{it}^{up} is the upstream drought indicator of Section 4.2; like $U_{it}^{>d}$, it varies across plots within the same canal network and year, so its level term b_3 survives the fixed effects and is estimated, though suppressed from the reported tables. α_i denotes plot fixed effects, absorbing all time-invariant characteristics (soil, elevation, baseline productivity, farmer type). $\gamma_{\text{crop},t}$ denotes crop $\times$ year fixed effects. NDMI, GCVI, and ET all enter as $\ln(1+x)$; standard errors are clustered two-way by crop-year and plot unless stated otherwise. The estimation sample at cutoff d retains plot-years with at least one upstream neighbor beyond d, dropping rather than zero-coding the rest.

A causal reading would require that, conditional on plot fixed effects, crop-by-year shocks, the controls, and nearby fallow shares, upstream cropped acreage and its drought interaction reach the focal plot only through the canal. The paper tests two implications of that condition rather than establishing it: the interaction should strengthen as nearby ancestors are excluded, and it should appear for upstream but not downstream cropped acreage. Appendix Table A10 also shows a lead that rules out strict causal timing.

The coefficient b_1 is the association between distant upstream cropped acreage and downstream outcomes outside drought years; b_2 tests whether that association shifts in sustained-drought years. A negative b_2 says the association is more negative in drought years than outside them: a shift in the slope, not a drought-year level, for which $b_1 + b_2$ is the relevant quantity (Appendix A20). It supports a directional interpretation rather than a claim of random assignment.

Two features qualify that reading. First, both regressors are built from the same upstream *active* set, so the interaction captures upstream pressure exerted while the upstream path is itself surface-water-scarce, rather than the product of two independently randomized shocks. Appendix Table A25 re-estimates it with drought measured at the plot centroid and at the district-year level

(neither of which conditions on the active set), and the far-cutoff GCVI interaction survives both.[18]

Second, equation (5) is estimated on active plot-years only, and scarcity can push a plot into fallow and out of the GCVI sample. The balanced-panel check shows that exit does not generate the pooled coefficient, so remaining in production is a condition of the sample. The outcome-year fallowing profile (Appendix A22, Supplement Table S2) uses a May–August indicator that cannot establish a pre-season response, and the all-plot comparison around the announced CAP cut in Section 7.3 is descriptive.

The model of Section 3 clarifies the economic object these coefficients approximate. Bringing a fallow upstream plot back into production can raise potential demand through both its entitlement and any implementation wedge, shrinking the focal plot's cushion. The CDL indicator observes whether a plot is in crop rather than what it diverts, so $U_{it}^{>d}$ proxies mapped upstream demand, not water volume or the residual supply reaching the focal plot. What b_1 and b_2 deliver is therefore a reduced-form conditional association rather than an estimate of the structural crop-water parameter β, and the design is not an instrumental-variables strategy for the shortfall share (q_{it}/A_{it}^0).[19]

Two clarifications follow. Districts divert at the headgate against orders, so in normal years water left unordered by a fallowed upstream plot need not flow down-canal. The fixed-supply rivalry logic applies when district allocations bind, precisely the drought years that identify b_2, and scheduling congestion works the same way otherwise. And realized deliveries, irrigation intensity, and crop switching are mediators of the water channel rather than omitted controls. Conditioning on them would block part of what the design is built to detect, and the crop × year fixed effects are a conditioning choice in the same sense. Because $U_{it}^{>d}$ counts active plots beyond d directly, no separate distant fallow-share control is included.

I estimate the specification for the full integer grid $d \in \{0, 1, 2, \ldots, 10\}$ km. If the estimates are driven primarily by spatially correlated local shocks, excluding nearby variation should weaken or destabilize the coefficients.

Inference is where the far cutoff is thinnest. Plots sharing upstream ancestors within a canal network can have correlated $U_{it}^{>d}$ in a year, so the effective variation is closer to network-year than plot-year. Appendix Table A21 re-estimates Table A22 clustering by WCC, canal block, and WCC-by-year: the >10 km GCVI interaction stays significant at 5% under WCC clustering ($p = 0.045$) and is marginal under canal-block clustering ($p = 0.067$), on 114 and 81 clusters. At nearer cutoffs the network-clustered interaction is -0.015 to -0.017 with standard errors of 0.010–0.014, so under network clustering significance is confined to the >10 km GCVI cell. That cell is informative rather than decisive; the argument rests on the pattern across cutoffs, outcomes, and designs rather than on any single p-value.

6.3 The Descendant Exposure: A Directional Negative Control

Two validation tests evaluate whether the specification captures directional water competition rather than symmetric spatial shocks.

Fallow shares symmetry test. If fallow spatial correlation reflects directional water com-

[18] -0.023 ($p = 0.002$) and -0.016 ($p = 0.027$) respectively.

[19] The model writes the crop link in terms of the shortfall share q_{it}/A_{it}^0. The regressor is $\ln(1 + \text{count})$; the log-count specification is an empirical convention rather than a functional form derived from the model.

petition (activity above a plot reducing what remains for it), upstream fallow should predict a plot's own fallow status more strongly than downstream fallow. If instead it reflects symmetric local shocks (shared weather or market conditions), upstream and downstream fallow shares should have similar predictive power. Upstream and downstream non-rotation fallow coefficients are of comparable magnitude (Appendix Table A2), consistent with fallow clustering being driven by local shocks rather than directional water competition. That is why near fallow variation is controlled for rather than used as the regressor of interest.

Downstream cropped acreage as a directional negative control. I construct a mirror measure, $D_{it}^{>d}$, equal to the log count of active downstream plots beyond distance d, and include both upstream and downstream cropped acreage interactions with drought in the same regression. If the identifying variation reflects water competition, the upstream interaction should remain negative and the downstream interaction should fade: by the directional-asymmetry identity of Section 3.2, downstream diversions cannot enter plot i's residual supply through the physical delivery channel. Because downstream cropped acreage can still correlate with outcomes through the local-shock channel, the prediction is a directional contrast rather than a strict downstream null. What would contradict the mechanism is a persistent, same-signed downstream interaction that *strengthens* with distance, mirroring the upstream profile. The strengthening clause rests on an empirical premise outside the model, that local-shock correlation decays with distance; Section 7.1 confronts it with the estimates. If the pattern instead reflects broad spatial shocks, upstream and downstream coefficients should be similar. This is a negative control rather than a placebo in the strict sense. Downstream cropped acreage has no guaranteed zero coefficient, since it responds to the same supply, expectations, and district management as upstream cropped acreage, and the comparison bounds shocks symmetric with respect to graph orientation rather than everything that could confound the upstream estimate. Section 7.1 reports the estimates; the ET version of the joint regression is weaker than the GCVI one, so the directional evidence is strongest for GCVI, the yield proxy, rather than uniformly sharp across outcomes.

7 Results

This section reports the results in the order of the argument. Section 7.1 establishes the statewide drought-year shift, compares upstream and downstream exposure, and checks robustness to how the graph, drought, and sample are built. Section 7.2 establishes the junior-rights setting, separates inherited upstream exposure from changing cropping, and places the association alongside the drought and delivery calendars. Section 7.3 turns to the four CAP districts, whose CAP deliveries fell from the late summer of 2021 and were cut in 2022, and asks whether a field's water use depended on the land that takes water before it, and compares them with the Upper Gila canals, which run short in spring. The junior-district decomposition treats 5 km as primary because it retains substantially more supported plots; the directional diagnostic uses 10 km because its purpose is to separate distant ancestor cropped acreage from the descendant mirror. These cutoffs serve different design tasks rather than imposing one universal radius.

Throughout, the reported quantity is the drought-year shift in the slope, the interaction b_2 in Equation (5), so a negative interaction does not by itself imply a negative slope in drought years; Appendix A20 reports the total drought-year slopes $b_1 + b_2$ separately. The static head-to-tail pattern of Section 5.7 is descriptive motivation and is not re-estimated here, because plot fixed effects absorb the main effect of a time-invariant regressor. Table 4 collects the central statewide

estimates with network-level inference.

Table 4: Statewide Drought Amplification and Canal Direction

Panel A. Upstream exposure: selected exclusion cutoffs			
	Full path	>5 km	>10 km
Outside drought	+0.005 [0.749]	−0.004 [0.857]	+0.013 [0.635]
Change in drought	−0.015 [0.156]	−0.017 [0.191]	−0.035 [0.045]
Total in drought	−0.010 [0.631]	−0.021 [0.477]	−0.022 [0.553]
Field-years	326,732	137,229	75,826
Networks	1,812	355	114
Panel B. Direction: joint regression beyond 10 km			
	Outside drought	Change in drought	Total in drought
Upstream	−0.016 [0.678]	−0.030 [0.026]	−0.046 [0.316]
Downstream	−0.069 [0.069]	+0.002 [0.871]	−0.067 [0.117]
Upstream − downstream	—	−0.032 [0.075]	+0.021 [0.755]
Field-years; networks		21,396; 34	

Notes: Active fields statewide; outcome ln(1 + GCVI). Exposure is ln(1 + number of cropped fields) beyond the stated distance. Entries are coefficients; brackets give two-sided p-values clustered by connected canal network. Models include field and crop-by-year fixed effects. *Change* is the interaction with the upstream 24-month drought indicator; *total* adds this interaction to the outside-drought slope. Panel B includes both exposures jointly. Samples change across cutoffs and panels. Full cutoff and standard-error results remain in the appendix.

7.1 Statewide Evidence of Directional Drought Amplification

Drought shifts the statewide association between upstream cropping and a field's crop performance downward. Two comparisons establish the pattern: it survives the exclusion of nearby fields, and it differs between land upstream and downstream of the field. Throughout, "amplification" refers to the change in the slope between drought and other years. Table 4, Panels A and B, reports this change alongside the outside-drought and total drought-year slopes.

The upstream GCVI interaction is negative at every exclusion radius, moving from −0.015 over the full upstream path to −0.035 beyond 10 km (Panel A). Excluding nearby fields removes the neighbors most likely to share local shocks while retaining canal-connected fields. Within large canal systems, the >10 km estimate is −0.037, close to the pooled estimate (Appendix Table A8). Wider cutoffs change the sample and concentrate it in large systems, so the increasingly negative estimates describe a profile rather than a tested trend. The upstream-only profiles are retained in Appendix Figure A1.

The joint regression asks whether the drought-related change follows canal direction. Beyond 10 km, the upstream and downstream GCVI interactions are −0.030 and 0.002, giving a contrast of −0.032 (Panel B). Figure 5 shows both sides and their difference across cutoffs. Nearby downstream interactions are also negative, but fade as nearby fields are excluded while upstream interactions remain negative. The distant comparison uses 21,396 plot-years across 34 network components; requiring active fields on both sides preserves the contrast (Appendix Table A7).

The distant GCVI contrast has $p = 0.026$ with field and crop-year clustering, as displayed in Figure 5. With network-component and canal-block clustering, the corresponding values are 0.075

and 0.106 (Appendix Table A21). These estimates concern changes in slopes: the total drought-year directional gap is imprecise and does not establish a negative level difference (Panel B). GCVI supplies the strongest directional evidence. The ET contrast is significant only at intermediate cutoffs, and the NDMI contrast nowhere. Seasonal ET combines months of low and high demand; Section 7.3.2 examines monthly ET and crop demand inside the CAP districts.

The statewide pattern survives changes in its construction and sample. Canal-structure links alone retain a negative GCVI interaction, and drought measures built without the active upstream set retain the distant interaction (Appendix Tables S5 and A25). Fields farmed in every year give similar or larger negative estimates (Appendix A23). Across six specification variants, the >1 km interaction remains negative and significant for GCVI in all six and ET in four (Appendix Table A20).

The contrast identifies a directional pattern without identifying the actor: operators rationing or scheduling deliveries in flow order can generate the same divergence as decentralized upstream competition. Together, the statewide results are consistent with delivery conditions making upstream cropped land more consequential during drought. They can combine persistent exposure differences with annual changes; Section 7.2 separates these components inside junior-rights districts.

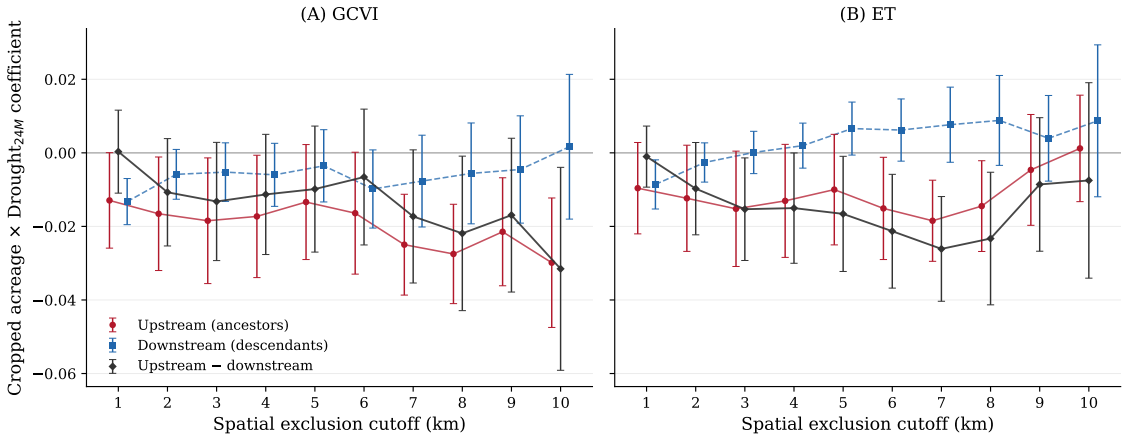


Figure 5: Upstream vs. downstream cropped acreage in the joint regression across spatial exclusion cutoffs, the continuous version of Table A23. Both upstream and downstream cropped acreage × drought enter the same regression, so each coefficient is conditional on the other. Red circles show upstream ancestors and blue squares downstream descendants; dark diamonds show the upstream-minus-downstream difference, estimated as a single coefficient by reparameterizing the joint regression in terms of the sum and difference of the two exposure counts. This is an algebraically exact transform with identical sample and fit, so the difference carries its own two-way clustered standard error. Bars are 95% confidence intervals (Crop×Year + Plot); the three series are offset horizontally so their intervals do not overlap. Each cutoff is a separate regression on its own nested sample, which shrinks from 218,597 plot-years at 1 km to 21,396 at 10 km. Panels share a vertical axis.

7.2 Junior Rights, Upstream Exposure, and Delivery Shortages

The statewide results establish an association along the canal network. The question about formal allocation requires looking inside the districts that distribute water to fields. Junior-rights districts provide that setting: their source claims are curtailed early, but district rules still govern delivery

below the headgate. This section establishes the relevant population, separates inherited upstream exposure from changing cropping, and examines when the association appears. These comparisons prepare the monthly analysis of delivery shortages.

7.2.1 Why Junior-Rights Districts?

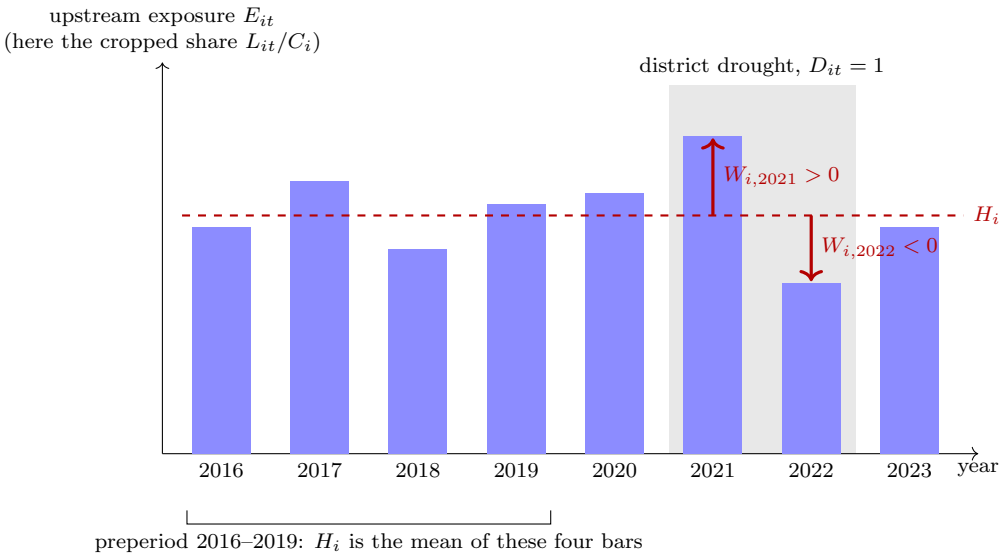
A cut at the source does not describe how its burden is distributed within a district. Even when a district reduces farmers' allocations in equal proportion, fields share a branching delivery network, and groundwater and delivery timing can differ. The relevant question is whether those rules leave a field's crop performance dependent on the demand of land that receives water before it.

The adverse upstream association appears inside mapped junior-rights district boundaries, rather than only outside organized systems. At 5 km, the GCVI interaction is −0.028 inside junior districts and −0.006 inside senior districts, holding eligible upstream acreage fixed (Appendix Table A11). The senior estimate is imprecise and there is no direct junior–senior equality test. Boundaries proxy for service geography; this comparison establishes the setting, rather than an effect of legal priority.

The ten surface-connected junior-rights districts comprise the four CAP districts—Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala—Cibola Valley on the Colorado River, and five Upper Gila canal companies: Fort Thomas, Graham, Montezuma, San Jose, and Union. The four CAP districts are a subset, whose recorded monthly deliveries allow the closer within-season test in Section 7.3.

7.2.2 Upstream Exposure and Changing Cropping

A field's position is fixed, but the cropped acreage connected above it changes each year. The decomposition uses 22,105 active field-year observations inside these ten districts, with ancestors more than 5 km away. For exposure E_{it}, measured as logged upstream cropped acreage or the cropped share of eligible upstream land, H_i is the field's 2016–2019 mean and $W_{it} = E_{it} - H_i$ its annual departure. Plot effects absorb H_i itself, but not its interaction with drought. Annual departures can contain lasting post-2019 shifts as well as one-year changes. Figure 6 defines the construction.



Symbols (focal plot i, year t; canal ancestors more than 5 km away)

L_{it}	upstream cropped acreage: the acreage of those ancestors in crop in year t
C_i	eligible upstream acreage: the acreage of all of them; fixed over time
E_{it}	upstream exposure, on one of two scales: the cropped share L_{it}/C_i or $\log(1 + L_{it})$
H_i	preperiod baseline: the 2016–2019 mean of E_{it}; differs across plots, constant over time
W_{it}	$E_{it} - H_i$, the preperiod-centered residual; moves over time within a plot
D_{it}	district drought indicator: one when the district's 24-month SPEI is below -0.6

Estimating equation (Table 5):

$$y_{it} = \beta_H (H_i \times D_{it}) + \beta_W (W_{it} \times D_{it}) + \beta_1 W_{it} + \beta_D D_{it} + X'_{it}\pi + \alpha_i + \gamma_{\text{crop},t} + e_{it},$$

where the plot effect α_i absorbs H_i itself. β_H is the drought-year association with the exposure a plot inherits; β_W is the drought-year shift in the slope on annual departures from it.

Figure 6: Construction of the junior-district exposure decomposition. The bars are a stylized plot's upstream exposure by year; the dashed line is its 2016–2019 mean, the preperiod baseline H_i; the arrows are the preperiod-centered residuals W_{it} in two drought years. The box defines each symbol and the equation estimated in Table 5.

Table 5 enters both components interacted with district drought. One between-field standard deviation of baseline share has a −0.034 drought-related shift in ln(1 + GCVI); this persistent between-field association is less precise under district or WCC clustering. A ten-percentage-point annual rise above baseline has a −0.014 drought-related slope shift (two-way $p = 0.033$; district wild-bootstrap $p = 0.039$), with an approximately zero slope outside drought. These units differ and the coefficients should not be added. Both estimates condition on the focal field remaining in production.

Table 5: Drought-Revealed Spatial Baselines and Preperiod-Centered Upstream Activation in Junior-Rights Districts

Exposure component	Log active	Activation share
Panel A: Exposure components × drought		
Preperiod 2016–2019 spatial baseline × drought	−0.028	−0.034***
	(0.017)	(0.013)
Preperiod-centered activation residual × drought	−0.013	−0.014**
	(0.009)	(0.007)
Panel B: Preperiod-centered residual slope		
Outside drought	+0.001	−0.000
	(0.004)	(0.004)
During drought	−0.013	−0.014**
	(0.008)	(0.006)
Equal baseline and residual interactions: p	0.362	0.191
Observations	22,105	

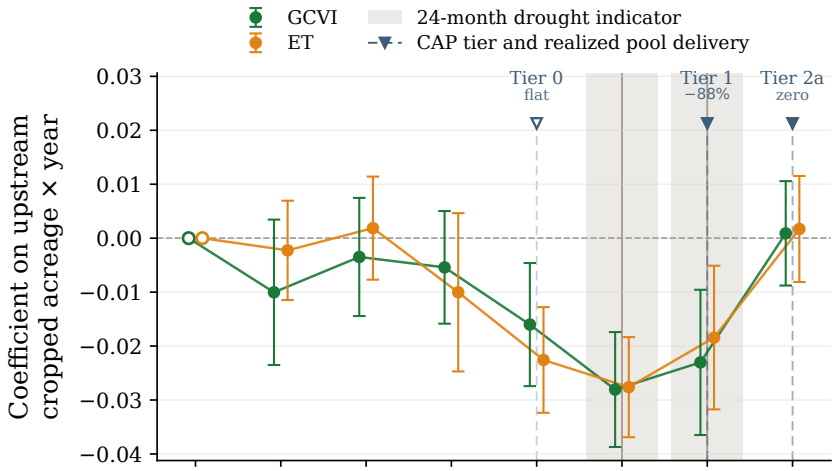
Notes: Active focal plots inside mapped boundaries of the ten surface-connected junior-rights districts; ancestors are connected through the directed transitive closure and more than 5 km away in Euclidean distance. L_{it} is active upstream acreage and C_i is fixed eligible upstream acreage. For each exposure scale, H_i is its plot-specific 2016–2019 mean and $W_{it} = E_{it} - H_i$ is the preperiod-centered activation residual. Both $H_i \times D_{it}$ and $W_{it} \times D_{it}$ enter jointly with the residual and drought main effects, plot and Crop×Year fixed effects, the cutoff-matched broad-fallow control, and the paper's control set; the level of H_i is absorbed by plot fixed effects. The drought indicator equals one when the acreage-weighted SPEI-24 mean over the fixed set of all plots mapped inside the district is below −0.6; Appendix Table A13 adds the columns in which a continuous scarcity index replaces it. Displayed standard errors and two-sided stars use the paper's baseline two-way clustering by plot and Crop×Year. Under the activation share, the residual interaction has $p = 0.025$ with district clustering and $p = 0.039$ under the exact district wild-cluster Rademacher test (versus $p = 0.033$ displayed); its total drought-year slope has district-clustered $p = 0.073$ (versus $p = 0.012$ displayed). The baseline is per one between-plot SD (13.6 percentage points for the share); logged residual exposure is per one full-sample within-plot SD and the share residual is per 0.10. Component magnitudes use different scales and should not be added; equality tests use native common units. The exposure measures upstream cropped acreage, not diversion, delivery, or water flow. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The annual interaction stays negative with trends and newly linked rights and well controls, but weakens with legacy-rights controls. It is near zero under district-by-year effects and continuous scarcity; the matched 5 km directional comparison is inconclusive. Appendices A11 and A13 report the checks and conditional revenue equivalents. The decomposition is descriptive.

7.2.3 Drought and the Timing of Delivery Constraints

The yearly pattern does not follow the drought indicator mechanically. Figure 7 returns to the statewide >5 km specification, with upstream interactions relative to 2016. For GCVI and ET, these are indistinguishable from zero in 2017–2019, negative and significant in 2020–2022, and back at zero in 2023. The 2020 interaction is already negative while the 24-month drought indicator is almost entirely off. The year-specific interaction returns to zero in 2023 although the CAP pool

receives no water. Declared shortage tiers therefore do not mechanically switch the relationship on; the monthly CAP evidence for 2023 is mixed (Section 7.3.10).



Drought (%)	2016	2017	2018	2019	2020	2021	2022	2023
24-month	1	10	8	14	0	**46**	**42**	0
3-month	0	2	30	0	44	70	72	28

Figure 7: Year-specific upstream cropping interactions. GCVI and ET coefficients are relative to 2016, with 95% confidence intervals, using the statewide $d > 5$ km specification of Table A22; two-way clustered SEs (Crop×Year + Plot), $N = 137{,}229$. Shading marks years when the 24-month upstream drought indicator is materially on. Rows below the axis give the percentage of active field-years flagged by the 24- and 3-month indices at the same −0.6 cutoff; the 3-month index is descriptive. Dashed markers show shortage tiers; open markers indicate a declaration without a realized CAP Agricultural Settlement Pool cut. Appendix Table S16 gives the delivery record.

A separately constructed 2008–2023 panel provides the historical comparison. During the meteorologically harsher 2010–2014 drought, major delivery schedules largely held and no comparable negative upstream interaction appears. Negative interactions emerge in the later era. The eras differ in field vintage and each contains one principal drought episode; the extension has no directional comparison or field-level deliveries (Appendix S17). Appendix A12 reports the separate early-ET check.

The four CAP districts allow a closer test. With neighbors' planting set before irrigation, Section 7.3 combines recorded monthly deliveries, fixed crop-demand calendars, and flow direction. It asks whether upstream demand becomes more consequential than downstream demand when deliveries are low and the focal crop needs water.

7.3 Competition for Canal Water in the CAP Districts

The timing evidence says the upstream relationship follows delivery conditions rather than rainfall. Four of the six districts that draw on the Central Arizona Project's Agricultural Settlement Pool give the sharpest test of that reading. Their CAP deliveries fell in steps that are recorded by month, and the shortfall fell on fields whose neighbors had already decided how much land to farm that year. The district drought indicator of the junior-district estimates misses the 2022 cut in three of these districts (Section A13), so those estimates draw little on 2022 there. The design

below uses the recorded monthly cut. This subsection asks whether, inside these districts, a field's water use depended on the land that takes canal water before it, and what produces the pattern. It ends by comparing them with the Upper Gila canals, which run short at another time of year.

7.3.1 The CAP Shortage

The 2022 Tier 1 cut is the sharpest removal of surface supply in the sample. In 2022 the Agricultural Settlement Pool districts lost about two-thirds of their CAP water even after replacement water, and in 2023 the pool received none. CAP deliveries fell in steps (Figure 8, panel a). In 2021, before the pool was cut, CAP deliveries to the four districts fell to 40 to 60 percent of normal from July on, in a late summer of heavy rain: 149 mm fell at Coolidge in July and August (Arizona Meteorological Network, n.d.). In 2022 they were 15 to 40 percent of normal in every month, and in 2023 they were near zero. Two of the districts, Central Arizona and Maricopa–Stanfield, also pump district groundwater into their canals (Irrigation Leader, n.d.). In Hohokam and Harquahala, farms pump their own wells.

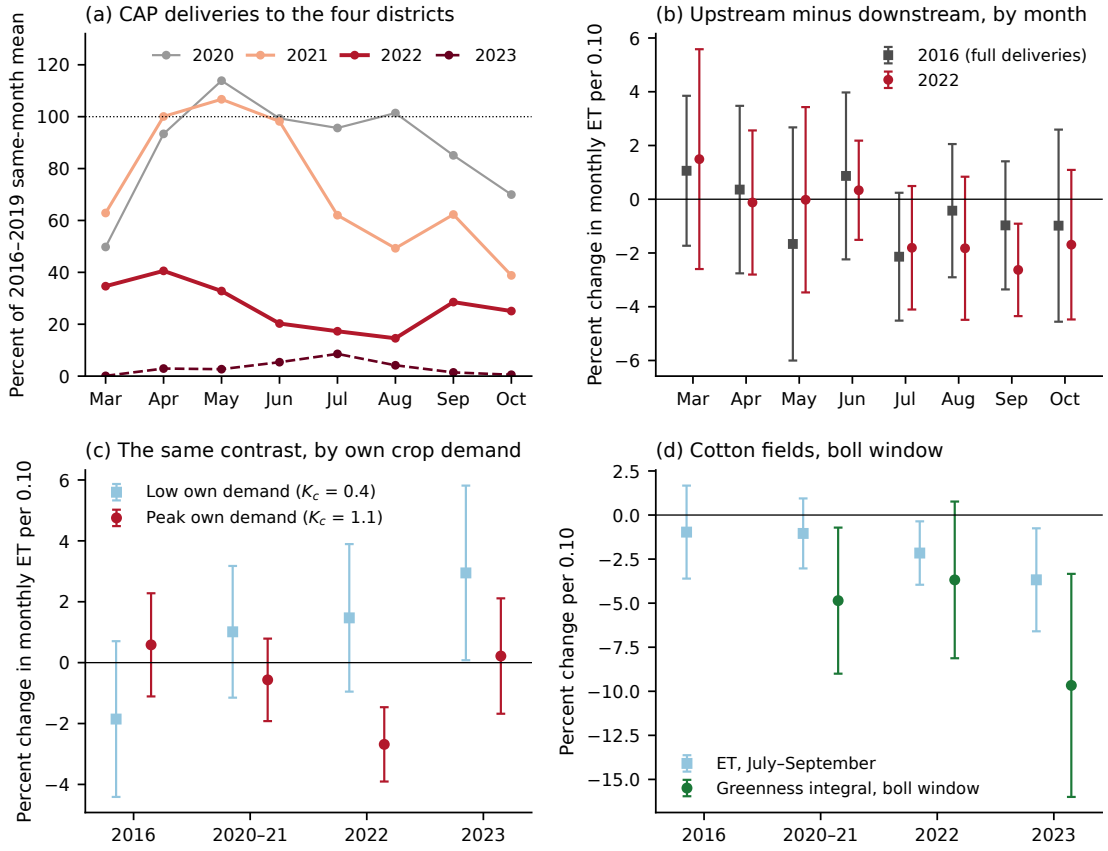
Farmers decide early in the year, before irrigation starts, whether to plant a field (Appendix A30), and many did not. Relative to comparison districts, fallowing rises from 2019 to 2022 by 13.3 percentage points under the nearest-district assignment and 18.9 points inside the mapped boundaries, and average satellite outcomes fall. Pre-trend and comparison-pool sensitivity make these level changes descriptive (Appendix A28). The pattern is consistent with adjustment through fallowing and pumping, as after earlier CAP cuts (Harris 2024) and in recent Maricopa-Stanfield shortages, where groundwater reliance is inferred from a water-balance deficit rather than measured pumping (Sharma 2025). In a balanced panel of monitoring wells, depth to groundwater in the districts that lost CAP water rose 8.33 feet more than in comparison districts by 2023 ($p = 0.008$). Section 7.3.9 shows which fields stopped farming. The rest of this subsection follows the fields that kept farming. Their neighbors' planting was set before the season, so the question is how the water that reached them during the season depended on it.

7.3.2 Crop Water Demand Inside the CAP Districts

If upstream farmed land matters because fields share canal water, its effect should be largest when the canal is short and the crops along it need the most water. Crop water need follows the crop's growth stage, not only the weather. I measure it with the monthly crop coefficient (K_c) of the FAO-56 method. The stage lengths and coefficients are those FAO-56 gives for the closest arid regions, and each crop's calendar is anchored on its 2017–2019 timing in its district. These calendars track normal-year ET closely, with correlations of 0.82 to 0.97 across crops. The exposure is the crop water demand of the land linked to a field through its own district's canal network: the farmed share of that land, with each farmed field weighted by the crop coefficient of its crop in that month, as a departure from the field's 2016–2019 mean for the same month. It is built separately for linked land upstream and for linked land downstream of the field. Both leave out land within a distance cut of the field. The cut is set from district size, at 5 percent of the square root of each district's area. It is about 1 km in Central Arizona and Maricopa–Stanfield, 0.5 km in Hohokam, and 0.8 km in Harquahala. The sample is the farmed field-years with a calendar crop of every field inside the four districts' boundaries that has land both upstream and downstream beyond the cut, 4,641 fields. Fields near the head of a canal have few linked fields upstream, so one neighbor's change can move their share across its whole range; each monthly departure is therefore capped at ± 0.5, a

choice made after the first all-field estimates showed the problem; the estimate with the departure as measured is reported beside it. A uniform 5 km cut would keep only 1,099 of these fields, and it would drop most fields near the tail of each network.

Figure 8: Competition for Canal Water Inside the CAP Districts



Notes: Panel (a): CAP deliveries to Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala, summed by month, as a percent of the 2016–2019 same-month mean. Panels (b) to (d): upstream minus downstream estimates per ten points of the crop water demand of linked land, with 95 percent intervals from two-way clustering by field and crop-year, for the 4,641 CAP fields inside the district boundaries; periods are measured against 2017–2019. Panel (b): monthly ET, slope by month. Panel (c): monthly ET at the field's own crop coefficient. Panel (d): the cotton fields among them (2,817 with ET, 2,717 with the greenness integral), ET over July–September and the Sentinel-2 greenness integral over July 15 to September 30, with the exposure averaged over July–September; the integral starts in 2017.

Figure 8 and Table 6 report the contrast between upstream and downstream demand, per ten points of the exposure, with 2017–2019 as the base. In 2022 the contrast follows the shortage through the season. It is close to zero from March through June and negative from July on, precisely so in September and over August–October. Ten points more demand upstream lower monthly ET by 2.1 percent more than the same demand downstream over August–October (95 percent interval −3.8 to −0.3), and by 2.6 percent in September (interval −4.3 to −0.9). In 2016, when deliveries were full, there is no late-season dip (panel b). The contrast also follows the field's own demand. At a crop coefficient of 1.1 it is −2.7 percent in 2022 (interval −3.9 to −1.5) and −1.6 percent over 2020–2022 (interval −2.8 to −0.5). At a crop coefficient of 0.4 it is +1.5 percent in 2022 (interval −1.0 to +3.9). In 2016 it is +0.6 percent and in 2023 +0.2 percent (panel c). Bootstrap p-values by district-year are below 0.05 for the 2022 ET contrasts; by canal block, where one block holds two-thirds of the fields, the same contrasts are marginal (table notes).

Table 6: Upstream Minus Downstream Crop Water Demand and Field Outcomes in the CAP Districts

	2016	2020–21	2022	2020–22	2023
1. Monthly ET at peak crop demand ($K_c = 1.1$)	0.58 (0.85)	−0.57 (0.68)	−2.68 (0.61)	−1.63 (0.57)	0.22 (0.95)
2. Monthly ET at low crop demand ($K_c = 0.4$)	−1.85 (1.28)	1.01 (1.09)	1.47 (1.22)	1.24 (0.94)	2.95 (1.44)
3. Monthly ET, August–October	−0.79 (1.15)	−1.28 (0.82)	−2.05 (0.87)	−1.67 (0.72)	0.27 (1.19)
4. Cotton: ET, July–September	−0.97 (1.12)	−1.04 (0.84)	−2.16 (0.76)	−1.60 (0.64)	−3.68 (1.24)
5. Cotton: greenness, boll window	–	−4.86 (1.69)	−3.68 (1.82)	−4.27 (1.49)	−9.67 (2.59)

Notes: Each cell gives the upstream minus downstream estimate per ten points of the crop water demand of linked land (farmed share weighted by crop coefficients, departure from the 2016–2019 same-month mean, capped at ±0.5), with the two-way standard error (field and crop-year) in parentheses. Rows 1 to 3: percent change in monthly ET, all 4,641 CAP fields inside the district boundaries, with field by month and crop by year by month effects and month-specific controls. Rows 4 and 5: cotton fields (2,817 with ET, 2,717 with the greenness integral), exposure averaged over July–September, with field and crop by year effects; the greenness integral is from Sentinel-2 five-day composites over July 15 to September 30 and starts in 2017. Periods are measured against 2017–2019; 2020–22 is the average of the 2020–21 and 2022 estimates. Restricted wild cluster bootstrap p-values by district-year (Webb weights, 9,999 draws): row 1: 2016 0.727, 2022 0.017, 2020–22 0.041; row 3: 2022 0.025; row 4: 2022 0.015, 2020–22 0.006; row 5: 2022 0.206, 2020–22 0.012. The same p-values by canal block (64 clusters): row 1: 2016 0.507, 2022 0.121, 2020–22 0.140; row 3: 2022 0.144; by canal block by year (469 clusters): row 1: 2016 0.731, 2022 0.052, 2020–22 0.081; row 3: 2022 0.115. Appendix Table A36 gives the upstream and downstream estimates separately, the fields with 5 km of canal above them and the head-end fields, and the other crops.

The same timing appears a year earlier, when CAP deliveries fell from July. With 2020 and 2021 as separate periods, the contrast in 2021 is positive from March through June, +1.7 percent on average (standard error 0.8), much as in 2020, and negative from July on, most clearly from August, after the rains of July: −1.6 percent over July–October (standard error 0.8; district-year $p = 0.03$) and −2.4 percent in September (standard error 1.1), close to its 2022 values of −2.0 and −2.6 (Table 7). In 2020, with near-normal deliveries, there is no late-season dip (−0.05 over July–October). Over the whole 2021 season the contrast at peak demand is −1.2 percent (standard error 0.8), since it averages the months before and after deliveries fell. The gap opens when the canal water falls short of what the crops need, in 2021 as in 2022.

Fewer of these fields were farmed in 2022 than in 2021. 2,735 grew a crop of the calendar, against 3,529 in 2021 (2,977 against 3,732 with all crops), so the 2022 estimates rest on a smaller group. On the 2,460 fields farmed in both years, the 2022 contrast at peak demand is −3.5 percent (interval −4.8 to −2.2), and on the 1,709 fields farmed in every year from 2016 to 2022 it is −1.9 percent (interval −3.6 to −0.1). The change in which fields were farmed does not produce the 2022 gap.

Table 7: The Upstream–Downstream Contrast by Month in 2020, 2021, and 2022

	2020	2021	2022
March	4.21 (1.14)	2.09 (1.16)	1.49 (2.05)
April	0.32 (1.04)	0.70 (1.01)	−0.11 (1.35)
May	−1.57 (1.24)	3.57 (1.94)	0.10 (1.75)
June	2.29 (1.74)	0.22 (1.21)	0.34 (0.93)
July	1.06 (1.52)	−0.58 (0.84)	−1.80 (1.15)
August	0.08 (1.12)	−1.78 (1.10)	−1.81 (1.34)
September	−0.72 (1.16)	−2.41 (1.05)	−2.63 (0.86)
October	−0.61 (1.35)	−1.74 (1.69)	−1.72 (1.41)
March–June mean	1.31 (0.84)	1.65 (0.79)	0.45 (1.17)
July–October mean	−0.05 (0.84)	−1.63 (0.83)	−1.99 (0.80)

Notes: Each cell gives the slope of monthly ET on the crop water demand of linked land upstream minus the same slope downstream, in that month, as a change against 2017–2019, in percent per ten points, with the two-way standard error (field and crop-year) in parentheses. All 4,641 CAP fields; field by month and crop by year by month effects and month-specific controls; 2016, 2020, 2021, 2022, and 2023 enter as separate periods. CAP deliveries to the four districts were near normal in 2020 and fell to 40 to 60 percent of normal from July 2021.

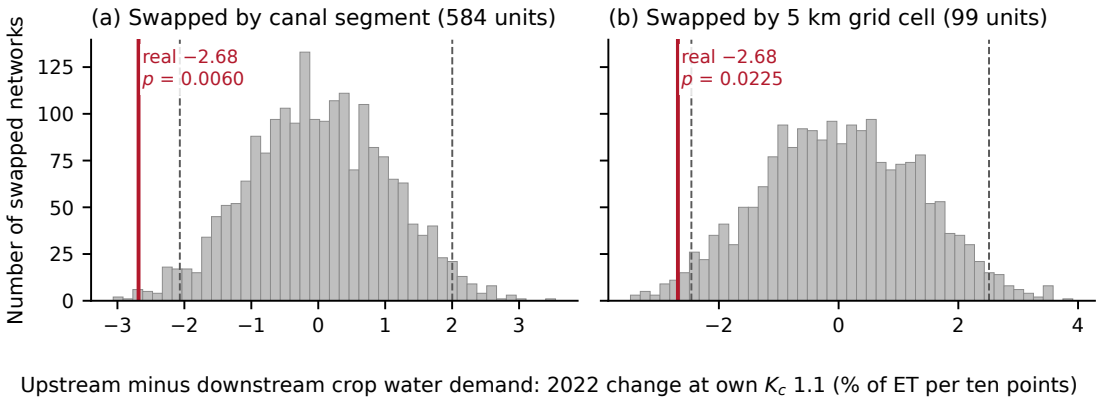
Adding canal block by year effects, and by month as well in the monthly ET model, keeps the signs of the main contrasts and takes away part of their size. On all fields the peak-demand contrast in 2022 is −1.8 percent (interval −3.3 to −0.2) and the 2020–2022 average is −0.7 percent (interval −2.2 to +0.7). The cotton contrasts in July–September ET and in the boll window keep most of their size. The upstream slope itself turns positive, so what remains comes from a larger downstream slope (Appendix Table A36). These effects also absorb any shortage that all fields on a canal block share in a month. The comparison that remains is between fields on the same block in the same month. On the 2,608 fields with 5 km of canal above them, where the design was first run with the farmed share of linked land in place of its crop water demand, the pattern is the same with K_c times reference ET in place of K_c and with a longer desert calendar for cotton. It weakens when the cut is halved or doubled, or when network links may cross district boundaries. On the same fields, holding each district's monthly crop demand fixed, a one standard deviation lower CAP delivery deepens the upstream penalty in peak-demand months by 1.4 percent (interval 0.0 to 2.8), and downstream farmed land does not respond (Appendix Table A35). CAP deliveries measure canal supply with error, because the canals also carry district-pumped groundwater that is not recorded by month.

Without the crop demand interaction, a dose in monthly CAP deliveries alone is not found. These deliveries leave out the district well water that the canals also carry, and districts ration by seasonal allotment rather than by month.

7.3.3 The Direction of Flow

If the gap came from conditions that a field shares with the land on both sides of it, such as soils, weather, pests, or a local supply of water, then calling a field's upstream land downstream and its downstream land upstream would not change it. The test does exactly this. It keeps every field's linked land as built and, in each of 1,999 draws, swaps the two sides for the fields of randomly chosen canal segments, each with probability one half, re-estimates the model, and records the 2022 contrast at peak demand. A swap leaves the sum of the two exposures unchanged and only reverses their difference, so the fake networks keep every condition the two sides share. Only 11 of the 1,999 fake networks give a t statistic as large in absolute value as the real contrast of −2.7 ($p = 0.006$; Figure 9); the middle 95 percent of the fake contrasts runs from −2.1 to +2.0. Swapping by grid cells of 5 km, which keeps larger patches of land together, gives $p = 0.0225$. The contrasts of 2016, 2020–21, and 2023 lie well inside their own fake distributions (p of 0.63, 0.49, and 0.86 by segment). In the swap test, the monthly contrasts of Table 7 follow the same pattern (Table 11). The July–October means of 2021 and 2022, when the canals were short, lie in the lower tail of the fake networks. July–October 2020 lies in the middle. Swapping by segment, p is 0.14 in 2021, 0.10 in 2022, and 0.97 in 2020; swapping by grid cell, it is 0.18, 0.17, and 0.97. The September contrast of 2022 has p of 0.07 and 0.12, and the positive March–June contrast of 2021 has 0.07 and 0.10. These monthly contrasts average over all fields, whatever their own crop's demand. In 2022 the July–October mean is −2.0 percent, against −2.7 percent at peak demand, and the fake contrasts spread about as widely for both. The test does not cover shocks that line up with the direction of flow over areas larger than a segment or a grid cell. District by year by month effects absorb such shocks at the district level, and with them the 2022 contrast is −2.5 percent.

Figure 9: Swapping Upstream and Downstream at Random



Notes: Each histogram shows the 2022 upstream minus downstream contrast at peak crop demand (own crop coefficient 1.1; percent of ET per ten points, as a change against 2017–2019) in 1,999 fake networks. In each fake network the upstream and downstream land of the fields in each unit is swapped with probability one half, which leaves the sum of the two exposures unchanged and reverses their difference, and the model is re-estimated. Units: (a) the 584 canal segments; (b) the 99 grid cells of 5 km. The red line is the real contrast; the dashed lines bound the middle 95 percent of the fake contrasts. The p-value is the share of networks, the real one included, whose t statistic, clustered by unit, is at least as large in absolute value as the real one.

7.3.4 Where the Gap Sits

The result is the contrast between the two directions. On all fields at peak demand in 2022 the upstream slope is −1.0 percent (interval −2.1 to +0.1) and the downstream slope is +1.7 percent (interval +0.7 to +2.7). Delivering water to upstream fields first produces both signs (Section 7.3.6). Among the 2,608 fields with at least 5 km of canal above them, the upstream slope alone carries it: −4.8 percent over August–October 2022 (interval −7.2 to −2.4), against a downstream slope of −0.6. Among the 2,033 fields nearer the head of a canal, both slopes are positive, +1.4 percent upstream and +2.3 percent downstream, and the contrast is −1.0 (interval −3.5 to +1.6). There, fields whose neighbors farm more do better in both directions, as fields that share local conditions would. The difference between the two directions is the part that depends on the direction of flow. Entering the two exposures as their mean and their difference separates the two parts in one regression. The slope on the mean, which carries anything common to both sides, such as how busy the canal around the field is, is +0.7 percent per ten points on all fields at peak demand in 2022 (interval −1.0 to +2.5), and the slope on the difference is the −2.7 percent above. Over August–October 2022 the mean slope is −5.4 among the fields with 5 km of canal above them (interval −9.1 to −1.8) and +3.7 among the head-end fields (interval +0.7 to +6.6), so the common part differs in sign between the two groups and is close to zero on all fields. The difference is negative in both groups, −4.2 and −1.0. With the departure as measured rather than capped, the peak-demand contrast on all fields is −2.0 percent instead of −2.7 (Appendix Table A36).

Splitting each side's linked land into rings by distance shows where the two parts sit (Table 8). In 2022 the contrast is −1.7 percent for land 1 to 2 km from the field and −1.6 percent at 2 to 5 km, and about zero within 1 km and beyond 5 km. A field's ET rises with the farmed land within 1 km in every year: in 2017–2019 the sum of the two slopes is +1.3 at peak demand and +3.1 at low demand, and this land has no 2022 contrast. This is the shared local variation the distance cut is meant to leave out. The positive downstream slope of 2022 sits at 2 to 5 km, where both sides rise together (sum +2.8) at peak demand and not at low demand. No single ring passes every

check. The 1 to 2 km contrast passes the swap test ($p = 0.021$) but not the district-year bootstrap ($p = 0.34$), and the 2 to 5 km contrast does the reverse (0.084 and 0.025). The whole set beyond the cut passes both.

Table 8: The 2022 Slopes by Distance Ring

	2022 change				$U + D$,	p for $U - D$, 2022	
Ring	U	D	$U - D$	$U + D$	2017–19	swap test	district-year
0–1 km	0.21 (0.40)	0.21 (0.53)	0.00 (0.47)	0.42 (0.81)	1.32 (0.51)	0.994	0.997
1–2 km	−1.12 (0.36)	0.62 (0.45)	−1.74 (0.57)	−0.50 (0.57)	−0.16 (0.32)	0.021	0.336
2–5 km	0.59 (0.59)	2.20 (0.44)	−1.61 (0.77)	2.79 (0.70)	−0.83 (0.44)	0.084	0.025
beyond 5 km	−0.39 (0.59)	−0.23 (0.98)	−0.16 (1.21)	−0.62 (1.07)	1.36 (0.71)	0.907	0.918

Notes: Each side's linked land is split into rings by the distance between field centroids, with no distance cut, and the four rings of both sides enter one regression; a field with no land in a ring gets zero and an indicator. All 4,641 CAP fields. Cells: percent change in monthly ET per ten points of the ring's crop water demand at peak crop demand (own crop coefficient 1.1), as the 2022 change against 2017–2019, with the two-way standard error in parentheses; the column 2017–19 gives the sum of the two slopes in 2017–2019 itself. Swap test: share of 1,000 networks (the real one and 999 fake ones, swapped by canal segment) whose t statistic is at least as large in absolute value as the real one. District-year: restricted wild cluster bootstrap with 32 district-year clusters and Webb weights, 9,999 draws.

7.3.5 District Pumps

District wells are a second source along the canals of Central Arizona and Maricopa–Stanfield. Upstream land with a district pump between it and the field takes water that the pump can replace. Upstream land with no pump in between takes water the field would otherwise get. I place on the canal network every active non-exempt well within 250 m of a mapped canal with at least 1,000 gallons per minute of registered capacity: 111 wells in Central Arizona and 144 in Maricopa–Stanfield, with 348 and 509 cubic feet per second of capacity. Each well enters the network at the nearest field intake within 500 m and serves the fields to which water can flow from it within 2 km. Each field's upstream land beyond the cut is then split into land with no serving pump between it and the field and land behind such a pump, and its downstream land is split the same way. The four parts enter one regression on the 3,444 fields of the two districts (Table 9).

In 2022 the upstream penalty sits in the land with no pump in between. Its 2022 change at peak demand is −2.2 percent per ten points (standard error 0.6; district-year p below 0.001), against −0.6 in 2016 and in 2020–21, all measured against 2017–2019. The 2022 change holds with district by year by month effects (−2.2) and with canal block by year by month effects (−2.3), in each district (−1.9 in Central Arizona and −2.6 in Maricopa–Stanfield), and with other rules for which wells count (−1.6 to −2.1). Land behind a pump shows no penalty in the two districts together: its 2022 change is +0.5 (standard error 0.8), and its slope in 2022 itself is −0.7 (standard error 0.6). In Central Arizona alone it carries a penalty as well (−2.2, standard error 1.5). The split holds within the 1,932 fields that a district pump serves. There, upstream land with no pump in between has a 2022 change of −1.9 percent per ten points (standard error 0.7), against −0.8 in 2016 and 0.0 in 2020–21, and land behind the pump has +0.7 (standard error 0.9). For land with no pump in between, the upstream minus downstream difference is −3.1 (standard error 1.1), against −0.2 in 2016 and −1.0 in 2020–21. In Hohokam and Harquahala, where farms pump private wells, upstream land with no pump in between shows no 2022 penalty (+0.4, standard error 0.8). Their canals carry only CAP water, which in 2022 was a small fraction of normal, and farms there ran their own wells. In the other two districts, district wells kept water in the canals and fallowing cut the land to be served, so the canals remained the fields' main source.

The split says where the 2022 penalty sits more than how much a pump reduces it. In 2022 itself the difference between the two parts is −1.4 (standard error 0.9), smaller than the change of −2.7, because land behind a pump already had a negative slope in normal years (−1.3 in 2017–2019). Moving the pumps at random to other field intakes, keeping their number and capacities, already gives a negative change for land with no pump in between: −1.2 on average when the pumps move anywhere in their district and −1.6 when they stay on their own lateral, because the land just above a field seldom has a pump in between, wherever the pumps sit (Table 9, Panel B). The real pump sites give a sharper penalty than most placements. The margin is not one that chance placements rarely reach (one-sided p of 0.09 and 0.15). Fields served by more registered pump capacity did not do better in 2022 either. Against 2017–2019, one standard deviation more log capacity goes with 2.5 percent less ET at peak demand (standard error 1.0). The registry lists capacity, not the pumping of 2022, which no public record gives by well or by month.

Fields served by the same district pump also compete for its water. Each served field takes its nearest serving pump as primary. The other fields served by that pump are its sharers, split into those nearer the pump than the field and those farther from it. Each part's planting departure is built as for the canal sets and enters the regression with the canal terms, on the 1,728 fields that have sharers on both sides. In 2022, at peak demand, more planting by nearer sharers lowers a field's ET more than the same planting by farther sharers. The difference is −3.3 percent per ten points (standard error 1.0), against −1.0 in 2016 and −0.7 in 2020–21, all measured against 2017–2019. In 2023 it is −2.5 (standard error 1.2). At low demand in 2022 it is +1.3 (standard error 1.4). It holds with controls for the field's own distance to its pump and for the number of plots upstream of it on the network (−3.7, standard error 1.0), and with sharers weighted down by their distance from the pump (−3.1, standard error 1.0). The strict unit for this test is the pump, and the 1,728 fields share 175 of them. Swapping the nearer and farther sharers of each pump at random, 999 times, gives a p of 0.09 for 2022, against 0.59 for 2016 and 0.55 for 2020–21. The pump contrast is separate from the canal contrast. Within the fixed effects the two correlate at 0.2, and the canal exposures explain 4 percent of the pump contrast. The canal result does not come from fixed position on the network either. With the number of plots upstream of each field as a control, its 2022 change for all CAP fields stays at −2.7 (standard error 0.6).

7.3.6 What Produces the Pattern

To see which ways of moving water can produce this pattern, I keep a district's real network, crops, and fallowing, replace only its ET with water delivered under a stated rule, and estimate the same model on it (Appendix A34, Table A41). When a shortage is shared equally, every estimate is exactly zero: the model responds to who gets the water, not to how much the district has. Conditions that a field shares with the land around it in every direction raise the upstream and downstream slopes together and leave the contrast near zero. When water is delivered upstream first, the upstream slope turns negative and the downstream slope positive, with no shared condition at all. Part of the positive downstream slope is water following orders: when the farms below a field plant more, more water is sent down its branch past its turnout. A shortage that binds only at the canal's peak capacity reproduces the rest of the pattern: the gap appears at peak crop demand only, is largest at 1 to 2 km, and is absent beyond 5 km. In Maricopa–Stanfield, with the district's wells and reported deliveries, letting well owners take part of their wells' water first gives the shape of the pump split of Table 9: a penalty for upstream land with no pump in between and

Table 9: Upstream Land With and Without a District Pump in Between

Panel A. The 2022 slopes

	District wells feed the canals		Private wells
	Change against 2017–19	Slope in 2022	Change against 2017–19
Upstream land, no pump in between	−2.17 (0.59)	−2.11 (0.56)	0.37 (0.77)
Upstream land behind a pump	0.54 (0.80)	−0.72 (0.61)	2.37 (1.37)
Difference (no pump minus behind a pump)	−2.71 (1.07)	−1.39 (0.93)	−2.00 (1.81)
$U - D$, land with no pump in between	−3.31 (0.82)	−3.15 (0.60)	0.01 (1.22)
$U - D$, land behind a pump	−1.14 (1.40)	−2.77 (1.11)	−2.51 (2.59)

Panel B. Pumps moved at random

		Pumps moved within the district		Pumps moved within their lateral	
2022 change	Real	Mean (SD)	One-sided p	Mean (SD)	One-sided p
Upstream land, no pump in between	−2.17	−1.21 (0.57)	0.087	−1.57 (0.54)	0.147
Difference (no pump minus behind a pump)	−2.71	−1.09 (1.80)	0.167	−2.19 (1.47)	0.387

Notes: Panel A: each side's linked land beyond the cut is split into land with no district pump between it and the field and land behind such a pump; the four parts enter one regression, with an indicator for a field that has no land in a part. Pumps: active non-exempt wells within 250 m of a mapped canal with at least 1,000 gallons per minute of registered capacity, placed at the nearest field intake within 500 m; a pump serves the fields to which water can flow from it within 2 km. Percent change in monthly ET per ten points at peak crop demand (own crop coefficient 1.1), with the two-way standard error in parentheses. "Change against 2017–19" is the 2022 change; "slope in 2022" adds the 2017–2019 slope. District wells feed the canals in Central Arizona and Maricopa–Stanfield (3,444 fields); farms pump private wells in Hohokam and Harquahala (1,197 fields). Panel B: the pump sites of each district are moved at random, 149 times, to other field intakes of the district or of the same lateral (a connected part of the district's canal network), keeping their number and capacities; the exposures and the model are rebuilt each time. Mean and standard deviation of the fake 2022 changes, and the share of placements, the real one included, at least as negative as the real one.

none behind a pump. The simulated district runs shorter in 2022 than its records show, so only the shape carries over, not the size. No rule reproduces the shared rise at 2 to 5 km or the sign of the capacity term. Appendix A2 adds these ways of moving water to the model of Section 3 and gives the sign each one implies (Table A1).

7.3.7 Output

The loss carries through to output in the weeks that form yield, more precisely over 2020–2022 than in 2022 alone. For cotton, the exposure averaged over July–September lowers ET in those months by 2.2 percent per ten points in 2022 (interval −4.0 to −0.4) and by 1.6 percent over 2020–2022 (interval −3.1 to −0.1). The integral of Sentinel-2 greenness over the boll window, July 15 to September 30, falls by 3.7 percent in 2022 (interval −8.1 to +0.8) and by 4.3 percent over 2020–2022 (interval −7.9 to −0.6; Figure 8, panel d). Within fields, this integral moves by 1.5 percent for each percent of window ET, so the ET loss implies a loss of 3.3 percent in 2022 and 2.4 percent over 2020–2022, close to the direct estimates. The two estimates come from the same fields and months, so this is a check of consistency, not a second measurement. For alfalfa, the season ends 7 days earlier per ten points in 2022 (interval −11.6 to −2.4), and September–October greenness is 4.0 percent lower (interval −8.4 to +0.4). Across all crops, each in its own window, the integral falls by 3.5 percent in 2022 (interval −6.5 to −0.4). The season maximum of the index, the yield proxy used elsewhere in the paper, moves less. For cotton it falls by 0.9 percent in 2022 (interval −2.7 to +0.9). It is the field mean of each pixel's May–September maximum. Within fields it moves by 0.9 percent for each percent of window ET, so the ET loss would imply a fall of about 2 percent, which the interval includes.

7.3.8 Size of the Effect

I size the effect the way the statewide estimate is sized (Appendix A13), comparing fields at the 25th and 75th percentiles of the predicted 2022 shift, and fields at the 10th and 90th (Table 10). The shift is the part of the 2022 estimates that depends on the direction of flow, the slope on the difference between the two exposures, applied to each field's own departures at its crop coefficient and summed over March to October. A typical difference costs 2.1 percent of seasonal ET, about two-thirds of an inch of water per acre. Through the seasonal yield response factors of FAO-33 (Doorenbos and Kassam 1979) and 2021–22 NASS Arizona revenue per acre (USDA-NASS 2026), that is 1.8 percent of revenue, $28 per acre of cotton. For cotton, alfalfa, and corn fields taken separately it is 2.1 to 2.4 percent. The yield response can also be estimated in these fields, as the within-field response of each crop's greenness yield proxy to its seasonal water use relative to crop demand: 1.03 for cotton, 1.80 for alfalfa, and 1.28 for corn and sorghum. With these estimates a typical difference costs 1.75 percent of revenue when greenness passes to revenue at the rate assumed in Appendix S13, and 2.2 percent if greenness moves one for one with yield. The wide gap is about twice as large, 3.8 percent of revenue or $60 per acre of cotton. The 2022–23 Lower Basin conservation offers paid $330 to $400 per acre-foot (Reclamation 2023), and the average revenue per acre-foot of ET in 2022 was about $700. At these values the typical gap is worth $18 to $39 per acre. With returns over total costs at about a tenth of gross income (Evancho et al. 2023a, 2023b), a typical difference takes about a fifth of a field's net return in a shortage year, and the wide one more than a third. The statewide design gives 1.4 percent of revenue within the CAP-served and the other districts and 2.7 percent pooled (Appendix A13). The two designs share neither the exposure, the outcome, nor the conversion to revenue, and they agree in size. These are gaps between fields. In 2022, 1,455 of the 2,735 fields farmed with a crop of the calendar, with about 49,000 acres, were on the losing side of their neighbors' planting. Their predicted loss sums to about 2,200 acre-feet of ET and $1.3 million of revenue, 1.6 percent of their revenue and about a sixth of their net return. With the yield responses estimated in these fields, the loss is $1.5 to $1.9 million. At $330 to $700 per acre-foot the water is worth $0.7 to $1.5 million. I do not net the other fields' gains against these losses. With a yield response that is linear in water they cancel by construction, and the data do not pin down how yield bends near full water. For the fields on the wrong side of their neighbors' planting the losses are a large share of profit.

Table 10: Size of the 2022 Effect

	Typical gap (25th–75th)			Wide gap (10th–90th)		
	ET (%)	Revenue (%)	Per acre	ET (%)	Revenue (%)	Per acre
Through the FAO-33 yield response						
All fields, per acre of cotton	2.1	1.8	$28	4.5	3.8	$60
Cotton fields	2.9	2.4	$38	5.5	4.7	$72
Alfalfa fields	2.2	2.4	$54	4.5	4.9	$109
Corn fields	1.7	2.1	$34	3.0	3.8	$62
Water valued at $330–700 per acre-foot						
All fields			$18–39			$39–83

Notes: Fields at the 25th and 75th (10th and 90th) percentiles of the predicted 2022 shift in seasonal ET from the part of the estimates that depends on the direction of flow (the model of Table A36, all 4,641 fields; 2,735 farmed with a crop of the calendar in 2022), each month at the field's own crop coefficient, summed over March–October with ET weights; percentiles among fields of all crops in the first row and among fields of the crop in the next three. Revenue: the share of seasonal ET times the FAO-33 seasonal yield response factor (cotton 0.85, alfalfa 1.1, corn 1.25); dollars at 2021–2022 NASS Arizona revenue per acre (cotton lint). Water: the ET gap at the median seasonal ET of cotton fields, in acre-feet per acre, at $330 to $700 per acre-foot.

7.3.9 Fields Left Fallow

Most of the CAP cut was absorbed by fallowing. Among the 4,641 fields, the fallow share rose from 12 to 17 percent in 2017–2019 to 36 percent in 2022 and 49 percent in 2023. The decision was made before the season (Appendix A30). Which fields were left fallow followed the crop and groundwater (Appendix A32). In 2022, 41 percent of the cotton fields farmed the year before and 31 percent of the corn and sorghum fields were left fallow, against 11 and 5 percent in 2017–2019, and 14 percent of the alfalfa fields. More old wells in a field's groundwater right lowered fallowing in 2023, when CAP deliveries were near zero. Competition for canal water did not move fallowing. Neither the expected nor the previous year's planting of the land competing for canal water changed which fields were farmed. With allotments per acre, the season's water is fixed, and competition moves water between months among the fields that are farmed. The fields still farmed are therefore not selected on the contrast the estimates use. Position did not favor the head either. Fields near where CAP water enters the district were left fallow more often in 2022 and 2023, and none of the measured traits of the land explains it. In the Upper Gila canals fallowing followed the river. A quarter of the fields farmed the year before were left fallow in the short springs of 2018 and 2021, against 5 to 10 percent in the other years, and the same annual crops left first. In 2022, also short in spring, 8 percent stopped.

7.3.10 When the Pool Was Empty

In 2023 the pool received no water. CAP deliveries were near zero, so the canals no longer carried a shared supply for the land upstream to take first. The canal contrast is gone. On all fields the peak-demand contrast is +0.2 percent (interval −1.7 to +2.1) and the late-season profile is flat. Cotton fields still show a loss in the boll window. For land with no district pump in between, where the 2022 penalty sat, the upstream minus downstream difference in Central Arizona and Maricopa–Stanfield is −0.3 (standard error 0.8), against −3.3 (standard error 0.8) in 2022 (Table 9). In Hohokam and Harquahala, where the canals carry only CAP water and farms pump their own wells, the contrast is +0.3 (standard error 1.2).

The canals of Central Arizona and Maricopa–Stanfield still ran in 2023. They carried district-pumped groundwater, 98 percent of Maricopa–Stanfield's deliveries. The competition shows up at the pumps instead. Among fields served by the same district pump, more planting by the sharers nearer the pump again lowers a field's ET more than the same planting by those farther away: −2.5 percent per ten points at peak demand (standard error 1.2), against −3.3 for the same sharers in 2022 (Section 7.3.5). The swap test by pump gives a p of 0.16 for 2023. The inequality follows the source that still has water. With the canal source empty, upstream land had no shared water to take first. The district pumps kept running, and the fields that shared a pump competed for it as canal fields had competed for canal water the year before. The year is also hard to read with this design. Half of the 4,641 fields were fallow in 2023, against a fifth in 2021, so the fields still farmed were a smaller group that farmers chose to keep. Which wells ran, and for how much, is not observed.

Inside the CAP districts, the loss shows up where competition for canal water says it should. The contrast follows the direction of flow, opens in the months the canals ran short, in 2021 as in 2022, is largest when the field's own crop needs the most water, and sits in upstream land with no district pump between it and the field. It carries through to output in the weeks that form yield, and in a shortage year a typical difference in neighbors' planting costs a field about a fifth of its

net return.

7.3.11 Shortage in Spring: The Upper Gila Canals

The Upper Gila canals of the Safford Valley run short at another time of year and for another reason. Their water is the natural flow of the Gila River, which the Gila Water Commissioner divides among the canals by decree priority (Doyle 2022). When the river is low, a priority call shuts junior rights out for weeks. Inside a canal, farms pay per acre and may take the water in rotation (Arizona Supreme Court 1955), and wells supply more than half of the farm water (Gila Water Commissioner, n.d.). The river failed in the springs of 2018, 2021, and 2022. April diversions fell by a third to two-thirds, and canals ran dry for weeks in May. In the summers of 2021 and 2022 the monsoon filled the canals, and in the dry summers of 2019, 2020, and 2023 wells made up for the low river.

I apply the design of this section to the 717 fields inside the five canal polygons, with the Gila's own shortage calendar in place of the CAP years. A short month has less canal water than the 2016–2019 normal for that month, and a dry field-month has SPEI-3 below zero at the field. The contrast compares the slope in short and dry months with the slope in dry months with normal canal water, with district by year by month effects. Table 11 sets the timing of the gap in the two systems side by side. In the CAP districts the contrast is about zero or positive in 2020 and in the spring of 2021, when deliveries were near normal. It is +0.5 in the spring of 2022, when deliveries were already low and the crops were early in the season, and negative from July in 2021 and 2022, when the canals ran short at peak crop demand. In the Upper Gila canals the contrast is negative in the short and dry months of spring: −2.3 percent per ten points on average over March–June (standard error 1.3). Of these field-months, 78 percent fall in the springs of 2018, 2021, and 2022. In the short and dry months of July–October, which the monsoon or the wells filled, the contrast is +3.0 (standard error 1.5).

In spring most Gila crops are at low demand. Cotton makes up 57 percent of the short and dry spring field-months, and from March to May it is at its planting stage, with a crop coefficient of 0.30 to 0.42. At a crop coefficient of 0.4 the spring contrast is −3.2 percent (standard error 1.4; district-year bootstrap $p = 0.09$), and cotton fields show it in April, the month of planting (−4.5, standard error 1.8), though not over the spring as a whole (−0.8, standard error 1.3). Over all short and dry months the low-demand contrast is −2.6 percent. As in the CAP districts, it comes from both sides, an upstream slope of −1.1 and a downstream slope of +1.4. In the swap test of Section 7.3.3 it has $p = 0.052$ by canal segment and 0.11 by grid cell. For land 2 to 5 km away, where the upstream side carries it, the contrast is −4.4 and lies beyond almost all of the fake networks ($p = 0.003$ and 0.011). The spring gap reaches cotton output. For each short and dry month in spring, ten points more spring demand upstream lower the greenness integral over July 15 to September 30 by 2.0 percent more than the same demand downstream (bootstrap $p = 0.033$), and July–September ET by 1.1 percent more ($p = 0.045$). At peak crop demand there is no gap (+0.8 percent, standard error 1.0), which is what rotation inside a canal predicts; no simulated rule, however, gives a gap at low demand without a larger one at peak (Appendix A33).

7.3.12 What Farmers and Canal Operators Describe

The districts and farmers describe the CAP shortage the way the estimates show it. In its 2022 application for federal funds, Central Arizona wrote that it had "to proportionally reduce the

Table 11: The Gap Follows the Shortage: CAP Districts and Upper Gila Canals

Months	Canal water	Upstream minus downstream	District-year p	Swap p (segment / grid)
Panel A. CAP districts (change against 2017–2019)				
2020, March–June	Near normal	1.31 (0.84)	0.175	0.139 / 0.239
2020, July–October	Near normal	−0.05 (0.84)	0.958	0.968 / 0.968
2021, March–June	Near normal	1.65 (0.79)	0.017	0.065 / 0.100
2021, July–October	40–60% of normal from July	−1.63 (0.83)	0.029	0.138 / 0.182
2022, March–June	15–40% of normal; crops early in the season	0.45 (1.17)	0.445	0.664 / 0.759
2022, July–October	15–40% of normal; peak crop demand	−1.99 (0.80)	0.008	0.097 / 0.171
Panel B. Upper Gila canals (short and dry months against dry months with normal canal water)				
Short and dry, March–June	River short; mostly the springs of 2018, 2021, 2022	−2.32 (1.26)	0.085	–
Short and dry, July–October	Short in the records; monsoon water or wells filled it	2.97 (1.47)	0.112	–

Notes: Each cell gives the slope of monthly ET on the crop water demand of linked land upstream minus the same slope downstream, in percent per ten points, averaged over the months shown, with the two-way standard error (field and crop-year) in parentheses and the district-year p-value. Panel A: all 4,641 CAP fields; change against 2017–2019 (Table 7); canal water is CAP deliveries to the four districts relative to normal (Figure 8, panel a). Panel B: all 717 fields inside the five Upper Gila canal polygons (Fort Thomas, Graham, Montezuma, San Jose, and Union); change in short and dry months against dry months with normal canal water, with field by month, crop by year by month, and district by year by month effects. A short month has canal water below its 2016–2019 same-month mean, and a dry month has SPEI-3 below zero at the field. Upper Gila canal water from the Gila Water Commissioner's records, which also report the district's pumping (Gila Water Commissioner, n.d.). Swap p: the test of Section 7.3.3 for each Panel A cell, with 1,999 fake networks swapped by canal segment and by 5 km grid cell; for the Upper Gila, Section 7.3.11 gives it for the low-demand contrast over all short and dry months.

water delivery allocation for each farmer," and that shortages "usually occur during periods of above normal temperatures in the growing season when there is high demand, or at the end of the year when the mandated reduced allotment has been used" (CAIDD 2022). That is when the gap of Section 7.3.2 opens: from July, at peak crop demand. A common cut did not mean a common supply. In Maricopa–Stanfield the usual allotment of 4 acre-feet per entitled acre became 1.7 to 2.8 acre-feet "depending on where the grower's fields are in the district" (Evancho 2022, 12). Local reporting put much of the difference in wells: "Some have good well systems that can supplement the CAP water and some farms do not" (InMaricopa 2021). Central Arizona carries both district well water and CAP water in its canals, and one of its turnouts "serves an area with restricted groundwater pumping, making the surface water it delivers vital to the water users" (CAIDD 2022). Its former manager described gates "that are leaking and leaking downstream" (Horn-Muller 2026). The penalty of Section 7.3.5 sits in upstream land with no district pump between it and the field, which fits these accounts, though fields served by more pump capacity did not do better.

In the Upper Gila canals the river, not an allotment, set the shortage. In 2021 the Gila was dry at Cosper Crossing, between the Duncan and Safford valleys, from March 11 until July 26, and its spring water was very salty (Doyle 2022, 3, 17). In scarce water the canals deliver by turns. Testimony from the Graham canal in the 1950s describes how a turn shrank along the ditch, from about 400 inches at the head to 250 at the lower end: "It affects more on the lower end" (Arizona Supreme Court 1955, 217–218). The district still reports gates that leak 10 to 50 miner's inches out of a typical 300-inch stream (AWPF 2022b), and its modernization plan aims to make field deliveries more equitable (ITRC 2019). When the river fails, farmers pump more: in the short springs the district pumped about a third more groundwater than usual (Gila Water Commissioner, n.d.). The Gila estimates follow the same lines: a gap in the short months of spring and in months of partial flow, and, by an imprecise margin, less of it for fields with their own well.

The two systems run short at different times, and in each the gap appears when its canals are short (Table 11). In the CAP districts that is the late season of 2021 and 2022, after CAP deliveries fell, when crops need the most water. In the Upper Gila canals it is the spring, when the river fails. Months that were short in the Gila canal records and filled by the monsoon or by wells

show no such gap; there the contrast is positive. Weather, pests, or conditions that neighboring fields share would not follow two different water calendars. In both, a cut that is shared on paper reaches fields unequally along the canal.

8 Conclusion

This paper asks whether farms with the same water rights share the shortage from a long drought equally. Arizona is a hard case for finding a gap, because its water rules are well developed. Water rights and contracts decide which users lose water first, and the 2022 cut in Colorado River water was carried out as the rules required. To answer the question, I rebuild Arizona's canal network from public maps and measure, for each field and year, how much land upstream of it is farmed.

Farms with the same rights do not share the shortage equally. In drought years, fields with more farmed land upstream have weaker crops. If shared local conditions drove this result, leaving out nearby upstream fields would weaken it. Instead, the effect grows. The effect also follows the direction of water flow. Farmed land far upstream matters in drought years. Farmed land far downstream does not. Inside the junior-rights districts, the effect has two parts. One is long-standing and depends on where a field sits in the canal network. The other changes each year with how much land upstream is farmed. In 2021–22, the gap between fields with a lot of farmed land upstream and fields with little is worth about 1.4–2.7 percent of crop revenue, or roughly 14–28 percent of a field's net return.

The effect appears in 2020–2022. An earlier, drier drought, when major water deliveries largely held, shows no such effect. When CAP water was cut in 2022, farms left much more land fallow.

The four districts that lost CAP water show how the gap arises. Among the fields still farmed, the loss in crop water use opened in the months the canals ran short: from August in 2021 and from July in 2022, when crops needed the most water. In 2016 and 2020, with near-normal deliveries, there was no such gap. In 2022, the loss at peak demand follows the direction of flow: when the upstream and downstream land of fields is swapped at random, few of the fake networks give a gap as large. In the two districts whose wells feed the canals, it sits in upstream land with no district pump between it and the field. It carries through to cotton greenness in the weeks when bolls form, and a typical difference costs a field about 1.8 percent of its revenue, close to the statewide estimate. The Upper Gila canals, which run short in spring when the river fails, show their gap in spring. Weather or conditions that neighboring fields share would not follow two different water calendars.

The findings do not show who causes the gap. A district may serve the upstream farms first when water is short, or farms upstream may use more water than the rules intend. Both would produce the same pattern. So would a canal that cannot carry what every field needs at peak demand. A shortage shared in equal proportion at every turnout would leave no gap. The study also has limits. The crop measures come from satellites and are proxies for yield. I do not observe the water delivered to each field, so I cannot separate delivery schedules, losses along the canal, and gate operations. District maps stand in for the areas each district serves. Most fields in the sample sit in the largest canal systems of one state and are observed over one main drought. The revenue figures assume a link between the satellite index and revenue, and they measure lost revenue, not lost profit or welfare.

These results matter for how shortage rules are judged. In 2022 Arizona's rules worked as written at the source. Inside the canals, farms with the same claims did not lose the same amount.

Programs that pay farms to leave land fallow might consider where along the canal the fallowed land lies, although water saved upstream does not always reach the fields below. The method in this paper makes such questions possible to study. It uses only public data on canals, land elevation, and field boundaries, so it can be applied to other irrigated regions and to other networks that deliver a resource to users one after another.

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Appendix A. Supplementary Results and Model Details

This appendix groups the robustness evidence by its role in the argument. Appendix A1 collects the model's derivations, auxiliary assumptions, and its map to the estimating equations. Appendix A2 extends the model to the months of a season for the in-season tests of Section 7.3. Appendices A3–A8 contain the local-shock and directional evidence: the fallow-share symmetry diagnostic, the same-block local association, the near/far timing profile, the separate final-status decomposition, the joint regression without fallow controls, and the common-support downstream negative control. Appendices A9–A10 report the two location and cutoff checks: canal-size bins and the extended 15 and 20 km grid. Appendices A14–A23 carry the priority profile, formal eligible-acreage and activation-gap definitions, unit-free normalization, alternative specifications and inference, total drought-year slopes, the timing check, and the balanced-plot panel. Appendix A25 documents non-CAP delivery constraints and their source coverage. Appendix A26 reports the exploratory measured-flow evidence and its limits. Appendix A27 adds rights and well controls to the junior-district estimates. Appendix A28 replaces the drought proxy with the administrative CAP curtailment schedule. Appendix A29 reports the crop water demand test inside the junior districts, Appendix A30 the timing of planting and fallowing, Appendix A32 which fields stopped farming, Appendix A33 the same design in the Upper Gila canals, and Appendix A34 the simulated mechanisms. Appendix A35 reports the graph validation statistics. The full fallowing covariance matrix and the outcome-year fallowing profile join secondary data construction, the ET-member sensitivity, economic magnitudes and the reallocation screen, equal-weighted district estimators, and the exploratory 2008–2023 crop-outcome panel in the online supplement.

A1 Model Details

This appendix collects the derivations and auxiliary assumptions behind the model of Section 3. Notation follows Section 3.2: S_t is system supply, w_i^* the formal entitlement, ε_{jt} the upstream leak with monitoring intensity μ, ρ the return-flow fraction, λ_{ij} the recovery share, Δ_{it} the positional wedge of equation (2), $\ell_i(S_t) = A_{it}^0 - w_i^*$ the cushion, and $q_{it} = [\Delta_{it} - \ell_i(S_t)]_+$ the realized shortfall.

Monotonicity of the shortfall in position. The head-to-tail level-gradient prediction of Section 3.2 rests on two maintained assumptions: recovery λ_{ij} constant or weakly decreasing in hydrologic distance (the standard case for tailwater and shallow-percolation pathways) and entitlements that do not fall toward the tail faster than adjacent return-flow recovery, $w_{i+1}^* \geq \rho\lambda_{i+1,i}\, w_i^*$ (automatic when entitlements are equal or rising, since $\rho\lambda_{i+1,i} \leq 1$, and mild when they decline, because only a small share of applied water is recoverable at the immediately downstream turnout). Together the two assumptions sign the cushion increment,

$$\ell_{i+1}(S_t) - \ell_i(S_t) \;=\; \rho\lambda_{i+1,i}\, w_i^* \;-\; w_{i+1}^* \;-\; \rho \sum_{j<i} \big(\lambda_{ij} - \lambda_{i+1,j}\big)\, w_j^* \;\leq\; 0,$$

so the cushion falls, rather than rises, along the canal. Without these assumptions, monotonicity depends on primitives. Comparing adjacent users, $q_{i+1,t} \geq q_{it}$ holds whenever

$$\big(1 - \rho\lambda_{i+1,i}\big)\, \varepsilon_{it} \;+\; \rho \sum_{j<i} \big(\lambda_{ij} - \lambda_{i+1,j}\big)\, \varepsilon_{jt} \;\geq\; \ell_{i+1}(S_t) - \ell_i(S_t), \tag{6}$$

where the left-hand side is the wedge increment $\Delta_{i+1,t} - \Delta_{it}$ and the right-hand side is the cushion increment; the condition is sufficient rather than necessary (when neither user's constraint binds, both shortfalls are zero and the comparison holds trivially). Two limits clarify the primitives: as monitoring intensity $\mu \to \infty$, all ε terms vanish and the left-hand side collapses; when S_t is high, the cushion is large and $q_{it} = 0$ regardless of the wedge. The left-hand side shrinks (and monotonicity can fail) when ρ is large or when recovery $\lambda_{i+1,j}$ rises with hydrologic distance, because returned water is recaptured before it compounds into a downstream wedge; the right-hand side turns positive (and monotonicity can likewise fail) when entitlements fall toward the tail faster than adjacent recovery, because a small own entitlement enlarges the next user's cushion. Under the maintained assumptions the left-hand side of (6) is non-negative and the right-hand side non-positive, so the inequality holds; this is the prediction behind the negative head-to-tail gradient documented in Section 5.7. Groundwater substitution can still flatten the wedge in theory. Estimates split by district water source do not follow the order that buffering predicts (Appendix Table S7). Evidence for buffering comes instead from the district pumps of the CAP districts. The 2022 penalty sits in upstream land with no district pump between it and the field (Section 7.3.5).

Drought amplification. Define the binding margin $x_{it} = \Delta_{it} - \ell_i(S_t)$ and the adjacent-user increment

$$\delta_{it} \;=\; \big(\Delta_{i+1,t} - \Delta_{it}\big) \;-\; \big(\ell_{i+1}(S_t) - \ell_i(S_t)\big),$$

which is non-negative under the maintained assumptions of the monotonicity paragraph. Then

$$q_{i+1,t} - q_{it} \;=\; [x_{it} + \delta_{it}]_+ - [x_{it}]_+,$$

which is non-decreasing in x_{it} for fixed $\delta_{it} \geq 0$. Since Δ_{it} does not depend on S_t and the cushion moves one-for-one with supply ($d\ell_i/dS_t = 1$ from equation (1) at $\varepsilon = 0$), we have $dx_{it}/dS_t = -1$: a lower supply draw raises the binding margin for every user, so the positional shortfall gap $q_{i+1,t} - q_{it}$ weakly widens as S_t falls. The outcome gradient is amplified further, on the domain where $A^0_{it} > 0$, because the denominator A^0_{it} in the outcome equation also falls with S_t. The increment δ_{it} is invariant to S_t under the maintained assumptions (both the Δ-increment and the ℓ-increment are supply-invariant), so the amplification operates purely through the binding margin. This is the comparative static behind the panel drought interaction in Section 7.1 (Table A22).

Pricing. Index usage-based pricing by τ and let monitoring intensity rise with it, $\mu'(\tau) > 0$, so the leak shrinks: $\partial\varepsilon_{jt}/\partial\tau < 0$. The direct effect on the wedge is

$$\frac{\partial \Delta_{it}}{\partial \tau} \;=\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big)\, \frac{\partial \varepsilon_{jt}}{\partial \tau} \;\leq\; 0,$$

strict for $i > 1$ whenever $\rho\lambda_{ij} < 1$ for some $j < i$. If pricing also reduces runoff and recharge, $\rho'(\tau) < 0$ adds an offsetting positive term through the recovery share, and the net theoretical effect is ambiguous. The district pricing records do not resolve this ambiguity empirically, and the two channels are not separately identified by the design (Section 7.2.1).

Canal size: level versus interaction. Taking the leak as i.i.d. across upstream users, $E[\Delta_{it}] = \sum_{j<i}(1-\rho\lambda_{ij})\, E[\varepsilon]$ scales with the number of upstream extractors: the wedge is mechanically larger in bigger systems. The empirical test in Appendix Table A8, however, is on the drought *interaction*

by canal-size bin, which requires more: the wedge must also *bind* more readily in larger systems during drought. A sufficient auxiliary assumption is that the cushion per user is tighter in larger systems. Holding season supply S_t and the entitlement distribution fixed, this follows within the model: upstream net extraction $\sum_{j<i}(1-\rho\lambda_{ij})\,w_j^*$ is mechanically larger where more users sit upstream, so residual supply A_{it}^0, and with it the cushion, is lower for a larger share of users in big systems. Because supply itself tends to scale with system size, the cross-system version is an assumption rather than a derivation: it requires that cushions per user not scale up one-for-one with capacity. Under it, a low S_t pushes a larger share of users past the binding threshold, and the wedge itself is larger. Together these produce the stronger interaction the data show. The assumption is auxiliary to the core identity and is not tested directly.

Priority queue and enforcement. The seniority comparative static of Section 3.3 rests on asymmetric enforcement. Under an enforced queue, the entitlement margin is served in priority order, and the leak margin retains the physical ordering. Abstracting from return-flow recovery of entitlement claims for readability (the general case nets those claims by the same recovery logic), senior user i's residual supply for entitlement purposes is

$$A_{it}^{\text{pri}} \;=\; S_t \;-\; \sum_{j\in P(i)} w_j^* \;-\; \sum_{j<i}\big(1-\rho\lambda_{ij}\big)\,\varepsilon_{jt},$$

where $P(i)$ is the set of users senior to i by priority date, not position; the middle sum is entitlement claims served ahead of i in the queue; and the last sum is the physically accumulated leak net of recovery, which no legal queue can reorder. Delivery is $w_{it} = \big[\min\{w_i^*, A_{it}^{\text{pri}}\}\big]_+$. For senior users, $P(i)$ is small, so $A_{it}^{\text{pri}} \geq w_i^*$ except for the leak term, giving $q_{it}^{\text{sen}} \approx 0$; the approximation fails exactly through the leak sum and mid-season shocks, which is the asymmetric-enforcement point. Juniors have large $P(i)$ and compete over the residual pool. The priority queue is enforceable ex post through curtailment authority (the "call on the river" of Section 3.1, with the Bureau of Reclamation as watermaster, curtailing Central Arizona Project deliveries first under the 2007 Interim Guidelines). The leak is not detectable in real time within a season. Because curtailment operates only ex post, realized mid-season supply shocks can reach senior users before the queue reorders delivery, which is why the senior shortfall is approximately rather than identically zero. The empirically relevant queue runs at the river level: the Bureau administers Colorado River priorities across districts. Within districts the order of service is an operating matter; where parcel-level priorities exist (Salt River's Kent classes, the San Carlos and Upper Gila decree schedules), they attach to land, not to canal position. The comparative static therefore treats the river as the canal and districts as its users.

Parameterization choices. Five reduced-form choices are implicit in Section 3. First, λ_{ij} is treated as constant or weakly decreasing in hydrologic distance; if recovery instead rose steeply with distance, non-monotone wedges become possible (see the monotonicity discussion above). Second, the leak is additive rather than proportional to entitlements; a proportional alternative ($\varepsilon_j \propto w_j^*$) would generate additional cross-district predictions not pursued here. Third, the leak is exogenous undetected over-diversion rather than a strategic choice; endogenizing it would require equilibrium monitoring-and-sanctions machinery without changing the sign predictions the paper uses. The empirical counterpart of monitoring intensity μ is heterogeneous across districts: metered Bureau of Reclamation turnouts (e.g., Yuma, Wellton-Mohawk) report per-device accuracy near

±3% (Stringam, Sauer, and Pugh 2003). Smaller systems allocate per acre with no volumetric measurement, and Arizona imposes no statewide metering requirement. Fourth, equation (1) treats upstream diversions as $w_j^* + \varepsilon_{jt}$ regardless of upstream availability. It is exact when upstream constraints are slack ($A_{jt} \geq w_j^* + \varepsilon_{jt}$ for all $j < i$) and physically coherent at i when additionally $A_{it} \geq 0$, which caps the shortfall at the entitlement ($q_{it} \leq w_i^*$); $A_{it}^0 > 0$ is required separately for the outcome equation's denominator. When intermediate users are themselves rationed, realized diversions fall short of entitlements and the shortfall cascades toward the tail, preserving the level and directional predictions; the drought-amplification comparative static, by contrast, is derived on the domain where upstream diversions remain feasible. In deep enough drought, rationed upstream users can no longer divert entitlement plus leak, and the leak-attributable wedge itself shrinks. Fifth, the comparative statics hold the primitives (μ, ρ, λ_{ij}, and the leak distribution) fixed across drought and non-drought conditions. The scaffold is deliberately static and non-strategic: entitlements w_i^* do not adjust through transfers or rights markets within the model.

Pro-rata rationing. The institutions described in Section 3 include pro-rata shortage reductions, under which districts scale allocations by a common ration factor $\zeta_t \in [0, 1]$ rather than letting the tail absorb the aggregate shortfall. Equation (1)'s residual-claiming rule is the opposite polar case, in which administration cannot adjust within the season. The model's predictions survive under faithful pro-rata rationing, through a slightly different channel. With scaled entitlements $\zeta_t w_j^*$, the leak is extraction beyond the scaled entitlement (its absolute scale is governed by monitoring, not by the entitlement), and the wedge retains the form of equation (2), so the level and directional predictions are unchanged. For drought amplification, write the cushion under rationing as $\ell_i^\zeta = S_t - \zeta_t C_i$, where $C_i = \sum_{j<i}(1 - \rho\lambda_{ij})\, w_j^* + w_i^*$ cumulates claims toward the tail. A faithful ration ties the factor to supply, $\zeta_t = S_t / C_N$, so $\ell_i^\zeta = S_t\,(C_N - C_i)/C_N$, which shrinks proportionally with S_t for every user; under the entitlement condition of the monotonicity paragraph above, C_i rises toward the tail, so the cushion also falls toward the tail (reaching zero at the last user, where any leak binds immediately) and $C_N = \max_i C_i$, making the ration feasible for every user. Drought therefore pushes users past the binding threshold from the tail upward, and the wedge is unchanged. This is the same amplification logic as the main text, with the aggregate cut shared rather than concentrated at the tail. The residual-claiming rule of the main text thus sharpens, rather than creates, the positional predictions; the empirically relevant regime (administered scheduling ex ante, physical residual claiming for realized within-season shocks) lies between the two poles, as the timing discussion in Section 3.2 states. The administrative test in Appendix Table A34 instantiates this pole: announced tier cuts are anticipated shortfalls, so the aggregate reduction is shared, and the upstream cropped-acreage interaction it induces may vary with surface dependence and groundwater conditions.

Model-to-empirics map. Table 1 maps the model's five sign predictions to the tables that confront them. Three refinements matter. First, the institutional-scope row bundles distinct channels: the junior-inside GCVI interaction is negative, and senior-inside estimates are imprecise. The priority grid does not test equality and therefore does not identify priority-based protection. The physical-supply split does not order with source, and the pricing comparative static is unestimable under the strict static audit (Tables A11, S7, and S3). Second, an administrative pool cut enters as a reduction in district headgate supply S_t, net of well injection (Appendix Table A34). In the CAP districts, district pumps supply that injection. The 2022 penalty sits in upstream land with

no district pump between it and the field, and fields that share a pump compete for its water (Section 7.3.5). Third, upstream claims inside municipal service areas enter cumulated claims C_i, predicting a weakly negative urban-adjacency interaction alongside an unchanged upstream cropped-acreage drought interaction (Appendix Table S10). The spatial-exclusion machinery of Section 6 has no counterpart in the model, which contains no spatial-correlation structure for the leak.

Because the upstream count is measured exactly, the omitted sibling-lateral competition acts as a correlated omitted regressor rather than classical measurement error, and its sign turns on the within-plot covariance between sibling and ancestor cropped acreage.

A2 The Model Within the Season

This appendix extends the identity of Section 3.2 to the months of a season, for the in-season tests of Section 7.3. Let S_{mt} be the water that enters the canal in month m of season t, w^*_{jm} user j's scheduled delivery in that month, with $\sum_m w^*_{jm} = w^*_j$, and ε_{jmt} the amount by which j's diversion departs from that schedule, with $\sum_m \varepsilon_{jmt} = \varepsilon_{jt}$. Equation (1) for one month is

$$A_{imt} \;=\; S_{mt} \;-\; \sum_{j<i}\big(w^*_{jm} + \varepsilon_{jmt}\big) \;+\; \rho \sum_{j<i} \lambda_{ij}\big(w^*_{jm} + \varepsilon_{jmt}\big), \tag{7}$$

and the wedge, cushion, shortfall, and outcome take the month index: $\Delta_{imt} = \sum_{j<i}(1-\rho\lambda_{ij})\,\varepsilon_{jmt}$, $\ell_{im}(S_{mt}) = A^0_{imt} - w^*_{im}$ with A^0_{imt} the value of equation (7) at $\varepsilon_{jmt} = 0$, $q_{imt} = [\Delta_{imt} - \ell_{im}(S_{mt})]_+$, and $Y_{imt} = Y^0_{im} - \beta\, q_{imt}/A^0_{imt} + u_{imt}$. A monthly departure can be negative. The exposure of Section 7.3.2 moves with the part of ε_{jmt} that follows what the land above i planted that year. Planting is set before the season (Appendix A30) and is taken as given. The four cases below can hold together.

(a) A canal capacity that binds at the peak. The canal carries at most $\bar{S}_t$ in a month, and deliveries are scheduled within it. Below capacity, the district releases what the farms divert, so $q_{imt} = 0$. At capacity, $S_{mt} = \bar{S}_t$ cannot rise. A departure above i then takes its water during delivery, before the canal reaches i, which is the residual claiming of Section 3.2 within one month. Once the cushion is used up,

$$\frac{\partial q_{imt}}{\partial \varepsilon_{jmt}} \;=\; 1 - \rho\lambda_{ij} \;\geq\; 0 \quad (j < i), \qquad \frac{\partial q_{imt}}{\partial \varepsilon_{kmt}} \;=\; 0 \quad (k > i). \tag{8}$$

Schedules follow crop need, so the canal reaches capacity in the peak months. The cushion is thinner where the field's own crop needs more ($\partial \ell_{im}/\partial w^*_{im} = -1$) and when the capacity is cut ($\partial \ell_{im}/\partial \bar{S}_t = 1$). So the upstream slope is negative only in months at capacity and strongest at the field's own peak demand, and the downstream slope is zero. The gap opens when the canal water is cut, which is the in-season form of drought amplification. Without an account (case d), the seasonal leak $\varepsilon_{jt} \geq 0$ still makes $\Delta_{it} = \sum_m \Delta_{imt}$ rise toward the tail, and case (a) alone keeps the level gradient.

(b) Branch releases routed by orders. Let flow F_{mt} reach the junction above i, short of the orders O_{mt} of the branches below it, and let the district split it in proportion to orders. With

$O_{bmt} = \sum_{k \in b}(w^*_{km} + \varepsilon_{kmt})$ the orders on i's branch b, which include the farms below i,

$$A_{imt} \;=\; F_{mt}\,\frac{O_{bmt}}{O_{mt}} \;-\; \sum_{j<i,\; j \in b} \big(1 - \rho\lambda_{ij}\big)\big(w^*_{jm} + \varepsilon_{jmt}\big),$$

$$\frac{\partial A_{imt}}{\partial \varepsilon_{kmt}} \;=\; F_{mt}\,\frac{O_{mt} - O_{bmt}}{O^2_{mt}} \;>\; 0 \quad (k > i,\; k \in b).$$

More orders below i send more water down its branch, and a field that is short takes the added water as it passes. Downstream demand then raises a field's water. The same term enters for the farms above i on its branch, so the contrast in the field's water between the two directions stays $-(1 - \rho\lambda_{ij})$, as in case (a). With splits fixed in advance, the term and the downstream slope are zero.

(c) District wells along the canal. Let district pump p add $g_{pmt} \le \bar{g}_p$ to the canal in month m, where $\bar{g}_p$ is its registered capacity, so that $\sum_{p<i} g_{pmt}$ joins the right side of equation (7). A pump runs on demand. If it sits between upstream land j and field i and runs below capacity, it replaces what j diverts, and $\partial A_{imt}/\partial \varepsilon_{jmt} = 0$. Land with no pump in between keeps $\partial A_{imt}/\partial \varepsilon_{jmt} = -(1 - \rho\lambda_{ij})$, so the penalty should sit in that land. Capacity can only add water ($\partial A_{imt}/\partial \bar{g}_p \ge 0$), so a field served by more capacity should lose less. A pump at capacity is a source like the canal head in case (a). The fields it serves draw on it in order, and those nearer the pump take first. When the canal carries only pumped water, as in Central Arizona and Maricopa–Stanfield in 2023, each pump is the source for the fields it serves, and the competition runs among them. A farm with its own well replaces a shortfall by pumping at cost c_g (Section 3.3), so its ET should fall little.

(d) Seasonal allotments per acre. Central Arizona and Maricopa–Stanfield ration water by a seasonal allotment per acre, and Central Arizona cut each farmer's allotment in proportion (Section 7.3.12). Deliveries are charged against the allotment, so a farm that diverts more in one month diverts less later. When every farm uses its whole allotment, $\sum_m \varepsilon_{jmt} = 0$ for each j, and

$$\Delta_{it} \;=\; \sum_m \Delta_{imt} \;=\; \sum_{j<i} \big(1 - \rho\lambda_{ij}\big) \sum_m \varepsilon_{jmt} \;=\; 0. \tag{9}$$

The pro-rata case of Appendix A1 keeps a leak beyond the scaled entitlement, which cumulates toward the tail. The account leaves none. The seasonal shortfall then does not rise toward the tail, as long as the canal falls below capacity later in the season so that each farm can take the rest of its allotment. If schedules also match what farms expect to divert in each month, $E[\varepsilon_{jmt}] = 0$, and then $E[\Delta_{imt}] = 0$ at every position. The fixed head-to-tail gradient of the first prediction in Section 3.2, which comes from the mean of the wedge, disappears. Only the spread of Δ_{imt} still grows toward the tail, which can leave a second-order gradient through the kink in q_{imt}. The monthly wedge itself remains. A farm above i that diverts more than its schedule in a month at capacity leaves i short then, by equation (8), and i gets the water back later, when its crop needs less. Central Arizona wrote that shortages "usually occur during periods of above normal temperatures in the growing season when there is high demand, or at the end of the year when the mandated reduced allotment has been used" (CAIDD 2022). Diversions that escape the meter are a seasonal leak again, and the gradient returns in proportion to them. The allotments themselves need not be equal, as long as they do not fall toward the tail. In Maricopa–Stanfield they ranged

from 1.7 to 2.8 acre-feet per entitled acre, by where the grower's fields were (Section 7.3.12). The CAP districts show no fixed gradient. Water use among farmed fields does not fall toward the end of the canal network, and the head fallows most (Table A1). The statewide cross-section shows lower greenness toward the tail (Section 5.7).

Simulations. The simulations of Section 7.3.6 (Appendix A34) are the quantitative illustration of these cases, with planting fixed as in the model. Equal rationing, which gives every field the same share of its need in each month, leaves no in-season competition, and every estimate is exactly zero. Upstream first under a flat capacity that binds at the peak is case (a), with a gap at peak demand only (−1.86 at a crop coefficient of 1.1, +0.04 at 0.4). Fixing lateral shares and branch splits at 2017–2019 demand switches off case (b) and lowers the downstream slope from +4.94 to +2.26. The Maricopa–Stanfield runs with district wells are case (c). Their capacity term is positive in every rule, unlike the data, and no rule reproduces the rise of both sides together at 2 to 5 km (Table 8).

A3 Fallow-Share Symmetry Diagnostic

Table A2: Upstream and Downstream Fallow Shares Predict Own Fallowing

	Upstream only		*Downstream only*		*Joint*	
	Non-Rot. (1)	Rotation (2)	Non-Rot. (3)	Rotation (4)	Non-Rot. (5)	Rotation (6)
Upstream non-rot. fallow	0.568***	−0.007			0.423***	−0.009
	(0.032)	(0.012)			(0.015)	(0.006)
Upstream rot. fallow	−0.018	0.659***			−0.012	0.451***
	(0.014)	(0.024)			(0.010)	(0.013)
Downstream non-rot. fallow			0.556***	−0.012	0.415***	−0.011
			(0.031)	(0.010)	(0.015)	(0.007)
Downstream rot. fallow			−0.011	0.642***	−0.005	0.441***
			(0.010)	(0.021)	(0.005)	(0.014)
Observations	412,144	412,144	407,056	407,056	388,744	388,744
R^2	0.109	0.219	0.109	0.221	0.146	0.271
Year FE + Controls	✓	✓	✓	✓	✓	✓

Notes: Year FE. Two-way clustered SE (Year + plot). Sample: full plot-year panel (fallow status observed; no active-plot restriction). The comparable upstream and downstream fallow coefficients support the interpretation that local fallow variation reflects spatially correlated shocks, motivating the local-shock controls.

Table A1: In-Season Predictions and What the Data Show

Prediction (case)	**Test in Section 7.3**	**Supported?**	**Result**
Gap opens in the months the canal is short (a)	Contrast by month, Table 7, Fig. 8	✓	2022: about zero March–June, negative from July. 2021, cut from July: +1.7 (0.8) March–June, as in 2020; −1.6 (0.8) July–October. No late dip in 2020
Gap largest at own peak demand (a)	Own crop coefficient, Fig. 8	✓	−2.7 (0.6) at a crop coefficient of 1.1; +1.5 at 0.4
Negative upstream minus downstream contrast (a, b)	Swap test, Fig. 9	✓	$p = 0.006$ in 1,999 swaps by canal segment
Positive downstream slope (b)	Slopes by direction, Sec. 7.3.4	✓	Upstream −1.0, downstream +1.7; in simulation, routing gives about half of the downstream slope
Penalty only from upstream land with no district pump in between (c)	Pump split, Table 9	partial	No pump in between −2.2 (0.6); behind a pump +0.5 (0.8), slope in 2022 −0.7 (0.6). Within fields a pump serves, upstream minus downstream for land with no pump in between −3.1 (1.1). Pumps moved at random give −1.2 and −1.6 on average
Fields sharing a pump: those nearer it take first (a, c)	Near and far sharers, Sec. 7.3.5	✓	Near minus far −3.3 (1.0) in 2022; −1.0 in 2016, −0.7 in 2020–21; swap test by pump $p = 0.09$
With the canal source empty, the gap moves to the pumps (a, c)	2023, Sec. 7.3.10	✓	Canal contrast +0.2 on all fields; land with no pump in between −0.3 (0.8), against −3.3 in 2022; near minus far sharers −2.5 (1.2)
Little penalty where farms pump their own wells (c)	Hohokam and Harquahala, Table 9	✓	No pump in between +0.4 (0.8)
More pump capacity, smaller loss (c)	Capacity term, Sec. 7.3.5	No	−2.5 (1.0) per standard deviation of log capacity; +7 to +13 in every simulated rule of Table A41. Allotments set by where the fields are, which local reporting ties to wells (Sec. 7.3.12), lie outside the four cases
Fixed head-to-tail gradient (Sec. 3.2, first prediction)	Statewide cross-section, Sec. 5.7; CAP fields by fixed position	statewide, descriptive; not in CAP	Statewide: lower greenness toward the tail, ET null (Table 3). CAP: water use of farmed fields rises toward the end, +1.8 per standard deviation in 2017–19 and +2.2 in 2022; 2022 change +0.5 (0.6). In 2022 the head fallows 45 percent, the end 29, so the fields farmed at the head are a selected group
No fixed gradient, competition within the season (d)	Rows 1–3 and 10	✓	A gap in the months the canal is short; no fixed gradient in the CAP districts

Notes: Unless stated, percent change in monthly ET per ten points of the crop water demand of linked land at peak crop demand (own crop coefficient 1.1), as the 2022 change against 2017–2019, with standard errors in parentheses. Rows 6 and 7: the sharers are the other fields served by the field's nearest serving pump, split into those nearer the pump than the field and those farther; the swap test exchanges the two groups at random by pump. Row 7: the 2023 change against 2017–2019; land with no pump in between in Central Arizona and Maricopa–Stanfield, upstream minus downstream. Row 10: relative position along the canal network, 0 at the head and 1 at the end, fields in the three Pinal County districts; water use of farmed fields relative to the 2017–2019 normal of their crop and district, per standard deviation of position toward the end; the 2022 change compares each field with its own normal; fallow shares among fields farmed in 2017–2019, first and last fifth of position. Simulated values from Table A41. ✓: supported; partial: supported in sign, with the caveat shown; No: contradicted.

A4 Same-Block Local Fallowing Association

Table A3: Same-Block vs. Other-Block Neighbors and Fallowing

Distance band	Shared block	Other block	Ratio	N
$[0, 1)$ km	0.679^{***}	0.131^{***}	5.2×	120,512
	(0.012)	(0.006)		
$[1, 2)$ km	0.325^{***}	0.161^{***}	2.0×	198,288
	(0.014)	(0.008)		

Notes: LPM. DV: any strict fallowing indicator (stress or rotational). Shared-block neighbors have the focal plot's canal-block identifier; other-block neighbors are Euclidean neighbors in the same distance band without that shared identifier. Sample is restricted to boundary plot-years with both neighbor types in the band, so N is band-specific. Controls include the > 1 km upstream/downstream network terms interacted with SPEI_{24M}, distant strict-fallow shares, plot, distance, topography, soil, water-right, market-access, groundwater, and graph controls, plus flowline-type × distance-to-canal. FE: year + plot. Two-way clustered SE (year, plot). Tier-1 base sample restrictions apply. The contrast is a descriptive same-block association; the block identifier does not by itself verify a hydraulic link or distinguish shared operations, programs, or classification. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A5 Near/Far Common-Directional Fallowing Timing Diagnostic

Table A4 decomposes mapped activation on both sides of the focal plot into Common $= (Z^{up} + Z^{down})/2$ and Direction $= (Z^{up} - Z^{down})/2$, separately for connected plots within 5 km and beyond 5 km. Common is a symmetric log-gap mode, not pooled total acreage. Each timing column is estimated separately on the same 2017–2022 outcome-year support. The same-year close-common association is −6.89 percentage points per full-panel within-plot SD; it differs from the far-common slope under year-plus-plot inference ($p = 0.003$). The same-year close-directional and both far-mode estimates are small and imprecise. The lagged close estimates are suggestive but sensitive to the clustering level. The lead estimates are generally null under baseline, district, and WCC inference. With only six outcome-year clusters and twelve exploratory timing coefficients, the unstarred table is a local-co-movement and distance-exclusion diagnostic, not an extensive-margin mechanism test. Moreover, 1,963 of 3,601 plots have no eligible far descendants, so far common-directional separation comes from the remaining supported plots. No cross-timing equality test is reported because the three columns are separate overlapping regressions.

Supplement Table S1 holds each coefficient fixed and changes only the clustering structure. District-only and WCC-only clustering already permit arbitrary within-plot serial correlation because plots are nested in stable districts and WCCs; the complementary crossed estimators instead combine plot clustering with district-by-year or WCC-by-year clusters.

A6 Final-Status Fallowing Diagnostic Adjusted for Eligible Upstream Acreage

Table A5 applies the eligible-acreage and 2016-anchored activation-gap accounting decomposition of Table A18 to final broad-fallow status. It is separated from the active-crop table because same-year upstream active-crop status and focal fallowing are jointly realized and because the outcome-year May–August drought indicator is not information known when all planting decisions were made. District-by-boundary-side-by-year effects remove common annual changes in fallowing

Table A4: Near/far common-directional fallowing timing diagnostic

	Lagged exposure $(t-1)$	Current exposure (t)	Future exposure $(t+1)$
Close common mode	−2.406	−6.886	−0.705
	(1.209)	(1.114)	(0.743)
Close directional mode (Up − Down)	−1.795	−0.480	+0.065
	(0.813)	(0.782)	(0.399)
Far common mode	−0.167	+0.066	−0.096
	(0.603)	(0.577)	(0.520)
Far directional mode (Up − Down)	+0.562	−0.436	+0.065
	(0.353)	(0.417)	(0.557)

Notes: Each column is a separate regression on the common 2017–2022 outcome-year sample. Effects are fallowing percentage points per one full-panel within-plot SD of the displayed eligible-acreage-adjusted activation gap; two-way year and plot clustered standard errors are in parentheses. All models include outcome-year fixed-upstream drought, the final-status controls used in Appendix Table A5, plot fixed effects, and district-by-boundary-side-by-year fixed effects. Close is $0 < d \leq 5$ km and far is $d > 5$ km using Euclidean centroid distance. Common is the symmetric log-gap mode, not pooled total acreage. The future exposure is a negative-control association, not a causal placebo. There are only six outcome-year clusters, and the weaker lag estimates are sensitive to the clustering level. The table is an exploratory 12-coefficient timing family and therefore omits significance stars; two-sided p-values under six covariance estimators are reported separately. $N = 21,606$ in every column.

within those cells. Thus the remaining comparison concerns differential exposure within cells, not the total district fallowing response; failure to reject does not rule out coordinated acreage reductions. Network-year effects would likewise absorb a common network-year component. Neither set of effects necessarily absorbs heterogeneous responses across focal upstream sets. Clustering is a separate inference choice: changing clusters holds the coefficient and identifying variation fixed.

Table A5: Junior-Inside Final-Status Fallowing Decomposition

Outcome	All-active eligible acreage $\times D$	2016 gap $\times D$	Trend $\times D$	Residual $\times D$	Total gap $\times D$	$p(T = R = 0)$	N
Panel A: Ancestors beyond 5 km							
Fallowing (pp)	+0.499	+0.705	−0.143	+0.758	+0.350	0.612	28,808
	(0.669)	(1.518)	(0.876)	(0.742)	(0.726)		
Panel B: Ancestors beyond 10 km							
Fallowing (pp)	−0.433	+1.933	+1.049	+0.491	+0.738	0.523	18,856
	(1.234)	(1.064)	(0.889)	(0.844)	(0.933)		

Notes: The outcome is final broad-fallow status in percentage points for plots inside mapped boundaries of the ten surface-connected junior-rights districts. Exposure definitions, cutoff-specific eligible sets, eligible-acreage adjustment, component scaling, and the plot-specific fixed-upstream-set drought indicator match the 2016-anchored accounting decomposition in Table A18, not the district-drought decomposition in Table 5. Models use plot and district-by-boundary-side-by-year fixed effects, omit spatial-fallow controls because fallowing is the dependent variable, and cluster two ways by plot and calendar year. There are eight year clusters. Stars, if any, use two-sided unadjusted p-values; no directional one-sided test is used. The two time-varying components are jointly null at both cutoffs. These are same-year final-status associations, not pre-season behavioral responses. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A7 Joint Upstream–Downstream Regression Without Fallow Controls

Table A6: Joint Upstream–Downstream Regression Without Fallow Controls

	>1 km		>5 km		>10 km	
	UP×D	DN×D	UP×D	DN×D	UP×D	DN×D
GCVI, with fallow	−0.013*	−0.013***	−0.013*	−0.004	−0.030***	0.002
GCVI, no fallow	−0.013*	−0.013***	−0.014	−0.004	−0.030***	0.003
ET, with fallow	−0.010	−0.009**	−0.010	0.007*	0.001	0.009
ET, no fallow	−0.010	−0.008**	−0.011	0.005	0.000	0.010
N (with/no fallow)	218,597 / 218,597		62,048 / 62,048		21,396 / 21,396	
Plot + Crop×Year FE	✓		✓		✓	

Notes: UP = upstream cropped acreage; DN = downstream mirror, both entered in the same regression. Rows compare the with-fallow specification (matched spatial fallow-share controls included) to a no-fallow variant that drops only those controls, holding sample, FE, and cluster fixed. This table uses the same count-based scaling as Table A23 (log(1+active count) × Drought_{24M}); the with-fallow rows reproduce that table's joint-regression estimates, so coefficients are directly comparable in magnitude to Table A23. Active plots only. FE: Crop×Year + Plot. Full controls. Two-way clustered SE (Crop×Year + plot). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A8 Downstream Negative Control: Support and Common-Support Joint Regression

Table A7: Downstream Negative Control: Support and Common-Support Joint Regression

Cutoff	DN support	SD(UP)	SD(DN)	GCVI UP	GCVI DN	ET UP	ET DN
Full	0.930	0.187	0.200	−0.002**	−0.002***	−0.002	−0.001*
>1km	0.789	0.218	0.212	−0.003**	−0.002***	−0.002	−0.002**
>2km	0.652	0.224	0.200	−0.004**	−0.001	−0.003*	−0.000
>3km	0.557	0.222	0.191	−0.005**	−0.001	−0.004**	−0.000
>5km	0.435	0.211	0.187	−0.003	−0.001	−0.002	0.001
>10km	0.261	0.205	0.132	−0.007***	0.000	−0.001	0.002
Mid-canal N (first four cutoffs): Full: 289,482; >1km: 208,729; >2km: 141,701; >3km: 101,155							
Mid-canal N (wide cutoffs): >5km: 57,060; >10km: 18,607							

Notes: Standardized coefficients are per within-plot SD of the log active upstream or downstream count. Mid-canal denotes plots with nonzero upstream and downstream counts at the cutoff. DN support is the share of joint-regression sample plot-years with nonzero active downstream count. FE, controls, and two-way clustering match the joint-regression specification. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A9 Canal Size and Distant Upstream Cropped Acreage

Table A8: Canal Size × Spatial Exclusion Cutoff, Table A22 Specification

	Small (≤25 plots)	Medium (25–300)	Large (>300)
>1km upstream cropped acreage (spatial fallow controls, 1km)			
NDMI	0.002	0.002	−0.004*
	(0.002)	(0.001)	(0.002)
GCVI	0.002	0.012*	−0.017**
	(0.010)	(0.007)	(0.008)
ET	0.007	−0.001	−0.013
	(0.006)	(0.004)	(0.008)
>3km upstream cropped acreage (spatial fallow controls, 3km)			
NDMI	−0.004*	−0.002	−0.004
	(0.003)	(0.001)	(0.003)
GCVI	−0.008	−0.000	−0.013
	(0.011)	(0.007)	(0.009)
ET	−0.002	−0.009	−0.012
	(0.008)	(0.007)	(0.008)
>5km upstream cropped acreage (spatial fallow controls, 5km)			
NDMI	−0.005	−0.000	−0.004
	(0.005)	(0.002)	(0.003)
GCVI	−0.018	0.006	−0.015
	(0.020)	(0.009)	(0.010)
ET	0.008	−0.007	−0.014
	(0.016)	(0.007)	(0.009)
>10km upstream cropped acreage (spatial fallow controls, 10km)			
NDMI	−0.005	0.002	−0.008**
	(0.008)	(0.005)	(0.004)
GCVI	−0.008	0.011	−0.037***
	(0.034)	(0.027)	(0.012)
ET	−0.012	−0.010	−0.025*
	(0.025)	(0.012)	(0.013)
N (>1km)	14,934	66,307	188,629
N (>3km)	4,505	35,347	148,236
N (>5km)	2,102	18,649	116,478
N (>10km)	582	3,586	71,658
Plot + Crop×Year FE	✓	✓	✓

Notes: Active plots only. Entries report the distant upstream cropped acreage × Drought_{24M} interaction by canal-size bin, estimated under Table A22's exact specification: treatment is the unweighted count $\ln(1 +$ active upstream plots beyond d km), near-fallow controls scale with the cutoff, and no distant fallow shares are added. Coefficients are therefore directly comparable in magnitude to Table A22. FE: Crop×Year + Plot. Two-way clustered SE. Small- and medium-bin cells at wide cutoffs rest on small samples.

A10 Extended Distance Cutoffs (15 and 20 km)

Table A9: Distant Upstream Cropped Acreage × Drought at Extended Cutoffs

	>10km	>15km	>20km
Upstream cropped acreage × *Drought*$_{24M}$			
GCVI	−0.035***	−0.042***	−0.026
	(0.012)	(0.012)	(0.017)
ET	−0.023*	−0.031**	−0.022
	(0.012)	(0.015)	(0.023)
N	75,826	47,182	30,848

Notes: Upstream counts at 15 and 20 km are recomputed from the pair-level network data. Same specification as Table A22, except that matched-radius spatial fallow shares are not constructed beyond 10 km, so the 15 and 20 km columns reuse the 10 km local-shock controls. Samples at wide cutoffs increasingly select plots on long canal networks (7,412 plots retain an upstream ancestor beyond 15 km; 4,915 beyond 20 km); these columns are a diagnostic, not a preferred specification. Two-way clustered SE. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A11 Annual decomposition, robustness, and adjustment

Appendix Table A12 locates the baseline: WCC identity and fixed eligible upstream acreage explain 40.2% of its cross-plot variation, ordinal position adds 1.22 percentage points, and neither the incremental position component nor the unexplained remainder carries a detectable interaction. The attribution is descriptive and order-dependent.

When a continuous scarcity index replaces the indicator (Appendix Table A13), the residual interaction is approximately zero on both exposure scales, and the preperiod baseline remains negative. The annual pattern therefore belongs to drought years rather than rising smoothly with scarcity.

The annual interaction stays negative, with smaller magnitudes and weaker precision, under district- or network-specific linear trends, which allow gradual group trajectories without resolving abrupt common shocks (Appendix Table A15). It also stays negative when drought responses may vary with newly linked priority dates and preperiod wells, and shrinks to −0.006 ($p = 0.22$) once the loosely linked legacy rights bundle is added (Appendix Table A32).

Because drought is common within each district-year, district-by-year or network-by-year effects absorb it together with any shared district-year response, and the within-district interaction that remains is close to zero. Appendix Table A15 reports it as a within-system diagnostic rather than a test of the total district response to drought. District-by-period effects locate the variation the estimate uses. When the periods keep each district's drought years together with non-drought years, the annual interaction keeps its size (−0.011 to −0.015 per ten points; district wild-bootstrap p from 0.035 to 0.133); it disappears when the periods group the drought years 2021–22 together, as district-by-year effects do by construction (Appendix Table A16, all alignments). The annual result is therefore a comparison of each district's drought and non-drought years, not a difference between fields in the same district-year.

At the 5 km cut, a matched junior upstream–downstream comparison is inconclusive. Requiring farmland downstream beyond 5 km keeps 10,084 of the 22,105 plot-years and eight of the ten districts. Restricting the upstream-only model to these fields already reduces the interaction to −0.004 before any downstream term enters. At this cut, statewide directionality therefore does not validate the junior decomposition (Appendix Table A17). Section 7.3.2 compares upstream and downstream land in a different design, which uses monthly ET and a cut set from district size.

In the CAP districts, this cut keeps most of the fields that the 5 km cut drops.

Lags and leads of annual cropping

If the annual association were the effect of this year's upstream cropping on this year's downstream yield, upstream cropping in later years should carry nothing and earlier years little. Lags and leads test that reading directly.

Table A10 asks whether the residual association is confined to the outcome year. The exposure histories are built on the complete panel before selecting active GCVI rows, and every horizon is estimated on the same 2019–2022 support.

Table A10: Lag, Current, and Lead Profiles of Preperiod-Centered Upstream Activation Residuals

Exposure year	Log active	Activation share
Panel A: Separate marginal exposure × current-drought associations		
$t-3$	-0.010	-0.019
	(0.020)	(0.013)
$t-2$	$+0.009$	-0.011
	(0.014)	(0.010)
$t-1$	-0.014^{**}	-0.015^{*}
	(0.005)	(0.007)
t	-0.017^{**}	-0.020^{**}
	(0.005)	(0.007)
$t+1$	$+0.016^{**}$	$+0.008$
	(0.006)	(0.006)
Panel B: Shared-covariance timing tests (p-values)		
Past three years and current equal	0.442	0.880
Current equals one-year lead	< 0.001	0.003
Lead equals past/current average	0.029	0.004
Panel C: Joint distributed-lag exposure × drought associations		
Current t	-0.032^{***}	-0.024^{**}
	(0.009)	(0.010)
One-year lead $t+1$	$+0.032^{**}$	$+0.027^{**}$
	(0.011)	(0.009)
Joint five-horizon zero test: p	0.013	0.051
Observations per equation	10,980	

Notes: Exact common support consists of 10,980 active GCVI plot-years from 2019–2022 (3,356 plots, 10 districts, 33 WCCs). $t+h$ indexes only the preperiod-centered activation residual; the outcome, district drought indicator, preperiod spatial baseline, controls, and fixed effects remain at t. Panel A reports separate marginal regressions. Panel B reports district-clustered shared-covariance Wald tests from horizon-stacked equations that reproduce every Panel A coefficient. Panel C enters all five residual histories and interactions together. Log-active effects use the fixed full-sample within-plot SD; activation-share effects are per ten percentage points. Appendix Table A14 adds the continuous-scarcity columns. All displayed standard errors and stars use two-sided district-clustered inference. Drought is absent in 2019–2020, classifies 97% of rows in 2021, and 19% in 2022. A lead association cannot be a causal effect of future active-crop status on current GCVI. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The exposure relationship changes over the cropping cycle. Past and current departures in upstream cropped acreage carry similar negative drought interactions, whereas next year's depar-

ture has the opposite sign: entered jointly, the current interaction is −0.024 and the lead +0.027 ($p = 0.014$). Both signs survive every leave-one-district and leave-one-outcome-year run, the logged exposure scale, and the continuous scarcity index (Appendix Table A14).

The pattern therefore contains temporal adjustment beyond a fixed exposure ranking. The positive lead is compatible with subsequent adjustment or feedback, so the coefficients describe a dynamic association rather than the causal effect of an isolated annual planting decision. In the clock-time design of Appendix A12, whose exposure window does not overlap the outcome season, a genuinely future exposure carries no such lead.

Appendix Table A18 reports a level, trend, and residual decomposition of the full exposure path, including ET, and Appendix Table A19 the cutoff-specific cropped-share and priority profiles conditioned on eligible upstream acreage. The final-status fallowing diagnostic in Appendix Table A5 finds no detectable time-varying relationship. Same-year upstream cropped acreage and final fallow status are jointly realized. In the CAP districts, competition for canal water did not move fallowing either (Section 7.3.9).

A12 Seasonally ordered upstream water use

The annual measures are taken in the same year as the outcome: upstream cropping and downstream yield in the same year are realized together, so their co-movement could run in either direction or reflect a common within-season shock. Measuring upstream water use before the outcome season imposes the temporal order the mechanism requires. The January–April window, followed by an unused May–June buffer, records upstream decisions to farm and the water those fields consume before the July–October crop outcome, not at the same time. Satellite ET also measures water actually consumed rather than an annual crop-or-fallow label.

Upstream water use measured before the outcome season carries the same negative drought-year relationship. For all 3,601 plots inside the ten junior-rights district boundaries in 2020–2023, this design measures January–April OpenET among each plot's fixed upstream plots beyond 5 km and the plot's own July–October ET, with the district drought indicator lagged one year. Exposure therefore precedes outcome, and the sample never conditions on outcome-year crop status. In an additional specification without area-by-year effects, the total slope of late ET on early upstream water use in years following district drought is −0.224, and the drought-year shift is −0.288 (Appendix Figure A2). Full-refit ten-district wild-bootstrap p-values are 0.121 and 0.104. Next year's early upstream water use, which should carry nothing if the ordering is right, is small and indistinguishable from zero.

The two models with district-by-year or WCC-by-year effects are within-system comparisons that exclude the common district response, since those effects absorb the lagged drought indicator and the shared response by construction. Appendix S15 reports them.

The vegetation outcomes of the same design agree. Replacing late ET with late GCVI or NDMI, all twelve drought contrasts across outcomes, scales, and specifications are negative. In the specification without area-by-year effects the logged total drought-year slopes are −0.265 for GCVI and −0.073 for NDMI (wild $p = 0.139$ and 0.170; Appendix Table S11). This is consistent supporting evidence on a sample that retains fallowed focal plots, with precision throughout limited by ten districts and essentially one drought year.

A13 Economic calculation and benchmarks

Moving a field to a position with more upstream demand changes both its baseline and its annual departures, so an economic summary uses both coefficients, each multiplied by its own exposure. For every field-year I form the predicted drought-related shift in $\ln(1 + \text{GCVI})$, $\hat{b}_H H_i + \hat{b}_W W_{it}$, and compare fields at the 25th and 75th percentiles of that index. Comparing percentiles of the combined index respects the negative correlation between the two components (-0.37 in 2021–22); adding each component's own interquartile range would overstate the gap by about half.

The magnitudes come from a companion specification that interacts the same components with a common 2021–22 scarcity window, relative to 2016–2020 and with 2023 entered separately. The district SPEI-24 indicator used above misses the 2022 delivery cut in three of the four CAP-served districts and flags 2017 in one of them, so a common window compares the same years in every district; everything else in the specification is unchanged (Appendix S13). The departure and baseline interactions are -0.158 and -0.184 per unit share, against -0.14 and -0.25 with the district indicator of Table 5. I value a proportional change in $1 + \text{GCVI}$ as 0.8 times the same proportional change in revenue, using 2021–22 NASS Arizona revenue per acre.

A typical difference in network position then corresponds to 2.7 percent of revenue in a scarcity year, \$42 per acre of cotton (95 percent coefficient range \$14–73), and to \$37 with the well and rights controls. With separate crop-by-year effects for the CAP-served and the other junior districts, so that fields are compared only within a water source, the gap is \$21 per acre (\$6–41), 1.4 percent of revenue. Because the gap is a share of revenue, it applies to every crop: for alfalfa it is \$59 per acre pooled and \$30 within groups, and for corn \$44 and \$22. Cotton, alfalfa and corn cover 81 percent of cropped acres in these districts in 2021–22. Within those groups the GCVI response is close to what the accompanying response of consumptive use (ET) implies along the fields' own relationship between GCVI and ET, whereas the pooled figure also reflects differences between the two groups of districts. These are revenue-equivalent magnitudes of an association rather than losses in profit or welfare, and they scale with the assumed pass-through.

The gap is a difference between fields that face the same drought, so it comes on top of the drought's common effect on every field, which the crop-by-year effects absorb; it is not a loss shared by all farms. Two comparisons give it scale. Relative to margins it is large: University of Arizona Extension budgets for flood-irrigated alfalfa and cotton in southern Arizona report returns over total costs of \$203 and \$181 per acre, about a tenth of gross income (Evancho et al. 2023a, 2023b), so a gap of 1.4–2.7 percent of revenue corresponds to roughly 14–28 percent of the net return, before any savings in harvest costs that scale with yield. The gap is also of the same order as the drought's aggregate toll: California's statewide crop-revenue losses in the 2021 and 2022 droughts were 3.5 and 4.3 percent (Medellín-Azuara et al. 2022), although those include fallowing and are not differences between fields that remained in crop.[20]

A14 Priority and Service-Geography Profiles

The full priority-by-geography grid is retained here as an institutional-incidence diagnostic. Its same-year fallowing coefficients are not primary evidence: same-year upstream activation and focal

[20] Arizona revenue per acre is not a usable benchmark for a common drought loss: between 2016–2020 and 2021–22, NASS revenue per acre for cotton, alfalfa and corn rose by 42–53 percent as prices rose. Yields changed by -6 to $+7$ percent.

fallowing are jointly realized, and the separate timing check in Appendix A22 does not validate a pre-season network response.

Table A11: Activation-Gap Profiles, Adjusted for Eligible Upstream Acreage, by Priority and Service Geography

		GCVI		Fallowing (pp)		
Cutoff	Priority / geography	$Z \times D$	p	$Z \times D$	p	Plots / districts
5 km	Senior, inside	-0.006	0.530	+0.211	0.532	5,254 / 6
		(0.009)		(0.320)		
	Junior, inside	-0.028***	< 0.001	+0.350	0.645	3,600 / 10
		(0.007)		(0.726)		
	Junior, outside	+0.009	0.536	-0.426	0.822	2,755 / 10
		(0.015)		(1.819)		
10 km	Senior, inside	-0.028	0.134	+2.171**	0.026	2,759 / 6
		(0.019)		(0.772)		
	Junior, inside	-0.034*	0.084	+0.738	0.455	2,357 / 9
		(0.020)		(0.933)		
	Junior, outside	-0.044**	0.038	-2.796	0.208	1,910 / 10
		(0.021)		(2.014)		

Notes: For each plot and cutoff, $K_i = \log(1 + C_i)$ is logged all-active eligible upstream source acreage and $Z_{it} = \log(1 + L_{it}) - K_i$ is the cutoff-matched activation gap, equal to zero under full activation. Entries report total $Z_{it} \times D_t$ per one within-plot SD from models that also include $K_i \times D_t$; they combine persistent, trending, and residual activation rather than isolating annual innovations. This is the eligible-acreage-adjusted exposure used in Table A18, not the two-component baseline/residual decomposition in Table 5. GCVI uses active plots, plot and Crop×Year fixed effects, one cutoff-matched broad CDL fallow share, and two-way Crop×Year/plot clustering. Fallowing uses all observed broad-fallow rows, no spatial-fallow control, plot and district-by-inside/outside-by-year fixed effects, and two-way year/plot clustering; it is displayed in percentage points and has eight year clusters. The support column reports active-crop plots/districts from the GCVI model. Supply and priority come from the independently verified 30-district audits. "Inside" means the field lies inside the mapped district boundary; "outside" is canal-connected nearest-system geography, not verified service. Stars and p-values are two-sided and unadjusted across the displayed grid. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A15 Exposure Decompositions and Eligible-Acreage Definitions

At cutoff d, let $\mathcal{A}_i(d) = \{j : j \text{ is a mapped ancestor of } i,\ \text{Dist}(i,j) > d\}$, with the strict exclusion defined by Euclidean centroid distance. If w_j is source acreage and I_{jt} indicates mapped active-crop status, define active upstream acreage $L_{it}(d) = \sum_{j \in \mathcal{A}_i(d)} w_j I_{jt}$, all-active source acreage $C_i(d) = \sum_{j \in \mathcal{A}_i(d)} w_j$, its logged eligible acreage $K_i(d) = \log[1 + C_i(d)]$, and the zero-safe activation gap

$$Z_{it}(d) = \log[1 + L_{it}(d)] - K_i(d) = \log\left(\frac{1 + L_{it}(d)}{1 + C_i(d)}\right) \leq 0. \tag{10}$$

$Z_{it}(d) = 0$ under full activation of the same cutoff-eligible set. $K_i(d)$ is a cardinal measure of all-active eligible acreage, not hydraulic capacity or ordinal network position; the latter remains the signed-hop rank used in Table 3. Plot fixed effects absorb the level of $K_i(d)$, not its interaction with a drought indicator that changes over time, so a conventional $Z_{it} \times D_t$ coefficient can still combine fixed and time-varying exposure differences.

The main decomposition therefore begins with either logged active upstream acreage $E^Q_{it} = \log(1 + L_{it})$ or the unit-free activation share $E^U_{it} = L_{it}/C_i$. For scale $m \in \{Q, U\}$, define

$$H^m_i = \frac{1}{4} \sum_{s=2016}^{2019} E^m_{is}, \qquad W^m_{it} = E^m_{it} - H^m_i. \tag{11}$$

Here W^m_{it} is the preperiod-centered activation residual used in main Table 5. It is current exposure after removing the fixed plot-specific preperiod mean, not a detrended or unexpected innovation.

The appendix then projects the preperiod baseline sequentially. The first projection uses WCC indicators and a quadratic in standardized K_i; after centering its fitted value, call that component A_i^m. A cubic Legendre basis in normalized signed-hop position is residualized against the first-stage regressors, and its fit to the first-stage residual is P_i^m. The remainder is R_i^m, giving the exact identity

$$E_{it}^m = c^m + A_i^m + P_i^m + R_i^m + W_{it}^m. \tag{12}$$

Main Table 5 interacts H_i^m and W_{it}^m with the district drought indicator. Appendix Table A12 instead enters all four nonconstant components of Equation 12 jointly. The attribution is order-dependent: network identity and eligible acreage receive shared variation before ordinal position. A_i^m is therefore a network-and-eligible-acreage fitted exposure baseline rather than a hydraulic-structure effect; P_i^m is position-predicted exposure conditional on that first block rather than a direct positional effect on GCVI; and R_i^m is unexplained baseline heterogeneity, not dynamic persistence.

Table A12: Detailed Spatial Decomposition of the Preperiod Baseline

	Binary district drought		Continuous district scarcity	
Component × drought	Log active	Activation share	Log active	Activation share
Network-and-eligible-acreage fitted baseline	−0.028*	−0.047***	−0.017***	−0.019***
	(0.017)	(0.012)	(0.006)	(0.006)
Incremental ordinal-position component	−0.006	−0.008	−0.003	−0.003
	(0.009)	(0.008)	(0.004)	(0.004)
Unexplained baseline heterogeneity	−0.002	−0.005	+0.002	−0.000
	(0.010)	(0.011)	(0.005)	(0.005)
Preperiod-centered activation residual	−0.013	−0.013**	−0.000	+0.001
	(0.009)	(0.006)	(0.003)	(0.003)
Preperiod-centered residual slope during drought / at +1 SD scarcity	−0.013	−0.013**	−0.001	−0.001
	(0.008)	(0.005)	(0.006)	(0.004)
Equal component slopes: p	0.668	0.008	0.526	0.006
Observations	22,105			

Notes: The sample, 5 km ancestor set, outcome, district drought indicators, controls, fixed effects, and displayed plot/Crop×Year clustering match main Table 5. The 2016–2019 preperiod baseline is decomposed sequentially into a network-and-eligible-acreage fitted component, an incremental ordinal-position component, and unexplained baseline heterogeneity; the preperiod-centered activation residual completes the exact identity. All four component-by-drought terms enter jointly with the residual and drought main effects. Baseline components are displayed per their own cross-plot SD. Logged residual exposure is per one full-sample within-plot SD and the share residual is per 0.10. Displayed magnitudes use different scales and should not be added; equality tests use native common units. The projection is descriptive and order-dependent, and none of its components measures hydraulic capacity, diversion, delivery, or water flow. $^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Tables A13 and A14 extend main Table 5 and Appendix Table A10 with two further columns in which a continuous district scarcity index, the negative of the district SPEI-24 index standardized over the model sample, replaces the binary drought indicator. Under the continuous index the preperiod baseline interaction remains negative on both exposure scales. The residual interaction is approximately zero, and the lead reversal of the timing profile is unchanged.

Table A13: Preperiod Baseline and Annual Residual: Binary Drought and Continuous Scarcity

	Binary district drought		Continuous district scarcity	
Exposure component	Log active	Activation share	Log active	Activation share
Panel A: Exposure components × drought				
Preperiod 2016–2019 spatial baseline × drought	−0.028	−0.034***	−0.017***	−0.013**
	(0.017)	(0.013)	(0.006)	(0.006)
Preperiod-centered activation residual × drought	−0.013	−0.014**	−0.001	+0.000
	(0.009)	(0.007)	(0.003)	(0.003)
Panel B: Preperiod-centered residual slope				
Outside drought / at mean scarcity	+0.001	−0.000	−0.001	−0.002
	(0.004)	(0.004)	(0.004)	(0.003)
During drought / at +1 SD scarcity	−0.013	−0.014**	−0.002	−0.002
	(0.008)	(0.006)	(0.005)	(0.004)
Equal baseline and residual interactions: p	0.362	0.191	0.378	0.037
Observations	22,105			

Notes: Sample, exposure scales, controls, fixed effects and displayed clustering as in main Table 5, whose first two columns are repeated here. The binary indicator equals one when the acreage-weighted SPEI-24 mean over the fixed set of all plots mapped inside the district is below −0.6; continuous scarcity is the negative of that district index, standardized over the full model sample, and its interactions are per one-SD increase in scarcity. The baseline is per one between-plot SD; logged residual exposure is per one full-sample within-plot SD and the share residual is per 0.10. Component magnitudes use different scales and should not be added; equality tests use native common units. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A14: Lag, Current, and Lead Profiles: Binary Drought and Continuous Scarcity

	Binary district drought		Continuous district scarcity	
Exposure year	Log active	Activation share	Log active	Activation share
Panel A: Separate marginal exposure × current-drought associations				
$t-3$	−0.010	−0.019	−0.005	−0.010**
	(0.020)	(0.013)	(0.006)	(0.004)
$t-2$	+0.009	−0.011	−0.009*	−0.008**
	(0.014)	(0.010)	(0.004)	(0.003)
$t-1$	−0.014**	−0.015*	+0.000	−0.001
	(0.005)	(0.007)	(0.003)	(0.002)
t	−0.017**	−0.020**	−0.003	−0.004**
	(0.005)	(0.007)	(0.002)	(0.002)
$t+1$	+0.016**	+0.008	+0.008**	+0.006**
	(0.006)	(0.006)	(0.003)	(0.002)
Panel B: Shared-covariance timing tests (p-values)				
Past three years and current equal	0.442	0.880	0.439	0.141
Current equals one-year lead	< 0.001	0.003	0.018	< 0.001
Lead equals past/current average	0.029	0.004	0.003	< 0.001
Panel C: Joint distributed-lag exposure × drought associations				
Current t	−0.032***	−0.024**	−0.004	−0.005
	(0.009)	(0.010)	(0.002)	(0.003)
One-year lead $t+1$	+0.032**	+0.027**	+0.011**	+0.010**
	(0.011)	(0.009)	(0.004)	(0.003)
Joint five-horizon zero test: p	0.013	0.051	0.050	0.021
Observations per equation	10,980			

Notes: Support, construction, panels and district-clustered inference as in Appendix Table A10, whose two columns are repeated here. Continuous columns represent a one-SD increase in district scarcity in place of the binary indicator. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Linear trends and area-by-year comparisons. Table A15 reports an exploratory sensitivity of the binary-drought activation-share model, on the same 22,105 active plot-years, 3,600 plots, ten districts, and 35 WCCs. District or network linear outcome trends are fitted jointly over

2016–2023, not estimated only before drought. They leave H_i, W_{it}, the district drought definition, and the remaining controls unchanged. These are not the exposure trends in Table A18, and the coefficients are not those of the undecomposed statewide count model.

Shared ancestors make activation changes move together across the focal plots of a network, and the drought indicator is constant within a district-year. Writing $W_{it} = \overline{W}_{gt} + v_{it}$ for area g and year t, the interaction is $\overline{W}_{gt}D_{gt} + v_{it}D_{gt}$. Because drought is common within each district-year, district-by-year effects absorb its main effect together with all shared district-year outcome changes, including any common supply and cropping response to drought, so the specification cannot recover that component of system adjustment. The remaining interaction asks whether the relationship between relative upstream exposure and relative GCVI differs between drought and other years within districts. That estimate is close to zero. Table A15 therefore provides a within-system diagnostic rather than a test of the total district response to drought.

After residualizing the interaction against all other regressors and fixed effects, district- and network-year specifications retain 75.5% and 56.5% of the baseline residual variance, all of it plot-specific. Linear trends retain 98.4% and 94.0%, with smaller negative coefficients; these are related sensitivity estimates, not independent replications or tests of cross-model coefficient differences. Changing the clustering level, in contrast to adding fixed effects, changes uncertainty without removing variation or changing the fitted coefficient.

Table A15: Annual Activation: Linear-Trend Sensitivity and Within-Area-Year Diagnostics

	Drought interaction	Total slope in drought	Wild p interaction	Wild p total	Residual variance (%)
A. Baseline and smooth-trend sensitivity					
Baseline	−0.01415	−0.01422	0.039	0.150	100.0
	(0.00527)	(0.00700)			
District linear trends	−0.01057	−0.01162	0.064	0.221	98.4
	(0.00436)	(0.00707)			
Network linear trends	−0.00836	−0.00954	0.184	0.283	94.0
	(0.00588)	(0.00807)			
B. Area-by-year effects (within-system diagnostic)					
District×year FE	−0.00081	0.00001	—	—	75.5
	(0.00449)	(0.00815)			
Network×year FE	0.00035	0.00344	—	—	56.5
	(0.00302)	(0.00462)			

Notes: All rows use the same junior-inside sample, >5 km Euclidean ancestor exclusion, outcome $\ln(1 + \text{GCVI})$, plot and current Crop×Year effects, cutoff-matched broad-fallow control, and nuisance controls. H_i is the 2016–2019 upstream activation-share mean and W_{it} its annual departure; $H_i \times D_{it}$ remains included. Both displayed effects are per 0.10 share. The drought interaction is the coefficient on $W_{it} \times D_{it}$; the total drought-year slope adds the W_{it} main coefficient, with covariance retained. District-clustered standard errors are in parentheses. Wild p-values impose each null, refit all nuisance slopes and fixed effects, and enumerate 1,024 Rademacher assignments across ten districts; this is exhaustive computation, not a distribution-free test. Dashes mark bootstrap tests not rerun for the area-year benchmarks. Residual variance is relative to the baseline after all other regressors and fixed effects, not raw drought variance. For baseline, district trends, and network trends, respectively, interaction p-values are 0.033, 0.052, 0.108 under plot/Crop×Year clustering; 0.025, 0.038, 0.189 under district clustering; and 0.036, 0.068, 0.203 under WCC clustering.

Table A16 replaces the district-by-year effects with district-by-period effects of two and three years, in every alignment of the period boundaries.

Table A16: Annual Activation: District-by-Period Fixed Effects

		Annual departure × drought			Baseline × drought		
Added fixed effects	Periods	Estimate	District p	Wild p	Estimate	Variance kept	Drought-varying cells
None (main specification)	—	−0.014 (0.007)	0.025	0.039	−0.025 (0.009)	100.0%	—
District linear trends	—	−0.011 (0.005)	0.038	0.064	−0.023 (0.009)	98.4%	—
District × 3-year period	2016–18, 2019–21, 2022–23	−0.015 (0.006)	0.016	0.035	−0.025 (0.009)	94.3%	15 of 30
District × 3-year period	2016–17, 2018–20, 2021–23	−0.005 (0.005)	0.226	0.236	−0.017 (0.009)	92.6%	10 of 30
District × 3-year period	2016, 2017–19, 2020–22, 2023	−0.011 (0.006)	0.069	0.078	−0.016 (0.009)	96.2%	10 of 40
District × 2-year period	2016–17, 2018–19, 2020–21, 2022–23	−0.012 (0.006)	0.103	0.133	−0.019 (0.010)	88.9%	15 of 40
District × 2-year period	2016, 2017–18, 2019–20, 2021–22, 2023	−0.002 (0.005)	0.518	0.508	−0.008 (0.009)	90.7%	5 of 50
District × year	each year	−0.001 (0.005)	0.861	—	−0.006 (0.010)	75.5%	0 of 80

Notes: Same junior-inside sample and specification as Table A15: >5 km Euclidean ancestor exclusion, outcome $\ln(1+\text{GCVI})$, plot and crop×year effects, and nuisance controls, with the listed effects added. Estimates are per 0.10 of upstream cropped share; parentheses are two-way standard errors; district p uses district clustering; wild p is the null-imposed full-refit district wild bootstrap over all 1,024 Rademacher assignments, not computed for district-by-year effects. Variance kept is the residual variance of the annual interaction after all other regressors and fixed effects, relative to the main specification. Drought-varying cells count the district-periods that contain both drought and non-drought years. Every alignment of two- and three-year periods over 2016–2023 is reported. The main-specification, district-trend, and district-by-year rows reproduce the coefficients of Table A15.

Matched junior directionality and the change in support. Table A17 separates sample restriction from adding downstream exposure. The first column repeats the preferred >5 km annual-share model. The second changes only the outcome sample, retaining fields with positive eligible descendant acreage beyond the same cutoff. This removes 54.4% of active plot-years and two districts: the sample falls from 22,105 to 10,084 observations, and the upstream drought interaction moves from −0.01415 to −0.00388. The exposure baseline, drought definition, and upstream-only formula are unchanged. The third column then adds the downstream annual departure, its drought interaction, and its inherited-baseline interaction on exactly that restricted sample; the upstream interaction is −0.00455.

The joint upstream-minus-downstream interaction is +0.00373, with a two-way-clustered 95% interval of [−0.01656, 0.02401] and a district-clustered interval of [−0.04286, 0.05032]. This is inconclusive about junior directionality, not evidence of equal effects. Most of the fall in the upstream coefficient occurs before downstream controls enter, so it cannot be described as downstream adjustment explaining away the full-sample result. Conversely, the statewide directional test cannot supply the missing junior contrast. The comparison is descriptive across different support; it does not test equality of the full- and restricted-sample coefficients.

Table A17: Junior Annual Activation: Directional Comparison and Sample Restriction

	Full sample Upstream only	Matched sample Upstream only	Matched sample Joint U/D
Upstream $W \times D$	−0.01415	−0.00388	−0.00455
Two-way p	0.033	0.583	0.541
District p	0.025	0.425	0.404
Downstream $W \times D$	—	—	−0.00828
District p	—	—	0.698
Upstream minus downstream	—	—	+0.00373
Two-way p	—	—	0.717
District p	—	—	0.855
Active plot-years	22,105	10,084	10,084
Fields	3,600	1,638	1,638
Districts	10	8	8
Networks (WCCs)	35	18	18

Notes: Outcome: ln(1 + GCVI) on active plot-years inside the ten junior districts, 2016–2023. Upstream and downstream shares use fixed eligible acreage beyond 5 km Euclidean distance along their respective directed graph sets; each is split into its 2016–2019 mean H and annual departure W. All columns retain the same district drought indicator, plot and Crop×Year effects, original controls, cutoff-matched broad-fallow control, and no observation weights. The matched sample additionally requires positive eligible downstream acreage; it contains eight districts and 18 WCCs. Joint U/D adds downstream W, $H \times D$, and $W \times D$. Displayed coefficients are drought amplification per 0.10 share, not total drought-year slopes. Contrast inference retains the upstream–downstream covariance. Two-way p-values cluster by plot and Crop×Year; district p-values use conventional cluster inference (ten or eight districts), not the wild bootstrap in Table A32. No stars or cross-sample equality test are reported.

Single-district influence in the shared/within decomposition. A supplementary model separates W_{it} into the equally weighted district- or network-year mean over a fixed recipient roster and the plot-specific departure from that mean. Both main terms and drought interactions enter jointly with the original controls; each grouping is estimated with and without its corresponding linear outcome trends. Freezing the recipient roster, means, and upstream sets before deleting outcome rows, I omit each of ten districts from each of the four models. Both drought interactions remain negative in all 40 omission fits. Thus no single district creates their negative signs; the omissions change the estimation sample, not the graph or upstream exposure. Magnitudes and inference remain qualified: full-sample district wild-bootstrap p-values for the shared interaction range from 0.121 to 0.322, and the shared-minus-within contrasts from 0.248 to 0.660. Omitting Montezuma substantially shrinks the network-trend shared coefficient. These overlapping influence checks narrow a single-district explanation, not shared confounding, and are not independent replications of the original annual coefficient.

Table A18 reports an alternative decomposition of the activation gap into its observed 2016 level, a plot-specific linear trend fitted to all 2017–2023 changes and constrained through 2016, and the exact residual. This is useful descriptive accounting, including for ET. It is not the preferred temporal decomposition: because the fitted slope uses the complete path, early residuals incorporate later exposure observations.

Table A18: 2016-Anchored Full-Path Accounting Decomposition

Outcome	All-active eligible acreage $\times D$	2016 gap $\times D$	Trend $\times D$	Residual $\times D$	Total gap $\times D$	$p(T = R = 0)$	N
Panel A: Ancestors beyond 5 km							
GCVI	-0.035^{**}	-0.012	-0.008	-0.033^{***}	-0.028^{***}	< 0.001	22,105
	(0.017)	(0.018)	(0.008)	(0.008)	(0.007)		
					$p_{\text{dir}} < 0.001$		
ET	$+0.004$	-0.012	$+0.001$	-0.007	-0.006	0.388	22,105
	(0.009)	(0.010)	(0.005)	(0.005)	(0.004)		
					$p_{\text{dir}} = 0.090$		
Panel B: Ancestors beyond 10 km							
GCVI	-0.048^{**}	-0.008	-0.043^{***}	-0.054^{***}	-0.034^{*}	< 0.001	14,530
	(0.023)	(0.022)	(0.014)	(0.016)	(0.020)		
					$p_{\text{dir}} = 0.042$		
ET	-0.014	$+0.006$	-0.002	-0.006	-0.003	0.720	14,530
	(0.009)	(0.017)	(0.008)	(0.008)	(0.008)		
					$p_{\text{dir}} = 0.362$		

Notes: Active focal plots inside mapped boundaries of the ten surface-connected junior-rights districts. $K_i = \log(1 + C_i)$ is fixed logged eligible upstream source acreage and $Z_{it} = \log(1 + L_{it}) - K_i$. The table writes $Z_{it} = B_i + T_{it} + R_{it}$, where $B_i = Z_{i,2016}$, T_{it} is the plot-specific linear trend fitted with the complete 2017–2023 exposure path, and R_{it} is the ex-post remainder. $K_i \times D_t$ and all three component interactions enter jointly. "Total gap" comes from a separate undecomposed model retaining $K_i \times D_t$. GCVI and ET use plot and Crop×Year fixed effects, the cutoff-matched broad CDL fallow share, and two-way plot/Crop×Year clustering. Component-specific scaling differs, so displayed magnitudes should not be added. The residual is not an as-of annual innovation, and lagging or leading it does not restore temporal ordering because its fitted trend uses future observations. Stars use two-sided unadjusted p-values; p_{dir} is a one-sided directional diagnostic. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

The downstream construction is identical on the cutoff-eligible descendant set. The raw directional acreage contrast decomposes exactly as

$$\log\left(\frac{1 + L_{it}^{U}}{1 + L_{it}^{D}}\right) = \left(K_i^{U} - K_i^{D}\right) + \left(Z_{it}^{U} - Z_{it}^{D}\right). \tag{13}$$

Thus upstream minus downstream separates a fixed all-active eligible-acreage balance from a same-year directional activation balance; it does not turn mapped cropping activity into theoretical water availability. Direction and timing are distinct axes: upstream versus downstream asks whether the pattern follows the mapped graph rather than symmetric surrounding activity, whereas exposure fixed in advance versus same-year exposure asks what is known when acreage is chosen. Count-based definitions replace acreage sums with counts and use the same cutoff-matched eligible set.

A16 Unit-Free Activation-Share Normalization

The activation gap used in the priority and full-path accounting tables is zero-safe and uses an identical cutoff-specific set in its numerator and benchmark. Its acreage version contains a one-acre offset. Table A19 therefore reports the eligible-acreage-conditioned grid using $G_{it} = L_{it}/C_i - 1 \in [-1, 0]$ on $C_i > 0$ support. A 0.10 increase means ten percentage points more of eligible upstream source acreage is active, independent of whether acreage is recorded in acres or another unit. That grid continues to include $\log(1 + C_i) \times D_t$. The main spatial–temporal decomposition instead uses L_{it}/C_i directly; subtracting the fixed 2016–2019 mean removes the constant difference between L/C and $L/C - 1$ from its preperiod-centered activation residual.

Table A19: Unit-Free Activation-Share Robustness for the Junior-Inside and Priority Profiles

Outcome	Cutoff	$K \times D$	$B \times D$	$T \times D$	$R \times D$	Total $G \times D$	N
Panel A: Junior-inside decomposition							
GCVI	5 km	-0.032^{**}	-0.011	-0.018	-0.031^{***}	-0.024^{***}	22,105
		(0.014)	(0.010)	(0.013)	(0.007)	(0.006)	
Fallowing (pp)	5 km	$+0.504$	$+0.594$	$+0.011$	$+0.302$	$+0.091$	28,808
		(0.532)	(0.874)	(1.029)	(0.304)	(0.409)	
GCVI	10 km	-0.040^{*}	-0.021^{*}	-0.074^{***}	-0.034^{***}	-0.032^{**}	14,530
		(0.020)	(0.012)	(0.017)	(0.010)	(0.013)	
Fallowing (pp)	10 km	-0.392	$+1.951^{**}$	$+1.726$	$+1.368^{*}$	$+1.177$	18,856
		(1.299)	(0.690)	(0.951)	(0.688)	(0.707)	

Panel B: Total activation-share gap by priority and geography						
Cutoff	Sample	GCVI $G \times D$ / p		Fallowing $G \times D$ / p		N (GCVI / fallow)
5 km	Senior, inside	-0.008	0.389	$+0.174$	0.677	39,060 / 42,040
	Junior, inside	-0.024^{***}	< 0.001	$+0.091$	0.831	22,105 / 28,808
	Junior, outside	$+0.001$	0.922	-0.692	0.644	17,146 / 22,072
10 km	Senior, inside	-0.021	0.146	$+1.564^{**}$	0.036	20,937 / 22,080
	Junior, inside	-0.032^{**}	0.016	$+1.177$	0.140	14,530 / 18,856
	Junior, outside	-0.080^{**}	0.011	-2.403	0.277	12,253 / 15,296

Notes: $G = L/C - 1$, where L is active upstream acreage and C is all-active upstream source acreage in the same strict Euclidean-distance-eligible set. Panel A repeats the 2016-anchored full-path accounting decomposition of Table A18 on the unit-free scale. $K \times D$ is per one cross-plot SD of $\log(1 + C)$; $B \times D$, $T \times D$, $R \times D$, and total $G \times D$ are effects of a 0.10 increase in the corresponding activation-share component. Panel B reports total-gap effects for a 0.10 increase in G. Fallowing is in percentage points. These models use the plot-specific fixed-upstream-set drought indicator, not the district-level indicator of Table 5 and Appendix Table A10. Controls, fixed effects, and clustering otherwise match the corresponding active-crop and fallowing specifications; fallowing inference has eight year clusters. Each cutoff redefines the eligible set, C_i, and supported plots, so 5 and 10 km are distinct nested-set quantities. Stars use two-sided unadjusted p-values. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A17 Alternative Specifications

Table A20: Robustness to Alternative Fixed Effects, Controls, and Clustering

	Baseline	Simpler FE	Crop×Yr×Cty FE	Canal×Yr FE	Min. Ctrl	Cluster: Plot
	(1)	(2)	(3)	(4)	(5)	(6)
Cross-Section Position × Drought						
Position × Drought$_{24M}$						
GCVI	−0.025	0.004	−0.028	−0.067**	−0.034	−0.025**
	(0.020)	(0.023)	(0.020)	(0.030)	(0.024)	(0.011)
ET	−0.029	0.018	−0.015	−0.015	−0.034	−0.029***
	(0.019)	(0.021)	(0.019)	(0.029)	(0.024)	(0.009)
Position level						
GCVI	−0.032***	−0.021	−0.006	−0.021	−0.009	−0.032***
	(0.011)	(0.013)	(0.011)	(0.014)	(0.010)	(0.008)
ET	−0.003	−0.004	0.011	0.015	0.004	−0.003
	(0.012)	(0.014)	(0.012)	(0.014)	(0.010)	(0.007)
N	326,732	326,732	326,732	326,732	327,145	326,732
FE	Baseline	Crop×Year	County-rich	Canal×Year	Baseline	Baseline
Cluster	Two-way	Two-way	Two-way	Two-way	Two-way	Plot
Controls	Full	Full	Full	Full	Minimal	Full

Notes: This panel replicates the cross-section position × Drought specification of Table 3 across six FE, clustering, and control variants. Baseline FE are Crop×Year + Irrigation District×WCC×Year; "county-rich" denotes Crop×Year×County FE. The position level effect is significant only under the baseline and plot-clustered specifications; the interaction term, insignificant in Table 3 itself, is significant in two of six variants and changes sign in one. Read this as showing FE sensitivity, not uniform robustness.

	Baseline	Simpler FE	Crop×Yr×Cty FE	Canal×Yr FE	Min. Ctrl	Cluster: Plot
	(1)	(2)	(3)	(4)	(5)	(6)
Distant Upstream Cropped Acreage, >1km Cutoff						
Distant upstream cropped acreage × Drought$_{24M}$						
GCVI	−0.015**	−0.018***	−0.011**	−0.010*	−0.015**	−0.015***
	(0.006)	(0.007)	(0.005)	(0.005)	(0.006)	(0.001)
ET	−0.011	−0.016**	−0.011**	−0.011**	−0.011	−0.011***
	(0.007)	(0.006)	(0.006)	(0.005)	(0.007)	(0.001)
Distant upstream cropped acreage, level						
GCVI	−0.009	0.005	0.002	0.015	−0.009	−0.009**
	(0.012)	(0.013)	(0.011)	(0.011)	(0.012)	(0.004)
ET	−0.022**	−0.016	−0.005	−0.005	−0.022**	−0.022***
	(0.010)	(0.010)	(0.012)	(0.009)	(0.009)	(0.004)
N	269,870	269,870	269,870	269,870	270,139	269,870
FE	Baseline	Year + Plot	County-rich	Canal×Year	Baseline	Baseline
Cluster	Two-way	Two-way	Two-way	Two-way	Two-way	Plot
Controls	Full	Full	Full	Full	Minimal	Full

Notes: This panel replicates Table A22's > 1 km distant upstream cropped acreage × Drought interaction across the same six variants. Baseline FE are Crop×Year + Plot; "county-rich" denotes Crop×Year×County + Plot FE. The GCVI interaction remains negative and significant across all six columns. The ET interaction is significant in four of six. Coefficients and standard errors are unchanged from the combined robustness table; the split is for readability.

A18 Inference at the Network Level

The treatment varies at the canal-network level rather than the plot level, so this appendix re-estimates the Table A22 interaction under variance estimators that cluster where assignment is correlated: by weakly connected component, by canal block, and by WCC × year. Coefficients are identical across rows within a cutoff, since only the variance estimator changes. The >10 km GCVI interaction remains significant at the 5% level under WCC clustering and is marginal under canal-block clustering, on 114 and 81 clusters respectively. The cell is informative rather than individually decisive (Section 6.2).

Design-based randomization inference cannot adjudicate this cell: with only 34 exchangeable components on the both-support subsample at >10 km, the minimum detectable effect of a component-level reassignment test exceeds the estimate it would adjudicate. Cluster-robust inference at the component level is therefore the informative alternative, and it is what the table reports.

Table A21: Inference at the Network Level

	Full		>1km		>5km		>10km	
Inference	GCVI	ET	GCVI	ET	GCVI	ET	GCVI	ET
Two-way: crop-year + plot	−0.015** (0.007)	−0.011 (0.007)	−0.015** (0.006)	−0.011 (0.007)	−0.017** (0.008)	−0.014 (0.009)	−0.035*** (0.012)	−0.023* (0.012)
One-way: WCC	−0.015 (0.010)	−0.011 (0.009)	−0.015 (0.010)	−0.011 (0.009)	−0.017 (0.013)	−0.014 (0.013)	−0.035** (0.017)	−0.023 (0.018)
One-way: canal block	−0.015 (0.010)	−0.011 (0.010)	−0.015 (0.010)	−0.011 (0.010)	−0.017 (0.014)	−0.014 (0.013)	−0.035* (0.019)	−0.023 (0.019)
Two-way: WCC + year	−0.015 (0.010)	−0.011 (0.011)	−0.015 (0.010)	−0.011 (0.010)	−0.017 (0.013)	−0.014 (0.013)	−0.035* (0.017)	−0.023 (0.018)
N	326,732	326,732	269,870	269,870	137,229	137,229	75,826	75,826

Notes: Active plots only. The specification matches Table A22; treatment varies at the canal-network level, so these variants cluster where assignment is correlated. Coefficients are identical across rows within a cutoff because only the variance estimator changes. Cluster counts (Full, >1 km, >5 km, >10 km): crop-year/plot = 425/54,043; 420/44,184; 393/22,160; 355/12,054; WCC = 1,812; 1,086; 355; 114; canal block = 1,220; 794; 262; 81; WCC + year uses the same WCC counts plus 8 years. Year-level bootstrap inference was unstable with 8 clusters and is omitted. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A19 Distance-Cutoff and Joint Directional Estimates

Table A22 reports all three outcomes and the level coefficients at each exclusion cutoff, and Table A23 reports the joint upstream–downstream regression at each cutoff. Table 4 in the main text summarizes the GCVI estimates.

Table A22: Distant Upstream Cropped Acreage by Spatial Exclusion Cutoff

	Full	>1km	>2km	>3km	>5km	>10km
Distant upstream cropped acreage × *Drought*$_{24M}$						
NDMI	−0.004*	−0.004**	−0.004**	−0.005**	−0.005**	−0.008**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)
GCVI	−0.015**	−0.015**	−0.014**	−0.017**	−0.017**	−0.035***
	(0.007)	(0.006)	(0.007)	(0.008)	(0.008)	(0.012)
ET	−0.011	−0.011	−0.011	−0.013*	−0.014	−0.023*
	(0.007)	(0.007)	(0.007)	(0.007)	(0.009)	(0.012)
Distant upstream cropped acreage, level						
NDMI	−0.000	−0.005**	−0.005**	−0.004	−0.004	−0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.005)
GCVI	0.005	−0.009	−0.013	−0.009	−0.004	0.013
	(0.012)	(0.012)	(0.011)	(0.012)	(0.013)	(0.020)
ET	0.004	−0.022**	−0.022**	−0.018*	−0.013	0.012
	(0.010)	(0.010)	(0.009)	(0.010)	(0.010)	(0.015)
N	326,732	269,870	224,068	188,088	137,229	75,826

Notes: Active plots only. Treatment is $U_{it}^{>d} = \ln(1 + \text{active upstream plots beyond } d \text{ km})$, computed over plots upstream of plot i in the directed water graph. Columns vary the Euclidean spatial exclusion cutoff d. Nearby fallow controls are the non-rotation and rotation fallow shares within radius d of plot i, computed in geographic space regardless of canal direction to absorb spatially correlated local shocks; for the Full column, where no exclusion distance d applies, these controls use the 1 km radius. FE: Crop×Year + Plot. Two-way clustered SE (Crop×Year + Plot). $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table A23: Joint Ancestor–Descendant Regression: Distant Upstream vs. Downstream Cropped Acreage

	Full path (1)	>1 km (2)	>5 km (3)	>10 km (4)
Panel A. GCVI				
UP × Drought_{24M}	−0.012*	−0.013*	−0.013*	−0.030***
	(0.007)	(0.007)	(0.008)	(0.009)
DN × Drought_{24M}	−0.013***	−0.013***	−0.004	0.002
	(0.003)	(0.003)	(0.005)	(0.010)
Panel B. ET				
UP × Drought_{24M}	−0.009	−0.010	−0.010	0.001
	(0.007)	(0.006)	(0.008)	(0.007)
DN × Drought_{24M}	−0.009**	−0.009**	0.007*	0.009
	(0.003)	(0.003)	(0.004)	(0.011)
N	259,368	218,597	62,048	21,396
Plot + Crop×Year FE	✓	✓	✓	✓
Controls + spatial fallow	✓	✓	✓	✓

Notes: Active plots only; UP (DN) is the log active upstream (downstream mirror) count beyond the column cutoff. Both enter the same regression, so each coefficient is conditional on the other; the corresponding single-equation upstream estimates are in Table A22, which is estimated on a larger sample. FE: Crop×Year + Plot. Controls and spatial fallow shares included. Two-way clustered SE (Crop×Year + Plot). No full-path downstream count exists in the panel, so the full-path column uses the >1 km downstream count as a proxy and downstream cells in columns (1)–(2) coincide. Upstream-minus-downstream differences and their standard errors are reported in the text. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

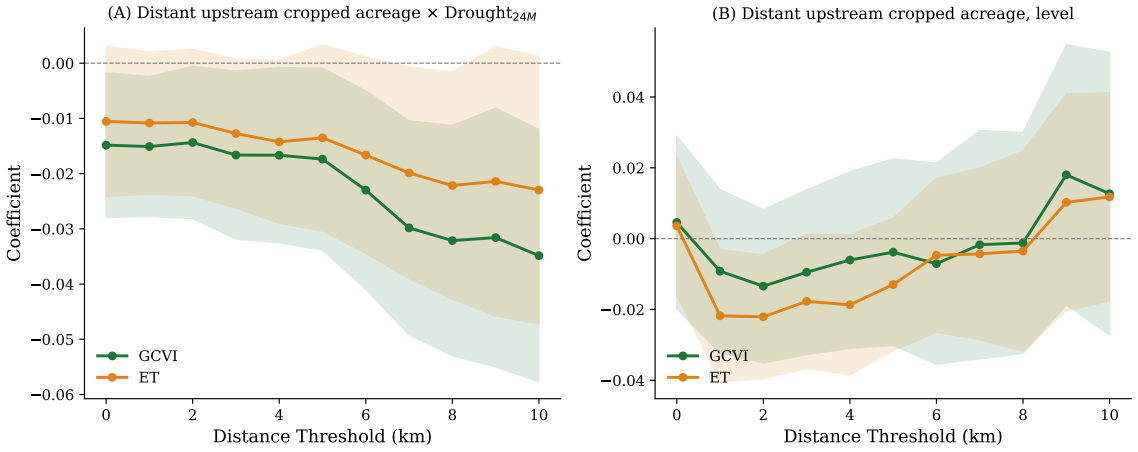


Figure A1: Distant upstream cropped acreage coefficients across distance thresholds. Panel A shows the upstream cropped acreage × drought interaction coefficient at threshold d; Panel B shows the level effect of upstream cropped acreage at threshold d. Green marks GCVI and orange marks ET; bands show 95% CI. Same specification as Table A22.

A20 Total Drought-Year Slopes and Directional Gaps

The baseline distance-cutoff and directional tables report interactions, b_2, or differences between two interactions. This appendix reports the two quantities those coefficients are most often mistaken for: the total drought-year slope $b_1 + b_2$, and the difference between the ancestor and descendant total drought-year slopes. Both are linear combinations of coefficients the paper already estimates, so obtaining them requires no new specification, only the full coefficient covariance matrix. The decomposition table in Section 7.2 already reports the corresponding preperiod-centered residual slope in drought and in other years and is not repeated here. The baseline distance-cutoff and joint ancestor–descendant regressions were refit for this appendix. The refit reproduces all 36 upstream coefficients and standard errors of Table A22, and all 15 cutoff-matched amplification contrasts, to numerical precision. The directional grid excludes the full-path proxy cell, whose descendant variable uses a 1 km exclusion rather than the same full-path support.

Table A24 reports the decisive cell. Three readings follow. The total drought-year slope on ancestor cropped acreage is negative but not distinguishable from zero, so the paper's interaction results should not be restated as an absolute drought-year association. The total drought-year directional gap is likewise null, and its point estimate is *positive*: the descendant total slope is more negative than the ancestor total slope, though neither is estimated precisely. What survives is the amplification contrast, which is significant under the two-way clustering used throughout the paper, marginal under WCC clustering, and insignificant under canal-block clustering. This is the empirical basis for describing the far-cutoff result as a directional diagnostic of differential drought amplification rather than as a drought-year level comparison.

The pattern is not confined to that cell. Across the full grid all eighteen upstream-only total drought-year slopes are negative in point estimate, and eleven of eighteen are significant at 5% under two-way clustering. None is significant under either network-level scheme, and no total drought-year ancestor-minus-descendant contrast reaches 10% in any of the forty-five cutoff-matched cells (five distances, three outcomes, and three inference schemes). Rejection is confined to the amplification margin.

Table A24: Total Drought-Year Slopes, Directional Gaps, and the Amplification Contrast

	Estimate	SE	p-value by inference scheme		
		(two-way)	Two-way	WCC	Canal block
Panel A. Beyond 10 km, GCVI					
Upstream-only total drought slope, $b_1 + b_2$	−0.0222	(0.0255)	0.385	0.553	0.528
Joint-regression ancestor total drought slope	−0.0457	(0.0323)	0.158	0.316	0.383
Joint-regression descendant total drought slope	−0.0669	(0.0453)	0.141	0.117	0.113
Total drought-year directional gap	+0.0212	(0.0616)	0.731	0.755	0.770
Drought-amplification contrast, $a_2 - c_2$	−0.0315	(0.0141)	0.026**	0.075*	0.106
Panel B. Full grid: upstream totals at 6 cutoffs; directional tests at 5 matched cutoffs × 3 outcomes × 3 inference schemes					
All 18 of 18 upstream-only total drought-year slopes are negative in point estimate. Under two-way clustering 11 of 18 are significant at 5%; under WCC and canal-block clustering, 0 and 0 are. None of the 45 total drought-year ancestor-minus-descendant contrasts is significant even at 10%. The amplification contrast is significant at 10% in 10 of 45 cells: GCVI at 10 km, ET at 3 and 5 km, and NDMI at 3 km.					

Notes: Post-estimation linear combinations of the baseline specifications, not new regressions: Panel A's upstream-only row uses the Table A22 regression at >10 km (N = 75,826; 355 crop-year, 114 WCC, and 81 canal-block clusters), and the joint-regression rows use the Table A23 regression at the same cutoff (N = 21,396; 266, 34, and 28 clusters). b_1 and b_2 are the level and interaction coefficients of equation (5); (a_1, a_2) and (c_1, c_2) are the corresponding ancestor and descendant coefficients in the joint regression. Point estimates are invariant to the covariance estimator; only standard errors and p-values change across the three columns. One-restriction p-values use a t reference with each estimator's cluster degrees of freedom. The joint null $H_0 : b_1 = b_2 = 0$ for the upstream-only cell is rejected under two-way ($p = 0.005$) and WCC ($p = 0.049$) inference and is marginal under canal-block inference ($p = 0.090$); that test says the two coefficients are not jointly zero, not that the total drought-year slope is nonzero. Statistical rejection here remains conditional and observational. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A21 Drought Measures Independent of the Active Upstream Set

Table A25: Drought Measures Independent of the Active Upstream Set

		>1km		>5km		>10km	
Drought variant	Outcome	GCVI	ET	GCVI	ET	GCVI	ET
Plot SPEI-24 dummy	UP x drought	−0.008**	−0.004	−0.009*	−0.006	−0.023***	−0.011*
		(0.004)	(0.003)	(0.005)	(0.004)	(0.007)	(0.006)
District-year SPEI-24 dummy	UP x drought	−0.003	−0.003	−0.005	−0.003	−0.016**	−0.008*
		(0.004)	(0.003)	(0.005)	(0.003)	(0.007)	(0.005)
N (plot variant)		269,870	269,870	137,229	137,229	75,826	75,826

Notes: Active plots only. The table replaces upstream-path drought with plot-centroid and district-year SPEI-24 dummies. These measures address the shared-active-set construction of $U_{it}^{>d}$ and Drought_{it}^{up}. FE, controls, spatial fallow shares, and two-way clustering match Table A22. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

A22 Extensive-Margin Timing Audit

The outcome-year activation-share diagnostic in Supplement Table S2 fixes 2016–2019 crop strata and the upstream activation share before measuring 2020–2023 active/nonfallow status. Its drought indicator and focal weather controls use May–August conditions from the outcome year. Its positive 5 km coefficient therefore describes how the fallowing–activation-share slope differs with the drought indicator observed during the outcome growing season; it cannot show that farmers responded to information available when acreage was chosen. Broad fallowing can also include failed planting, abandonment, and other noncrop classifications. Supplement Table S2 reports the full profile.

Timing-corrected junior-inside 5 km variants use information available before the outcome season and impose year-specific exposure slopes, a network-relative upstream-minus-focal scarcity contrast with a separate focal-scarcity interaction, and source weights matched to the preperiod activation share. Across these variants, the estimate is approximately −0.8 percentage points per standard deviation of the preperiod activation share for a one-standard-deviation increase in scarcity and is imprecise; it also shrinks or changes sign under alternative baseline windows and

leave-one-year-out checks. The binary lag is nearly a calendar indicator (its acreage-weighted dry share is 0, 0, 0.965, and 0.166 across 2020–2023), so it is not corroborating evidence. These checks validate neither a positive pre-season reading nor a simple opposite-signed lag association. The paper therefore finds no validated ex-ante within-network fallowing response.

A23 Balanced-Plot Panel: Exit-Selection Check

Re-estimating the Table A22 specification on the panel of plots active in every sample year (23,840 plots, so 190,720 nominal plot-years; the Full-column regression is estimated on the 190,480 of these with complete outcome and regressor support, 58.1% of active rows) does not move the drought interaction: the primary > 5 km GCVI interaction is -0.028 on the balanced panel against -0.017 on the full active sample, a difference of -0.0105 against a combined standard error of 0.0134. ET behaves the same way, -0.023 against -0.014 at the same cutoff. Drought-year exit from the active panel therefore does not generate the headline drought interaction.

Table A26: Balanced-Plot Panel: Exit-Selection Check

Sample	Outcome	Full	>5km	>10km
Full active	GCVI	−0.015**	−0.017**	−0.035***
		(0.007)	(0.008)	(0.012)
Full active	ET	−0.011	−0.014	−0.023*
		(0.007)	(0.009)	(0.012)
Full active	N	326,732	137,229	75,826
Balanced plots	GCVI	−0.022***	−0.028***	−0.042***
		(0.008)	(0.010)	(0.014)
Balanced plots	ET	−0.015*	−0.023**	−0.030**
		(0.008)	(0.011)	(0.015)
Balanced plots	N	190,480	83,944	47,504

Notes: Exact Table A22 specification: active plots only; fixed effects are Crop×Year and Plot; 42 controls; matched spatial fallow shares; two-way cluster-robust standard errors, clustered by Crop×Year and plot. The check is motivated by the Appendix A28 exit-composition diagnostic. Balanced active-filter panel: 23,840 plots, 190,720 nominal plot-years; the displayed N is the complete-case estimation sample (Full column: 190,480), 58.1% of active rows. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Balanced-panel re-estimation strengthens the pattern at every displayed cutoff: GCVI moves from -0.015, -0.017, and -0.035 to -0.022, -0.028, and -0.042 over the full path, >5 km, and >10 km, and ET from -0.011, -0.014, and -0.023 to -0.015, -0.023, and -0.030. The primary >5 km GCVI difference stays within one combined standard error, so this is sign-and-magnitude corroboration rather than evidence that the two samples differ (Table A26).

A24 Static Position and Within–Between Exposure Diagnostic

Table A27 asks whether the baseline interaction is confined to the tail and whether its negative component is associated with annual deviations or persistent mapped exposure. Position terciles are fixed on the full plot universe, and the saturated regressions allow the upstream cropped-acreage level, drought intercept, and interaction to differ across terciles. For GCVI, the raw interaction is negative in all three terciles at every cutoff. The tail point estimate is generally more negative. Equality across terciles is not rejected under baseline inference, and the far-cutoff tail–head contrast is not robust to network clustering.

The second part is a Mundlak-style diagnostic. Writing $U_{it} = \bar{U}_i + \widetilde{U}_{it}$, the unrestricted regression assigns separate coefficients to $\bar{U}_i D_{it}$ and $\widetilde{U}_{it} D_{it}$. Plot fixed effects absorb $\bar{U}_i$ itself. They do not absorb its interaction with the drought indicator. The persistent component reproduces the negative baseline coefficient. The annual-deviation component is imprecise. Equality of the two components is generally not rejected, so the exercise is not a unique causal partition and does not replace the baseline specification. It establishes a narrower boundary: the fixed-effects interaction should not be described as arising exclusively from annual deviations in ancestor cropped acreage.

Table A27: Static Position and Within–Between Upstream Exposure: GCVI

	Full	>1 km	>5 km	>10 km
GCVI				
Baseline $U \times D$	−0.015**	−0.015**	−0.017**	−0.035***
	(0.007)	(0.006)	(0.008)	(0.012)
Tercile-adjusted common $U \times D$	−0.015**	−0.015**	−0.016*	−0.034***
	(0.007)	(0.006)	(0.008)	(0.011)
Raw: Head tercile	−0.011**	−0.011**	−0.010	−0.009
	(0.006)	(0.005)	(0.007)	(0.011)
Raw: Middle tercile	−0.010	−0.011*	−0.011	−0.028**
	(0.007)	(0.006)	(0.008)	(0.012)
Raw: Tail tercile	−0.020*	−0.021**	−0.023*	−0.044***
	(0.010)	(0.010)	(0.013)	(0.016)
Annual deviation (8-year mean) $\widetilde{U} \times D$	0.031	0.042	−0.030	−0.011
	(0.030)	(0.031)	(0.036)	(0.062)
Annual deviation (estimation mean) $\widetilde{U} \times D$	0.027	0.036	−0.022	0.004
	(0.034)	(0.031)	(0.034)	(0.058)
Static 8-year mean exposure $\overline{U} \times D$	−0.015**	−0.016**	−0.017**	−0.035***
	(0.007)	(0.007)	(0.009)	(0.012)
Tercile-adjusted within $\widetilde{U} \times D$	0.032	0.042	−0.030	−0.004
	(0.030)	(0.031)	(0.036)	(0.065)
Head tercile	0.016	0.027	−0.031	−0.042
	(0.042)	(0.038)	(0.062)	(0.110)
Middle tercile	0.080**	0.114**	−0.003	0.046
	(0.039)	(0.045)	(0.055)	(0.095)
Tail tercile	0.003	−0.007	−0.049	−0.025
	(0.038)	(0.035)	(0.035)	(0.070)
p: equal raw interactions	0.446	0.420	0.495	0.189
Raw Tail − Head interaction	−0.009	−0.010	−0.013	−0.034*
p: equal annual-deviation interactions	0.154	0.022	0.665	0.680
Tail − Head within interaction	−0.013	−0.034	−0.018	0.017
Position gradient of within interaction	−0.037	−0.072	−0.110	−0.027
p: within = static component	0.129	0.075	0.742	0.715
N	326,732	269,870	137,229	75,826

Notes: Exact Table A22 outcome, sample, controls, fixed effects, weights, and two-way Crop×Year plus Plot clustering. U is log active ancestor count and D is Drought_{24M}. Head, middle, and tail terciles are fixed using one observation per plot from the full 54,946-plot universe. "Within activity" is $U_{it} - \bar{U}_i$; the table reports both the raw eight-year plot mean and an estimation-sample centering check. Static position main effects are absorbed by Plot fixed effects. Group-specific drought intercepts and exposure levels are included in the saturated models. At 10 km the raw tail–head interaction contrast has $p = 0.069$ under the displayed inference, $p = 0.209$ under WCC clustering, and $p = 0.246$ under canal-block clustering. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Equality of the GCVI interaction across head, middle, and tail terciles is not rejected at any cutoff ($p = 0.189$–0.495). In the within–between decomposition, the negative GCVI component loads on each plot's eight-year mean upstream cropped acreage interacted with drought (−0.0149 to −0.0349 across cutoffs). The annual-deviation interactions are imprecise and equality between the two components is generally not rejected.

A25 Documented Delivery Constraints in Non-CAP Systems

Table A28 collects documentary evidence from the Gila Valley, where Graham, Fort Thomas, Union, Montezuma, and San Jose are represented in the junior-rights sample. For a fixed irrigation-turn duration, a smaller stream delivers less water; additional delivery time or supplemental supply can compensate. The search was directed by the study geography and results. These cases document the existence of delivery constraints; they do not measure their prevalence or independently identify the crop-outcome mechanism.

Table A28: Non-CAP Delivery Constraints: Documentary Evidence and Sample Match

System and date	Documentary record and source	Match and interpretation
Upper Gila / Safford Valley, 2021	The Commissioner recorded Cosper Crossing as dry on March 11 and next wet on July 26. When dry, the Duncan Valley's Gila flow was issued to Duncan/Virden canals; flowing conditions placed Duncan and Safford diversions under the same priority year. (Doyle 2022, p. 3; PDF p. 7)	Dated inter-valley flow and administration evidence in the region containing the five focal Gila companies. It does not imply zero supply throughout Safford or measure unequal deliveries within a canal.
Graham Canal, testimony reproduced in 1955	During scarce-water rotations, the reported stream diminished from roughly 400 "inches" at the head to 250 at the old Oregon ditch's end. Water boss Emery B. Hubbard testified: "It affects more on the lower end." (Arizona Supreme Court 1955, pp. 217–218, Udall dissent)	Direct historical testimony about downstream conveyance losses in a named sample company. The account concerns a river-and-pump system and predates the study; it is reproduced in the dissent, not a contemporary delivery measurement.
Gila Valley Irrigation District, May 2019	The modernization plan recommended equalizing basins, turnout measurement and control, and operational changes to improve the flexibility and equity of field deliveries. (ITRC 2019, recommendations 6–8)	The plan's map includes the five focal Gila companies. A case-specific engineering assessment of delivery needs; the recommendations do not measure downstream deficits or completed improvements.
Gila Valley Irrigation District, February 8, 2022	Meeting minutes attribute to GVID representative Justin Layton losses from aging gates across the eleven-canal system: existing gates reportedly leaked 10–50 miner's inches, against a typical 300 miner's-inch stream. (AWPF 2022b, attached February draft minutes, p. 4; PDF p. 13)	Contemporary operator evidence of hydraulic losses at system scale, with no individual focal canal or tail farm identified. The revised gate project covered Smithville and Dodge-Nevada; Union had been removed.

Notes: Dates identify the reported event, plan, or court opinion; the 2021 Commissioner's report was submitted August 10, 2022. GVID-wide statements are not five separate canal observations. The detailed February 2022 account is reproduced as draft minutes in the June meeting packet; the approved scope revision is recorded separately (AWPF 2022a, PDF pp. 2–3, 27–28). Historical "inches" and modern miner's inches describe stream-flow rates, retained as reported without conversion. These records concern non-CAP systems; groundwater supplementation can still occur within them.

Table A29 places the dated account beside reported diversions. Safford Valley's May–June total was 2,112 acre-feet in 2021, compared with 8,675 in 2020 and 3,975 in 2022. In 2021, diversions recovered to 5,603 acre-feet in July and 12,135 in August; the annual total slightly exceeded 2020's. The record therefore locates scarcity within the growing season. Among the five focal companies, annual changes are heterogeneous: Fort Thomas reported much lower river diversion in 2021 than its 2016–2019 mean, whereas Union's reported diversion was higher. Calendar-year diversion totals cannot reveal how water was distributed among farms when supply was tight.

Table A29: Reported Gila River Diversions and Seasonal Flow Conditions

Panel A. Seasonal conditions, entire Safford Valley

Year	Cosper dry	Next wet	May (AF)	June (AF)
2020	June 5	Dec. 31	5,659	3,016
2021	March 11	July 26	1,096	1,016
2022	April 13	June 30	1,351	2,624

Panel B. Calendar-year diversions, focal Gila companies (AF)

Canal	2016–2019 mean	2020	2021	2022
Graham	5,598	5,927†	5,204	4,812†
Fort Thomas	3,874	2,406	1,554	2,747
Union	11,846	11,320	12,555	13,237
Montezuma	8,158	7,565	6,479	7,648
San Jose	8,900	8,992	8,499	8,897

Sources: Gila Water Commissioner annual reports (Gila Water Commissioner, n.d.). Year labels link to the original reports. Panel A: printed p. 3 (PDF p. 7), Cosper status changes; printed p. 5 (PDF p. 10), Safford monthly diversions. Panel B: Plate 3, "Gila River Decreed Acreages and Diversions," Safford Valley rows; PDF pp. 28, 30, 31, 31, 28, 31, and 32 in the 2016–2022 reports, respectively. The preperiod mean uses the 2016, 2017, 2018, and 2019 reports and is rounded to whole acre-feet.

Notes: Panel A covers the entire Safford Valley reporting area, not only the five junior companies. Dates are reported status changes, not continuous daily observations; 2021 had the earliest listed dry onset but not the longest interval. Panel B includes all five Gila companies in the junior sample. Values are published calendar-year river diversions, not farm receipts, orders, or a complete groundwater-inclusive water budget. San Jose, Montezuma, and Union share a described measuring flume with Fourness; separate published totals are not independently gauged measurements. † Graham's 2020 and 2022 annual totals are reproduced as published despite unresolved unit conflicts in their monthly records; these totals do not repair the seasonal-data exclusions described in Appendix A26. No blank cell has been treated as zero.

Additional Evidence Outside the Junior-Rights Group

Additional searches covered senior and mixed-priority systems with negative coefficients in exploratory separate-district regressions, and systems with unresolved priority classifications and negative validation estimates. Table A30 reports both delivery constraints and a qualifying account of adequate supply. North Gila Valley and Yuma Irrigation District are near Yuma, distinct from the Upper Gila companies in Table A28.

Table A30: Additional Non-CAP Delivery Records: Senior, Mixed, and Unclassified Systems

System, priority, and date	District or operator record and source	Interpretation and limits
North Gila Valley, senior, July 2018	The McPherson Lateral application reports deteriorated lining, leakage, and seepage. Enlarging and relining the lateral was proposed to increase delivery capacity and shorten irrigation turns. (NGVIDD 2018, pp. 2–4)	A delivery constraint in a senior Colorado River system. Proposed gains are not measured outcomes. Earlier phases were completed in 2016–2017; the record does not establish unchanged conditions in later drought years.
Yuma Irrigation District, senior, May 2021	General manager Rex Green described year-round demand frequently testing operational capacity, while reporting no drought-induced diversion limitations yet. The district's undated ordering policy allows delayed service when system capacity is constrained. (Irrigation Leader 2021; YID, n.d.)	Capacity and scheduling pressure despite continued access to source water. Neither account measures a head-to-tail volume difference. The undated policy cannot establish when a particular delivery was delayed.
Roosevelt Irrigation District, unclassified, April 2017	The district reports summer prorating and difficulty matching supply to demand because of canal travel time. At an existing reservoir near the canal's downstream end, stored water is pumped back when downstream demand exceeds supply. (RID 2017, pp. 1, 3)	Direct downstream supply-demand mismatch and an operating response. The record does not establish a remaining seasonal deficit after compensation. This is Roosevelt Irrigation District, not Roosevelt Water Conservation District.
Buckeye, unclassified, July 2022	The district reports that changing main-canal levels alter farm delivery rates when manual gates remain fixed during an irrigation set. It also reports longer delivery durations associated with inadequate water levels and multiple users. (BWCDD 2022, pp. 15–16)	Contemporary farm-delivery rate and timing problems in a shared system. No head-to-tail comparison or drought attribution is provided; anticipated modernization benefits are not observed delivery gains.
Mohave Valley, mixed, 2019	The district's conservation plan states that, since formation in 1963, it had never been unable to provide a contract holder's requested amount. The district does not operate the private conveyance facilities. (MVIDD 2019, pp. 38–39)	A qualifying record against a district-wide supply-shortage interpretation. It does not audit distribution within individual private systems. A negative crop coefficient alone cannot establish unmet water orders.

Notes: Priority labels follow the study's audited classifications; "unclassified" is not a senior designation. The full-period North Gila and Yuma fits use nearest-district assignment; the stricter inside-boundary screen beyond 5 km lacks drought variation for both. Roosevelt and Buckeye have negative validation estimates but positive discovery estimates; Mohave's sign changes in the other direction. These exploratory matches guide source discovery, not independent causal validation. Grant applications report existing conditions alongside proposed improvements; the table distinguishes the two. No source here measures a dated, uncompensated seasonal water deficit at a named downstream farm.

A26 Measured Canal Flow and Network Position

This exploratory comparison relates the outcome gradient along the mapped Wellton-Mohawk network to measured water availability. The original fixed geographic cohort contains 4,562 fields observed in all eight years, 2016–2023; restricting it to the paper's inside-district flag retains 4,418. Neither cohort is a verified historical turnout-service roster. Position is fixed in 2016 and standardized once using the original cohort. Summer flow is the May–September mean at USGS

main-canal gauge 09522700. The original outcome is seasonal maximum NDMI in *raw units*; Appendix S16 reports the same specifications in $\ln(1+\text{NDMI})$ on identical observations. Seasonal maximum GCVI, the paper's main outcome, and seasonal OpenET ensemble ET are reported in raw units on the same field-years and specifications; neither has a missing value in either cohort. Fallowed field-years remain included for every outcome and scale. Equal-field-weighted models include field effects, year or baseline-network-by-year effects, continuous SPEI-24 and its position interaction. The trend specification adds position × calendar year, allowing the outcome gradient itself to drift linearly.

Table A31 reports the summer-flow comparison across both cohorts and fixed-effect structures for NDMI (Panel A), GCVI (Panel B), and ET (Panel C). For NDMI, every coefficient is positive, and every individual-year omission retains that sign. For the inside cohort with network-by-year effects and a gradient trend, omitting 2020 reduces the estimate from 0.01153 to 0.00663: the magnitude is sensitive to that year. The sign is not created by it. GCVI shows the same pattern: every coefficient is positive, every year omission keeps the sign, the gradient trend leaves the estimate essentially unchanged, and the shift is about the same size as the NDMI shift once each is scaled by the standard deviations of flow and of the outcome. ET is positive in every specification but small and imprecise with field and year effects alone, where the year-omission range crosses zero; with network-by-year effects it is larger and positive in every omission, though still imprecise. These are changes in the outcome–position slope associated with higher flow, not mean effects of water or drought-amplification coefficients.

Table A31: Wellton-Mohawk: Summer Canal Flow and the Downstream Gradient

Cohort / fixed effects	No gradient trend	Gradient trend	Trend-model omission range
Panel A: Raw NDMI (moisture)			
Original / field + year	0.01325	0.00952	[0.00462, 0.01642]
	(0.117)	(0.055)	
Inside / field + year	0.01363	0.00961	[0.00465, 0.01682]
	(0.117)	(0.039)	
Inside / field + network-year	0.01439	0.01153	[0.00663, 0.01607]
	(0.023)	(0.023)	
Panel B: Raw GCVI (satellite yield proxy)			
Original / field + year	0.180	0.182	[0.090, 0.261]
	(0.070)	(0.109)	
Inside / field + year	0.183	0.186	[0.082, 0.282]
	(0.070)	(0.125)	
Inside / field + network-year	0.178	0.184	[0.064, 0.287]
	(0.250)	(0.227)	
Panel C: Raw OpenET ensemble ET (mm)			
Original / field + year	4.79	1.95	[−6.77, 6.54]
	(0.766)	(0.727)	
Inside / field + year	4.71	1.79	[−7.22, 6.61]
	(0.773)	(0.742)	
Inside / field + network-year	11.04	13.21	[9.01, 17.53]
	(0.234)	(0.133)	

Notes: Interaction of fixed position with summer flow, per one fixed position SD farther downstream and 100 cfs higher May–September mean flow, in each panel's raw outcome units: seasonal maximum NDMI, seasonal maximum GCVI, and seasonal (May–October) OpenET ensemble ET in mm. All panels use the same field-years: 36,496 in the original cohort and 35,344 in the inside cohort. Scaled by the standard deviations of flow and of the outcome, the NDMI interactions are 0.018–0.028, the GCVI interactions 0.017–0.018, and the ET interactions 0.003–0.021. In $\ln(1+x)$, the GCVI interaction is positive in all six specifications and all 48 year omissions with wild p-values from 0.008 to 0.078; logged ET follows the raw pattern. All models retain weather adjustment. Parentheses are null-imposed year-cluster wild-bootstrap p-values, not standard errors, enumerating all 256 Rademacher assignments with nuisance and fixed-effect refits. Calibration relies on independence/symmetry assumptions across eight years; many fields do not supply additional independent flow shocks. Ranges cover all eight individual-year omissions and are influence diagnostics, not confidence intervals. The table is exploratory.

Annual gross diversion at Imperial Dam supplies a different temporal and measurement comparison, not an instrument for summer flow. Its inside/network-year moisture interaction is also positive, 0.00324 per position SD and 10,000 acre-feet, or 0.00270 with the gradient trend (year wild $p = 0.234$ for both); every year omission retains the positive sign. The fallowing counterpart is weaker: the summer-flow interaction in the inside/network-year trend model is -0.117 percentage points ($p = 0.734$), changes sign in three year omissions, and is positive but imprecise with annual diversion. Thus the moisture evidence does not establish an extensive-margin response. Realized diversion can respond to agricultural demand, nonlinear spatial shocks remain possible, and mapped service does not reveal field receipts or private pumping. Wellton is a separate senior/mixed-supply system, so this is supporting physical evidence rather than identification of the junior annual-activation coefficient.

A seasonal extension applies the same design to seven non-CAP agricultural entities with monthly Imperial Dam diversion records: Wellton-Mohawk, Yuma County WUA, Yuma Mesa, Yuma Irrigation District, North Gila Valley, Unit B, and Gila Monster Farms (11,020 fields, 2016–2023). The water measure is reported May–September gross diversion rather than a gauge. The primary model keeps field and baseline-network-by-year effects, weather adjustment, and a position trend, with position and water in within-entity standard deviations and raw outcomes. The two largest entities are positive for GCVI and NDMI in all four fixed-effect and trend designs and every year omission: Wellton-Mohawk at $+0.060$ and $+0.0040$ (year-cluster wild $p = 0.35$ and 0.031) and Yuma County WUA at $+0.077$ and $+0.0072$ ($p = 0.30$ and 0.070). Adding Yuma County WUA's reported well pumping leaves its estimates essentially unchanged. Yuma Mesa and Yuma Irrigation District are negative in every design, North Gila Valley and Unit B are positive but imprecise and unstable across year omissions, and Gila Monster Farms is mixed. None of the 14 primary GCVI and NDMI tests survives Holm adjustment (smallest adjusted $p = 0.44$). With annual rather than May–September totals, the Yuma County WUA estimates are negative and imprecise, so its pattern appears only with the summer window.

The exploratory Gila comparisons do not supply uniform corroboration. In a four-company, 323-field cohort, adjusting reported diversion by administrative irrigated acreage or a proxy based on changing cultivation does not yield a robust positive moisture response farther downstream. The broader series, conditional on treating blank diversion entries as zero, gives interactions opposite to the simple buffering prediction, including after company and year omissions; the stricter water-quality sample is unstable. Unverified service linkage, differences between administrative and cultivated acreage, and unobserved private groundwater limit interpretation. Matched raw/log comparisons preserve the negative conditional-series moisture estimates. They do not resolve the strict-sample instability (Appendix S16). These findings bound generalization of the Wellton-Mohawk and Yuma County WUA association rather than confirm it through a different specification.

A27 Rights and Pre-Period Well Controls in the Junior Districts

Table A32 asks whether observed access differences account for the junior annual-share association. It adds drought interactions with pre-2016 on-field wells and well proximity, geographically matched irrigation points-of-use (POU) priority dates and match status, and both bundles jointly. The original plot and Crop×Year effects and controls remain included, and all rows retain the same 22,105 active plot-years. The new bundles leave negative annual interactions: -0.01374 for wells,

−0.01081 for rights, and −0.01055 jointly (full-refit district wild $p = 0.031$, 0.039, and 0.039). The joint-control estimate remains negative with district linear trends (−0.00886, $p = 0.053$) and after every individual-district omission (range −0.01316 to −0.00799).

These checks narrow a specific observed-access confounding explanation, rather than establish a causal rights or groundwater effect. Adding the broader legacy priority and link-quality drought interactions shrinks the coefficient to −0.00620 ($p = 0.223$) without losing observations. That bundle combines direct POU buffers, irrigation-grandfathered-right bridges, and nearest-neighbor dates, so the extra shrinkage is a sensitivity to less securely linked information, not proof that legal priority explains the association. With district-by-year effects every control specification is near zero; those columns are a within-district diagnostic, since the effects absorb the shared district-year response.

Table A32: Junior Annual Activation: Rights and Pre-Period Well Control Sensitivities

	Baseline FE		District×Year FE	
Additional drought-response controls	Interaction	Wild p	Interaction	Wild p
None (original annual-share model)	−0.01415	0.039	−0.00081	0.861
Old on-field well and old-well proximity	−0.01374	0.031	−0.00086	0.852
POU match status and priority-date proxy	−0.01081	0.039	−0.00077	0.877
Both new bundles	−0.01055	0.039	−0.00083	0.865
Both plus legacy rights/link-quality bundle	−0.00620	0.223	−0.00081	0.846
Trend sensitivity (baseline FE plus district linear trends)				
Both new bundles	−0.00886	0.053	—	—

Notes: Every row uses 22,105 active plot-years, 3,600 fields, ten junior districts and 35 WCCs, with unchanged outcome $\ln(1 + \text{GCVI})$, >5 km upstream activation share, 2016–2019 baseline H, annual departure W, district drought D, original controls, plot and Crop×Year effects, and no weights. Entries are coefficients on $W \times D$ per 0.10 share, not total drought-year slopes or interaction contrasts. New controls enter only through their interactions with D; their time-invariant levels are absorbed by plot effects. Wells use the full dated pre-2016 non-exempt registry stock. Priority dates use irrigation POU matches within 200 m, centered within district; unmatched rights have a separate status indicator and are not assigned a junior priority. The legacy bundle additionally interacts its date and direct/bridge/nearest linkage indicators with drought. District-by-year columns are a within-district diagnostic; they absorb the shared district-year response. Wild p-values impose the null, refit all nuisance slopes and fixed effects, and enumerate 1,024 district Rademacher assignments; enumeration does not imply random assignment or distribution-free inference. Dashes indicate no corresponding combined trend/area-year fit.

A28 Administrative Allocation Cuts: The CAP Curtailment Schedule

This appendix separates three objects that should not be conflated: the administrative first stage, average plot outcomes, and the upstream cropped-acreage gradient. Central Arizona Project shortage tiers curtailed the Agricultural Settlement Pool: realized pool deliveries were unchanged under Tier 0 because 2019 deliveries already sat below the cap, fell 65% net of mitigation and 88% in realized deliveries under Tier 1 in 2022, and went to zero under Tier 2a in 2023. Six high-use districts are classified from CAP delivery accounting. Pool juniority was assigned by the 2004 settlement. The treated districts were not randomly selected and possess heterogeneous groundwater access, so the exercise is a mechanism-scope check rather than a natural experiment.

Average outcomes. Table A33 shows a large extensive-margin break. Relative to 2019 and the comparison districts, 2022 fallowing rises 13.3 percentage points under the nearest-district assignment and 18.9 points when treatment is restricted to plots inside the actual district polygons. The all-plot satellite indices decline as well. The boundary restriction strengthens these level estimates. It also worsens pre-trend performance; estimates are sensitive to the comparison pool, and the nearest assignment labels 31% of treated plots outside the corresponding district polygon.

These results establish that the delivery collapse coincides with a large fallowing and production break in the treated geography, not that the CAP cut alone caused the reported magnitudes. Conditioning on plots that remain active removes the extensive-margin response and changes crop composition, so it cannot recover a clean intensive-margin effect.

Table A33: Tier 1 and Average Plot Outcomes

Treatment geography	Outcome	Estimate	District SE	Webb p
2022 treated–control change relative to 2019				
Nearest-district assignment	Fallowing	+0.133	0.032	0.015
	$\ln(1 + \text{GCVI})$	−0.164	0.071	0.052
	$\ln(1 + \text{NDMI})$	−0.035	0.018	0.120
	$\ln(1 + \text{ET})$	−0.140	0.081	0.249
Distance-zero boundary proxy	Fallowing	+0.189	0.031	0.001
	$\ln(1 + \text{GCVI})$	−0.205	0.071	0.003
	$\ln(1 + \text{NDMI})$	−0.048	0.021	0.036
	$\ln(1 + \text{ET})$	−0.210	0.075	0.024

Notes: Plot-weighted district-year event-study comparisons with District and Year fixed effects; 2019 is the reference year. The nearest-district assignment uses the paper's nearest-district label; the boundary restriction retains only plots geographically inside the named district polygon. Fallowing is in probability units; satellite outcomes enter as $\ln(1 + x)$. Standard errors cluster by irrigation district. Webb wild-cluster p-values use 9,999 null-imposed draws. The table is descriptive: joint pre-trend tests fail for multiple outcomes, particularly under the boundary restriction, and estimates vary with the comparison pool.

Gradient response. Adding the administrative controls leaves the SPEI-drought interaction essentially unchanged (−0.021 to −0.018 for GCVI). The curtailed districts show no robust additional deepening when the schedule binds. The 5 km GCVI triple interaction is +0.0235 under plot-level inference; because treatment varies by district-year, the relevant district-clustered and Webb p-values are 0.286 and 0.335, and the equal-weighted treated-district estimate is +0.0055 ($p = 0.853$). At 10 km the estimate is −0.0030 ($p = 0.846$). The district-clustered confidence interval remains wide enough to include deepening of the size of the main drought interaction, and 2017–2019 interaction deviations are jointly marginal ($p = 0.050$). This is an inconclusive gradient result, not a demonstrated null.

The active-sample comparisons also differ in composition. Fallowing can change the measured gradient. Differences across estimates with different samples, exposure definitions, or reference periods are not an exact accounting of exit's contribution. Matched fixed-exposure accounting does not establish a uniformly positive selection effect. Among plots active in both 2021 and 2022 the separate gradient estimate is mildly negative (−0.013), although that survivor comparison conditions on continued activity. The CAP evidence shows a large surface-delivery loss, worsening average outcomes, groundwater drawdown (Section 7.3), and no reliably steeper canal gradient under the logged-count exposure. The crop water demand design of Section 7.3.2 weights the farmed share of linked land by crop coefficients and uses a cut set from district size. It does show a 2022 upstream penalty (Appendix Tables A35 and A36).

Table A34: Administrative Allocation Cuts: CAP Curtailment Schedule Interactions

Panel A: Schedule interactions, $d > 5$ km

	GCVI S1(a)	GCVI S1(b)	GCVI S1(c)	ET S1(a)	ET S1(b)	ET S1(c)
U x Tier 0 x CAP Ag Pool		0.043***	0.048***		0.036***	0.039***
		(0.008)	(0.009)		(0.007)	(0.007)
U x 2022 x CAP Ag Pool		0.030***	0.024**		0.027***	0.024**
		(0.011)	(0.010)		(0.010)	(0.009)
U x 2023 x CAP Ag Pool		0.011	0.011		−0.007	−0.007
		(0.015)	(0.015)		(0.011)	(0.011)
U x Drought	−0.021**		−0.018***	−0.014		−0.010
	(0.009)		(0.007)	(0.009)		(0.008)
N	119,570	119,570	119,570	119,570	119,570	119,570

Panel B: Inference for the primary cell

Inference variant	Coef.	SE / stat	p
Collapsed cells, cluster-robust standard errors by district	0.024	0.022	0.286
Collapsed cells, Webb score bootstrap p	0.024		0.335
Wald test: upstream cropped acreage × CAP Ag Pool × year, 2017–2019 jointly zero	2.768		0.050
Wald test: upstream cropped acreage × CAP Ag Pool × year, Tier 0 (2020–2021) jointly zero	3.002		0.058
Wald test: upstream cropped acreage × CAP Ag Pool × year, 2023 vs. 2022	1.462		0.232
Ibragimov-Mueller treated-district t	0.005	0.028	0.853
Plot-level two-way cluster-robust standard errors: crop×year and plot	0.024	0.010	0.021
Plot-level cluster-robust standard errors: weakly connected component	0.024	0.022	0.276
Plot-level two-way cluster-robust standard errors: weakly connected component and year	0.024	0.018	0.230
Plot-level cluster-robust standard errors: canal block	0.024	0.022	0.296
Plot-level cluster-robust standard errors: irrigation district	0.024	0.022	0.286

Notes: Treated = the six CAP Agricultural Settlement Pool districts with mean 2016–2019 pool deliveries ≥1,000 AF, classified from CAP delivery accounting (replication package); eight CAP-exposed districts (San Carlos IDD and GSF-only recipients) are excluded from the sample entirely. Primary cell: U×2022×CAP Ag Pool, GCVI, $d > 5$ km, column (c). Two-way plot-level stars are anti-conservative for a district-year treatment; assignment-level inference (Panel B) is primary: effective treated clusters ≈3.4, MDE ≈0.06, and the district-clustered 95% CI $[-0.02, 0.07]$ does not exclude interaction deepening of the size of the main drought interaction. The placebo-permutation test re-assigns the treated label to placebo district sets on the residualized district-year cells (1,999 draws; true-treated interactions retained as controls); its p-values are reported in the text and in the replication package rather than as rows of this table. 5 km is primary because treated support at 10 km rests on three districts; both cutoffs and all robustness variants (leave-one-out, no-fallow-mediators, composition-free upstream measure, district×year fixed effects, junior-vs-junior and senior-only comparisons, SPEI-3 confrontation) are in the replication CSV.

A29 Crop Water Demand Test in the Junior Districts

This appendix reports the crop water demand test of Section 7.3.2. Table A35 is the first form of the design, on the 2,608 CAP fields and 467 Upper Gila fields of the 3,600-field sample built with the statewide rule of at least 5 km of canal above the field, with the farmed share as measured. Rows are field-months from March to October. Upstream and downstream farmed shares are the annual departures from each field's 2016–2019 baseline. They cover land linked to the field through its own district's canal network and farther away than the district cut, 5 percent of the square root of the district's area. Each share enters interacted with the period and with the monthly crop coefficient of the field's crop. Fixed effects are field by month and crop by year by month; the Upper Gila model adds district by year by month. Crop calendars use FAO-56 stage lengths and coefficients for the closest arid regions, anchored on 2017–2019 Sentinel-2 timing by crop and district. Table A36 is the form reported in the text: all 4,641 CAP fields inside the district boundaries with land upstream and downstream beyond the cut, and the crop water demand of linked land in place of its farmed share, each farmed field weighted by its monthly crop coefficient and each monthly departure capped at ±0.5. Panel A interacts the demand of linked land with the field's own crop coefficient. Panel B lets its slope differ by month. Panel C compares the

departure as measured with the capped departure. Panel D replaces monthly ET with Sentinel-2 greenness integrated over each crop's yield-forming window, and with ET summed over the same months, with the exposure averaged over the window months (cotton July–September, corn and sorghum May–August, small grains March–May, alfalfa March–October). The windows come from the crop stage calendars: cotton days 196 to 273 of the year, flowering through boll development; corn and sorghum days 150 to 230; small grains days 60 to 135; alfalfa from green-up to the end of the season. The season maximum of the index in Panel B of Table A35 is the field mean of each pixel's May–September maximum. Panel E repeats the Upper Gila model of Table A35 on all 717 fields inside the five canal polygons, with the field-month drought index extracted at the centroid of every plot; its 5 km subset is the 467 fields of that table, and its head-end fields are the 250 others, most of them with few plots above them on their canal.

Table A35: Crop Water Demand, Supply, and Upstream versus Downstream Farmed Land

	Upstream	Downstream	Up−Down	Wild bootstrap p
Panel A. CAP, monthly ET (Mar–Oct): penalty at peak minus low crop demand (K_c 1.1 vs 0.4)				
2016	−1.81 (3.20) [0.573]	+2.64 (1.81) [0.148]	−4.46 (3.68) [0.230]	
2020–21	−0.86 (1.73) [0.620]	+1.73 (1.20) [0.154]	−2.59 (1.89) [0.175]	
2022	−2.35 (1.41) [0.100]	+1.73 (1.42) [0.227]	−4.08 (1.70) [0.019]	0.159
2020–22 average	−1.61 (1.45) [0.270]	+1.73 (1.04) [0.100]	−3.34 (1.55) [0.035]	0.126
2023	−6.31 (2.50) [0.014]	+0.03 (0.95) [0.978]	−6.34 (2.84) [0.029]	0.001
Panel B. CAP, $\ln(1+\text{GCVI})$, annual peak (Sentinel-2): crops at peak summer demand				
2016	+1.88 (2.75) [0.496]	+2.07 (1.92) [0.283]	−0.19 (2.50) [0.939]	
2020–21	−2.27 (1.04) [0.033]	+2.87 (0.80) [0.001]	−5.14 (1.44) [0.001]	
2022	−2.52 (1.16) [0.033]	+1.60 (0.86) [0.066]	−4.12 (1.42) [0.005]	
2020–22 average	−2.40 (0.99) [0.018]	+2.24 (0.69) [0.002]	−4.63 (1.24) [0.000]	0.001
2023	−3.31 (1.36) [0.017]	+1.53 (1.06) [0.155]	−4.84 (1.88) [0.012]	
Panel C. CAP, monthly ET, 2016–22: per one SD lower CAP delivery, district crop demand held fixed				
Peak minus low demand	−1.41 (0.70) [0.050]	−0.14 (0.50) [0.781]	−1.26 (0.99) [0.207]	0.097
Panel D. CAP, monthly ET, 2023: Up−Down, peak minus low demand, by how groundwater reaches fields				
	Canal-delivered wells	Private wells	Difference	
Lean model	−4.93 (3.41) [0.152]	−1.04 (2.71) [0.703]	−3.89 (3.95) [0.328]	
Fully split model	−4.75 (4.03) [0.242]	+0.04 (3.40) [0.991]	−4.79 (5.51) [0.387]	
Panel E. Upper Gila canals, monthly ET: peak minus low demand, by kind of month				
	Upstream	Downstream	Up−Down	
Short and dry	−3.09 (1.58) [0.056]	+0.27 (0.90) [0.770]	−3.36 (1.92) [0.088]	
Short, not dry			+2.13 (2.41) [0.380]	
Dry minus not dry			−5.49 (2.50) [0.033]	
Panel F. CAP, monthly ET, Up−Down at peak minus low demand, 2020–22 average: declared checks				
$K_c\times$ reference ET in place of K_c			−3.56 (1.85) [0.057]	
Desert stage lengths for cotton			−3.23 (1.60) [0.048]	
Cut 2.5% of $\sqrt{\text{area}}$			−2.81 (1.61) [0.085]	
Cut 10% of $\sqrt{\text{area}}$			−1.49 (1.74) [0.396]	
Links may cross district boundaries			−2.91 (2.92) [0.323]	

Notes: Each cell gives the estimate, the two-way standard error (field and crop-year) in parentheses, and the two-way p-value in brackets. The last column gives the restricted wild cluster bootstrap p-value for the Up−Down estimate (district-year clusters, Webb weights, 9,999 draws, fixed effects projected out before the draws and again within each draw). Panels A and C to F: percent change in monthly ET per ten percentage points of upstream or downstream farmed share, at a crop coefficient of 1.1 minus the same at 0.4. Panel B: percent change in $\ln(1+\text{GCVI})$ per ten percentage points, at a June–September crop coefficient of 1.1. Periods are measured against 2017–2019. Panel C enters CAP deliveries and district crop demand (crop coefficient times reference ET times area, summed over the district's fields) separately, each per standard deviation; CAP deliveries leave out groundwater pumped into the canals. Panel D: canal-delivered wells are Central Arizona and Maricopa–Stanfield; private wells are Hohokam and Harquahala. Panel E: a short month has less canal water than the 2016–2019 same-month mean, and a dry month has SPEI-3 below zero at the field. CAP: 2,608 fields; Upper Gila: 467 fields.

Table A36: Competition for Canal Water on All CAP Fields Inside the District Boundaries

	Upstream	Downstream	Up−Down	Wild p
Panel A. Monthly ET, slope of the demand of linked land at the field's own K_c; all 4,641 fields unless noted				
2016, own K_c 1.1	1.11 (0.83) [0.189]	0.52 (0.73) [0.480]	0.58 (0.85) [0.494]	0.727
2020–21, own K_c 1.1	−0.20 (0.41) [0.626]	0.36 (0.50) [0.473]	−0.57 (0.68) [0.408]	
2022, own K_c 1.1	−0.99 (0.56) [0.082]	1.70 (0.52) [0.001]	−2.68 (0.61) [0.000]	0.017
2020–22, own K_c 1.1	−0.59 (0.41) [0.150]	1.03 (0.43) [0.018]	−1.63 (0.57) [0.005]	0.041
2023, own K_c 1.1	0.27 (0.82) [0.744]	0.05 (0.56) [0.926]	0.22 (0.95) [0.821]	
2022, own K_c 0.4	0.11 (0.93) [0.902]	−1.36 (1.25) [0.281]	1.47 (1.22) [0.231]	
2020–22, own K_c 0.4	0.25 (0.82) [0.764]	−0.99 (0.85) [0.248]	1.24 (0.94) [0.192]	
2022, own K_c 1.1, 5 km subset	−1.86 (0.94) [0.051]	1.22 (0.64) [0.060]	−3.08 (0.90) [0.001]	
2020–22, own K_c 1.1, 5 km subset	−1.36 (0.81) [0.096]	1.06 (0.63) [0.095]	−2.42 (0.78) [0.003]	
2022, own K_c 1.1, head-end	−0.19 (0.85) [0.828]	1.28 (0.68) [0.064]	−1.46 (0.99) [0.142]	
2020–22, own K_c 1.1, head-end	−0.03 (0.62) [0.960]	0.43 (0.57) [0.449]	−0.47 (0.89) [0.601]	
With canal block by year by month effects added				
2022, own K_c 1.1	−0.62 (0.59) [0.298]	1.16 (0.49) [0.021]	−1.78 (0.77) [0.023]	
2020–22, own K_c 1.1	−0.18 (0.46) [0.704]	0.55 (0.47) [0.243]	−0.73 (0.73) [0.321]	
Sum of the two slopes, Up+Down (the part common to both sides)				
2022, own K_c 1.1		0.71 (0.88) [0.421]		
2020–22, own K_c 1.1		0.44 (0.61) [0.477]		
2022, own K_c 1.1, 5 km subset		−0.64 (1.33) [0.630]		
2022, own K_c 1.1, head-end		1.09 (1.18) [0.359]		
August–October 2022, all fields		0.57 (1.25) [0.650]		
August–October 2022, 5 km subset		−5.42 (1.84) [0.004]		
August–October 2022, head-end		3.69 (1.48) [0.015]		
Panel B. Monthly ET, slope of the demand of linked land by month, 2022; all 4,641 fields unless noted				
March	−1.99 (0.94) [0.037]	−3.49 (1.58) [0.030]	1.49 (2.05) [0.470]	
April	−1.75 (0.80) [0.032]	−1.63 (1.13) [0.153]	−0.12 (1.35) [0.928]	
May	0.20 (1.06) [0.851]	0.22 (1.18) [0.853]	−0.02 (1.73) [0.991]	
June	1.52 (0.73) [0.040]	1.18 (0.69) [0.092]	0.33 (0.93) [0.719]	
July	−0.12 (0.75) [0.878]	1.69 (0.78) [0.033]	−1.81 (1.15) [0.122]	
August	−0.57 (1.19) [0.637]	1.26 (0.72) [0.083]	−1.82 (1.34) [0.176]	
September	−1.03 (0.82) [0.212]	1.60 (0.73) [0.032]	−2.63 (0.86) [0.003]	0.042
October	−0.62 (1.08) [0.567]	1.07 (1.01) [0.291]	−1.69 (1.40) [0.230]	
March–June mean	−0.51 (0.59) [0.389]	−0.93 (0.95) [0.329]	0.42 (1.17) [0.720]	
August–October mean	−0.74 (0.85) [0.387]	1.31 (0.65) [0.049]	−2.05 (0.87) [0.020]	0.025
August–October mean, 2016	0.31 (1.00) [0.756]	1.11 (0.68) [0.108]	−0.79 (1.15) [0.493]	
August–October mean, 2020–21	−0.20 (0.73) [0.782]	1.08 (0.46) [0.022]	−1.28 (0.82) [0.122]	
August–October mean, 2023	0.57 (1.16) [0.626]	0.30 (0.68) [0.665]	0.27 (1.19) [0.821]	
August–October mean, 5 km subset	−4.83 (1.21) [0.000]	−0.59 (1.07) [0.581]	−4.24 (1.34) [0.002]	
August–October mean, head-end	1.36 (1.17) [0.247]	2.33 (0.72) [0.002]	−0.96 (1.26) [0.446]	
Panel C. Monthly ET at own K_c 1.1, 2022: the departure as measured versus capped at ±0.5				
as measured, all fields	−0.83 (0.53) [0.117]	1.19 (0.43) [0.008]	−2.02 (0.57) [0.001]	
as measured, 5 km subset	−1.66 (0.90) [0.068]	1.05 (0.58) [0.076]	−2.71 (0.87) [0.003]	
as measured, head-end	−0.18 (0.77) [0.816]	0.64 (0.55) [0.254]	−0.82 (0.92) [0.376]	
capped, all fields	−0.99 (0.56) [0.082]	1.70 (0.52) [0.001]	−2.68 (0.61) [0.000]	
capped, 5 km subset	−1.86 (0.94) [0.051]	1.22 (0.64) [0.060]	−3.08 (0.90) [0.001]	
capped, head-end	−0.19 (0.85) [0.828]	1.28 (0.68) [0.064]	−1.46 (0.99) [0.142]	

Notes: Each cell gives the estimate per ten points of the crop water demand of linked land, the two-way standard error (field and crop-year) in parentheses, and the two-way p-value in brackets; the last column gives the restricted wild cluster bootstrap p-value for the Up−Down estimate where it was run (district-year clusters, Webb weights, 9,999 draws, fixed effects projected out before the draws and again within each draw). Panels A to C: percent change in monthly ET, with field by month and crop by year by month effects and month-specific controls. Periods are measured against 2017–2019; 2020–22 is the average of the 2020–21 and 2022 estimates. Rows name the sample only when it is not all 4,641 fields: the 5 km subset is the 2,608 fields with 5 km of canal above them, and head-end is the other 2,033 fields, nearer the head of a canal. The table continues with Panels D and E.

Table A36: Competition for Canal Water on All CAP Fields Inside the District Boundaries (continued)

	Upstream	Downstream	Up−Down	Wild p
Panel D. Sentinel-2 greenness and ET in the crop's yield-forming window, 2017–2023; all fields unless noted				
Cotton, greenness integral over the boll window (July 15 to September 30)				
2022	−1.60 (0.92) [0.132]	2.08 (1.70) [0.268]	−3.68 (1.82) [0.089]	0.206
2020–22	−2.03 (0.57) [0.012]	2.23 (1.43) [0.168]	−4.27 (1.49) [0.028]	0.012
2023	−3.86 (1.45) [0.037]	5.80 (2.07) [0.031]	−9.67 (2.59) [0.010]	
2022, 5 km subset	−5.67 (1.91) [0.025]	0.74 (1.10) [0.528]	−6.40 (1.73) [0.010]	
2020–22, 5 km subset	−4.98 (1.39) [0.012]	0.61 (1.07) [0.588]	−5.60 (1.59) [0.013]	
2022, head-end	3.46 (1.80) [0.103]	1.79 (2.50) [0.501]	1.67 (2.87) [0.581]	
2020–22, head-end	0.57 (1.12) [0.625]	2.54 (2.07) [0.267]	−1.96 (2.58) [0.476]	
Cotton, greenness integral over the boll window, with canal block by year effects				
2022	1.09 (1.19) [0.393]	3.47 (1.56) [0.068]	−2.38 (2.38) [0.356]	
2020–22	−0.64 (0.83) [0.473]	2.77 (1.31) [0.079]	−3.40 (1.88) [0.120]	
Cotton, boll-window integral implied by the ET loss (elasticity 1.523 (0.065))				
2022			−3.28 (1.17)	
2020–22			−2.44 (0.99)	
Cotton, ET July–September				
2022	−0.39 (0.50) [0.459]	1.76 (0.71) [0.042]	−2.16 (0.76) [0.025]	0.015
2020–22	−0.25 (0.37) [0.514]	1.35 (0.62) [0.067]	−1.60 (0.64) [0.042]	0.006
2016	0.53 (0.41) [0.236]	1.49 (0.95) [0.159]	−0.97 (1.12) [0.414]	
Cotton, ET July–September, with canal block by year effects				
2022	0.06 (0.63) [0.925]	1.92 (0.65) [0.021]	−1.85 (1.03) [0.114]	
2020–22	−0.07 (0.44) [0.876]	1.45 (0.58) [0.041]	−1.52 (0.80) [0.098]	
Cotton, season maximum of the index (Table A35, Panel B)				
2022	−0.01 (0.79) [0.987]	0.91 (0.99) [0.387]	−0.92 (0.76) [0.264]	
2016	2.37 (1.19) [0.087]	2.47 (1.48) [0.138]	−0.11 (1.56) [0.947]	
Alfalfa, September–October greenness				
2022	−0.70 (1.16) [0.568]	3.31 (0.91) [0.011]	−4.01 (1.80) [0.067]	
Alfalfa, date the season ends (days per 0.10)				
2022	−2.09 (1.16) [0.123]	4.94 (1.37) [0.011]	−7.03 (1.88) [0.010]	
2020–22	−1.85 (1.08) [0.139]	4.32 (1.33) [0.018]	−6.16 (1.78) [0.013]	
All crops, greenness integral over the crop's own window				
2022	−1.12 (0.93) [0.233]	2.34 (1.16) [0.048]	−3.45 (1.53) [0.027]	
2020–22	−1.16 (0.72) [0.113]	1.65 (1.12) [0.146]	−2.81 (1.51) [0.067]	
Corn and sorghum, greenness integral from silking to maturity				
2022	−1.41 (4.84) [0.775]	9.97 (2.97) [0.005]	−11.38 (5.84) [0.073]	
Small grains and wheat, greenness integral over heading and grain fill				
2022	−5.86 (3.23) [0.079]	6.61 (3.68) [0.082]	−12.47 (5.71) [0.036]	
2020–21	−3.01 (2.77) [0.285]	1.41 (2.95) [0.636]	−4.42 (4.36) [0.319]	
Panel E. Upper Gila canals, monthly ET: peak minus low demand by kind of month; all 717 fields unless noted				
Short and dry	−0.25 (0.97) [0.795]	−0.21 (0.83) [0.803]	−0.05 (1.34) [0.972]	
Short and dry, 5 km subset	−3.09 (1.58) [0.056]	0.27 (0.90) [0.770]	−3.36 (1.92) [0.088]	
Short and dry, head-end	2.00 (1.14) [0.087]	−0.08 (0.94) [0.930]	2.08 (1.34) [0.127]	
Short and dry, capped departure	0.00 (0.95) [0.997]	−0.16 (1.02) [0.875]	0.17 (1.44) [0.909]	
Short, not dry	0.65 (0.92) [0.482]	−0.90 (1.06) [0.400]	1.55 (1.22) [0.211]	
Short, not dry, capped departure	1.13 (1.13) [0.324]	−1.41 (1.24) [0.262]	2.53 (1.54) [0.107]	
Dry minus not dry	−0.91 (1.20) [0.451]	0.69 (0.94) [0.466]	−1.60 (1.29) [0.221]	
Dry minus not dry, 5 km subset	−4.10 (2.15) [0.062]	1.39 (1.22) [0.261]	−5.49 (2.50) [0.033]	
Dry minus not dry, head-end	1.52 (1.03) [0.147]	−0.44 (1.46) [0.767]	1.95 (1.77) [0.276]	
Dry minus not dry, capped departure	−1.12 (1.38) [0.421]	1.24 (1.07) [0.251]	−2.37 (1.53) [0.129]	

Notes: Panel D: percent change in the outcome per ten points of the exposure averaged over the crop's window months, with field and crop by year effects; the alfalfa season-end rows are in days. Cells, periods and samples as in the first part of the table. The cotton boll window is July 15 to September 30; the implied rows multiply the cotton ET loss by the within-field elasticity of the greenness integral to window ET. Cotton: 2,717 fields with the greenness integral and 2,817 with ET. Panel E: percent change in monthly ET per ten points of upstream or downstream farmed share, at a crop coefficient of 1.1 minus the same at 0.4, with field by month, crop by year by month, and district by year by month effects; a short month has less canal water than the 2016–2019 same-month mean, and a dry month has SPEI-3 below zero at the field; the rows marked capped departure clip each monthly departure of the farmed share at ±0.5, as the demand exposure of Panels A to D is clipped, and the other rows use the departures as measured, as in Table A35.

A30 Planting Decided Before the Season

The crop water demand exposure of Section 7.3.2 counts what each linked field planted that year, from the Cropland Data Layer, with the calendar of its crop. The Cropland Data Layer gives one status per field and year, assigned after the season. If farmers had decided during the season whether to farm a field, in response to the water they received, upstream planting and a field's water in the same months would be set together. Sentinel-2 greenness shows when the status was set. I use the five-day GCVI composites of all 6,818 fields inside the boundaries of the four CAP districts in 2017–2023 and call a field green in a window when its GCVI is at least 3.0 (Table A37). Thresholds of 2.5 and 4.0 give the same pattern.

Land fallowed in the shortage years was not planted first. Of the fields fallow in 2022 and farmed in 2021, 11.7 percent were green at some time from March through June. Among newly fallowed fields, the share was 12.9 to 19.0 percent in 2017–2019 and 10.3 percent in 2021. The median ET of the newly fallowed fields from January through June 2022 was 177 mm, close to that of newly fallowed fields in other years and far below the 477 mm of farmed fields. Fields that were planted stayed planted. Of the cotton fields green by July 14, 2.2 percent had no green window from July 15 through October 31 in 2022 and 1.5 percent in 2021, against 0.5 to 2.1 percent in 2017–2019. The share of all fields that are green fell more after mid-July than in spring in 2021 and 2022, by 5.6 and 7.3 points relative to 2017–2019. It fell even more in 2020, by 10.5 points, when deliveries were near normal. This drop does not follow the shortage. Corn is a large part of it: in every year from 2017 to 2022, half or more of the corn fields green by July 14 had no green window after it.

In 2021 and 2022, what the linked fields planted was set before the canal water fell short. The response within the season is in water use, the outcome of the design. The year 2023 differs: 24.1 percent of its newly fallowed fields were green in spring and idle afterwards, and 4.1 percent of its green cotton fields had no green window after mid-July.

Table A37: Planting and Fallowing Were Set Before the Season

	2017	2018	2019	2020	2021	2022	2023
Panel A. Fields fallow in the year							
Newly fallowed fields	416	498	636	601	551	1,407	1,545
percent green at any time March–June	19.0	12.9	15.4	16.6	10.3	11.7	24.1
median ET, January–June (mm)	150	161	168	180	143	177	227
Fallow two years in a row: median ET, January–June (mm)	114	98	188	190	129	121	212
Farmed fields: median ET, January–June (mm)	459	444	461	494	493	477	487
Panel B. Fields planted in the year							
Cotton fields green by July 14	1,314	1,141	698	674	931	626	314
percent with no green window July 15–October 31	0.5	2.1	1.4	0.4	1.5	2.2	4.1
Corn fields green by July 14	338	546	623	643	740	408	446
percent with no green window July 15–October 31	50.6	51.8	66.1	80.7	83.0	61.8	42.4
All fields: change in the share green, late season minus spring (points)	−0.6	3.4	−2.9	−10.5	−5.6	−7.3	−14.5

Notes: All 6,818 fields inside the boundaries of Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala, in every year 2017–2023. Status in the year (fallow or farmed, and the crop) from the Cropland Data Layer; newly fallowed: fallow in the year and farmed the year before. Green: Sentinel-2 GCVI of at least 3.0 in a five-day composite with at least half of the field observed. ET: OpenET ensemble, sum over January–June, median over fields. Panel B, last row: the share of fields green in each five-day window minus its 2017–2019 mean, averaged over July 15–October 31, minus the same average over March–June, in percentage points. CAP deliveries were near normal in 2020, fell to 40 to 60 percent of normal from July 2021, and were 15 to 40 percent of normal through 2022.

A31 The Farmed Status Against Greenness and Water Use

The exposures of the CAP design and the fallowing of Appendix A32 count fields by one label per field and year, farmed or fallow, from the Cropland Data Layer. Two errors are possible. A crop label on land that grew nothing hides fallowing. A fallow label on land that grew a crop adds fallowing that did not happen. Satellite records can find both. I check the label of all 6,818 fields inside the four CAP districts in 2017–2023 against two of them: the Sentinel-2 green windows of Appendix A30 and annual OpenET (Table A38).

Crop labels on land that grew nothing are rare. Over the seven years, 3.4 percent of farmed acres have no green window, between 2.2 and 4.5 percent in each year (Panel A). Grain labels fail most often, on 6 to 7 percent of grain acres in 2019, 2020 and 2022. Cotton labels fail on at most 1.8 percent of cotton acres. Farmed fields with no green window used a median of 512 mm of water a year. That is about half the 959 mm of green farmed fields and more than the 328 mm of fallow fields with no green window. Some are fallow land with a crop label, and some are thin crops. At a GCVI threshold of 2.0, 1.7 percent of farmed acres have no green window, and their median of 376 mm is close to the 311 mm of fallow land.

Fallow labels on green land are more common. Over the seven years, 28 percent of fallow field-years have a green window. Most turn green in winter. Of these fields, 68 percent peak between December and March, and the median one stays green for 35 days. Fallow fields green for less than 30 days used a median of 425 mm of water a year, about 100 mm more than fallow fields that never turn green. They are weeds after winter rain. One fallow field-year in six stays green for 30 days or more, with a median of 605 mm, nearly two thirds of a farmed field (Panel B). In these records a short winter crop and a dense stand of weeds look alike.

Counted from satellite records, farmed land fell as much in 2022 as the Cropland Data Layer says, and less in 2023 (Panel C). In 2022, farmed acres were 75 percent of their 2017–2019 level by the Cropland Data Layer, 77 percent counting fields green for 30 days or more, and 77 percent counting fields with at least 600 mm of annual ET. In 2023 the three counts are 57, 67 and 64 percent. The gap is widest in Maricopa–Stanfield, 57 percent against 67 percent by water use. Water use per farmed acre stays between 3.2 and 3.4 acre-feet under every count and in every year. The Cropland Data Layer and the water-use count agree on 90 percent of field-years. The 2022 fallowing and the flat water use per farmed acre hold when farmed land is counted from satellite records. The Cropland Data Layer may overstate the 2023 fallowing by 7 to 10 points of the 2017–2019 level.

Table A38: The 2022 Fallowing Holds Under Satellite Counts

	2017	2018	2019	2020	2021	2022	2023
Panel A. Fields the crop map calls farmed							
Farmed fields	5,860	5,858	5,597	5,471	5,409	4,304	3,328
percent of farmed acres with no green window	2.2	3.5	4.2	3.4	3.0	4.5	3.5
percent of grain acres with no green window	0.8	2.1	6.0	6.4	3.6	7.2	2.1
percent of cotton acres with no green window	0.3	0.9	0.9	1.3	1.0	1.8	1.2
median annual ET, green fields (mm)	964	975	963	947	952	935	969
median annual ET, fields with no green window (mm)	525	503	493	469	485	501	641
Panel B. Fields the crop map calls fallow							
Fallow fields	958	960	1,221	1,347	1,409	2,514	3,490
percent with a green window	20.1	22.9	27.4	35.2	27.7	23.3	32.5
percent green for 30 days or more	12.1	13.0	12.0	18.3	13.1	15.0	22.2
median annual ET, fields with no green window (mm)	241	292	312	293	328	345	362
median annual ET, fields green for 30 days or more (mm)	870	651	633	546	663	608	587
Panel C. Farmed acres (2017–2019 = 1) and ET per farmed acre (acre-feet)							
Farmed acres, crop map	1.01	1.02	0.97	0.95	0.95	0.75	0.57
Farmed acres, green for 30 days or more	1.03	1.02	0.96	0.97	0.97	0.77	0.67
Farmed acres, annual ET of at least 600 mm	1.02	1.02	0.96	0.96	0.97	0.77	0.64
ET per farmed acre, crop map	3.29	3.29	3.26	3.25	3.28	3.22	3.32
ET per farmed acre, green for 30 days or more	3.34	3.36	3.35	3.29	3.33	3.25	3.18
ET per farmed acre, annual ET of at least 600 mm	3.35	3.37	3.35	3.33	3.34	3.28	3.31

Notes: All 6,818 fields inside the boundaries of Central Arizona, Maricopa–Stanfield, Hohokam, and Harquahala, in every year 2017–2023. Farmed or fallow from the Cropland Data Layer. Green window: Sentinel-2 GCVI of at least 3.0 in a five-day composite with at least half of the field observed, over the calendar year, as in Table A37. ET: OpenET ensemble, January–December sum. Panel C counts a field as farmed by the Cropland Data Layer, by at least six green windows, or by annual ET of at least 600 mm (about 2 acre-feet per acre). Farmed acres are relative to their 2017–2019 mean, and ET per farmed acre is the ET of the counted fields divided by their acres. GCVI thresholds of 2.5 and 4.0 leave 2.4 and 6.0 percent of farmed acres without a green window over the seven years.

A32 Who Left Fields Fallow

This appendix asks which CAP fields stopped farming when CAP water was cut. A field enters year t when it was farmed in $t-1$, and the outcome is whether it is fallow in t. The sample is the 4,641 fields of Section 7.3.2 in 2018–2023. Each field trait enters with its own slope in every year, with district by year effects and the crop of the year before. Table A39 reports how each slope changed in 2022 and in 2023 against its 2018–2021 mean. Standard errors are clustered by canal network component, of which there are 89, and 20 in the three Pinal districts.

The crop of the year before carries most of the change. Against alfalfa, a cotton field was 24 points more likely to stop in 2022 than in 2018–2021, and a corn or sorghum field 18 points (Panel A). Measured against 2017–2019 rates, about 27,000 acres were added to fallow land in 2022. Cotton fields account for 40 percent of them, corn and sorghum fields for 23 percent, fallow fields that did not return for 20 percent, and alfalfa for 11 percent. Smaller fields stop more in every year, and the gap does not widen in the shortage years.

In the three Pinal districts, more old wells in the field's groundwater right lowered stopping in 2023, when CAP deliveries were near zero, by 4.9 points per standard deviation. In 2022 the change is about zero. Deeper groundwater went with more stopping in 2022 (4.0 points per standard deviation, standard error 2.0). Service by a district pump changes sign between 2022 and 2023.

Fields near the head of the canal network stopped more in the shortage years. One standard deviation toward the end lowered stopping in 2023 by 4.0 points. The network head is the top of the mapped district canals, not the point where CAP water enters. I therefore place the six CAP turnouts named in the districts' delivery records at the points where district canal lines leave the aqueduct, and check each against public CAP, Reclamation, district and map sources.

From each turnout I follow the field flow graph down from the most upstream fields near it that lie no higher than the turnout. For the 1,469 fields downstream of a turnout on the graph, one standard deviation farther downstream lowered stopping by 3.4 points in 2022 and by 3.3 points in 2023 against 2018–2021 (Panel B). In 2022, 35 percent of the farmed fields in the fifth nearest the turnout stopped, against 21 percent in the farthest fifth. The result holds with other rules for the entry fields and with the network position in the model. Distance along the canals that ignores the direction of flow does not predict stopping, and neither does straight-line distance to the aqueduct. Fields lower below the aqueduct stopped less in 2022, and height and position overlap.

None of the measured traits explains why fields near the turnout stopped more. Normal-year fallowing, greenness and July–August water use in 2016–2019 barely change with position, and adding them leaves the turnout slope unchanged. District wells upstream at any distance leave it unchanged as well, although fields near the turnout have fewer of them. Depth to groundwater, soil carbon, distance to the nearest town and field size leave the slope on network position unchanged. Maricopa–Stanfield varied its 2022 allotments by where a grower's fields were (Section 7.3.12). A proxy for local well supply does not explain the slope there, and the allotments by grower are not public.

Neighbors' planting does not move fallowing through the canal. I build the exposure of Section 7.3.2 for every field, farmed or not, from two expectations of what the competing land will plant. One is its planting the year before. The other is the chance that each neighboring field is farmed, given its own history and what fields with the same history in its district did that year. For upstream minus downstream land with no district pump in between, neither changes fallowing in any period (Panel C). With allotments per acre the season's water is fixed, and competition moves water between months among the farmed fields. Among fields served by the same district pump, fields whose nearer sharers were expected to plant more were left fallow more often in 2022 and 2023, by 3.6 and 4.3 points per ten points (standard errors 1.7 and 2.0), against 2.3 in 2020–21. A swap test by pump, which trades the near and far sharers of a random half of the pumps, is a strict test. It gives $p = 0.21$ in 2022 and in 2023, so I treat the pattern as suggestive. Crop choice moves with the expected planting of the land upstream, already in 2020–21, which points to crop patterns shared along a canal, not to competition.

In the five Upper Gila canals (717 fields), fallowing follows the river. In the short springs of 2018 and 2021, 23 and 24 percent of the fields farmed the year before were left fallow, against 5 to 10 percent in the other years. Across canal-years the share that stopped rises with the share of March to June months with canal water below normal (correlation 0.53). In 2022, also short in spring, 8 percent stopped. The same crops left first. In 2021, 55 percent of the corn and sorghum fields and 57 percent of the grain fields were left fallow, against 15 percent of the alfalfa and other hay fields.

A33 Crop Water Demand in the Upper Gila Canals

The Upper Gila canals have their own shortage definition. There a short month has less canal water than the 2016–2019 normal, and a dry field-month has SPEI-3 below zero. The first form of the design, on the 467 Upper Gila fields of the 3,600-field sample, showed the CAP pattern. With the same cut rule (0.2 to 0.3 km here) and calendars built the same way, the upstream–downstream contrast lowered ET by 3.4 percent more at peak than at low demand in short and dry months (interval -7.2 to $+0.5$), and the difference between short and dry months and short

Table A39: Who Left Fields Fallow

	All fields		Pinal, with groundwater	
	2022	2023	2022	2023
Panel A. Field traits; change of the slope against 2018–2021, percentage points				
Cotton the year before (against alfalfa)	+23.6* (3.8)	+20.7* (3.3)	+23.9* (3.5)	+19.8* (3.3)
Corn or sorghum the year before	+17.6* (2.1)	+18.5* (4.7)	+19.4* (1.4)	+11.2* (3.8)
Small grains the year before	+1.3 (3.8)	+15.2* (3.9)	−2.4 (4.3)	+3.6 (3.8)
Toward the end of the canal network, per SD	−1.7 (1.4)	−4.0* (1.2)	−1.4 (2.1)	−3.6 (1.8)
Lower below the CAP aqueduct, per SD	−3.2* (1.4)	−3.0 (2.1)	−1.0 (2.0)	−1.2 (1.5)
Log acres, per SD	+0.5 (0.7)	+1.4 (0.9)	−0.7 (0.8)	+2.5* (0.6)
Served by a district pump	−5.5 (6.1)	+4.6 (6.7)	−16.4* (4.6)	+12.2* (4.4)
Old wells in the groundwater right, per SD	–	–	−0.7 (1.1)	−4.9* (1.5)
Depth to groundwater, per SD	–	–	+4.0 (2.0)	−1.3 (2.2)
Panel B. Farther downstream from the CAP turnout, per SD; change against 2018–2021				
Entries within 5 km of the nearest	−3.4* (0.5)	−3.3* (0.4)		
Entries within 2 km	−3.5* (0.6)	−2.3* (0.5)		
Entries within 10 km	−2.6* (0.5)	−1.9* (0.7)		
Within 5 km, with the network position	−2.5* (0.8)	−1.7* (0.8)		
Panel C. Neighbors' planting and fallowing, per ten points; change against 2017–2019				
Canal, upstream minus downstream, expected	+0.2 (1.0)	−0.6 (1.0)		
Canal, upstream minus downstream, previous year	+0.2 (0.8)	0.0 (0.7)		
Pump sharers, near minus far, expected	+3.6* (1.7)	+4.3* (2.0)		

Notes: Panels A and B: fields of the 4,641 farmed in year $t-1$; linear probability of being fallow in t, in percentage points. Each trait enters with its own slope in every year from 2018 to 2023, with district by year effects and the crop of $t-1$; a cell is the 2022 or 2023 slope minus the 2018–2021 mean slope. Continuous traits per standard deviation; crop against alfalfa or other hay. Panel A also controls for distance to the canal, a field above its canal, distance to the nearest town, soil organic carbon, normal upstream farming, and the pump capacity serving the field. The Pinal columns add depth to groundwater (2008–2014), old wells in the groundwater right and an own registered well, on the fields of Central Arizona, Maricopa–Stanfield and Hohokam with groundwater data. Panel B: position downstream of a CAP turnout on the field flow graph for the 1,469 fields that lie downstream of one; entry fields are the most upstream fields within the stated distance along the canals of the field nearest the turnout and no higher than the turnout; the other fields enter with indicators. Standard errors in Panels A and B clustered by canal network component. Panel C: fields with linked land farmed in 2016–2019 (6,335; 2,023 served by a district pump with sharers on both sides in Central Arizona and Maricopa–Stanfield); exposure of June to August; field and district by year effects; standard errors two-way by field and canal block by year. * $p < 0.05$.

but not dry months was −5.5 (interval −10.5 to −0.5). On all 717 fields inside the five canal polygons, with the drought index extracted for every plot, the pattern is not there. The contrast in short and dry months is −0.05 percent (interval −2.7 to +2.6), and the difference between the two kinds of months is −1.6 (interval −4.2 to +1.0). Capping each monthly departure at ±0.5, the measurement fix used for the CAP fields, changes nothing here: the two numbers become +0.2 (interval −2.7 to +3.1) and −2.4 (interval −5.4 to +0.7). The 250 fields that the move to all fields adds, most of them with few plots above them on their canal, have a contrast of +2.1 in short and dry months (interval −0.6 to +4.8). Downstream farmed land shows no gradient in either sample (Appendix Table A36, Panel E). With this first form, the Upper Gila result depends on which fields enter. The form compares peak with low crop demand, which in the Gila mostly compares summer with spring. Section 7.3.11 and Table A40 use the design of Section 7.3 with the Gila's shortage calendar.

Table A40 applies the design of Section 7.3 to all 717 fields: the capped crop water demand of linked land, the field's own crop coefficient from the Gila calendars, and the short and dry months above, with field by month, crop by year by month, and district by year by month effects. Over all short and dry months the gap appears at low crop demand (−2.6 percent at a crop coefficient of 0.4; bootstrap $p = 0.08$) and not at peak demand (+0.8; Panel A). By season, the low-demand gap is −3.2 in March–June and −0.9 in July–October. It is largest for land 2 to 5 km away (−4.4; bootstrap $p = 0.018$), it stays with canal block by year by month effects (−3.0), and the swap test gives p of 0.052 by canal segment and 0.11 by grid cell.

Panel B follows the rules of the Gila canals. Inside a canal, water is charged per acre and may be rotated, so a short month should lower every field's share rather than favor the head of the canal, except in weeks when the canal runs partly full. When the canal is dry, every field gets nothing. The low-demand gap sits in months of partial flow (−3.0; bootstrap $p = 0.065$), in the springs of 2018, 2021, and 2022 (−3.0), among fields without their own well (−2.8, against −2.0 with one), and in the canals with the most junior rights under the decree, Fort Thomas and Graham (−3.6, against −1.9 in the other three canals). Each comparison points the way the rules predict. None of the differences is precise. Months with a dry canal are 13 percent of the short and dry months, too few to test the rule that a dry canal leaves no gap. Fields with their own well cluster in the other three canals, so that comparison also compares canals.

Panel C repeats the simulations of Appendix A34 on the five canal networks. Each canal carries its recorded water relative to its 2016–2019 normal for the month, times its normal planted demand. When water rotates, every estimate is zero. When it goes to the head of the canal first, in every week or only in weeks of partial flow, the contrast is negative at peak demand (−3.4 and −2.1) and smaller at low demand. At peak demand the observed contrast matches rotation. No rule in the simulation gives a gap at low demand without a larger one at peak demand.

Panel D relates cotton output to the spring exposure: the crop water demand of linked land averaged over March–June, interacted with the share of the spring months that are short and dry, with field, crop by year, and district by year effects (512 cotton fields). For each short and dry month in spring, the gap is −2.0 percent in the greenness integral over July 15 to September 30 (bootstrap $p = 0.033$), −1.1 percent in July–September ET (0.045), −1.3 percent in the season maximum of greenness (0.14), and −0.7 percent in March–June ET (0.22). Using the three short springs as periods instead gives the same sign with less precision (−2.2 for the integral; $p = 0.36$).

Table A40: The Upper Gila Canals with the Design of Section 7.3

Panel A. Short and dry months

	Upstream minus downstream	Bootstrap p	District-year p	Swap p (segment / grid)
All short and dry months, own K_c 1.1	0.79 (0.96)	0.460	0.444	0.366 / 0.378
All short and dry months, own K_c 0.4	−2.59 (1.21)	0.083	0.056	0.052 / 0.113
March–June, own K_c 0.4	−3.17 (1.35)	0.088	0.051	–
March–June, own K_c 1.1	1.45 (2.39)	0.448	0.421	–
July–October, own K_c 1.1	0.62 (0.85)	–	0.591	–
July–October, own K_c 0.4	−0.94 (1.59)	–	0.708	–
Land 2–5 km away, own K_c 0.4	−4.37 (1.41)	0.018	0.003	0.003 / 0.011
With canal block × year × month effects, own K_c 0.4	−3.01 (1.22)	–	0.037	–
Cotton fields, April (by month, no own K_c)	−4.45 (1.76)	–	0.049	–

Panel B. The rules of the Gila canals

Short and dry months or fields	Own K_c 1.1	Own K_c 0.4	Bootstrap p, K_c 0.4
Canal dry all month	−1.35 (3.08)	−2.95 (2.94)	0.579
Partial flow	0.88 (0.93)	−2.97 (1.28)	0.065
Springs of 2018, 2021, 2022	1.24 (2.77)	−3.01 (1.45)	0.125
Other short and dry months	0.72 (0.82)	−1.47 (1.46)	–
Fields without their own well	0.61 (1.20)	−2.84 (1.55)	0.138
Fields with their own well	1.28 (0.95)	−1.99 (1.88)	0.282
Canals with the most junior rights (Fort Thomas, Graham)	−0.93 (2.27)	−3.64 (2.48)	0.286
Other canals (Montezuma, San Jose, Union)	1.03 (0.90)	−1.92 (1.35)	0.312

Panel C. Simulated water rules

Water in the simulated canals	Own K_c 1.1	Own K_c 0.4
Rotation: every field the same share	0.00	0.00
Head of the canal first in weeks of partial flow	−2.13	−0.85
Same, and fields with their own well fill their need	−2.89	−1.41
Head of the canal first in every week	−3.35	−2.09
Observed, same effects	0.74	−1.91

Panel D. Cotton output and the spring exposure

Cotton outcome	Per short and dry spring month	Bootstrap p
Greenness integral, July 15–September 30	−2.04 (1.19)	0.033
ET, July–September	−1.14 (0.50)	0.045
Season maximum of greenness	−1.29 (0.78)	0.144
ET, March–June	−0.74 (0.40)	0.216
Greenness integral, springs of 2018, 2021, 2022 as periods (per spring)	−2.25 (2.90)	0.358

Notes: All 717 fields inside the five Upper Gila canal polygons. Panels A to C: upstream minus downstream slope of monthly ET in percent per ten points of the crop water demand of linked land, in short and dry months against dry months with normal canal water, with the two-way standard error (field and crop-year) in parentheses. Bootstrap: restricted wild cluster bootstrap with 40 district-year clusters (35 for the greenness integral, which starts in 2017) and Webb weights, 9,999 draws. Swap: share of networks, the real one included, whose t statistic is at least as large in absolute value as the real one, among 1,999 fake networks (999 for the ring cell) swapped by canal segment or 5 km grid cell. Panel B: partial flow, canal water above zero in the month; own well, a registered well of at least 500 gallons per minute within 100 m of the field, installed by 2015 (227 of 717 fields). Panel C: noise-free estimates on simulated ET with field by month and crop by canal by year by month effects; ET is the smaller of the crop's need and a quarter of reference ET plus the delivered share of the need; in weeks of partial flow a share of the water equal to one minus the canal's water relative to normal goes to the head of the canal first. Panel D: cotton field-years; the outcome on the March–June exposure times the share of spring months that are short and dry, per short and dry spring month (a quarter of the slope at a share of one).

A34 Simulated Mechanisms

The simulations keep a district's real fields, crops, planting and fallowing, crop calendars, canal network, and every regressor of the model, and replace only the outcome. Planting stays as observed because farmers decide it before the season; the simulations make up only how water moves during the season. A field's need in a month is its crop coefficient times the 2016–2023 mean reference ET of a nearby AZMET station for that month (Coolidge for Hohokam, Maricopa for Maricopa–Stanfield). Its ET is the smaller of that need and a quarter of reference ET plus the delivered share of the need; the quarter stands for stored soil water, rain, and some pumping, so a shortfall bites more when the crop needs more. The paper's model, with its fixed effects and controls, is then estimated on the made-up outcome, without noise. A rule that gives every field the same share gives exactly zero for every estimate, which checks the code.

Panel A of Table A41 uses the Hohokam network (673 fields; 1,027 plots on its canals), whose fallowing rose in 2022 like that of the other districts. In 2022 and 2023 the district's supply in a month is 60 percent of its 2017–2019 planted demand in that month, or, as a canal capacity that binds only at the peak, 60 percent of its 2017–2019 demand in the peak month. The water is shared equally or delivered upstream first: each lateral (a connected part of the network) gets the same share of its demand as the district, water enters at the head plots, and each plot takes its need or what is left. One row instead adds, in 2022, conditions that a field shares with every plot around it, weighted by distance, with an arbitrary scale. Hohokam's own data, in the last row, show no gap; its farms pump private wells.

Panel B uses the Maricopa–Stanfield network (1,713 fields; 4,040 plots), the district's reported deliveries by source, and all its active wells within 500 m of a canal (300 wells). Water moves down a tree in which each plot takes water from its nearest upstream neighbor. District wells run on demand up to their registered capacity, and total pumping in a month is held to the month's reported well volume; the remaining need is served by the month's CAP water from the canal heads, upstream first. A share of each well's water can go first to the plots of its owner's farm, taken as the one-mile grid square that holds the well, with five random placements of the grid. With reported volumes the simulated district delivers 95 to 100 percent of planted demand in normal years and 66 to 76 percent in 2022. Its records show deliveries close to planted demand that year, so the simulated gaps are larger than the observed ones. The pattern across rows is what the panel shows.

Table A41: Simulated Mechanisms

Panel A. Hohokam network

	2022, own K_c 1.1			$U - D$,	$U - D$ by ring	
Water in the simulated canals	U	D	$U - D$	K_c 0.4	1–2 km	beyond 5 km
Equal rationing, supply 0.6 of normal demand	0.00	0.00	0.00	0.00	0.00	0.00
Shared conditions in 2022, all directions (2 km)†	1.32	1.31	0.01	0.09	0.57	−0.79
Upstream first, shortage in every month	−2.27	4.94	−7.21	−2.52	−4.55	−5.83
Upstream first, shortage at the peak (canal capacity)	−0.65	1.21	−1.86	0.04	−1.73	0.01
Observed, Hohokam	−0.89	−1.77	0.88	−0.06	0.38	−1.95

Panel B. Maricopa–Stanfield network with district wells

		Upstream land		Pump
Who gets the district wells' water	$U - D$	No pump in between	Behind a pump	Capacity (per SD)
Well water moves only down the canal	−8.80	−4.50	1.39	12.96
Owners take half of their wells' water first	−5.78	−2.41	−0.03	7.53
Owners take all of their wells' water first	−5.24	−2.24	−0.09	7.02
Observed, Maricopa–Stanfield	−3.73	−2.55	0.47	−4.55

Notes: Estimates of the paper's model on simulated ET, without noise: percent change in monthly ET per ten points of the crop water demand of linked land, as the 2022 change against 2017–2019; pump capacity in percent per standard deviation of its logarithm. Panel A: supply in 2022 and 2023 is 60 percent of the 2017–2019 planted demand. †Shared conditions have an arbitrary scale; only their pattern across columns matters. Panel B: averages over five placements of the square-mile farms.

A35 Graph Validation Statistics

NHDPlus VAA direction attributes cover 91.7% of the 547,842 raw NHDPlus flowline segments, of which 147,892 are retained as canals, ditches, or aqueducts; percentages below are shares of that retained set. Before any reconciliation, the DEM-inferred flow direction agrees with VAA ground truth in 95.5% of the 105,744 retained segments where both are defined, rising to 98.6% among the 88,653 segments with reliable elevation regressions ($R^2 \geq 0.6$), with the remaining disagreements concentrated in exactly the low-R^2 segments the reconciliation step targets. A manual audit of 100 random canals shows 100% agreement between the assigned flow direction and the start-point coordinates. The DEM regression also supplies directions for the 30,276 retained segments (20.5%) that lack valid VAA attributes, with 35.8% of them having reliable fits ($R^2 \geq 0.6$) and the remainder resolved by the landscape-gradient fallback and network-consistency corrections.

Canals with DEM-regression $R^2 < 0.6$ (32.1% of retained segments) are left unprotected for graph-based conflict correction. The separate spatial elevation-gradient fallback is recorded for 21.7% of retained segments. In the saved 132,971-edge plot graph with edge provenance, 29.9% of edges are taken directly from the canal map and 70.1% link spatially adjacent plot pairs whose upstream–downstream order is assigned from canal reachability, or from intake position along a shared canal. These statistics are computed by the graph-validation routine in the replication package. The graph-positioning step also flags 466 plots sitting in grid-loop irrigation structures (spread across Maricopa, Yuma, La Paz, and other counties) as lying on a cycle in the water graph; they receive valid positions, and excluding them leaves the Table A22 estimates essentially unchanged: the $d > 5\,\text{km}$ GCVI interaction is −0.0181 versus −0.0174 at baseline.

A36 CoreLogic Market Outcomes

CoreLogic tax, deed, and MLS records provide an exploratory check of market adjustment. The fixed 2016 cohort contains agricultural parcels of at least 10 acres without residential or dwelling land-use descriptions, linked by a unique parcel centroid within a mapped field and county agreement. Never-listed and never-transferred parcels remain in the annual risk set. Table A42 compares mean exposure gradients in 2020–2023 with those in 2016–2019; this calendar contrast does not isolate an exogenous intervention.

Panel A uses the preperiod upstream cropped share beyond 5 km. Panels B–C use simpler exposures within five audited junior irrigation districts: Central Arizona, Maricopa-Stanfield, Hohokam, Harquahala Valley, and Cibola Valley. Membership requires the linked field to be mapped inside or intersecting the district; junior canal-company service areas are excluded from these two panels.

Table A42: CoreLogic Listing Entry and Ownership Transfers

	Listing entry	Qualifying transfer
A. Baseline upstream cropped share, beyond 5 km		
Change per 10 percentage points	+0.045	−0.065
95% confidence interval	[−0.011, +0.102]	[−0.602, +0.472]
Network-clustered p	0.116	0.811
Area wild-bootstrap p	0.2005	0.9600
Parcels (parcel-years)	2,852 (22,816)	2,852 (22,816)
B. Junior irrigation districts: downstream position		
Change per 0.25 of the position scale	+0.141	−0.197
95% confidence interval	[−0.074, +0.356]	[−1.663, +1.269]
Network-clustered p	0.197	0.790
Area wild-bootstrap p	0.3125	0.5625
Parcels (parcel-years), panels B–C	1,060 (8,480)	1,060 (8,480)
C. Junior irrigation districts: upstream acreage		
Change per doubling of (1 + acres)	+0.067	−0.360
95% confidence interval	[−0.008, +0.142]	[−0.829, +0.109]
Network-clustered p	0.080	0.131
Area wild-bootstrap p	0.3125	0.3125

Notes: Effects and intervals are in percentage points. Each contrast is the mean 2020–2023 exposure slope minus its mean 2016–2019 slope; 2019 is the omitted year in the annual exposure profile. All models include parcel fixed effects and annually interacted baseline controls. Panel A uses local-area-by-year effects (district within county for mapped members, county otherwise); panels B–C use district-by-year effects. Controls include acreage, improvement missingness, city/highway distances, and soil water-storage capacity, with productivity where it varies. Panel A additionally controls for eligible upstream acreage, panels B–C for network size. Panel A exposure is the preperiod mean cropped share; panels B–C use fixed network position and all upstream acreage, without a distance cutoff or same-district restriction on upstream contributors. Confidence intervals use network clustering (142 labels in A, 104 in B–C). The null-imposed area bootstrap uses 1,999 draws over 47 district/county groups in A and all 32 sign vectors over five districts in B–C. Few districts make the latter inference coarse. Never-listed and never-transferred parcels remain in the samples. Transfer is unconditional on listing. No displayed contrast passes its recorded 5% multiplicity adjustment.

The listing coefficients are suggestive but imprecise, and uncertainty is substantial under area-level inference. Listings occur in only three of the five junior irrigation districts; adding soil organic carbon and slope leaves the results similar. Some transfer specifications also exhibit changing preperiod gradients. A lower annual transfer probability would not establish a lower probability of selling conditional on listing. Incomplete listing-to-deed linkage and sparse conditional-completion and selected-price samples prevent a firm conclusion about selling difficulty or capitalization. These diagnostics neither establish a land-market mechanism nor rule out economically meaningful effects.

Supplementary Material for Online Publication Only

This supplement collects data-construction detail, secondary sensitivity analyses, an alternative fallowing specification, and the exploratory long-panel extension. Section 4 and Appendix A carry the core robustness battery; the full fallowing covariance matrix, the outcome-year fallowing profile, the alternative drought measures, and two checks cited in Section 7 (the canal-edges-only recomputation and the urban-adjacency decomposition) are reported here rather than in Appendix A and are referenced as supplementary tables.

S1 Water-Graph Construction Details

Canal graph construction. Canal segments are selected from the NHDPlus flowline layer by feature code, retaining canals, ditches, and aqueducts and excluding ephemeral streams (code 46007). For each segment, flow direction is inferred by regressing elevation along the segment's coordinates using high-resolution DEM rasters. A statistically significant negative slope ($R^2 > 0.6$) indicates that the segment is digitized in the direction of flow; a positive slope triggers reversal. Where the NHDPlus Value Added Attributes (VAA) flow table records flow connections between segments, those connections correct the DEM-based inferences. Segments are connected into the directed graph G_{canal} by spatial proximity: if the downstream endpoint of segment A falls within 100 m of the upstream endpoint of segment B, an edge $A \to B$ is added. Two further steps address infrastructure complexity. Parallel canal segments (common in NHDPlus where a single physical canal is represented by multiple overlapping polylines) are merged into unified nodes, reducing redundancy without losing connectivity. Canals are split at T-junctions where lateral canals connect mid-segment, so that the graph routes water through branching points rather than treating the entire parent segment as a single node.

Plot–canal cost function. Each plot is matched to the canal segment minimizing

$$\text{Cost}_{ij} = \text{Distance}_{ij} + \max\big(0,\ \alpha_{pump} \times (\text{Elev}_{plot_i} - \text{Elev}_{canal_j})\big) \tag{14}$$

where i indexes plots, j canal segments, and $\alpha_{pump} = 30$ penalizes assignments requiring uphill pumping: each meter of lift adds 30 meters of equivalent distance. It is a tuning constant rather than an engineering estimate, chosen so a typical lift trades off against a few tens of meters of horizontal distance; no maximum plot–canal distance is imposed, and where the canal is at equal or higher elevation the cost reduces to Euclidean distance. In practice 43% of plots sit above their nearest canal (median lift 1.1 m), a median penalty of about 34 m against a median plot–canal distance of 398 m; at 5 m the penalty reaches 150 m, enough to redirect the assignment to a more elevation-compatible canal nearby.

Root plots, edge prices, and traversal. Within each WCC the root plot is chosen among source canals (in-degree = 0 in G_{canal}): the canal whose plots have the highest mean elevation, then the in-degree-0 plot on it with the most graph descendants. From that root, signed hop counts are computed along *least-hydrologic-cost* paths. Each edge admits a forward move (+1 hop) and a backward move (−1), priced by the edge's provenance: same-canal edges cost zero, since within-canal order is fixed by the intake projection and treated as a hard constraint; canal-structure edges cost 1 forward and 1.5 backward; spatial-neighbor augmentation edges cost 100 forward and 200

backward. A Dijkstra (shortest-path) traversal assigns each plot the signed sum of ± 1 steps along its minimum-cost path. Because augmentation edges are two orders of magnitude dearer, ordering follows canal structure wherever a canal path exists, and augmentation orders a pair only when none does. A first-visit breadth-first search would not have this property, since augmentation edges act as shortcuts and the sign can lock in backwards. Finally each component is oriented to minimize canal-structure edges running against the resulting order, and signed hops are min-max normalized to $[0, 1]$ within the WCC.

Canal blocks. Plots are grouped into 1,162 canal blocks. The canal graph holds 21,998 weakly connected canal networks, 1,060 of which contain at least one plot. For compact networks fitting a 50×50 km bounding box (1,054 of 1,060) the network forms one block. For the 6 largest, spanning multiple districts, DBSCAN clustering (a density-based method that groups points lying within a chosen radius of one another) on plot locations, with a 5 km radius and a minimum of 3 plots, separates distinct agricultural clusters, and a connectivity check ensures each block's canal subgraph is connected. A single block can contain several WCCs where intermediate canals without associated plots break the plot graph.

S2 Fallowing Inference and an Outcome-Year Fallowing Profile

Table S1 reports the complete covariance-sensitivity matrix for Appendix Table A4. The six estimators impose different dependence assumptions; they are not independent replications, and the table should not be read as a vote across significance markers.

Table S1: Clustering sensitivity of the near/far fallowing timing diagnostic

Exposure mode	Year + Plot	District only	Plot + District×Year	WCC only	Plot + WCC×Year	Canal block only
Lagged exposure ($t-1$)						
Close common mode	0.103	0.040	0.018	0.007	0.008	0.004
Close directional mode (Up − Down)	0.078	< 0.001	0.012	0.018	0.007	< 0.001
Far common mode	0.793	0.703	0.764	0.733	0.747	0.508
Far directional mode (Up − Down)	0.172	0.055	0.228	0.204	0.211	0.126
Current exposure (t)						
Close common mode	0.002	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Close directional mode (Up − Down)	0.567	0.239	0.404	0.316	0.525	0.363
Far common mode	0.913	0.846	0.908	0.834	0.899	0.723
Far directional mode (Up − Down)	0.344	0.129	0.299	0.237	0.361	0.248
Future exposure ($t+1$)						
Close common mode	0.386	0.164	0.309	0.227	0.343	0.020
Close directional mode (Up − Down)	0.877	0.922	0.887	0.910	0.900	0.903
Far common mode	0.860	0.852	0.821	0.894	0.866	0.871
Far directional mode (Up − Down)	0.911	0.901	0.898	0.863	0.885	0.794

Notes: Cells are two-sided, unadjusted p-values for the same coefficients reported in Table A4; only the covariance estimator changes. The year-plus-plot specification has six outcome-year clusters. District-only (10 clusters) and WCC-only (35 clusters) allow arbitrary dependence across years within the corresponding system and therefore already subsume within-plot correlation. The crossed alternatives combine plot clustering with 60 district-by-year or 210 WCC-by-year clusters; because every field in this sample is inside its mapped district boundary, district-by-boundary-side-by-year reduces to district-by-year. The last column uses 21 canal-block clusters. Different clustering assumptions can materially change the lagged close-mode inference and do not constitute independent replications.

Table S2 reports an outcome-year fallowing profile based on the preperiod upstream activation share. It is not part of the paper's affirmative evidence because its drought indicator and focal SPEI controls use May–August of the outcome year rather than information dated before the acreage decision.

Table S2: Preperiod Upstream Activation Share and Outcome-Year Fallowing: Alternative Specification

	>5 km	>10 km
Panel A: Estimated fallowing association per one SD (percentage points)		
Junior-rights, surface-connected	1.68	5.18**
	(1.06)	(2.20)
Inside mapped district	2.74***	2.47
	(0.83)	(1.61)
Outside mapped district	−0.05	5.74*
	(1.54)	(3.04)
Inside minus outside (common SD)	3.33**	−0.24
	(1.44)	(3.61)
N, all	25,408	17,056
N, inside	14,392	9,412
N, outside	11,000	7,624
Mean fallow, inside	0.255	0.267
Mean fallow, outside	0.118	0.087
Panel B: p-value profile (two-way / district / WCC / canal block)		
Inside association	0.002/0.005/0.142/0.002	0.129/0.047/0.267/0.001
Inside-minus-outside difference	0.021/0.212/0.290/0.119	0.946/0.928/0.951/0.835

Notes: The specification fixes each plot's 2016–2019 modal active-crop stratum and mean upstream activation share, then estimates active/nonfallow status in 2020–2023, with signs reversed so positive entries denote more fallowing. The drought indicator and focal SPEI controls use May–August of the outcome year. Entries are therefore outcome-year associations, not pre-season responses. "Junior-rights, surface-connected" pools fields mapped inside and outside assigned district boundaries; it is not an inside-service-area estimate. Panel B reports p-values under baseline two-way, district, WCC, and canal-block clustering. Appendix A22 gives the timing interpretation. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Fallowing is imprecise across the priority cells. The isolated senior-inside 10 km estimate is +2.17 percentage points ($p = 0.026$) and +1.56 points ($p = 0.036$) under the activation-share gap. Fallowing inference uses eight calendar-year clusters, the grid has no cross-cell equality test, and a positive senior cell does not identify priority-based protection of crops that remain active.

S3 Geospatial Sources and Plot Panel

- **Irrigation canal network.** The USGS National Hydrography Dataset Plus (NHDPlus) provides canal geometries, flow attributes, and unique identifiers (NHDPlusID). After filtering out ephemeral streams (NHD feature code 46007), 147,892 canal segments are retained from the 547,842 raw flowlines across Arizona.

- **Agricultural plot boundaries.** The USDA Crop Sequence Boundaries (CSB) dataset provides satellite-derived field boundaries spanning 2016–2023, each plot identified by a unique field identifier. I restrict to plots in Arizona (FIPS state code 04) across all 15 counties.

- **Digital elevation model (DEM).** USGS 1/3 arc-second elevation tiles provide the high-resolution elevation rasters used to determine water-flow directions within the canal network by calculating elevation gradients along canal segments.

- **Crop-type classification (CDL).** The USDA CropScape Cropland Data Layer provides annual satellite-based crop-type classification at 30 m resolution for 2016–2023. CDL is used for three purposes: constructing crop-specific fixed effects (crop × year), classifying plots as active or fallow in each year (which drives the time-varying upstream cropped-acreage measure), and identifying crop-rotation patterns for the fallow-type decomposition.

The final analytical dataset contains the 54,946 agricultural plots of Section 4.1, observed annually over 2016–2023. Every plot has a canal network position assigned through the water-graph algorithm of Section 5. Row-specific sample counts differ across variables because satellite outcomes, drought measures, and network-position variables are observed on different supports.

S4 Satellite Outcomes

The native panel preserves the original NDMI, GCVI, and ET values alongside working outcomes that equal $\ln(1+x)$ exactly. All three enter the native regressions and descriptive tables on this transformed scale. The retained Google Earth Engine code uses May 1 through September 30 for vegetation and January 1 through February's end for bare-soil indices; its October 1 and March 1 end dates are exclusive. It retains vegetation scenes with less than 20% cloud cover and additionally masks cloud shadow, medium/high-probability cloud, and cirrus pixels using the scene-classification layer. For NDMI and GCVI, a temporal maximum is taken at each pixel and then averaged over the plot at 20 m scale, rather than taking the maximum of date-specific plot means. ET uses a different seasonal window and aggregation, described below.

- **$\mathbf{NDMI_{max}}$.** The plot mean of pixelwise seasonal maxima of NDMI $= (B8-B11)/(B8+B11)$. The extraction code uses Sentinel-2 harmonized surface-reflectance imagery accessed through Google Earth Engine. NDMI captures the water content of vegetation and is the primary water-stress proxy. I also compute the within-season standard deviation (NDMI_{sd}), which is included in the replication package. It is not used in the analyses below.

- **$\mathbf{GCVI_{max}}$.** The plot mean of pixelwise seasonal maxima of GCVI $= (B8/B3)-1$, using the same imagery and vegetation window. GCVI is sensitive to chlorophyll content and serves as a yield and vigor proxy. Satellite-derived vegetation indices have been validated as reliable yield proxies against ground-truth harvest data (Lobell et al. 2020).

- **$\mathbf{ET_{ensemble}}$.** The seasonal sum (May–October) of monthly ET from the OpenET Ensemble product (v2.0), which combines multiple energy-balance models (SIMS, eeMETRIC, and others) at 30 m Landsat-scale resolution. I compute plot-mean ET depth at 30 m scale; this outcome is not total water volume per field. ET captures actual crop water consumption and complements the stress-based vegetation indices. I additionally extract ET from the individual SIMS and eeMETRIC models and construct a "water gap" measure ($\text{ET}_{sims} - \text{ET}_{eemetric}$), which is included in the replication package. It is not used in the analyses below.

For the bare-soil season (January–February), the retained code extracts three Sentinel-2 indices as time-varying soil controls: a salinity index ($\sqrt{B2 \times B4}$, median), a clay ratio ($B11/B12$, median), and a bare-soil index (BSI, median).

The early–late exercise of Appendix S15 uses separately extracted January–April upstream ET and July–October focal ET, with July–September focal vegetation outcomes. Its primary ET outcome is logged; the original secondary GCVI/NDMI specifications and the Wellton moisture comparison use raw indices. The original Gila follow-up uses raw NDMI and ET but logged GCVI. Appendix S16 reports matched raw and $\ln(1+x)$ versions of these supporting outcomes, keeping the original choices identifiable and holding observations and specifications fixed. The transformation check does not change the native annual panel.

S5 Drought Measures

I extract SPEI at two timescales from the SPEIbase global gridded dataset (Vicente-Serrano, Beguería, and López-Moreno 2010) at approximately 0.5° resolution, sampling at each plot's centroid and averaging over May–August. This SPEI window deliberately ends earlier than the May–September vegetation and May–October ET windows: it targets the months when irrigation demand peaks and surface-water allocation decisions bind.

- **SPEI-3M** is a 3-month index representing immediate meteorological conditions and soil-moisture pulses.
- **SPEI-24M** is a 24-month index proxying reservoir storage and long-term water reliability.

I construct binary drought indicators equal to one when the respective SPEI value falls below -0.6, an operational cutoff chosen to mark dry surface-water conditions in the empirical design; Appendix Table S6 reports the corresponding estimates under alternative drought definitions.

- **Plot-centroid drought.** The 3- and 24-month indicators threshold the centroid-sampled SPEI series and appear only in the descriptive statistics (Table 2); the underlying continuous centroid series enter the regressions as controls and, in Appendix Table S6, as alternative drought measures.
- **Upstream-path and district drought.** The baseline Table A22 drought variable is built from each plot's position in the water graph: for every active focal plot-year I take the acreage-weighted mean SPEI-24M across actively cropped plots on plot i's upstream path and apply the same -0.6 threshold. Its source set therefore changes with annual activation. Tables A11 and A18 instead use a plot-specific fixed-network counterpart that averages over every structurally eligible plot in the full upstream set. Table 5 and Appendix Table A10 use the acreage-weighted May–August SPEI-24 mean over a fixed set of all mapped plots inside the irrigation district. That indicator varies at the district-year level. Appendix A22 explains why an outcome-year May–August drought indicator is not a pre-decision fallowing measure. All three constructions are atmospheric supply proxies, not observed diversions or deliveries.

The two timescales flag different episodes. The 3-month index marks the acute dry spells of 2018, 2020, and 2021–2023, flagging roughly a third to three-quarters of plot-years in those years. The slower 24-month index (the basis for Drought_{it}^{up}, the primary drought measure in the analysis) concentrates in 2021–2022, after consecutive record-dry years, with modest incidence in 2017–2019 and essentially none in 2016, 2020, or 2023. The sample period captures the intensification of the ongoing Colorado River Basin megadrought, with both timescales flagging the 2021–2022 peak that coincides with the Tier 1 shortage declaration. As a robustness check, I also estimate specifications using the continuous SPEI measure to exploit finer variation in drought intensity across years.

SPEI-24 is the primary measure because it captures slower surface-water scarcity and reservoir/storage conditions (the drivers of canal-delivery reliability) better than short-run meteorological drought; Section 4.2 states how the resulting indicator should be read.

S6 Water Rights, Groundwater, and Irrigation Districts

- **Surface-water rights.** Surface-water rights data from the Arizona Department of Water Resources (ADWR) "Statement of Claimant Locator" database provide ownership, priority dates, and locations. These records represent Place of Use (POU) designations, the locations where water may legally be applied for beneficial use. Priority dates enable calculation of weighted seniority metrics for upstream and downstream neighbors.[21] A significant limitation is that water rights are recorded as point locations rather than polygons: a single point cannot precisely define which plots fall under a given right, and the service area may extend beyond the recorded coordinates. The water-rights controls therefore rely on distance-based matching from plot centroids to the nearest recorded right location, introducing measurement error that likely biases toward zero the estimates of how legal priority affects crop outcomes. The nearest-neighbor match imposes no maximum distance; matches beyond 0.2 km are flagged as too distant for the water-rights controls.

- **Irrigation districts.** I obtain irrigation-district boundary shapefiles from ADWR and spatially assign each plot to the nearest district, generating a district identifier, a within-district indicator, and distance to the nearest district. The nearest identifier is a geographic assignment for every field, not evidence of membership or service. The institutional profiles therefore distinguish fields inside the named boundary from outside fields and describe "inside" as a service proxy rather than a legal entitlement.

- **Grandfathered groundwater rights.** The Arizona Grandfathered Groundwater Rights database identifies historical plot boundaries with established groundwater-extraction rights (Irrigation Grandfathered Rights, IGFR). These rights, which predate Arizona's 1980 Groundwater Management Act, provide legal access to groundwater, an important alternative source during surface-water shortages.

- **Well registry.** The ADWR Wells55 registry provides well locations, use codes, application status, and reported installation dates. I calculate the number of wells within the grandfathered-rights boundaries associated with each plot as the primary access measure; the predetermined-capacity check separately counts non-canceled irrigation-well applications and reported installations through 2014 within 5 km.

- **Groundwater conditions.** The ADWR Groundwater Site Inventory provides dated depth-to-water readings and measurement-source metadata. The CAP scope check retains direct readings with blank measurement remarks and credible agency source codes, collapses duplicate site-date records by their median, and requires the same site in every year from 2018 through 2023. Depth is a groundwater-condition outcome: greater depth means a lower water table, not a metered withdrawal volume.

- **Physical supply.** The source audit verifies ordinary agricultural water used within mapped service areas from district, Reclamation, CAP, ADWR, and related primary records. Its fixed 30-district set covers 95.6% of service-supported member plots. The strict classification retains unresolved cases; the approved broader-system sensitivity classifies six Harquahala/Gila

[21] Plots with no matched water right are assigned a priority year of 2025, treating them as having the lowest possible seniority. This affects approximately 72% of observations, reflecting the coarse point-based matching described below.

Valley encompassing-system cases as mixed without imputing a groundwater share. The regression division is therefore groundwater only (3 districts), mixed (22), and surface water only (5).

- **Pricing and governance.** The audit records volumetric, per-acre, flat, tiered, time-based, and energy-pass-through charges, plus rotation schedules and shortage protocols. A descriptive code establishes that a mechanism is documented at some point in 2016–2023; a separate static code requires evidence suitable for a time-invariant interaction variable throughout the analysis period. Unknown, conflicted, and period-ineligible entries remain missing. The strict static view has too little yes–no support for pricing or governance interaction regressions.

- **Seniority.** Priority is classified within the district's operational allocation system rather than translated to a universal rank. The priority categories are senior (6 districts), junior (10), mixed (5), unknown (6), and not applicable (3). Binary regressions use only the affirmatively senior and junior categories.

The integrated audit workbook joins the 81 live panel identifiers to pricing/governance evidence and preserves the independently reviewed 30-district priority/supply scope used for analysis. Table S3 reports the complete category distribution and the yes/known denominators; it does not code undocumented mechanisms as zero.

Table S3: Irrigation-District Characteristics and Documentation Coverage

Category or mechanism	Districts	Service-supported plots	Plot share	Documented yes/known	Static yes/known
Panel A: Physical supply (broader-system sensitivity)					
Groundwater only	3	1,642	5.9%		
Mixed surface/groundwater	22	23,714	84.6%		
Surface water only	5	2,663	9.5%		
Panel B: Operational priority					
Senior	6	10,879	38.8%		
Mixed priority	5	3,588	12.8%		
Junior	10	7,870	28.1%		
Unknown	6	4,040	14.4%		
Not applicable	3	1,642	5.9%		
Panel C: Pricing and governance audit coverage					
Volumetric pricing	–	–	–	5/5	3/3
Per-acre assessment	–	–	–	26/26	18/18
Flat recurring fee	–	–	–	2/2	1/1
Tiered/progressive pricing	–	–	–	2/2	2/2
Time-based pricing	–	–	–	1/1	—
Energy pass-through	–	–	–	0/1	0/1
Rotation schedule	–	–	–	3/4	1/2
Shortage protocol	–	–	–	12/12	8/8

Notes: The fixed 30-district set accounts for 28,019 service-supported member plots (95.6% of the member-plot universe). Physical supply uses the approved broader-system sensitivity and does not impute a numeric groundwater share. Priority categories are operational within system; mixed, unknown, and not-applicable values are not forced into the senior/junior binary. "Documented" means the mechanism is established at some point in 2016–2023; "static" requires full-period eligibility for a time-invariant regression interaction variable. Entries are yes/known district counts, so 3/3 means three documented yes districts and no documented no districts, not a three-district balanced comparison.

S7 Soil, Market Access, and Secondary Controls

- **Soil quality (SSURGO).** The USDA NRCS Soil Survey Geographic Database provides time-invariant soil properties including clay content (water retention), salinity (EC), and soil organic carbon. These controls address soil heterogeneity that could confound estimates of water-delivery efficiency.

- **Market access (OpenStreetMap).** I calculate Euclidean distances from each plot centroid to the nearest urban centers, highways, and rural roads to control for market access and transportation costs. Beyond their direct economic relevance, these distances also proxy for water-rights seniority: under prior appropriation, early settlers who established farms first obtained senior rights, and these first farmers tended to locate near existing roads and emerging urban centers, so plots closer to roads and cities are more likely to hold senior rights even when priority dates are missing or imprecise. The distance controls thus help absorb variation in legal priority that might otherwise confound the physical network-position effects.

- **Community Water System service areas.** The ADEQ CWS service-area layer enters the panel in two ways. First, the common control set includes an indicator for whether the plot centroid lies inside any CWS polygon, absorbing broad municipal-service-area exposure. Second, Appendix Table S10 defines the narrower urban-adjacency decomposition from active CWS polygons with population at least 100,000, splitting upstream ancestors by whether their centroids fall inside that large-service-area union.

The regressions additionally include topographic and administrative controls (e.g., slope, network in-/out-degree and head/tail dummies, protected-area distances); the complete control list used in the cross-section is given in Section 5.7 and the replication package.

S8 Fallow Classification

A plot is classified as **active** in a year if it carries a crop, i.e., a CDL code outside a 24-code non-crop set (fallow/idle, water, developed, barren, forest, shrubland, pasture, wetlands, and similar categories, plus missing codes), and **fallow** otherwise. This active/fallow status drives the time-varying upstream cropped-acreage measure of Section 5.

Not all fallow is equally informative. Fallow type comes from a machine-learning classifier trained on 8-year crop-sequence patterns, which assigns each fallow plot-year a predicted rotation probability: **non-rotation (stress) fallow** requires probability < 0.3, likely fallowed due to water constraints or economic stress, and **rotation fallow** requires probability > 0.7, representing planned agronomic rotation. Fallow plot-years with predicted probabilities in the middle band $[0.3, 0.7]$ are excluded from both classifier-based shares. These two shares are the local controls in the Table A22 specification; they should not be conflated with the broad CDL fallow control used by the later main-exposure crop models.

For Table A22, the local-shock controls are computed in geographic space: for each plot-year, the shares of plots within Euclidean radius d that are stress-fallow and rotation-fallow, regardless of canal direction. The later active-crop models in Table 5 and Appendix Tables A10, A18, and A11 instead use one cutoff-matched broad CDL fallow share that includes all inactive classifications. Their own-fallowing models include no spatial-fallow control. Both choices aim to absorb spatially correlated local shocks without double-counting the canal-directional upstream cropped acreage that the exposure measures; they are different specifications and are labeled separately throughout.

S9 Full Summary Statistics

Table S4 reports the full summary statistics, including the soil, slope, upstream fallow-share, and secondary-control rows omitted from the compressed main-text panel (Table 2).

Table S4: Full Summary Statistics: Active vs. Fallow Plots

		Active Plots			**Fallow Plots**		
	TV	Mean	SD	N	Mean	SD	N
Crop Outcomes							
NDMI (max)	✓	0.253	0.156	327,999	0.022	0.138	107,154
GCVI (max)	✓	1.78	0.629	327,999	0.930	0.493	107,154
ET Ensemble (log)	✓	6.29	0.552	327,999	5.26	0.809	107,154
Network Position & Upstream Competition							
Network Position (0–1)		0.499	0.268	331,327	0.503	0.286	108,241
log(1 + upstream active >1km)	✓	3.49	1.65	272,464	2.79	1.77	85,112
log(1 + downstream active >1km)	✓	3.43	1.72	262,102	2.63	1.86	82,786
Canal Network Size (plots)		4330.8	4691.8	331,327	2692.8	3903.9	108,241
Drought							
Plot drought$_{24M}$ (binary)	✓	0.160	0.366	328,297	0.174	0.379	107,351
Upstream drought$_{24M}$ (binary)	✓	0.150	0.357	331,327	—	—	—
SPEI 24-Month	✓	-0.108	0.512	328,297	-0.092	0.601	107,351
SPEI 3-Month	✓	-0.303	0.545	328,297	-0.355	0.514	107,351
Fallow & Upstream Cropped Acreage							
Stress fallow (non-rotation)	✓	0.000	0.000	331,327	0.279	0.448	108,185
Rotation fallow	✓	0.000	0.000	331,327	0.260	0.439	108,185
Upstream non-rot. fallow share	✓	0.053	0.101	316,850	0.116	0.182	100,638
Upstream rot. fallow share	✓	0.037	0.093	316,850	0.141	0.235	100,638
Plot Characteristics							
Plot Size (acres)		21.39	36.12	331,327	14.58	40.89	108,241
Well Count		1.48	4.51	331,327	1.59	4.84	108,241
Distance to Canal (m)		727.1	970.2	331,327	1021.3	1355.0	108,241
Slope (%)		1.33	1.75	327,443	1.67	2.50	106,429
Elevation Diff. to Intake (m)		-0.103	4.84	331,327	0.463	7.16	108,241
Pumping (dummy)		0.418	0.493	331,327	0.503	0.500	108,241
Soil & Water Access							
Soil Salinity (EC)		6.81	0.499	327,471	7.25	0.427	106,863
Clay Content		0.849	0.116	327,471	0.776	0.071	106,863
GW Share (HUC12)		0.436	0.310	323,967	0.521	0.302	101,465

Notes: TV marks variables that vary by plot-year. Nominal panel: 2016–2023 (Section 4.1); all plots have canal network positions. Row-specific counts vary by variable support. Active ≈ 75%, Fallow ≈ 25%. NDMI, GCVI, and ET Ensemble are all transformed by $\ln(1 + x)$. Network position normalized 0–1 within each weakly connected component. The tabulated upstream drought$_{24M}$ row is the baseline active-set measure, so its fallow cells are omitted (—). The plot-specific fixed-upstream-set measure used in Tables A18 and A11, and the all-inside district measure used in Table 5 and Appendix Table A10, are constructed separately. This is the full panel; the main text (Table 2) shows a design-relevant subset.

Table S5: Canal-Structure-Edges-Only Upstream Sets: Table A22 Specification

	GCVI	ET
Spatial exclusion cutoff: $d > 1\,km$		
Baseline: upstream cropped acreage × Drought	-0.015^{**}	-0.011
	(0.006)	(0.007)
Canal-structure-edges-only: upstream cropped acreage × Drought	-0.023^{***}	-0.014^{*}
	(0.008)	(0.009)
N (baseline)	269,870	269,870
N (canal-structure-edges-only)	175,985	175,985
Spatial exclusion cutoff: $d > 5\,km$		
Baseline: upstream cropped acreage × Drought	-0.017^{**}	-0.014
	(0.008)	(0.009)
Canal-structure-edges-only: upstream cropped acreage × Drought	-0.025^{**}	-0.013
	(0.011)	(0.011)
N (baseline)	137,229	137,229
N (canal-structure-edges-only)	34,944	34,944
Fixed effects	Crop×Year + Plot	Crop×Year + Plot
SE clusters	Crop×Year + Plot	Crop×Year + Plot
Canal-only exposure corr. with deployed: 0.584 ($d > 1$ km); 0.574 ($d > 5$ km)		

Notes: Specifications match Table A22, except upstream ancestor sets are recomputed on the subgraph of canal-structure edges only (39,764 of 132,971 edges); the block-constrained spatial-neighbor augmentation edges are removed. The treatment is otherwise constructed identically. Samples shrink because fewer plot-years have nonzero distant upstream cropped acreage. SE are clustered two ways by Crop×Year and plot. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$. Restricting to canal-structure ancestors drops observations with none, cutting the sample by 34.8% at $d > 1$ km and 74.5% at $d > 5$ km (137,229 to 34,944); the canal-structure-only measure correlates 0.584 and 0.574 with the measure used in the main results at the two cutoffs.

S10 Alternative Drought Measures

Table S6: Distant Upstream Cropped Acreage Under Alternative Drought Measures

	Full Path (1)	>1km (2)	>5km (3)	>10km (4)
Panel A: Binary Upstream 24M (**1**[Upstream $SPEI_{24M} < -0.6$])				
NDMI	−0.003**	−0.030**	−0.045***	−0.068**
	(0.002)	(0.012)	(0.013)	(0.029)
GCVI	−0.011*	−0.107**	−0.170***	−0.403***
	(0.006)	(0.049)	(0.053)	(0.117)
ET	−0.009	−0.061	−0.125***	−0.189**
	(0.006)	(0.041)	(0.045)	(0.085)
Panel B: Cont Upstream 24M (−1× Upstream $SPEI_{24M}$)				
NDMI	−0.003***	−0.014	−0.018	−0.007
	(0.001)	(0.009)	(0.012)	(0.017)
GCVI	−0.014***	−0.075**	−0.080*	−0.038
	(0.003)	(0.035)	(0.046)	(0.069)
ET	−0.007***	−0.032	−0.041	0.000
	(0.003)	(0.032)	(0.045)	(0.073)
Panel C: Cont Plot 24M (−1× Plot $SPEI_{24M}$)				
NDMI	−0.002***	−0.011	−0.017	−0.002
	(0.001)	(0.009)	(0.011)	(0.017)
GCVI	−0.009***	−0.069**	−0.077*	−0.029
	(0.002)	(0.031)	(0.044)	(0.067)
ET	−0.004**	−0.024	−0.038	0.015
	(0.002)	(0.028)	(0.043)	(0.073)
Panel D: Binary Upstream 3M (**1**[Upstream $SPEI_{3M} < -0.6$])				
NDMI	0.002***	0.028***	0.013	0.005
	(0.001)	(0.007)	(0.012)	(0.018)
GCVI	0.007***	0.076**	0.001	−0.014
	(0.002)	(0.034)	(0.041)	(0.055)
ET	0.003*	0.045**	0.031	−0.009
	(0.002)	(0.021)	(0.043)	(0.072)
Panel E: Cont Upstream 3M (−1× Upstream $SPEI_{3M}$)				
NDMI	−0.000	0.006	−0.005	−0.021
	(0.001)	(0.008)	(0.010)	(0.015)
GCVI	−0.000	−0.005	−0.046	−0.066
	(0.003)	(0.035)	(0.032)	(0.048)
ET	−0.002	−0.014	−0.023	−0.070
	(0.003)	(0.023)	(0.037)	(0.054)
Panel F: Cont Plot 3M (−1× Plot $SPEI_{3M}$)				
NDMI	−0.000	0.006	−0.005	−0.022
	(0.001)	(0.008)	(0.011)	(0.016)
GCVI	−0.002	−0.003	−0.049	−0.078
	(0.002)	(0.034)	(0.034)	(0.049)
ET	−0.002	−0.012	−0.025	−0.084
	(0.002)	(0.022)	(0.039)	(0.057)
N (binary)	326,732	269,870	137,229	75,826
N (upstream cont.)	308,142	266,791	136,370	75,613
Plot + Crop×Year FE + Controls	✓	✓	✓	✓
Nearby fallow controls (<1km)		✓	✓	✓

Notes: Active plots only. Entries report the key distant upstream cropped acreage × Drought interaction under alternative drought definitions and time scales. Treatment here is the log of one plus the area- and hop-weighted active upstream plots beyond d km, not Table A22's log of one plus the unweighted active upstream count beyond d km; the local-shock controls are canal-directional upstream strict-inactive fallow shares computed along the water graph rather than Table A22's Euclidean geographic-radius shares; and the specification adds distant upstream fallow shares and fixes the near-control radius at 1 km instead of scaling it with the cutoff. Coefficients are therefore not directly comparable in magnitude to Table A22; the comparison of interest is across drought definitions within this table. FE: Crop×Year + Plot. Two-way clustered SE.

Under the 3-month index the >1 km GCVI interaction is positive (+0.076**), consistent with that index capturing a different scarcity horizon; the contrast does not identify a mechanism.

S11 Physical-Supply Profiles

Table S7 restricts fields to actual district boundaries, uses the documented broader-system physical-supply categories, and applies the selected upstream-count exposure and main GCVI/fallowing specifications at 5 and 10 km. The sparse groundwater-only 10 km GCVI row is retained as NA because its clustered covariance is nonfinite on two-district support.

Table S7: Physical Water Supply and Upstream-Activity Incidence

		GCVI		Fallowing (pp)		
Cutoff	Physical supply	$X \times D$	p	$X \times D$	p	Plots / districts
5 km	Groundwater only	+0.037***	< 0.001	+0.338	0.773	229 / 3
		(0.006)		(1.131)		
	Mixed surface/groundwater	-0.001	0.345	-0.200**	0.016	11,013 / 22
		(0.001)		(0.063)		
	Surface water only	+0.000	0.991	+0.904***	0.006	934 / 4
		(0.004)		(0.229)		
10 km	Groundwater only	–	–	-1.423	0.618	131 / 2
				(2.723)		
	Mixed surface/groundwater	-0.003**	0.032	+0.019	0.851	6,177 / 20
		(0.001)		(0.098)		
	Surface water only	-0.003	0.513	-0.819	0.166	97 / 4
		(0.004)		(0.530)		

Notes: Entries are upstream-count exposure × Drought_{24M} coefficients per one within-plot SD on the exact model sample. All fields lie inside the named mapped district boundary. Physical supply uses the independently verified 30-district audit's broader-system sensitivity: 3 groundwater-only, 22 mixed, and 5 surface-water-only systems before exposure-support losses. GCVI uses active plots, plot and Crop×Year fixed effects, cutoff-matched broad-fallow controls, and Crop×Year/plot clustering. Fallowing uses all observed broad-fallow rows, plot and district-by-inside/outside-by-year fixed effects, and year/plot clustering; coefficients are percentage points and inference has eight year clusters. The groundwater-only 10 km GCVI coefficient is shown as NA because the clustered covariance is nonfinite. Categories are estimated separately; no cross-category equality test is reported. $^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

On inside-boundary fields, 5 km GCVI interactions are +0.037 in groundwater-only systems, −0.001 in mixed systems, and approximately zero in surface-water-only systems; the groundwater-only cell holds 229 plots in three districts. At 10 km, mixed and surface-water-only estimates are both about −0.003. Groundwater-only inference is not finite on two-district support. Fallowing likewise does not order with source, so the split does not corroborate the groundwater-buffering prediction (Table S7).

S12 Year-by-Year Upstream-Activity Interactions

Table S8: Distant Upstream Cropped Acreage × Year (2016 Base), Table A22 Specification at >5 km

	GCVI	ET
Upstream cropped acreage (level)	0.005	−0.004
	(0.013)	(0.009)
× 2017	−0.010	−0.002
	(0.007)	(0.005)
× 2018	−0.003	0.002
	(0.006)	(0.005)
× 2019	−0.005	−0.010
	(0.005)	(0.007)
× 2020	−0.016***	−0.023***
	(0.006)	(0.005)
× 2021	−0.028***	−0.028***
	(0.005)	(0.005)
× 2022	−0.023***	−0.018***
	(0.007)	(0.007)
× 2023	0.001	0.002
	(0.005)	(0.005)
N	137,229	137,229

Notes: Same specification as Table A22 at the >5 km cutoff, with the upstream cropped acreage × Drought_{24M} interaction replaced by upstream cropped acreage × year indicators (2016 base; year main effects absorbed by Crop×Year fixed effects) and the Drought_{24M} main effect retained as a control (not displayed), so the year-interaction terms are not confounded with drought's non-varying-by-year component. The 24-month drought indicator loads on 2021–2022 and flags essentially no plot-year in 2020 (one of 39,709 active rows; mean upstream SPEI-24 +0.32), so the significant 2020 interaction is not identified off the 24-month drought indicator. Two readings of that year are not separated here: 2020 is dry on the shorter index, with the upstream 3-month indicator flagging 44% of active rows, and it is also the first Colorado River Tier 0 reduction year. The pooled 3-month interaction of Section 7.1 does not adjudicate between them, since it is estimated at the >1 km cutoff and is essentially zero at the >5 km cutoff used here. Two-way clustered SE. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

S13 Economic Magnitudes and an Illustrative Reallocation Screen

This appendix documents the magnitudes of Appendix A13 and uses them for an illustrative reallocation screen.

Companion specification. The sample (22,105 active plot-years on 3,600 fields in ten junior-rights districts), controls, plot and crop-by-year fixed effects, and two-way clustering by plot and crop-year are those of Table 5. The interactions use a common scarcity window $S_t = \mathbf{1}\{t \in \{2021, 2022\}\}$ relative to 2016–2020, with separate 2023 interactions, and the district drought indicator remains as a level control. The departure and baseline interactions are −0.158 ($p = 0.013$; district wild-bootstrap $p = 0.025$) and −0.184 ($p = 0.054$; $p = 0.090$) per unit share; with the well and rights controls interacted with the window they are −0.148 ($p = 0.003$) and −0.088 ($p = 0.28$). In the pooled year profile the window association is concentrated in 2021; inside the CAP-served districts both 2021 and 2022 are negative.

Conversion. The outcome is $\ln(1 + \text{GCVI})$, so a shift of Δ log points is a proportional change $1-e^{-\Delta}$ in 1+GCVI. Revenue is assumed to change by κ times that proportion, with $\kappa = 0.8$; this is a judgement rather than an estimate, applied equally to all crops, and entries scale approximately

linearly in it. Revenue per acre is the 2021–22 NASS Arizona mean (cotton including seed, $1,555). The evidence on κ points both ways. In irrigated cotton at Maricopa, the same near-infrared-to-green ratio measured at peak bloom explains 60 percent of the variation in lint yield across plots, with an implied elasticity of about 1.6 (Gutierrez et al. 2012); in these data, by contrast, the consumptive-use (ET) response to exposure is about a third of the GCVI response in percentage terms. A county-level comparison with NASS yields (USDA-NASS 2026) cannot settle the value: across 41 cotton county-years, the slope of log lint yield on county-mean $\ln(1 + \text{GCVI})$, with county and year effects, is 0.06 (standard error 0.06). Six counties, aggregation of noisy field indices, and the mismatch between mapped cotton, which includes Pima, and NASS upland yields make that slope uninformative. The dollar figures are therefore revenue equivalents at an assumed pass-through, not validated estimates of lost output.

Results. Panel A of Table S9 compares fields at the 25th and 75th (and 10th and 90th) percentiles of the combined index $\hat{b}_H H_i + \hat{b}_W W_{it}$ among 2021–22 field-years; intervals come from 4,000 joint draws of the two coefficients. The within-group rows add separate crop-by-year effects for the CAP-served and the other junior districts. Panel B uses the departure coefficient alone, because a change in upstream cropping within a year moves the departure and not the baseline. Upstream cropping fell below its 2016–2019 level in 2021–22 (revenue-weighted mean departure -0.061), and the model associates that fall with $1.06 million per scarcity year of downstream revenue in the ten districts ($0.55 million within groups). The second row of Panel B is a physical plausibility check from the same specification with ET as the outcome: each acre-foot of upstream May–October ET is associated with 0.035–0.042 acre-feet less downstream ET, far below the roughly 0.8 that a pure water-flow channel could deliver after conveyance losses.

Illustrative reallocation screen. Panel C asks what the estimates would imply if water freed upstream reached the fields below. For each upstream summer-crop field in 2021–22, the modeled downstream gain per acre is $-\kappa\,\hat{s}\sum_i R_i/C_i$, where $\hat{s}$ is the total scarcity-year departure slope, R_i the revenue of each downstream field and C_i its upstream area. The modeled gains are small relative to the donor's own crop and to the private value of its water at the 2022–23 Lower Basin conservation offers of $330–400 per acre-foot (Reclamation 2023), and paying the most favorable 5 or 10 percent of upstream acres to fallow would not cover a donor margin of half of revenue even if every modeled gain materialized. Moving water at the margin rather than fallowing whole fields does not change the picture: along the within-field relationship between GCVI and ET (eeMETRIC ET, plot and crop-by-year effects), the median upstream–downstream pair loses about $60 per acre-foot, and gains arise only for selected pairs.

The screen is illustrative. It assumes that freed water reaches the downstream fields, which proportional allocation cuts (CAIDD 2022) and turn-back rules that reassign unused orders by priority and timing (CAP 2019) do not guarantee. Its answer also hinges on the yield–water relationship. In consumptive use, agronomic evidence points to yield that is roughly linear above a threshold and flattens toward full irrigation (Doorenbos and Kassam 1979; Lindenmayer et al. 2011; Trout and DeJonge 2017); in these data GCVI rises with seasonal ET and flattens without a hard plateau. Under a linear response, reallocation between similar fields yields nothing, and only curvature or plateaus create gains, so I do not report transfer calculations that assume a particular quadratic yield function: their gains would come entirely from the assumed curvature.

Table S9: Economic Magnitudes of Upstream Exposure in Junior-Rights Districts, 2021–22

	Typical gap (25th–75th)		Wide gap (10th–90th)	
	$ per cotton acre	% of revenue	$ per cotton acre	% of revenue
Panel A. Difference in network position, per field in a scarcity year				
Pooled	$42 [$14, $73]	2.7	$77 [$28, $136]	5.0
Pooled, with well and rights controls	$37 [$15, $61]	2.4	$64 [$31, $110]	4.1
Within CAP and non-CAP groups	$21 [$6, $41]	1.4	$37 [$13, $80]	2.4
Within groups, with well and rights controls	$22 [$7, $41]	1.4	$40 [$15, $82]	2.6
Panel B. Ten junior districts, per scarcity year				
	Pooled		Within groups	
Revenue linked to the 2021–22 drop in upstream cropping	$1.06M [0.22, 1.98]		$0.55M [0.05, 1.09]	
Downstream ET per acre-foot of upstream ET	0.035		0.042	
Panel C. Illustrative reallocation screen, upstream summer-crop acres				
Modeled downstream gain per upstream acre, mean / median	$125 / $41		$48 / $16	
Value of that acre's ET at 2022–23 conservation offers	$765–$927		$765–$927	
Share of acres where the offer exceeds the modeled gain	97–98%		99–100%	
Share of modeled gain needed to break even, top 1 / 5 / 10%	72 / 114 / 152%		188 / 298 / 398%	
Moving water at the margin: median pair, per acre-foot	−$60 (gains for 40% of pairs)			

Notes: Companion specification of the junior decomposition with a common 2021–22 scarcity window relative to 2016–2020 (text). Revenue conversion at $\kappa = 0.8$ and 2021–22 NASS Arizona revenue per acre (cotton including seed). Alfalfa and corn amounts are the same percentages times their 2021–22 revenue per acre ($2,217 and $1,633). Panel A intervals are 95 percent ranges from 4,000 joint coefficient draws; Panel B intervals are two-way clustered 95 percent intervals mapped through the conversion. "Within groups" adds separate crop-by-year effects for the four CAP-served and the six other junior districts. Panel B's second row is from the same specification with May–October ET (mm) as the outcome. Panel C: donors are cropped upstream summer-crop fields more than 5 km from at least one junior field in the same year; the offer value multiplies the donor's May–October ET by $330–400 per acre-foot; break-even is the share of modeled downstream gains that must materialize to cover a donor margin of half of revenue for the top 1, 5 and 10 percent of acres ranked by modeled gain; the last row values an acre-foot moved from an upstream field to the fields below it at 0.8 times the recipients' marginal value minus the donor's, along the within-field relationship between GCVI and ET. All entries are revenue equivalents of associations, not profits or welfare.

S14 Urban Adjacency and Municipal Service Areas

This appendix is an urban-adjacency robustness check, not a municipal-demand test: it splits upstream canal-connected activity by whether ancestors fall inside large Community Water System service areas. These polygons bound urban-fringe claims and leakage. They do not measure municipal offtake, since intakes are not service polygons. In the primary GCVI $d > 5$ km cell the decomposition leaves the augmented drought interaction at -0.0172 (95% CI $[-0.0337, -0.0007]$; two-way $p = 0.042$; WCC $p = 0.198$), essentially the baseline -0.0174, and a stacked Hausman-style stability test does not reject equality with the baseline ($p = 0.371$; WCC $p = 0.778$). Downstream urban exposure, by contrast, carries its own negative drought loading, -0.1040 (95% CI $[-0.1749, -0.0332]$; $p = 0.004$), an urban-spatial correlation the canal channel cannot produce, and conditioning on it leaves the upstream coefficient at -0.0147 (95% CI $[-0.0305, 0.0010]$). That downstream loading departs from the expectation of no symmetric loading; I read it as spatial correlation rather than municipal-channel evidence. No curtailed district in Appendix A28 has an upstream ancestor inside a large service area at 5 km, so those estimates are invariant to the decomposition; municipal offtake itself remains unobservable at the plot level.

Table S10: Urban Adjacency: Municipal Service Areas and the Upstream-Activity Drought Interaction

Spec	Term	Cutoff	Outcome	Estimate	SE	N
Baseline	Upstream cropped acreage × Drought	>5 km	GCVI	−0.017**	(0.008)	137,229
Baseline	Upstream cropped acreage × Drought	>5 km	ET	−0.014	(0.009)	137,229
Baseline	Upstream cropped acreage × Drought	>10 km	GCVI	−0.035***	(0.012)	75,826
Baseline	Upstream cropped acreage × Drought	>10 km	ET	−0.023*	(0.012)	75,826
Primary decomposition	Upstream cropped acreage × Drought	>5 km	GCVI	−0.017**	(0.008)	137,229
Primary decomposition	Urban-fringe upstream × Drought	>5 km	GCVI	−0.015	(0.011)	137,229
Urban/non-urban upstream split	Upstream excl. urban × Drought	>5 km	GCVI	−0.017**	(0.008)	137,229
Urban/non-urban upstream split	Urban-fringe upstream × Drought	>5 km	GCVI	−0.032***	(0.012)	137,229
Downstream-city control	Upstream cropped acreage × Drought	>5 km	GCVI	−0.015*	(0.008)	62,048
Downstream-city control	Downstream urban × Drought	>5 km	GCVI	−0.104***	(0.036)	62,048
WCC-clustered inference	Upstream cropped acreage × Drought	>5 km	GCVI	−0.017	(0.013)	137,229
WCC-clustered inference	Urban-fringe upstream × Drought	>5 km	GCVI	−0.015	(0.013)	137,229
Primary decomposition	Upstream cropped acreage × Drought	>5 km	ET	−0.014	(0.009)	137,229
Primary decomposition	Urban-fringe upstream × Drought	>5 km	ET	0.007	(0.009)	137,229
Urban/non-urban upstream split	Upstream excl. urban × Drought	>5 km	ET	−0.014	(0.009)	137,229
Urban/non-urban upstream split	Urban-fringe upstream × Drought	>5 km	ET	−0.006	(0.010)	137,229
Downstream-city control	Upstream cropped acreage × Drought	>5 km	ET	−0.012	(0.008)	62,048
Downstream-city control	Downstream urban × Drought	>5 km	ET	−0.117	(0.081)	62,048
WCC-clustered inference	Upstream cropped acreage × Drought	>5 km	ET	−0.014	(0.013)	137,229
WCC-clustered inference	Urban-fringe upstream × Drought	>5 km	ET	0.007	(0.009)	137,229
Primary decomposition	Upstream cropped acreage × Drought	>10 km	GCVI	−0.034***	(0.012)	75,826
Primary decomposition	Urban-fringe upstream × Drought	>10 km	GCVI	−0.015	(0.016)	75,826
Urban/non-urban upstream split	Upstream excl. urban × Drought	>10 km	GCVI	−0.034***	(0.012)	75,826
Urban/non-urban upstream split	Urban-fringe upstream × Drought	>10 km	GCVI	−0.050***	(0.015)	75,826
Downstream-city control	Upstream cropped acreage × Drought	>10 km	GCVI	−0.027***	(0.009)	21,396
Downstream-city control	Downstream urban × Drought	>10 km	GCVI	0.011	(0.248)	21,396
WCC-clustered inference	Upstream cropped acreage × Drought	>10 km	GCVI	−0.034*	(0.018)	75,826
WCC-clustered inference	Urban-fringe upstream × Drought	>10 km	GCVI	−0.015	(0.020)	75,826
Primary decomposition	Upstream cropped acreage × Drought	>10 km	ET	−0.023*	(0.013)	75,826
Primary decomposition	Urban-fringe upstream × Drought	>10 km	ET	0.010	(0.014)	75,826
Urban/non-urban upstream split	Upstream excl. urban × Drought	>10 km	ET	−0.023*	(0.013)	75,826
Urban/non-urban upstream split	Urban-fringe upstream × Drought	>10 km	ET	−0.014	(0.012)	75,826
Downstream-city control	Upstream cropped acreage × Drought	>10 km	ET	0.004	(0.007)	21,396
Downstream-city control	Downstream urban × Drought	>10 km	ET	0.261	(0.278)	21,396
WCC-clustered inference	Upstream cropped acreage × Drought	>10 km	ET	−0.023	(0.019)	75,826
WCC-clustered inference	Urban-fringe upstream × Drought	>10 km	ET	0.010	(0.014)	75,826

Notes: Active plots only. Urban exposure is defined from the union of 10 active Arizona Community Water System service-area polygons with population at least 100,000. The decomposition uses the same active upstream counts as Table A22: $U_{it}^{>d}$ is the total log active upstream count, $U_{\text{out},it}^{>d}$ excludes active upstream ancestors inside those service areas, and $\Delta_{\text{city},it}^{>d} = U_{it}^{>d} - U_{\text{out},it}^{>d}$. The table reports five specifications: the Table A22 baseline; the primary decomposition; a secondary split of upstream cropped acreage into its urban-fringe and non-urban parts; a specification that adds downstream urban exposure; and the same rows re-estimated under WCC-clustered inference. Inference on the city terms is descriptive. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

S15 Early-Season Upstream Water Use and Late-Season Outcomes

For each of the 3,601 plots inside the ten junior-rights district boundaries and each year 2020–2023, exposure is January–April OpenET volume (acre-feet) summed over the plot's fixed upstream plots beyond 5 km, in $\ln(1+x)$ and centered on the plot's complete 2016–2019 mean; the outcome is the plot's July–October OpenET in $\ln(1+x)$; May–June is left out as a buffer; the drought indicator is the district SPEI-24 indicator of the preceding year; and no outcome-year crop classification enters the sample, exposure, or fixed effects. Two models add district-by-year or WCC-by-year effects to plot and preperiod-modal-crop-by-year effects. Because the lagged drought indicator is common within each area-year, those effects absorb it together with any shared response, so those models are a within-area diagnostic rather than a test of amplification.

The model that fits the design keeps plot and preperiod-modal-crop-by-year effects and omits area-by-year effects; additive district or WCC effects are nested in plot effects, so it reports district- and WCC-clustered uncertainty instead. Figure A2 shows it ($N = 14{,}404$): the other-years slope is 0.064, the interaction is −0.288, and the total drought-year slope is −0.224. Conventional

cluster-robust p-values are 0.018 and 0.003 for the interaction and 0.007 and 0.001 for the total; null-imposed ten-district wild-bootstrap p-values, refitting all nuisance terms and fixed effects, are 0.104 and 0.121, with all $2^{10} = 1{,}024$ Rademacher assignments enumerated. In the joint current and next-year model ($N = 10{,}803$), the current drought-year slope is -0.256 and the next-year slope is -0.098, indistinguishable from zero. With area-by-year effects, the total slope following district drought is -0.146 under district-by-year and -0.142 under WCC-by-year effects; the first assigns it to the interaction (-0.139) and the second to other years (interaction -0.001), and their full-refit wild-bootstrap p-values are 0.387 and 0.504. Ten districts and essentially one drought year limit precision throughout, and none of this makes upstream water use exogenous.

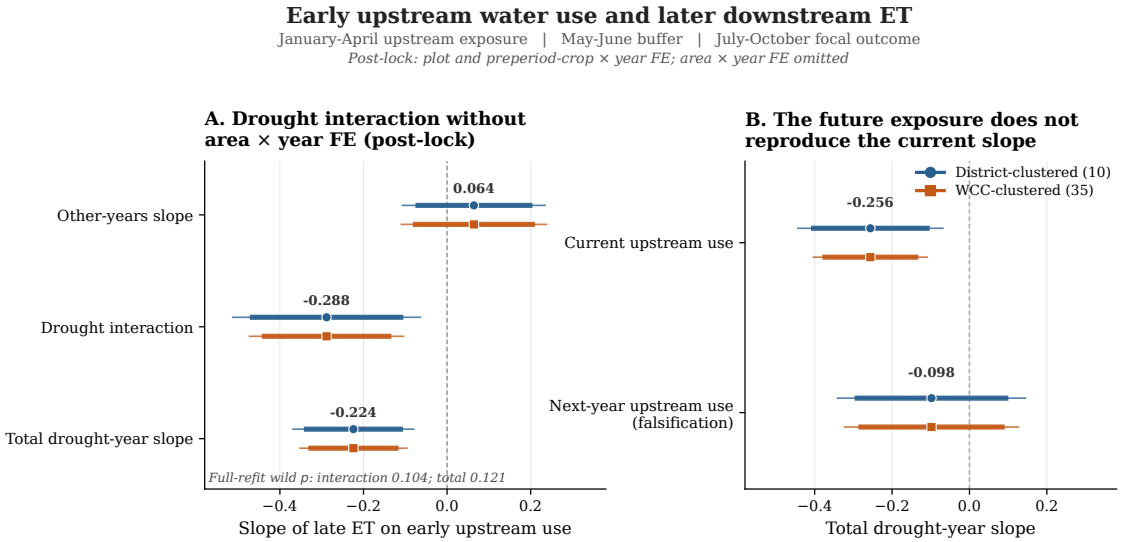


Figure A2: Early-season upstream water use and late-season downstream ET. Sample: all 3,601 plots inside the ten junior-rights district boundaries, 2020–2023, without conditioning on outcome-year crop status. Exposure is January–April OpenET volume among each plot's fixed upstream plots beyond 5 km, in $\ln(1+x)$ and centered on the plot's 2016–2019 mean; the outcome is the plot's July–October OpenET in $\ln(1+x)$; May–June is left out as a buffer. The drought indicator is lagged district SPEI-24. Models include plot and preperiod-modal-crop-by-year fixed effects and omit area-by-year effects; additive WCC membership is absorbed by plot effects, so the figure compares uncertainty clustered by district (10 clusters) and WCC (35 clusters). Panel A reports the other-years slope, drought interaction, and their total in drought years ($N = 14{,}404$). Panel B enters current and next-year early upstream water use jointly on 2020–2022 support ($N = 10{,}803$); the next-year term is a falsification. Whiskers are 90% (thick) and 95% (thin) confidence intervals. Null-imposed ten-district wild-bootstrap p-values are 0.104 for the interaction and 0.121 for the total drought-year slope, enumerating all 1,024 assignments with full nuisance and fixed-effect refits. This specification is supporting evidence.

S16 Matched Outcome-Scale Sensitivities

The native regressions use $\ln(1+x)$. Several supporting exercises specified raw indices. This appendix therefore estimates each selected specification on both scales, holding field-year observations, seasonal windows, exposures, controls, fixed effects, weights and clustering unchanged. No clipping or observation deletion is needed: every retained raw value is greater than -1. The grid contains nine seasonal-vegetation, sixteen Wellton and twenty-four Gila specification pairs. It

preserves the original specifications rather than selecting the more favorable transformation. The coefficients measure different outcome units and are not numerically interchangeable; the comparison concerns direction, inference and interpretation. The native junior annual result and the primary seasonal-ET results are unchanged.

Seasonal vegetation. Table S11 reports the drought interaction and total drought-year slope for each outcome and fixed-effect structure. Across the two all-plot outcomes and three structures, all twelve contrasts remain negative on both scales. With network-by-year effects, the logged GCVI and NDMI total slopes are -0.218 and -0.052 (full-refit district wild $p = 0.018$ and 0.031); their interactions have $p = 0.057$ and 0.660. The district-by-year and no-area-year total slopes are negative but imprecise under the same bootstrap. The alfalfa stratum does not provide uniform corroboration; its small network-year total changes sign across transformations and is imprecise on both. Thus the all-plot negative association is not specific to raw-index units. Logging does not establish a general drought-amplification result.

Table S11: Seasonal Vegetation: Matched Raw and Logged Outcomes

	Drought interaction		Total drought-state slope	
Area-year effects	Raw	$\ln(1+x)$	Raw	$\ln(1+x)$
GCVI, all plots; $N = 14,404$				
District × year	−1.180	−0.213	−0.810	−0.192
	(0.127)	(0.225)	(0.059)	(0.141)
Network × year	−0.489	−0.079	−0.857	−0.218
	(0.074)	(0.057)	(0.029)	(0.018)
No area-year (post-lock)	−0.519	−0.219	−0.978	−0.265
	(0.480)	(0.230)	(0.223)	(0.139)
NDMI, all plots; $N = 14,404$				
District × year	−0.054	−0.047	−0.050	−0.044
	(0.314)	(0.318)	(0.369)	(0.383)
Network × year	−0.011	−0.008	−0.058	−0.052
	(0.654)	(0.660)	(0.025)	(0.031)
No area-year (post-lock)	−0.091	−0.081	−0.082	−0.073
	(0.150)	(0.150)	(0.162)	(0.170)
GCVI, preperiod alfalfa; $N = 4,596$				
District × year	−0.410	−0.032	0.136	0.071
	(0.137)	(0.557)	(0.623)	(0.311)
Network × year	0.458	0.080	−0.008	0.041
	(0.496)	(0.680)	(0.957)	(0.490)
No area-year (post-lock)	1.029	0.089	0.194	0.033
	(0.271)	(0.432)	(0.654)	(0.492)

Notes: Each raw/log pair uses identical observations and the early–late design of Appendix S15, replacing only the focal outcome with July–September GCVI or NDMI. The all-plot samples contain 3,601 fields; the preperiod-modal-alfalfa stratum contains 1,149. Field and preperiod-crop-by-year effects remain in every model. Parentheses report null-imposed district wild-bootstrap p-values with full nuisance/FE refits and all 1,024 sign assignments, not standard errors. District- and network-clustered conventional inference and the other-years slopes are retained in the replication ledger. Both the transformation comparison and the no-area-year vegetation estimates are post-result sensitivities; displayed p-values are not adjusted across this grid. These all-plot land-surface indices are not harvested yield.

Wellton. Table S12 pairs the six weather-adjusted summer-flow specifications. For inside fields, the network-by-year model with a linear gradient trend gives a logged interaction of +0.00973 per fixed position SD and 100 cfs (year wild $p = 0.016$); its eight year-omission estimates range from +0.00564 to +0.01347. All sixteen full specifications in the retained grid, including annual

diversion and the unadjusted comparisons, are positive on both scales. Some significance thresholds change, so the raw and logged results are reported together. This strengthens the claim that the positive moisture association is not a raw-index artifact, without making realized flow exogenous or identifying field-level withdrawals.

Table S12: Wellton Summer Flow: Matched Raw and Logged NDMI

Cohort / time effects	Trend	Raw $\hat{\beta}$	Log $\hat{\beta}$	Raw p	Log p
Original / year	No	0.01325	0.01118	0.117	0.117
Original / year	Yes	0.00952	0.00794	0.055	0.047
Inside / year	No	0.01363	0.01157	0.117	0.117
Inside / year	Yes	0.00961	0.00804	0.039	0.031
Inside / network-year	No	0.01439	0.01229	0.023	0.023
Inside / network-year	Yes	0.01153	0.00973	0.023	0.016

Notes: Same cohorts, fixed position standardization, weather controls and equal-field weights as Table A31. All rows include field effects; *Trend* adds fixed position × calendar year. The original and inside cohorts contain 36,496 and 35,344 field-years. Coefficients are per one fixed position SD and 100 cfs higher May–September mean flow. Log means $\ln(1 + \text{NDMI})$. The p columns use the same full-refit year-cluster wild bootstrap with all 256 assignments on both scales. Annual-diversion and unadjusted pairs and all 256 individual-year omission fits are retained in the comparison ledger. These are assumption-dependent eight-year comparisons, not additional independent tests supplied by the number of fields.

Gila. Table S13 holds the two paper-facing denominator constructions fixed: administrative irrigated acreage and the existing cultivation-adjusted proxy. All twelve combinations of outcomes, denominators and controls using the conditional blank-as-zero water series are negative on both scales. The eight NDMI comparisons preserve their signs, including the mixed strict-sample pattern. Four of the twenty-two identified specification pairs change sign, all in strict-sample GCVI or ET cells; those cells are imprecise on both scales. The two strict-ET trend specifications remain unidentified on both scales. Logging therefore does not turn Gila into uniform buffering evidence. The comparison does not rerun the other denominator/decomposition grids or resolve service linkage, the blank-as-zero assumption, or unobserved groundwater.

Table S13: Gila Water per Acre: Matched Raw and Logged Outcomes

Water / controls	Area denominator	Raw: $\hat{\beta}$ (p_c, p_y)	Log: $\hat{\beta}$ (p_c, p_y)
NDMI			
Strict / weather	Administrative	-0.0082 $(0.250, 0.781)$	-0.0068 $(0.250, 0.812)$
Strict / weather	Cultivation proxy	0.0041 $(0.750, 0.844)$	0.0043 $(0.750, 0.828)$
Strict / + trend	Administrative	-0.0012 $(1.000, 0.953)$	-0.0012 $(1.000, 0.984)$
Strict / + trend	Cultivation proxy	0.0091 $(0.750, 0.609)$	0.0080 $(0.500, 0.609)$
Conditional / weather	Administrative	-0.0221 $(0.375, 0.328)$	-0.0195 $(0.375, 0.312)$
Conditional / weather	Cultivation proxy	-0.0289 $(0.125, 0.203)$	-0.0254 $(0.125, 0.234)$
Conditional / + trend	Administrative	-0.0271 $(0.125, 0.078)$	-0.0238 $(0.125, 0.047)$
Conditional / + trend	Cultivation proxy	-0.0328 $(0.125, 0.203)$	-0.0287 $(0.125, 0.188)$
GCVI			
Strict / weather	Administrative	-0.0927 $(0.500, 0.828)$	-0.0030 $(0.500, 0.953)$
Strict / weather	Cultivation proxy	-0.1313 $(0.500, 0.750)$	0.0391 $(0.750, 0.609)$
Strict / + trend	Administrative	-0.1603 $(0.750, 0.500)$	0.0088 $(0.750, 0.953)$
Strict / + trend	Cultivation proxy	-0.3629 $(0.500, 0.219)$	0.0372 $(0.500, 0.594)$
Conditional / weather	Administrative	-0.3679 $(0.375, 0.016)$	-0.0780 $(0.250, 0.172)$
Conditional / weather	Cultivation proxy	-0.5010 $(0.375, 0.031)$	-0.0963 $(0.125, 0.094)$
Conditional / + trend	Administrative	-0.4841 $(0.125, 0.031)$	-0.0957 $(0.125, 0.031)$
Conditional / + trend	Cultivation proxy	-0.6214 $(0.125, 0.062)$	-0.1115 $(0.125, 0.047)$
ET			
Strict / weather	Administrative	-114.3082 $(0.250, 0.500)$	-0.2969 $(0.250, 0.500)$
Strict / weather	Cultivation proxy	0.6660 $(1.000, 0.938)$	-0.0024 $(1.000, 0.938)$
Strict / + trend	Administrative	Not identified	Not identified
Strict / + trend	Cultivation proxy	Not identified	Not identified
Conditional / weather	Administrative	-19.8659 $(0.125, 0.141)$	-0.0542 $(0.500, 0.156)$
Conditional / weather	Cultivation proxy	-27.1844 $(0.125, 0.016)$	-0.0808 $(0.125, 0.047)$
Conditional / + trend	Administrative	-24.0359 $(0.125, 0.062)$	-0.0678 $(0.250, 0.078)$
Conditional / + trend	Cultivation proxy	-28.6222 $(0.125, 0.062)$	-0.0867 $(0.125, 0.047)$

Notes: Fixed 323-field cohort; each pair retains its original water-quality support, seasonal window and field weights. Vegetation uses May–September and ET May–October. Coefficients are per 0.10 higher fixed position and one unit of the relevant water-per-acre measure. Both specifications include field and company-year effects, year-specific position slopes, SPEI-24 and position × SPEI; *+ trend* adds company-specific linear position trends. Conditional water treats blank diversion entries as zero; it is not verified zero flow. The cultivation denominator is an assumption-based proxy, not measured water per served acre. Parentheses give separate full-refit company- and year-wild p-values, (p_c, p_y); they are not standard errors or multiplicity-adjusted tests. *Not identified* preserves the original rank failure. Raw ET is millimeters; logged columns use $\ln(1 + x)$. The original GCVI specifications were logged. Original NDMI and ET were raw. All company/year omission fits are retained in the comparison ledger; these closely related specifications are not independent replications.

S17 The 2008–2023 Extension Panel

This appendix reports an exploratory extension that re-estimates the panel design over a longer 2008–2023 horizon. Because the main sample's USDA Crop Sequence Boundaries begin in 2016, the extension is built as a cross-vintage *pair* panel: each current-vintage plot is matched to a spatially corresponding historical-vintage parcel, giving 78,169 keep-primary pairs observed over sixteen years, with upstream cropped-acreage exposure recomputed on the water-graph of Section 5. Estimation is on active plot-years with crop×year and current-vintage plot fixed effects (not pair fixed effects); inference is clustered two-way by crop×year and current-vintage plot ("cy"), or by weakly connected component ("wcc"). The design is corroborative, not identifying: exposure is constructed differently on either side of the 2016 CSB vintage boundary, and realized plot- or district-level water deliveries are not observed. Because the era boundary and the vintage boundary coincide, the seam is the central threat to the era contrast; it is examined directly at the end of this appendix. No directional (upstream-versus-downstream) claim is made from this panel.

Table S14 reports the era split itself, with the full grid of outcome × cutoff × support cells and both clusterings in the notes.

Table S14: Extension Panel: Drought × Upstream-Activity Interaction by Era (1 km)

	2008–2015		2016–2023	
Outcome	$\hat{\beta}_{\text{int}}$	p (cy / wcc)	$\hat{\beta}_{\text{int}}$	p (cy / wcc)
NDMI	+0.0006	0.45 / 0.43	−0.0039	0.038 / 0.038
GCVI	+0.0060	0.061 / 0.042	−0.0133	0.056 / 0.035
Post-2016 triple interaction (upstream cropped acreage × drought × $\mathbf{1}$[year ≥ 2016]):				
NDMI	$\hat{\beta} = -0.0053$;	$p = 0.019$ (cy) / 0.001 (wcc)		
GCVI	$\hat{\beta} = -0.0207$;	$p = 0.017$ (cy) / < 0.001 (wcc)		

Notes: Cross-vintage pair panel, 78,169 keep-primary pairs; active plot-years. Cells are the drought × log upstream cropped-acreage interaction at the 1 km cutoff, primary-support (≥90% coverage) variant, drought = $\mathbf{1}$[upstream SPEI-24 < −0.6]; FE crop×year and current-vintage plot; inference clustered two-way by crop×year and current-vintage plot (cy), or by weakly connected component (wcc). Over the full 2016–2023 grid (16 unique interaction estimates) all 16 are negative, significant at 5% in 12/16 (cy) and 7/16 (wcc); all 16 early-era (2008–2015) point estimates are positive, and only GCVI cells reach 5% significance. The triple-interaction rows test equality of the two era coefficients. On the same estimation support the earlier drought is the more severe one on the index that defines the treatment: acreage-weighted mean upstream SPEI-24 is −1.06 over 2010–2014 against −0.31 over 2020–2022, and the share of plot-years below the −0.6 cutoff is 83% against 33%. The positive early-era point estimates are carried by rows whose satellite outcomes were reused from the current-vintage parent; on freshly extracted rows the early-era estimate is an estimated zero (seam diagnostic below).

Table S15 reruns the design with the administrative CAP curtailment schedule in place of the SPEI drought indicator, mirroring Appendix Table A34. The exposure × 2022-cut interaction is positive, opposite in sign to the SPEI-drought interaction, in every headline cell (Panel A). In this extension sample, the plot-weighted district partition places the positive point estimates in Maricopa-Stanfield and Central Arizona, the two districts with the most post-2020 irrigation-well applications. The other district estimates are mostly negative and imprecise. That covariance is descriptive, not a groundwater test: the main decomposition uses different support, places the positive residual elsewhere, and the two district partitions have not been reconciled on a common sample with common weights and treatment definitions. I therefore do not use Panel B to infer pumping or buffering.

Table S16 records the institutional facts underlying the buffer reading, drawn from the delivery-fact audit that accompanies the replication package.

Table S15: Extension Panel: Administrative CAP-Cut Interaction and District Decomposition

Panel A. Exposure × 2022 CAP-cut interaction (headline cells)		
Outcome / cutoff	$\hat{\beta}$	p (cy / wcc / district)
NDMI, >1 km	+0.0054	<0.001 / 0.017 / 0.004
NDMI, >5 km	+0.0045	0.026 / 0.006 / 0.037
GCVI, >1 km	+0.0174	0.002 / 0.003 / 0.001
GCVI, >5 km	+0.0134	0.037 / 0.001 / 0.062

Panel B. District partition of the 2022 flattening (1 km, own-baseline reference)		
District (post-2020 well applications)	NDMI $\hat{\beta}$ (p)	GCVI $\hat{\beta}$ (p)
Maricopa-Stanfield Irrigation and Drainage District (13)	+0.0048 (0.003)	+0.0177 (0.001)
Central Arizona Irrigation and Drainage District (14)	+0.0057 (0.037)	+0.0153 (0.167)
Hohokam Irrigation District (0)	−0.0041 (0.178)	−0.0100 (0.224)
Harquahala Valley Irrigation District (1)	−0.0057 (0.142)	−0.0159 (0.212)
Queen Creek Irrigation District (1)	+0.0065 (0.559)	−0.0173 (0.588)

Notes: Active plot-years; FE crop×year and current-vintage plot. Restricted sample (six treated CAP-agricultural-pool districts + six junior on-river controls), era 2016–2023, event specification; treatment is exposure × (CAP-ag-pool ×$\mathbf{1}$[2022]), any-match support. Panel A p-values are clustered by crop×year + plot (cy), weakly connected component (wcc), and irrigation district. The interaction is positive (opposite-signed to the SPEI-drought interaction) in all eight headline cells (2 outcomes × 2 cutoffs × 2 support variants), significant at $p < 0.10$ in seven of eight under all three clusterings (five of eight at $p < 0.05$) and in all eight under both cy and wcc. Panel B partitions the pooled 2022 interaction into each district's deviation from its own non-cut baseline (positive = flatter gradient in the cut year), displaying cy-clustered p-values; well counts are post-2020 irrigation-well applications. District-clustered p-values are not reported because each district term is effectively a singleton cluster, so the district rows read as sign-consistency, not hard significance. The junior on-river controls themselves steepen in 2022 (own exposure × 2022 ≈ −0.057, GCVI 1 km). Panel A is estimated on active plot-years and therefore inherits the selection caveat in Appendix A28: fallowing changes the surviving sample. The sign and magnitude of its contribution to the gradient are not established. The separate −0.013 gradient among plots active in both 2021 and 2022 in that appendix conditions on survival; it is not a decomposition of this extension estimate, and no matched composition accounting has been run here. The two district decompositions also differ in support, period, weighting, and treatment definition. The post-2020 application counts are outcomes of an adjustment process, not interaction variables fixed in advance, so their covariance with Panel B cannot identify groundwater substitution.

Table S16: Institutional Water-Delivery Timeline, 2008–2023

Years	Delivery / buffer status	Source
2008–2014	CAP agricultural-pool deliveries within ≈0.9% of the 400 kAF schedule; no shortage declared	CAP delivery reports
2010–2014	Meteorological drought more severe than 2020–22 by SPEI; Lake Mead year-end 10.301–14.644 MAF (1,086.30–1,132.83 ft), above the 1,075 ft shortage trigger	Reclamation RISE
2015	CAP agricultural-pool deliveries fell to 318,279 AF; Forbearance I conservation of 80,922 AF, voluntary and compensated (not a shortage cut)	CAP delivery and conservation ledgers
2008–2023	SRP Salt–Verde allocation 3.0 AF/acre (2008–14), raised to 3.3 in 2015–23; available schedules show no allocation cut	Phoenix Water Resource Plan
2020	Tier 0 declared; realized CAP ag-pool deliveries flat against 2019 (255,415 versus 256,143 AF); the reduction was absorbed by credited forbearance (CAWCD DCP contribution 180,608 AF) and unused pool headroom, so orders sat near the reduced cap	Reclamation accounting; CAP delivery ledgers
2022	Tier 1 shortage (projected Mead 1,065.85 ft); CAP ag pool realized 30,825 AF (≈88% year-over-year fall)	Reclamation Tier 1 determination; CAP delivery report
2011–2022	San Carlos Reservoir below 1% of usable capacity (2011–13, 2015, 2018–19, 2021 zero in July, 2022)	USGS station 09469000
2023	Tier 2a; wet-year refill (San Carlos to 226 kAF); no mandatory Yuma senior (Priority 1–3) cut documented in 2021–23	USGS 09469000; Reclamation

Notes: Compiled from official records in the replication package's delivery-fact audit; CAP and Reclamation series are high-confidence, post-2019 SRP schedules and Yuma senior-priority status medium-confidence. Sources: CAP delivery and conservation ledgers (https://library.cap-az.com); Bureau of Reclamation RISE reservoir series and shortage determinations (https://data.usbr.gov); USGS daily storage, station 09469000 (https://waterservices.usgs.gov); City of Phoenix 2021 Water Resource Plan.

The 2016 seam. The era boundary coincides with the data-vintage boundary. Outcomes for 2016–2023 are taken from the current-vintage plot. Outcomes for 2008–2015 are either freshly extracted on the historical parcel geometry (25.5% of early active plot-years) or reused from the current-vintage parent where the parcel passed a crop-sequence homogeneity screen. A measurement change rather than an institutional one could therefore in principle generate the era contrast; three diagnostics, computed by the seam-diagnostic routine in the replication package, argue against that reading. Moving the era boundary to each year from 2010 to 2021 produces no discontinuity at 2016: at the 1 km cutoff the GCVI triple is −0.0199 with the boundary at 2015, −0.0207 at 2016, and −0.0202 at 2017, and the largest t-statistics occur at boundaries of 2013 and 2017–2018. A year-by-year interaction profile (base year 2014) locates the negative estimate away from the seam: the first current-vintage year, 2016, carries a *positive* level (+0.0129 GCVI, +0.0061 NDMI), indistinguishable from the base year, and the negative is concentrated in 2022, the Tier 1 curtailment year (−0.0467 GCVI and −0.0125 NDMI, each differing from the base year at $p < 0.001$); four years drop out because the upstream drought indicator is degenerate within them. And splitting the early era by how its outcomes were obtained makes the weakly positive early-era coefficient a property of the reused rows, +0.0066 ($p = 0.07$) on reused rows against −0.0009 ($p = 0.83$) on freshly extracted ones, on a split close to orthogonal to the design, correlating 0.013 with network position and 0.002 with upstream exposure, and the fresh rows are informative rather than merely silent (80%-power detectable effect 0.012 at 1 km GCVI, smaller in magnitude than the late-era estimate of −0.0133). These diagnostics cannot exclude a measurement change that binds only in 2022. The vintage seam is not the leading alternative explanation.

Two limitations bound the extension beyond the corroborative status conceded above: each era contains a single drought episode, so the contrast cannot separate era from episode, and the estimates span many outcome × cutoff × support cells, so individual significance markers should be read against that multiplicity. The treated-district decomposition additionally rests on five–six treated clusters, which makes district-level inference near-singleton.

S18 ET Measurement: Sensitivity to the OpenET Member

ET is the weakest of the three crop outcomes across the main designs. This appendix asks whether that weakness is a property of evapotranspiration as a construct or of the particular ET product the paper uses. The primary measure is the OpenET *ensemble*, an average across member models. Two individual members are also available at plot-season resolution and have materially different error structures: eeMETRIC, a surface-energy-balance model, and SIMS, a crop-coefficient model driven by vegetation indices. I substitute each member in the cross-section, distance-cutoff panel, and joint upstream–downstream regression and hold sample filters, controls, fixed effects, and clustering fixed. Table S17 reports the distance-cutoff panel member by member.

The drought-interaction point estimates barely move across members (at the full path −0.0105, −0.0129, and −0.0112 for the ensemble, eeMETRIC, and SIMS, and at the >10 km cutoff −0.0230, −0.0229, and −0.0190). Their standard errors differ systematically: SIMS runs at roughly 0.85–0.93 of the ensemble's, eeMETRIC at roughly 1.18–1.20, a spread of about 40% between the two members with the ensemble in between. Stable coefficients with varying dispersion is the signature of measurement error in the dependent variable, and the significance pattern follows mechanically: the ET drought interaction reaches $p < 0.10$ in two of six cutoffs under the ensemble, one of six under eeMETRIC, and five of six under SIMS. Invariance is confined to the interactions: the cross-

Table S17: Distant Upstream Cropped Acreage × Drought: ET by OpenET Member

Cutoff	Full	>1 km	>2 km	>3 km	>5 km	>10 km
Ensemble (published)	−0.0105	−0.0108	−0.0107	−0.0127*	−0.0135	−0.0230*
	(0.0069)	(0.0066)	(0.0068)	(0.0069)	(0.0086)	(0.0124)
eeMETRIC	−0.0129	−0.0127	−0.0124	−0.0139*	−0.0139	−0.0229
	(0.0082)	(0.0079)	(0.0080)	(0.0081)	(0.0101)	(0.0148)
SIMS	−0.0112*	−0.0115**	−0.0116*	−0.0129**	−0.0122	−0.0190*
	(0.0059)	(0.0057)	(0.0060)	(0.0064)	(0.0077)	(0.0105)
SE relative to ensemble (SIMS)	0.85	0.87	0.89	0.93	0.89	0.85
SE relative to ensemble (eeMETRIC)	1.19	1.20	1.18	1.18	1.18	1.20

Notes: Standard errors in parentheses. Each row re-estimates the exact Table A22 specification (active plot-years, Crop×Year and Plot fixed effects, full controls, matched spatial fallow shares, two-way clustering by Crop×Year and Plot), changing only the ET product used as the dependent variable. The ensemble row reproduces the published Table A22 ET coefficients. SIMS is estimated on 0.63% fewer active plot-years than the other two, which are on identical support; the relative-SE rows are essentially unchanged (median 0.99) when the ensemble is re-estimated on SIMS's common support. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

sectional position *level* moves from −0.003 ($p = 0.81$) under the ensemble to −0.025 ($p = 0.007$) under SIMS, so no level claim should be drawn from any single member.

One member-specific result is disclosed because it cuts against the delivery channel. In the joint regression at the >10 km cutoff, SIMS returns a positive downstream coefficient (+0.019, $p = 0.037$), the same sign the published ensemble estimates already carry at >5 km (+0.007*), and a sign the canal delivery channel cannot produce. It is a point estimate only: the upstream–downstream contrast is not tested for significance here, and the wide-cutoff joint regression rests on the reduced common-support sample discussed in Section 7.1.

I therefore retain the OpenET ensemble as the ET measure throughout; the directional and cross-sectional conclusions continue to rest on GCVI and NDMI.